On the Other Side of the Creek:

Historical Exclusion, Neighborhood Ties, and Human Capital Gaps*

L. Guillermo Woo-Mora†

Job Market Paper

Version: November 12, 2026

Click here for the most recent version.

Abstract

Can human capital gaps persist even after historically excluded communities gain equal access to resources? I show that they can, and that neighborhood ties left by exclusion help explain why. In 1542, colonial authorities in Guadalajara, Mexico, settled the conquered population east of a creek, opposite the Spanish city and the conquerors' Indigenous allies. Using census data from 1930 to 2020 and a boundary discontinuity design, I find that school attendance is 3 to 6 percentage points lower on the historically excluded side, with the gap emerging in adolescence, despite no detectable differences in measured school supply or quality. Surveyed parents on both sides have similar aspirations and expected returns to education, but east-side parents rely more on neighbors. In a preregistered experiment, parents are randomly assigned to draw their neighborhood before or after choosing between tutoring for their child and gift cards. Drawing it first lowers the share of choices favoring tutoring by 12 percentage points (21%) among east-side parents, with no detectable effect among west-side parents. I develop a framework of reciprocal exchange in which schooling limits children's ability to reciprocate with neighbors, delays earnings, and increases the likelihood of moving away. Under the framework's assumptions, survey evidence implies costs of about two-fifths of the perceived return to schooling, enough to generate an attendance gap of the observed magnitude. Historical exclusion may persist through the ties that helped excluded households cope, not only through the resources it denied them.

Keywords: Social Exclusion, Human Capital, Historical Persistence, Social Ties.
JEL: *C93*, *I24*, *N36*, *O15*, *R23*, *Z13*.

---

*I am grateful to my advisors, Suanna Oh and Thierry Verdier, for their guidance throughout this project. Ellora Derenoncourt, Rachel Kranton, Stelios Michalopoulos, Eduardo Montero, María Montoya-Aguirre, Felipe Valencia Caicedo, and Ekaterina Zhuravskaya commented on several versions of this paper, and it is much better for it. For comments and conversations, I thank Siwan Anderson, Eva O. Arceo-Gómez, Luz Marina Arias, Ragui A. Assaad, Celina Becerra Jiménez, Marie Beigelman, Davide Cantoni, Felipe Castro Gutiérrez, Denis Cogneau, James Fenske, Oded Galor, Paola Giuliano, Antoine Hubert de Fraisse, Peter Hull, Donia Kamel, Sylvie Lambert, John List, Sara Lowes, Karen Macours, David Margolis, Héctor Paredes Castro, Diego Ramos-Toro, Alex Rivadeneira, Desislava Tartova, Liam Wren-Lewis, Yves Zenou, and many others at seminars and conferences who gave me their time to discuss this project. This work benefited from the support of EHESS and the ANR (ANR-17-EURE-0001), and from financial support from ACES, EHA, UNU-WIDER, CEPREMAP, the PSE Stone Center, and AREUEA. All remaining errors are my own.

†PhD Candidate. Paris School of Economics, France. guillermo.woo-mora@psemail.eu.

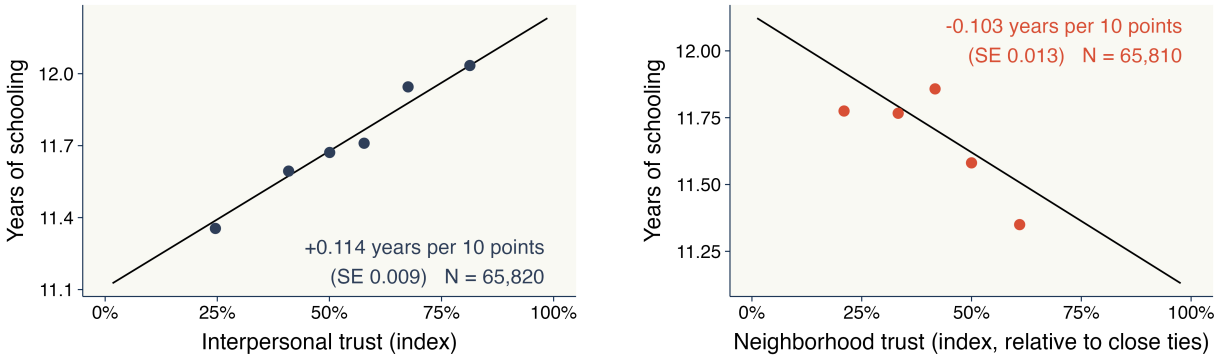
1 Introduction

Human capital gaps between economically excluded communities and the rest of society persist across generations and in many settings (Blanden et al., 2023). They affect African Americans, Scheduled Castes in India, Muslims in Africa, and people with darker skin in the Americas (Heckman and LaFontaine, 2010; Alesina et al., 2023; Althoff and Reichardt, 2024; Asher et al., 2024; Woo-Mora, 2026). Exclusion can also become attached to places, and marginalized neighborhoods can remain educationally disadvantaged across generations (Durlauf, 1996; Chyn and Katz, 2021). The persistence of these gaps is mostly attributed to what excluded groups and places did not get, such as schools, public goods, or educated parents (Dell, 2010; Valencia-Caicedo, 2019). We know much less about whether and why human capital gaps persist after resource equalization.

One reason why human capital gaps might persist even after the convergence of resources is that, in neighborhoods shaped by exclusion, neighbors become each other's most important resource. Historical exclusion can erode trust in outside institutions and groups, a legacy documented across regions and municipalities (Nunn and Wantchekon, 2011; Lowes and Montero, 2021; Ramos-Toro, 2023). In economically excluded neighborhoods, residents come to rely on neighbors to look after each other's children, lend money, and help in emergencies, as long as each household reciprocates (Stack, 1974; Adler Lomnitz, 1977; González de la Rocha, 1994). Whether these neighborhood ties complement or substitute for education is an open question. They can support education by sharing information and enforcing norms that favor school, but they can also compete with it, since time in school leaves less time to reciprocate and gives more reason to leave. Across countries, Figure 1 shows that trust concentrated in neighbors is associated with less schooling, while trust in people in general is associated with more. This correlation may reflect differences in household resources, or schooling itself weakening the ties. If neighborhood ties persist and sustain human capital gaps, equalizing school provision, household resources, or parents' education would leave that margin untouched.

This paper asks whether human capital gaps persist after resources converge, and whether the neighborhood ties that exclusion left behind sustain them. I study Guadalajara, Mexico, where colonial authorities settled the conquered population east of a creek in 1542, opposite the Spanish city and the conquerors' Indigenous allies. The creek was piped underground by 1911, leaving no physical barrier between the two sides for over a century. Using census data from 1930 to 2020 and a boundary discontinuity design (Cattaneo et al., 2026), I find that school attendance has been 3 to 6 percentage points lower on the historically excluded side since 2000, a gap that opens in adolescence, and that measured school supply and quality show no detectable difference. To understand why, I surveyed 239 parents on both sides. They have similar aspirations and expected returns to education, but east-side parents rely more on their neighbors. In a pre-registered experiment, parents are randomly assigned to draw their neighborhood before or after choosing between tutoring for their child and gift cards. Drawing it first lowers demand for tutoring by 12

Figure 1: Trust in neighbors and schooling



(a) Trust in people in general

(b) Trust concentrated in neighbors

Notes: Binned scatterplots (Cattaneo et al., 2024) of years of schooling for about 65,800 respondents in 42 countries in the World Values Survey, wave 7 (Haerpfer et al., 2022). Panel (a): principal component of six trust items. Panel (b): trust in neighbors minus average trust in family and acquaintances. Both indices are rescaled to range from 0 to 100. Dots are bin means net of country fixed effects, sex, age, and parental education, with the number of bins chosen optimally; lines are linear fits.

↩ Back to text

percentage points (21 percent) on the east side, with no detectable effect on the west. I propose a framework of reciprocal exchange in which schooling is costly for households that rely on their neighbors, since teenagers in school spend less time in the neighborhood and are less able to reciprocate, enter the labor market later, and are more likely to move out. Under the framework's assumptions, the survey implies a schooling cost of about two-fifths of the perceived return to education, and a cost of that size could generate an attendance gap of the observed order at the boundary.

At its founding in 1542, Guadalajara settled the conquered local population east of a creek, and the Spanish city and the conquerors' Indigenous allies west of it. The assignment denied the east both access and standing. No school appears east of the creek on the city's maps until 1908, eastern residents were long denied honorifics, and families married across the creek less than chance implies. Access later converged while standing did not. Differences in school provision stopped being detectable and the creek became an avenue that connects the two banks. The stigma did not fade, and the east is still described in the press and in everyday speech as *de la Calzada para allá*, beyond the avenue (Reguillo, 1998; Jaramillo-Molina and Saucedo, 2016; Jaramillo-Molina, 2022; Trujillo Bretón, 2018), and its residents today trust formal institutions less and their neighbors more.

Using census records geolocated at the block, I compare children living within about one kilometer of either side of the historical creek. The boundary design identifies the effect of living on the excluded side if, absent exclusion, school attendance would have been continuous at the creek. This is plausible because both sides share municipal institutions and a labor market, have similar topography, and the creek predates current residences and marks no other administrative or school

boundary. The estimate captures the total long-run effect, including residential sorting induced by exclusion. If exclusion made the east side a place where households relying on neighbors settle or stay, that sorting is part of its legacy rather than a confounder. Sorting unrelated to exclusion remains a concern, so I control for parental education and check sensitivity to household size, migration, and land values, which may themselves reflect exclusion.

Nearly five centuries later, children on the historically excluded side are still less likely to attend school, and that gap accumulates into fewer years of schooling. In 1930, 38.2 percent of 6-18 year olds attended school on the east side of the creek, compared to 53.8 percent on the west side, a raw difference of 15.6 percentage points. In each census since 2000, 6-22 year old residents living immediately east of the boundary are between 3 and 6 percentage points less likely to attend school than those living immediately west of it, a difference of between 4 percent and 8 percent of the mean at the boundary on the west side. The difference in attendance was much larger in 1930 but has not narrowed detectably since 2000. The gap is concentrated in adolescence, since differences are small for primary-school-aged children, for whom attendance is almost universal. In the 2020 census, residents aged 6 or older living just east of the boundary have received on average 0.8 fewer years of schooling than those on the west side.

The gap is not explained by the supply or quality of schools. Schools have no residential catchments, and households on both sides have access to the same number of schools within a thirty-minute walk. School quality was worse on the east side until the early 2000s, but administrative measures and test scores now show no detectable difference, although upper-secondary scores cover only students who remain enrolled.

Demand-side characteristics can only partially explain the gap. I find no evidence that the returns to schooling are different on either side of the boundary. Once I fix schooling, the gap in adult earnings is no longer statistically significant. Differences in parents' schooling can explain up to a third of the gap. I find no discontinuity in household income, but cannot rule out a difference in wealth.

Other place-based explanations do not account for the schooling gap either. One possibility is that less-educated parents sort into cheaper housing on the east side, mechanically generating the gap. Yet the gap survives controls for land values, which may themselves reflect exclusion. Nor are newcomers to the east necessarily less educated: after a gasoline leak triggered sewer explosions in 1992, households moving into rebuilt blocks had more schooling than their neighbors. The gap is also unlikely to reflect the avenue acting as a barrier or a Spanish–Indigenous divide. It is larger in the south, where the colonial settlements faced each other, than in the north, where the same avenue runs. It is also as large where the east faced the conquerors' Indigenous allies as where it faced the Spanish city. Finally, crime and most measures of urban form show no detectable discontinuity, and street infrastructure is, if anything, better on the east side.

The remaining gap goes with a greater reliance of east-side households on their neighbors. Ethnographers of Mexican cities find that adolescents in such households start working around

age 15 and that families who get ahead leave (Adler Lomnitz, 1977; González de la Rocha, 1999, 2001). In a 2022 survey of Guadalajara residents by a local NGO, *Jalisco Cómo Vamos*, which predates my own, residents on the east side trust their neighbors more and formal institutions less than those on the west side. There, and not detectably on the west side, greater trust in neighbors goes with fewer years of schooling, a correlation the framework expects where neighbors stand in for institutions and not where institutions do their job.

East-side parents rely more on their neighbors and require a higher return from schooling. I surveyed 239 parents on both sides of the boundary about their ties with neighbors and their children's schooling.[1] The sample reproduces the schooling gap of its census. Their aspirations for their children, the returns they expect from university, and their household incomes show no detectable difference across sides. East-side parents receive help from neighbors more often and spend more time with them. They also require a higher salary at age 30 before university is worthwhile: MXN 5,319 more per month, or 18 percent above west-side parents. Given the same expected returns, this suggests that they perceive a higher cost.

The survey cannot tell whether relying on neighbors reduces demand for schooling or whether schooling demand determines who lives on the east side, so I run an incentivized choice experiment. I cannot assign neighborhood ties, but I can vary whether parents have their neighborhood in mind when making an educational investment decision. If schooling puts exchange with neighbors at risk, making the neighborhood salient should reduce demand where this exchange is valuable, but not otherwise. Parents make 11 choices between an hour of online math tutoring for their child and gift cards of increasing value. The tutoring is identical for all parents and unrelated to local schools. All parents also draw their neighborhood on a satellite image, randomly before or after making their choices, a task that changes neither their constraints nor their information. Inspired by evidence that historical legacies can be reactivated (Fouka and Voth, 2023; Ochsner and Roesel, 2024; Ferlenga, 2025; Cantoni et al., 2026; Ghidoni and Schindler, 2026), the idea is that drawing makes salient a neighborhood marked by the colonial assignment.

Drawing their neighborhood before, rather than after, choosing between tutoring and gift cards reduces demand for tutoring among parents on the historically excluded side. Parents assigned to draw first choose tutoring 12 percentage points less often (21 percent), and a model of their choices puts the fall in willingness to pay at two-fifths. On the west side, drawing first has no statistically significant effect, but the difference in treatment effects between sides is statistically significant. Among parents assigned to draw after making their choices, demand for tutoring does not differ significantly across sides, although this comparison is imprecise. Because the order of the drawing task is randomized within each side, the experiment identifies the effect of making the neighborhood salient before the educational choice and how that effect differs across sides.

[1] A second wave of approximately 150 households, 120 on the west side and 30 on the east, goes to the field in November 2026. It is designed to tighten the west-side estimates, which rest on 46 households, and it draws neighborhoods farther from the boundary than the first wave. Its addition departs from the pre-registered sample size, so I will report results for the first wave alone and for both waves with a wave fixed effect.

It does not identify differences in underlying demand across sides, which may reflect residential sorting. The experiment thus shows that parents respond differently to their neighborhood across the historical boundary, but does not establish why.

Why does the map move east-side parents and not, detectably, west-side ones? Whatever explains it should differ between the two sides and predict parents' demand for tutoring on the east side. Parents' reliance on their neighbors is the one variable I measure that does both, and it is substantially higher on the east side. On the east side, parents who rely more on their neighbors demand less tutoring, controlling for when they draw the map, and not detectably on the west side. The map moves demand more among parents with fewer ties, which is consistent with the neighborhood already being on the minds of those who rely on it most, although the difference is imprecise. Parents' aspirations for their children's education, their beliefs about its return, and their beliefs about the relative importance of working and studying for young people show no detectable difference across sides and do not predict demand for tutoring. The response is not driven by the poorest households, which ethnographic evidence suggests are least able to fulfill their side of the exchanges (González de la Rocha, 1999). On the historically excluded side, neighborhood ties seem to be substitutes for, rather than complements to, schooling.

The map could work through any feature of the place, so a second randomized prime targets the history behind it. Before explaining why a young person from their neighborhood might stop studying, half of the respondents hear the expression *de la Calzada para allá*, which residents use for the east side and which nearly 80 percent of respondents recognize. On the east side, the expression lowers by 15 percentage points the share of explanations that refer to money or the need to work, and on the west side it shows no detectable change. This is a different outcome from the tutoring choice, and the shift away from money is consistent with a cost that runs through the exchange.

What is the cost of going to school for households embedded in networks of exchange with their neighbors? I interpret it as the cost of not exchanging with neighbors while at school, of delayed income, and of moving away. To assess its size and whether it can explain the gap, I combine the survey and experiment in a framework where exchange with neighbors is valuable when households lack outside options, such as standing in formal institutions (Kranton, 1996). In the survey, the MXN 5,319 gap equals 45 percent of the university premium parents expect. The experiment shows a fall in tutoring valuation up to two-fifths, depending on how choices beyond the price grid are treated. Under the framework's assumptions, both reflect the same cost: the survey puts it at around two-fifths of the expected return to schooling, while the experiment provides a lower bound. A cost of this size could generate an attendance gap of 3.9 to 7.6 percentage points at the boundary, against the 3.2 points in the census, and explain about a third of the larger gaps in the colonial core.

The framework also allows the gap to outlast the convergence of schools. Since parents transmit their place in the exchange to their children (Bisin and Verdier, 2000, 2001), the gap can persist

after the provision of schooling on both sides converges, although the framework does not derive its path. No source allows me to measure exchange with neighbors on both sides of the creek over time, so this remains a hypothesis. Using baptism registers between 1870 and 1919, I find that families were more likely to choose godparents from their parish in parishes where a lower share of fathers had the honorific *Don*, relative to other parishes in the same decade.

This paper provides the first evidence, to my knowledge, of the causal impact of historical exclusion on the persistence of the human capital divide even after the material resources it denied have converged. I develop and test the mechanism that historical exclusion shaped the strength of neighborhood ties, which in turn raise the opportunity cost of schooling. Under the framework's assumptions, the opportunity cost induced by neighborhood ties is about two-fifths of the expected return to schooling, and it can generate the order of magnitude of the attendance gap I find at the boundary. Historical exclusion may persist through the ties that helped excluded households cope, and not only through the resources it denied them.

Related literature. This paper contributes to work on long-run historical persistence (Nunn, 2014, 2020; Voth, 2021). Existing explanations emphasize institutions, geography, and human capital transmitted through public investment, families, and spillovers (Acemoglu et al., 2001; Dell, 2010; Davis and Weinstein, 2002; Bleakley and Lin, 2012; Huillery, 2009; Wantchekon et al., 2015; Valencia-Caicedo, 2019). Educational advantages persist across more generations than parent–child correlations imply, and the mechanisms behind this persistence are clearer in class and caste societies that restricted mobility than in contemporary high-income countries (Blanden et al., 2023). Within cities, neighborhood status often changes, but where it persists, historical shocks and exclusionary boundaries work through urban form and residential sorting (Rosenthal and Ross, 2015; Ambrus et al., 2020; Heblich et al., 2021; Aaronson et al., 2021; Hanlon and Heblich, 2022; Lin and Rauch, 2022; Baldomero-Quintana et al., 2025). History also shapes beliefs, norms, and trust, particularly after exclusion and mistreatment (Nunn, 2012; Giuliano and Nunn, 2021; Michalopoulos and Xue, 2021; Lowes, 2023; Nunn and Wantchekon, 2011; Grosfeld and Zhuravskaya, 2015; Alsan and Wanamaker, 2018; Lowes and Montero, 2021; Ramos-Toro, 2023). These legacies can be reactivated (Fouka and Voth, 2023; Ochsner and Roesel, 2024; Ferlenga, 2025; Cantoni et al., 2026; Ghidoni and Schindler, 2026) and transmitted across generations and between neighbors (Bisin and Verdier, 2000, 2001, 2026; Bezin and Moizeau, 2017). Unlike historical exchange between groups that can foster cooperation (Artiles, 2026), the exchange I study takes place within an excluded community and in place of outside institutions. I study individual human capital, not house values or neighborhood status, across blocks of a single city after differences in measured school supply and quality stopped being detectable. I show how the social organization inherited by an excluded place enters a contemporary economic decision.

The paper also contributes to work on how social ties shape economic decisions (Granovetter,

2005). In education, ties can support schooling through norms and information (Coleman, 1988). Beyond education, social ties and networks that provide insurance and jobs can restrict mobility (Kranton, 1996; Banerjee and Newman, 1998; Portes, 1998; Munshi, 2014; Munshi and Rosenzweig, 2016), although migration can also serve as insurance (Groeger and Zylberberg, 2016).[2] Recent experiments show that obligations to kin and neighbors change how much people work, save, and spend (Jakiela and Ozier, 2016; Goldberg, 2017; Boltz et al., 2019; Squires, 2024; Carranza et al., 2025; Goraya et al., 2025). There is little evidence on whether the costs of belonging to such networks extend to investment in human capital. Closest to this paper, Munshi and Rosenzweig (2006) show that caste networks in Bombay, India, kept boys in schooling for traditional occupations even as returns to other schooling rose. I provide experimental evidence on this margin, where the cost falls on children and can carry gaps across generations.[3]

Finally, the paper relates to work on neighborhood effects (Durlauf, 2004; Topa and Zenou, 2015; Chyn and Katz, 2021; Katz, 2026). A large literature, mostly from high-income countries, shows that where children grow up shapes their adult outcomes (Kling et al., 2007; Ludwig et al., 2013; Chetty et al., 2016; Bergman et al., 2024), with recent work emphasizing the social environment (Sampson et al., 2002; Chetty et al., 2026). In developing countries, housing lotteries and slum clearance that move poor urban households can sever informal insurance and other local ties, although networks may re-form when neighbors move together (Barnhardt et al., 2017; Franklin, 2020; Carrillo et al., 2023; Franklin, 2025; Belchior et al., 2026; Bryan et al., 2026; Rojas-Ampuero and Carrera, 2026). These studies ask what households lose when they move or leave their networks. I keep families where they are and, just before an educational choice, ask some of them to draw their subjective neighborhood (Aliprantis, 2023; McCartan et al., 2024). This does not identify the effect of exposure to a neighborhood, but it reveals a margin through which the neighborhood enters decisions, and one that differs across a historical boundary of exclusion.

The rest of the paper proceeds as follows. Section 2 describes the context and the historical natural experiment. Section 3 documents the persistent human capital gaps at the boundary and evaluates the standard channels. Section 4 reports the results of the survey and the experiment. Section 5 develops the framework, measures the cost, examines how the neighborhood ties could

[2] Neighbors and kin share risk within what they can observe and enforce (Townsend, 1994; Ligon et al., 2002; Ambrus et al., 2014, 2022; Laczó, 2015). Related work documents a trade-off between strong family ties, which Banfield (1958) called "amoral familism", and broader social capital and economic outcomes (Alesina and Giuliano, 2014; Alesina et al., 2015). In this context, ties are between neighbors, who are not necessarily related. Trust in family members and the share of close relatives living in the *colonia*, 19.6 percent on the east side and 14.1 on the west, are not detectably different on the two sides of the boundary.

[3] A related literature studies identity, social image, and peer pressure among students and their classmates (Akerlof and Kranton, 2002; Austen-Smith and Fryer, 2005; Sacerdote, 2011; Bursztyn and Jensen, 2015; Bursztyn et al., 2019). I instead study parents and the neighbors they interact with, in ties that come from place rather than caste or kinship. The experiment does not separate exchange from identity (Akerlof and Kranton, 2000; Shayo, 2020; Oh, 2023) or social image (Bursztyn and Jensen, 2017; Karing, 2024), but any of these channels can make neighborhood ties enter the cost of educational investment.

sustain the gap, and checks the prediction across countries. Section 6 concludes.

2 Context and Historical Natural Experiment

2.1 Guadalajara and the Colonial Boundary

Setting. Guadalajara, in western Mexico (Figure A.1), is a rich, highly schooled municipality by national standards and anchors the country's second-largest metropolitan area.[4] The gap I study lies within a wealthy, highly schooled city.

Much of the variation in school attendance in Mexico is across households within the same municipality.[5] Figure 2b shows that human capital in Guadalajara varies sharply across census blocks along a west–east divide, with more schooling west of Calzada Independencia, the avenue that runs north–south through the city center, and less east of it. This divide extends across the metropolitan area (Figure A.2), although residents on both sides share one labor market (Aldeco et al., 2024) and face no school catchment areas. Walton (1978) called this the divided city, with the working class concentrated in the eastern part and the middle and upper classes in the western one.

Foundation. Spanish authorities founded Guadalajara in the Atemajac Valley in February 1542 as the capital of New Galicia, on land with no existing settlement. Unlike central Mexico, where colonial rule co-opted the centralized Mexica (*Azteca*) state (Knight, 2002; Arias and Dirod, 2014), New Galicia was sparsely populated and politically fragmented, conquest proceeded group by group, and three earlier attempts to found Guadalajara failed amid Indigenous resistance and water shortages.[6] Because there was no town to inherit, the founding authorities decided where Spaniards, their Indigenous allies, and the conquered local population would live, and recorded these assignments in a single administrative act, implemented by 1543.

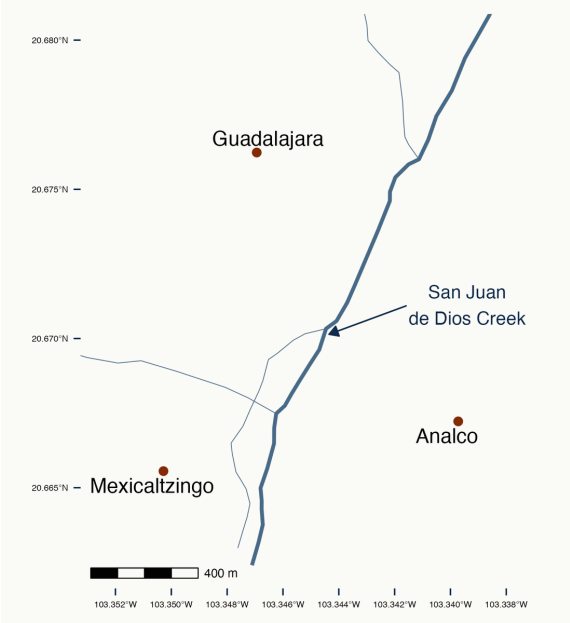
The foundation drew on a Crown-wide architecture of segregation. Beginning in 1538, the Crown established *Repúblicas de Indios*, corporate organizations for Indigenous populations with their

[4] In 2020, average quarterly household income was 68,308 pesos, 36 percent above the national average and at the 98th percentile among urban municipalities. Residents averaged 10.96 years of schooling, compared with 9.74 nationally, and attendance beyond primary has remained above the national rate for decades. Figure A.3 locates Guadalajara in the national distribution, and Figure A.4 shows attendance trends since 1970.

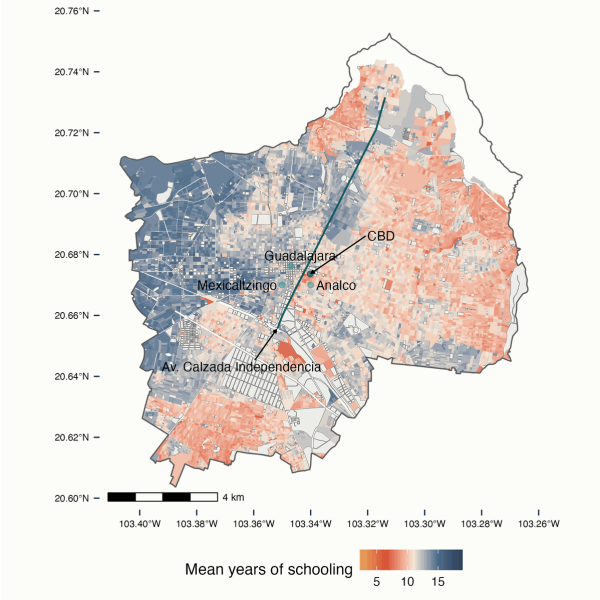
[5] Among Mexicans aged 12–17, municipalities account for only 3.7 percent of the variance in attendance in 2020, while households within municipalities account for 78.7 percent. Age, sex, parental schooling, birth order, and sibship size explain 13.7 percent of the total variance. Because the harmonized census records no geography below the municipality, the household component also contains neighborhood differences. Figure A.5 reports the decomposition; Figures A.6 to A.9 extend it to parental resources, schools, and adult schooling.

[6] The first settlement (1532, Nochistlán) collapsed after one year; attempts followed in Tonalá (1533) and Tlacotlán (1535). The final foundation took place in 1542 at the present-day site after the *Mixtón* War (1541) and an expedition led by Viceroy Antonio de Mendoza with participation by Tlaxcalteca and surrendered Mexica allies.

Figure 2: The foundational settlements in 1542 and schooling in 2020



(a) The three foundational settlements, 1542



(b) Mean years of schooling by census block, 2020

Notes: Panel (a) reconstructs the three foundational settlements from Tanck de Estrada (2005) and IMEPLAN (2016); the thick blue line is the SJdD creek as documented in 17th-century accounts (Lázaro de Arregui, 1946) and 19th-century cartography. Settlement boundaries are illustrative because jurisdictional delineations between *Pueblos* were not always recorded. Panel (b) maps mean years of schooling by census block in the 2020 *Censo de Población y Vivienda*, shaded from red (lower) to blue (higher), with Calzada Independencia, the avenue that replaced the piped-over creek, overlaid. Blank blocks are suppressed by INEGI because of small population counts.

↩ Back to text

own *cabildos* and communal land rights, distinct from *Pueblos de Indios*, the settlements where Indigenous populations lived (Castro Gutiérrez, 2010; Graham, 2013). The system sought spatial separation between Indigenous and Spanish populations, although enforcement in Guadalajara was partial (Anderson, 1988). These corporate distinctions ended in the 1820s (Tanck de Estrada, 1999).

Within Guadalajara, this architecture took a specific spatial form. Spanish Guadalajara was founded on the western bank of the San Juan de Dios (SJdD) creek. The Spaniards' Indigenous allies, Mexicas and Tlaxcaltecas, settled immediately south on the same bank in *Mexicaltzingo* ("house of the Mexicas"). In 1543, Franciscans established *Analco* ("on the other side of the creek" in Nahuatl) east of the creek as a *Pueblo de Indios* for the conquered local population (Solís Matías, 1986, 1992). Figure 2a maps the three settlements, all within roughly one kilometer of each other.

The two Indigenous settlements were not symmetric, and that distinction is central to interpreting the design. Mexicaltzingo's settlers had participated in the conquest and were recognized as allies, receiving communal land grants and recognition of their leaders as *principales*. They were "*not conquered, but among the conquerors*" (Castro Gutiérrez, 2010). Analco, the only

foundational settlement east of the creek, was populated by conquered local groups and later regional migrants. At the foundation, the east–west contrast was therefore one of colonial status, with conquerors and their Indigenous allies on the west and conquered groups on the east, rather than a Spanish-versus-Indigenous divide.[7]

A historical natural experiment in exclusion. The colonial assignment is a historical natural experiment in the sense of Cantoni and Yuchtman (2021), as is the colonial resettlement of native populations in Andean Peru (Artiles, 2026), a historical imposition of a treatment, here exclusion, evaluated against a control group, the settlements west of the creek, to understand an outcome today. Four features of the assignment support this interpretation, and Appendix A reports the supporting evidence.

First, where exclusion was assigned was unrelated to pre-colonial agglomerations, which can shape economic activity over long horizons (Davis and Weinstein, 2002; Bleakley and Lin, 2012; Maloney and Valencia-Caicedo, 2016; Lee and Lin, 2018). The documentary record and the coarse pre-colonial population data (Goldewijk et al., 2017) agree that the region had no urban centers before Spanish contact, and colonial-era urban density data (Buringh, 2018) show sparse population relative to central Mexico through the 16th–18th centuries (Figures A.10 and A.11).

Second, the excluded bank was not chosen for its measured geography. Altitude, slope, and climate are similar across banks, both banks had the creek, and a BD design on the grid cells covering the municipality finds no detectable discontinuity at the SJdD creek in measured locational characteristics (Figure A.12 and Table A.1).

Third, exclusion operated within shared political and municipal institutions (Acemoglu et al., 2001, 2005; Lowes et al., 2017). All three settlements fell under colonial authorities centered in Guadalajara and the *Audiencia* of New Galicia (Lázaro de Arregui, 1946; Solís Matías, 1992), and although both Indigenous settlements kept their own officers until 1821, their communal finances fell under the jurisdiction of the city's *ayuntamiento*.[8] The excluded and the favored settlements differed in status and privileges within these institutions, not in the institutions themselves, and they have remained within the same jurisdiction ever since.

Fourth, the assignment predates the industrial, infrastructural, and demographic shocks that transformed Guadalajara in the 19th and 20th centuries. Exclusion was part of the city's *seed*, rather than one later shock among many (Ahlfeldt et al., 2015; Hornbeck et al., 2025). In English cities, industrial pollution carried by prevailing winds left the east sides poorer (Heblich et al.,

[7] Reading the contrast through modern ethnic categories would be anachronistic. Alberro (2019) argues that colonial New Spain classified people through categories such as estate, *nación*, and corporate membership rather than race, and that *indio* was itself a porous category. The sources distinguish Mexicaltzingo and Analco by their origin and privileges at the foundation, not simply ethnic origin.

[8] Around 1800, Analco and Mexicaltzingo are two of four *pueblos de indios* near Guadalajara absent from the *propios y arbitrios* accounts submitted to the Contaduría. Tanck de Estrada (1999) interprets this absence as evidence that they fell under the city's *ayuntamiento*.

2021), and the assignment behind the west–east divide of Guadalajara predates its industry by three centuries. The shocks that followed deepened the divide along the same line, since from the 1890s wealthy families left the center for new *colonias*, subdivisions laid out on farmland west of the creek, rural migrants settled east and north of the creek, and public works followed the former (Walton, 1978; Jiménez Pelayo et al., 1995).

These features make the colonial assignment a credible source of variation in exclusion within a common physical, municipal, and market environment.

From the creek to the Calzada. Today the boundary is not a physical barrier. The SJdD creek was piped underground between 1903 and 1911 and replaced by Calzada Independencia, a surface-level avenue. Unlike the railroad tracks and urban freeways studied as infrastructural barriers (Ananat, 2011; Chyn et al., 2026; Bagagli, 2025; Valenzuela-Casasempere, 2025), the avenue connects the two banks rather than separating them (Figure A.13). Built blocks on its two sides stand 36 meters apart and it is crossed every 151 meters, against 253 meters and a crossing every 549 meters along the Dan Ryan Expressway in Chicago and 287 meters and a crossing every 406 meters along the Seine in Paris (Figure A.14). A discontinuity at the boundary is therefore unlikely to reflect the cost of crossing it.

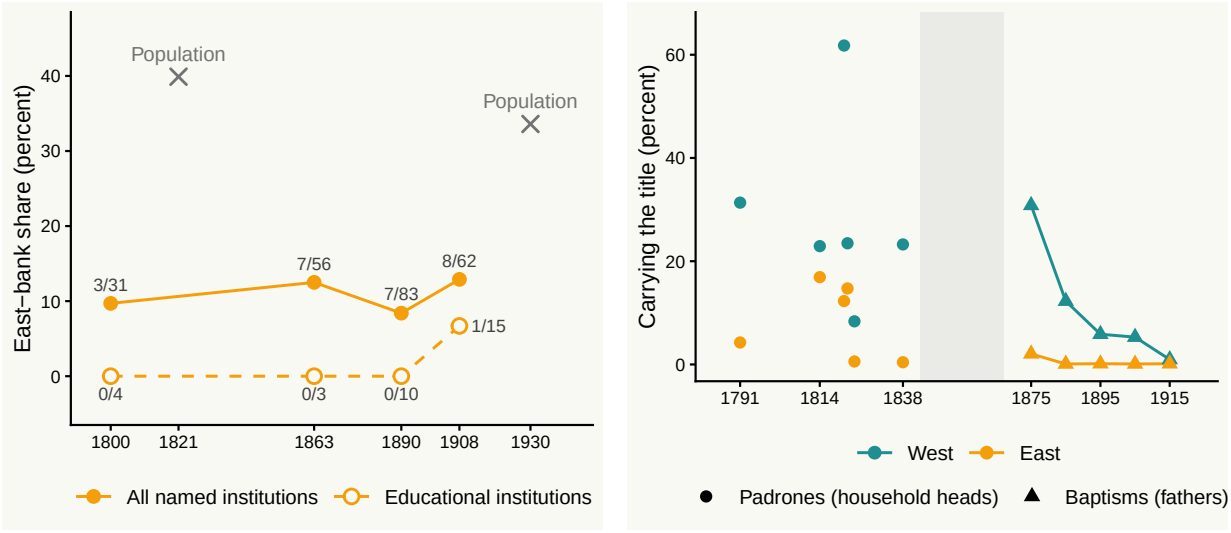
2.2 Two Margins of Exclusion

The colonial assignment denied the settlements east of the creek both the access and the standing that the Spanish city enjoyed.

Denied access. Figure 3a shows that on each of four general maps of the city, from 1800, 1863, 1890, and 1908, the east bank held between 8 and 13 percent of the named institutions, against 40 percent of the population of the central city in 1821 and 34 percent of the city's population in 1930 (Figure A.15). No named educational institution appears east of the creek until 1908, when a girls' school run by a religious order is the only eastern one among fifteen (Table A.2). In the newly digitized cadaster of 1882, land values per square meter within 150 to 500 meters of the creek were 48 to 67 percent lower on the east bank, and the western blocks in these windows lie mostly in the city center (Figures A.17 and A.18). With few institutions on the east bank, life there was organized in *barrios*, the old neighborhoods that had grown around a church and a market. Their residents shared trades and often a town of origin, and obtained basic services through their own effort (Jiménez Pelayo et al., 1995). The *colonias* built east of the creek in the twentieth century kept this form, with homes, workshops, markets, and churches on the same blocks, while those built west of it were laid out for residence alone (Jiménez Pelayo et al., 1995).

Denied standing. Figure 3b shows that from 1791 to 1919 census takers and priests wrote the honorific *Don* or *Doña* far less often for residents east of the creek. In 1821, 12 percent of

Figure 3: Access and standing on each bank of the creek



(a) Named institutions east of the creek, 1800–1908

(b) The honorific *Don*, 1791–1919

Notes: Panel (a) reports the share of the named institutions on the general maps of 1800, 1863, 1890, and 1908 that lie east of the creek, for all institutions and for educational ones, and the share of the population living east of the creek in the central city in 1821 and in the city in 1930 (crosses) (Appendix A.5). Labels give the eastern entries over all entries located on each map; in 1908, 62 of the 105 entries in the index could be located. Panel (b) reports the share carrying *Don* or *Doña* among household heads in the *padrones* (Anderson and Spike, 2007) (circles) and among fathers in the FamilySearch baptism registers, by decade (triangles). East is Analco, in the *padrones* and in the baptism registers of its parish. West is the central *cuarteles* of each enumeration, with Mexicaltzingo where it is recorded, and the parishes of Guadalupe and the Sagrario. Shaded years have no source for both banks.

↩ Back to text

household heads in Analco carried it, against 62 percent in the Spanish city, and in the baptisms of the 1870s, 2 percent of fathers in Analco did, against 31 percent in Guadalupe and the Sagrario. The gap holds among the baptisms of the ten priests who served both banks and closes only in the 1910s, when the title faded in the center as well. Where Mexicaltzingo is recorded separately, in 1791 and 1822, its household heads carried the title about as rarely as those of Analco.

Standing also shows in whom residents of each bank married. Among couples whose spouses had both been baptized in the city, most of whom married between 1888 and 1941, 31 percent joined the two banks, against 50 percent under random pairing, and at equal distance between parishes such marriages were half as frequent as marriages within the west bank (Appendix A.7).

The two banks also named their children differently. In the *padrones* of 1821–22 and 1824, which enumerated the same cohorts independently, and in the 1930 census, first names predict residence east of the creek for every cohort observed from those born in 1760 to those born in 1879, also within the ethnic categories recorded in 1821, and no longer for cohorts born after 1880 (Figure A.19).

The stigma attached to the eastern side survives in the press and in expressions such as *de la Calzada para allá* and *del otro lado de la Calzada*, which divide the city in two in its

residents' imaginary (De la Torre, 1998; Reguillo, 1998; Jaramillo-Molina and Saucedo, 2016; Trujillo Bretón, 2018; Jaramillo-Molina, 2022).

The gap in access has closed, and measured school provision no longer differs detectably at the boundary (Section 3). The gap in standing has not, as far as the honorifics, the marriages, and the stigma show, and no source measures standing continuously into the present.

3 Persistent Human Capital Gaps

3.1 Boundary Discontinuity Design and Identification

Geolocated census microdata. The main analysis uses the universe of individuals in the 2020 *Censo de Población y Vivienda*, geolocated to the census block, which lets me compare residents within meters of the historical boundary.[9] The census records parental education and household characteristics but not income. School attendance is what the household reports for each member aged 3 or older, whether the person currently attends school, and it is not linked to any school, so the census says whether a child is in school and not where. For income I turn to the 2015 *Encuesta Intercensal*, a survey representative at the municipality level with few observations near the boundary, which I use mostly for municipality-wide comparisons, and for BD estimates of household resources at the boundary (Table B.5).

Specification. I estimate the discontinuity at the SJdD creek, today Calzada Independencia, with a Boundary Discontinuity (BD) design (Cattaneo et al., 2026).[10] The baseline specification collapses location into the signed orthogonal distance to the boundary:

$$y_i = \alpha + \tau\, \mathbb{1}\{East_i\} + \delta s_i + \zeta\, \mathbb{1}\{East_i\} \cdot s_i + \gamma' X_i + \varepsilon_i. \tag{1}$$

The outcome y_i is the human capital variable of interest for individual i. The treatment indicator $\mathbb{1}\{East_i\}$ equals one for residents east of the boundary, and the running variable s_i is the orthogonal distance in meters to the boundary, positive on the eastern side. The interaction lets the outcome gradient differ across banks. The control vector X_i includes sex, age, longitude, latitude, and boundary-segment fixed effects, so that comparisons occur within the same segment (Dell, 2010). I follow Cattaneo et al. (2026), who show that the fit must be allowed to differ across sides and recommend MSE-optimal bandwidths with robust bias-corrected inference (Calonico et al.,

[9] These data are not publicly available and can be accessed only through INEGI's *Laboratorio de Microdatos*, which provides researchers access to anonymized census records on its secure servers.

[10] The BD design is the regression discontinuity design that applied work often calls a spatial or geographic RD design, in which units are assigned to treatment by the location of their residence relative to a boundary, so the score has two dimensions instead of one. Cattaneo et al. (2026) show that pooling observations by their distance to the boundary recovers a density-weighted average of effects that can vary along it, under conditions on how observations are distributed along the boundary, and they provide methods to estimate the effect at each point of the boundary.

2014). Because treatment changes along a line rather than at a point (Keele and Titiunik, 2015), τ pools the boundary. It equals a density-weighted average of the effects at each point when the residential distribution along the boundary is comparable on the two banks, or when the segment fixed effects and coordinates absorb the difference, and the continuity assumption is on the covariate-adjusted conditional means pooled by signed distance. The estimates at each point of the boundary in Figure 6, from the bivariate estimator of Cattaneo et al. (2025), do not rely on that condition and show where the pooled effect comes from.

I also estimate the attendance gap conditional on parents' education, household size, and migration status. Some of these controls may themselves reflect the historical assignment, so I read these estimates as the gap that remains among children from comparable households and not as the total effect of the assignment.

Identification and interpretation. The design identifies the effect of the historical assignment if, without it, average potential outcomes would be continuous at the creek. With a geographic boundary this is a strong assumption, since people choose where to live (e.g., Lee and Lemieux, 2010; Keele et al., 2017).

Three features of the setting leave residential sorting as the principal remaining identification concern. The boundary predates every current residence, since the area was not urban in 1542. The banks had access to the same creek, and the measured differences in altitude, slope, and climate are small, with intervals that exclude differences larger than 0.02 standard deviations in altitude and 0.12 in slope (Table A.1), which supports geographic comparability, while historical hydrological conditions are not directly measured. The boundary is not a compound treatment in the sense of Keele and Titiunik (2015). It delimits no administrative or school district,[11] both sides belong to the same municipality and the same court, fiscal, and police jurisdictions, they share one labor market, and the Calzada is no barrier to crossing (Figure A.21). Present-day ethnicity does not partition the city at the boundary either. Few residents of Guadalajara identify as Indigenous or speak an Indigenous language, and interviewer-assessed skin tone does not cluster by side (Figure A.22).

Sorting I cannot rule out. Residential density jumps at the creek, with more residents on the east side, although the gap falls to 4 percent within 500 meters (Figure B.1; McCrary, 2008; Cattaneo et al., 2018). The jump reflects the city's commercial core, which lies on the west side of the avenue, so I do not read it as manipulation of the running variable, but it leaves sorting open.

Under continuity, τ is the reduced-form long-run effect of the assignment. It bundles the initial difference in status and public provision with everything that followed from it, including intergenerational transmission, residential sorting, and any social mechanism that arose alongside. The

[11] Schools are organized into *zonas escolares* by the Secretaría de Educación del Estado de Jalisco for inspection, planning, and the assignment of new schools, and these zones do not impose a residential catchment, and I have not verified whether their boundaries follow the Calzada.

rest of the section examines these channels without separating them causally.

3.2 The Gap Today

The boundary still divides both completed schooling and current attendance, and the two tell different stories. Completed schooling is a stock, accumulated over many cohorts, and carries the history of school provision and of transmission within families. Attendance, as households report it in the census, is a decision that today's parents make with the schools that exist today.

Residents aged 6 and older just east of the Calzada have about 0.8 fewer years of schooling than residents just west, 0.17 standard deviations or 9 percent of the municipal mean (Figure 4, panel a). The jump is visible in 100-meter bins on either side of the boundary and is 0.14 standard deviations with covariates. Among residents aged 6–22, an age range wide enough to have observations near the boundary, attendance drops by 3.2 percentage points at the creek without covariates, with a p-value of 0.094 (panel b). Table B.1 shows that the estimates keep their sign across the polynomial order, the kernel, the variance estimator, and the covariate set, ranging from −0.08 to −0.17 standard deviations for schooling and from −0.08 to −0.14 for attendance.

3.3 The Gap over Time

The gap is old, it was much larger in 1930, and it has not narrowed detectably since 2000. I follow it with the 1930 census at the *cuartel* level, the 1990 census at the AGEB level, and the 2000, 2010, and 2020 censuses at the block level.[12]

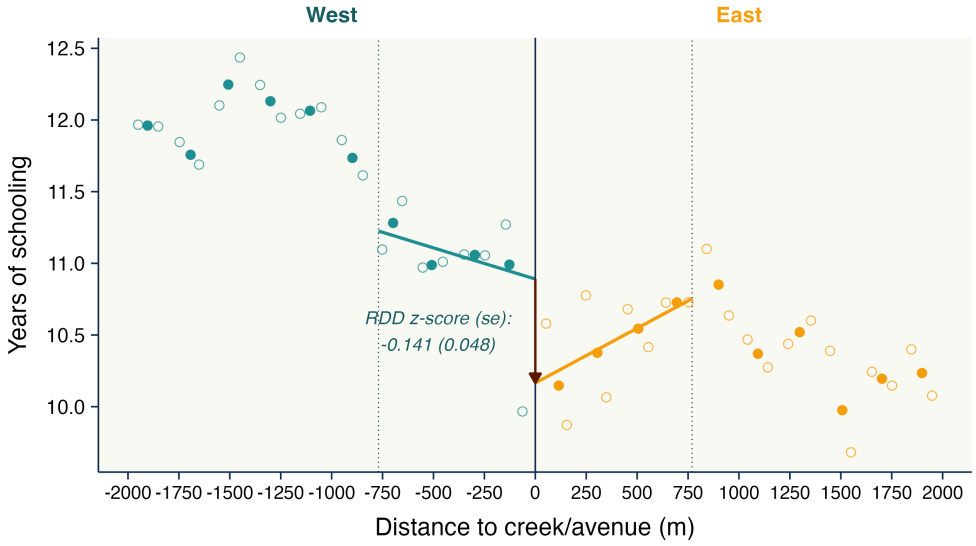
In 1930, two decades after the first named school appeared east of the creek, residents aged 6 to 18 in the *cuarteles* next to the boundary were 15.3 percentage points less likely to attend school on the east side than on the west, with age and sex fixed effects (15.6 points as a raw difference in means, Table B.2).[13] Literacy among adults aged 15 or older was 66.3 percent on the east against 78.8 percent on the west. The divide predates the expansion of public schooling in Mexico.

Figure 5 traces the divide in later censuses. The literacy gap closed as literacy approached universal coverage and shows no discontinuity from 1990 onward (Figure B.2). The attendance gap did not close. In the block series within five kilometers of the boundary, it fell to about 3 percentage points by 2000 and has since been between 3 and 6 percentage points, with no detectable narrowing, including the decade after differences in measured school quality stopped being detectable, and the series' 6.0 points in 2020 differ from the 3.2 of Table B.1, which uses

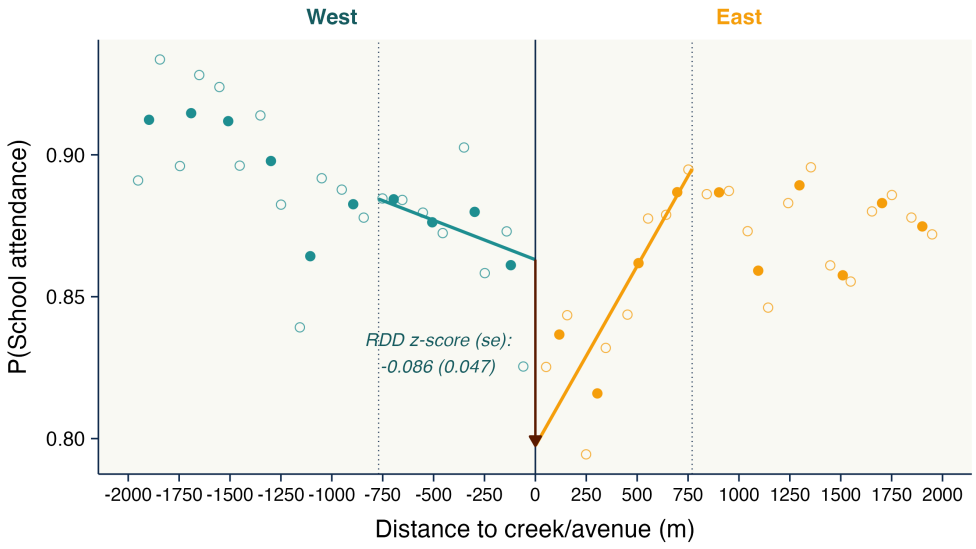
[12] For 1930, I compare the *cuarteles* immediately adjacent to the boundary, and for 1990, the AGEBs within 2.5 kilometers of it, using a parametric specification. For 2000–2020, I estimate the non-parametric BD specification of Equation (1) at the block level, restricting each census to blocks within five kilometers of the boundary.

[13] Analco comprises *cuarteles* 8 and 9 on the east; central Guadalajara and the historical Mexicaltzingo area comprise *cuarteles* 1, 2, and 7 on the west. The estimate depends on this choice, since excluding Mexicaltzingo raises the gap to 22.9 percentage points and including all *cuarteles* lowers it to 8.3.

Figure 4: BD estimates of human capital gaps at the historical boundary (2020)



(a) Years of schooling, ages 6+

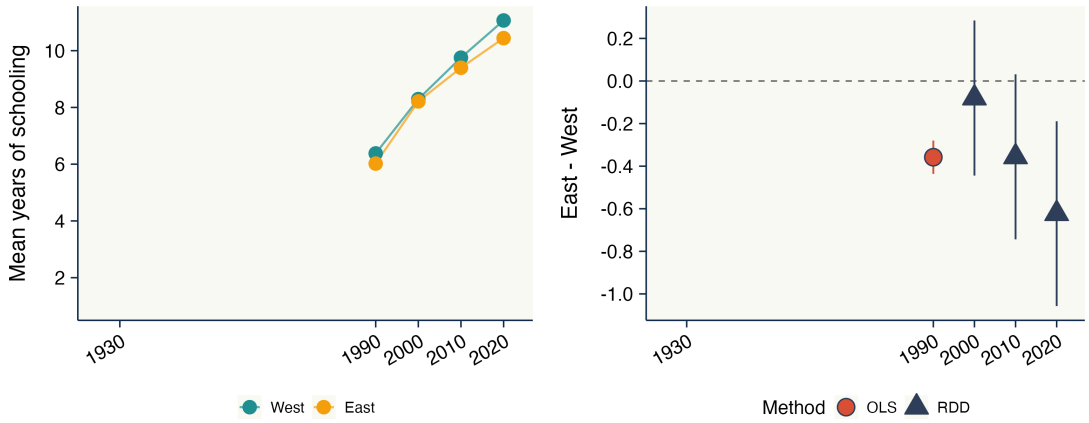


(b) School attendance, ages 6–22

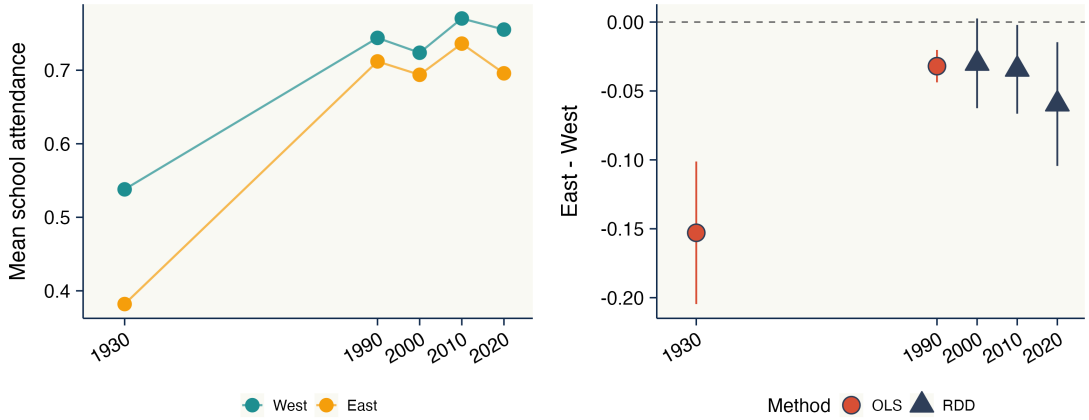
Notes: The figures display binned means and local linear fits of the outcome variables using individual-level data from the complete 2020 *Censo de Población y Vivienda* for the municipality of Guadalajara. The x-axis measures the (signed) orthogonal distance in meters to the historical boundary, where 0 indicates the current location of the Calzada Independencia, a former creek (San Juan de Dios); positive values are on the historical east side of the boundary. The y-axes measure the outcome variables in their original units: average years of schooling (Panel (a), for individuals aged 6 or above) and the probability of attending school (Panel (b), for individuals aged 6 to 22). Hollow (solid) circles correspond to the average outcome of individuals living within 100 (200) meters of the historical boundary; the x-axis binning is restricted to $[-2000, +2000]$ meters. The vertical dotted lines indicate the bias-corrected bandwidth $b = \pm 770$ meters for years of schooling chosen as in Calonico et al. (2014). The thick colored lines show local linear fits restricted to the same bandwidth on both sides of the boundary (and without covariates). The coefficients in the figures, instead, correspond to the covariate-adjusted, standardized estimates in Table B.1 with their respective MSE-optimal estimation bandwidth h (395 meters for Panel (a) and 895 meters for Panel (b)), controlling for sex, age, longitude, latitude, boundary segment fixed effects, and – in Panel (b) – parental education, household size, and migration. The local linear fits and the annotated estimates are thus not based on the same specifications. No additional curve or band is shown. The figure indicates a relative human capital disadvantage of individuals living on the east side of the boundary.

↩ Back to text

Figure 5: Human capital gaps across the historical boundary over time (1930–2020)



(a) Years of schooling



(b) School attendance

Notes: Evolution of the boundary gap in (a) years of schooling and (b) school attendance. Each panel shows, at left, the mean outcome on each side of the boundary, with the western *cuarteles* covering Spanish Guadalajara and Mexicaltzingo and the eastern side covering Analco, and, at right, the east-minus-west gap by census wave. Circles are OLS estimates for 1930 (*cuartel* level) and 1990 (AGEB level); triangles are non-parametric BD estimates for 2000–2020 at the block level following Calonico et al. (2014). Vertical bars show 95% confidence intervals. All estimates use individual-level census microdata for the municipality of Guadalajara. Panel (a) has no 1930 estimate because the census does not record years of schooling in that year; 1930 is retained on the axis for comparability with panel (b). In panel (b), the 1930 estimate covers residents aged 6 to 18 and the 1990–2020 estimates residents aged 6 to 22, since in 1930 only 26 of the 759 residents aged 19 to 22 in these *cuarteles* attended school. The 1930 estimate compares the *cuarteles* next to the boundary with sex and age fixed effects and no geographic controls, and the 1990 estimate is a parametric contrast with geographic controls among AGEBs within 2.5 kilometers of the boundary. For years of schooling, the 1990 estimate is measured among adults aged 18+, whereas the 2000–2020 estimates cover ages 6+, so the pre-2000 estimates are not directly comparable with the block-level series. The 1990 attendance estimate also conditions on parental schooling. The series uses a uniform kernel and restricts each census to blocks within five kilometers of the boundary, while Table B.1 uses all blocks of the municipality. The series is estimated without covariates, with bandwidths of 344, 465, and 390 m for schooling and 922, 1,132, and 844 m for attendance in 2000, 2010, and 2020, against 571 and 1,200 m in Table B.1, and in 2020 its gaps are −0.62 years and 6.0 points against −0.79 and 3.2 in that table's rows without covariates. Adult literacy is reported in Figure B.2.

↩ Back to text

all blocks of the municipality. The schooling gap, if anything, widens over the same period, from −0.08 years in 2000 to −0.62 years in 2020, as the cohorts affected by the attendance gap reach adulthood.[14] The earlier contrasts and the later boundary estimates differ in geographic support, age coverage, and conditioning. The much larger 1930 contrast suggests broad historical narrowing, and the series does not measure one harmonized boundary gap over time or isolate what expanding public schooling contributed. The comparable block-level estimates from 2000 onward document a continuing attendance disadvantage that has not narrowed detectably.

3.4 The Gap along the Boundary

The gap is largest where the colonial settlements faced each other and smallest where the boundary is only an avenue. This separates two readings of the east–west contrast. If it were ethnic, the gap would be narrower where Analco faced Mexicaltzingo, an Indigenous settlement whose residents arrived as allies of the conquerors. If the avenue itself divided the city, the conditions it creates would have to change along its length as the gap does, including in the northern stretch that lay outside the nineteenth-century city and was urbanized only in the twentieth (Figure B.3).[15]

Figure 6 estimates the effect at each point of the boundary with the location-based estimator of Cattaneo et al. (2025) and uniform confidence bands. Where Analco faced the Spanish city (point 4) and Mexicaltzingo (point 5), the gaps are −2.3 and −1.9 years of schooling and −12.6 and −11.8 percentage points in attendance (Table B.3). The discontinuity is larger at each of these two points than at each of the three northern ones, for years of schooling and for school attendance alike.[16] The two points show no detectable difference from each other, so the gap is not detectably smaller facing the conquerors' Indigenous allies than facing the Spanish city. Along the stretch urbanized in the twentieth century (points 1 and 2), the schooling gaps are smaller, −0.4 and −0.7 years, and no attendance gap is detectable, with intervals that exclude gaps larger than 10 percentage points against the east side. Point 3 borders the historic commercial core, where few people live on the western side, and its estimates are imprecise.

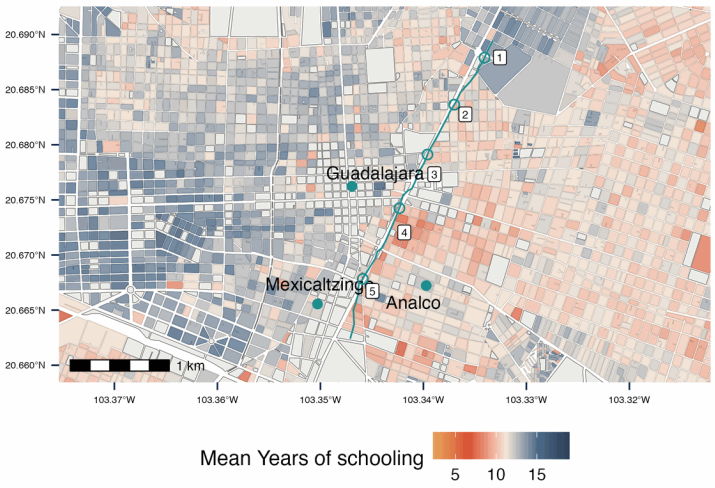
The Calzada is readily crossed (Section 2.1), and estimated at the same points of the boundary as the schooling gap, the discontinuities in land values, crime, sidewalks, and high schools within

[14] Spatial units and samples differ before 2000, so the 1930 and 1990 estimates establish that the divide is old, while the block-level 2000–2020 estimates are the ones comparable with each other. The near-zero schooling estimate in 2000 may partly reflect the 1992 Reforma explosions, which destroyed and repopulated part of Analco within the BD bandwidth shortly before the census. The paragraph on residential turnover in Section 3.5 and Appendix B.9 describe the episode.

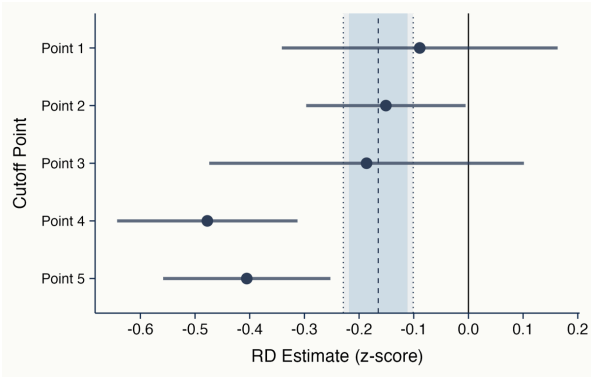
[15] The two northernmost boundary points lie more than 200 meters from the nearest block of the 1890 city map, and the other three lie within 60 meters of one.

[16] Five of the six pairwise contrasts are significant at the 5 percent level for each outcome and the sixth at the 10 percent level; all sharpened false-discovery-rate q-values are below 0.03 (Anderson, 2008). Because `rd2d` does not report covariance across boundary points, the standard errors of the contrasts treat the estimates as independent. One additional boundary point lacks a bias-corrected estimate and is omitted.

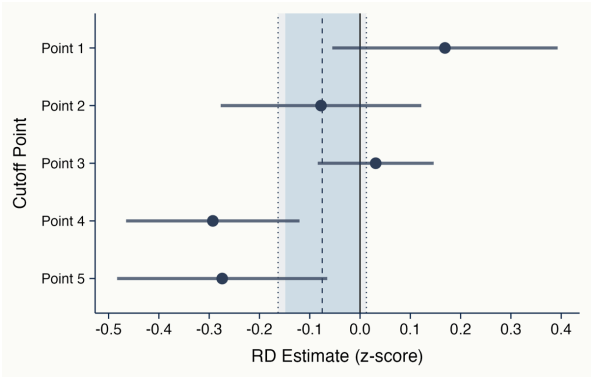
Figure 6: Boundary average treatment effect curve (2020)



(a) Study zone near the historical boundary



(b) Years of schooling



(c) School attendance

Notes: Panel (a) shows the study zone around Calzada Independencia; Figure 2b shows the full municipality on the same color scale. Panels (b) and (c) report the boundary average treatment effect curve and uniform confidence bands following Cattaneo et al. (2025), using individual-level data from the 2020 *Censo de Población y Vivienda*. The aggregated estimate is shown at the bottom of each panel. The vertical dashed line marks the distance-based BD estimate from Figure 4; shaded areas show the 95% and 90% confidence intervals.

↩ Back to text

reach show no detectable difference between the southern stretch, where the foundational settlements faced each other, and the northern stretch urbanized in the twentieth century (Table B.4). Paving, ramps, and the number of schools within reach favor the east side more in the south, and blocks there hold fewer residents and dwellings, while the schooling and attendance gaps are largest in the south. A present-day avenue effect would have to work through conditions that are worse on the east side where the schooling gap is largest, and none of the measured conditions is, so the pattern favors an interpretation tied to the historical assignment over one driven by the avenue's present form or by Indigenous ancestry. What the pattern does not do is separate colonial status from every other difference between the settlements that has accumulated since.

3.5 Standard Explanations

The standard determinants of educational investment account for part of the attendance gap and leave the rest. Parental human capital, household resources and credit, residential sorting, school quality, and local conditions can all sustain differences in investment (Becker and Tomes, 1986; Cunha and Heckman, 2007; Galor and Zeira, 1993; Galor, 2011; Blanden et al., 2023).

Parental human capital. Parental human capital explains at most a third of the attendance gap. In 1930 the census records literacy and not schooling, and conditioning on the household head's literacy lowers the gap among residents aged 6 to 18 from 15.7 to 13.8 percentage points, by 13 percent. In 2020, conditioning on parental schooling, household size, and migration status leaves an attendance discontinuity of 0.086 standard deviations, about 3.7 percentage points, against 0.075 standard deviations (3.2 points) without covariates (Table B.1). Near the boundary, where households on the two sides are more alike, parental schooling does not reduce the gap, and farther from it, where more differs between the sides, it accounts for about a third. In the 2015 Intercensal Survey, parental schooling reduces the gap among adolescents aged 12–18 from 8.1 to 5.1 percentage points, about a third, and adding household income produces no detectable further reduction (Table 1).[17] The gap opens in adolescence. Among children aged 6–11 with similarly educated parents it is 1.5 percentage points, and the public block tabulations of the 2020 Census place it at the transition to high school, with attendance 3.9 percentage points lower on the east side at ages 15–17 within three kilometers of the boundary and no lower at ages 6–14. Parental schooling may itself reflect the historical assignment, so these comparisons are not a mediation exercise, and they leave other forms of family transmission, such as skills or aspirations, in the gap (Cunha and Heckman, 2007; Genicot and Ray, 2020). What they show is that a substantial gap remains among children whose parents have the same schooling.

Household resources and returns to schooling. Household resources and the returns to schooling, as far as I can measure them, do not explain the remaining gap. In the Intercensal Survey, adding household size and current income leaves the adolescent attendance gap at about 5 percentage points. Household income shows no detectable discontinuity at the boundary, with an interval that admits an east-side shortfall of up to about half, and government transfers, remittances, and pensions show no detectable discontinuity either (Table B.5). In a municipality-wide comparison, the shares of workers with housing credit or retirement savings through their employment show no detectable difference between the two sides (Table B.6). Adults east of the boundary earn about 8 percent less per month, and the gap falls to 1 percent and loses significance once schooling is

[17] Because the Intercensal sample thins quickly near the boundary, I estimate a parametric east–west gap with side-specific linear trends in distance, so the coefficient on the east-side indicator is the fitted gap at the boundary. This specification relies on households farther from the creek, so it is more exposed than the local BD design to sorting, provision differences, and functional form.

Table 1: Human capital and income gaps in Guadalajara (2015)

	Years of schooling	$\mathbb{1}\{Attends\ school_i\}$						Monthly income (log)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
$\mathbb{1}\{East_i\}$	-1.31***	-0.052***	-0.081***	-0.054**	-0.051**	-0.052**	-0.015*	-0.084***	-0.010	0.075
	(0.118)	(0.015)	(0.024)	(0.024)	(0.023)	(0.025)	(0.008)	(0.027)	(0.024)	(0.068)
Years of schooling (YoS)									0.065***	0.068***
									(0.001)	(0.005)
$\mathbb{1}\{East_i\}\times$ YoS										-0.006
										(0.006)
Parental YoS				0.021***	0.018***	0.019***	0.003***			
				(0.001)	(0.001)	(0.001)	(0.0006)			
Household income (log)						0.0002	0.0000			
						(0.008)	(0.002)			
Sample (age in years)	6+	6–18	12–18	12–18	12–18	12–18	6–11	18+	18+	18+
Individual controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Geographic controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Household size					✓	✓	✓			
Observations	74,779	17,463	9,922	9,916	9,916	9,041	8,059	31,584	31,478	31,478
Dep. var. mean	9.18	0.884	0.807	0.807	0.807	0.803	0.984	8.65	8.65	8.65
Dep. var. SD	4.71	0.320	0.395	0.395	0.395	0.398	0.127	0.644	0.645	0.645
R^2	0.443	0.181	0.127	0.172	0.183	0.193	0.020	0.123	0.268	0.269

Notes: OLS estimates from individual-level data in the 2015 *Encuesta Intercensal* for the municipality of Guadalajara. The treatment indicator $\mathbb{1}\{East_i\}$ equals one if individual i resides east of the historical SJdD creek (today Calzada Independencia). *Individual controls* include sex and age fixed effects. *Geographic controls* include block longitude, latitude, log distance to the central business district, signed distance to the historical boundary, and the interaction of distance to the boundary with the east-side indicator. The first column reports years of schooling for individuals aged 6+. The next six columns report the probability of attending school for the indicated age range. The fourth through seventh columns additionally control for the maximum of mother's and father's years of schooling; the fifth through seventh for household size; and the sixth and seventh for log household income. Household income is the sum of the labor income of all household members aged 12 and older, so it includes the earnings of adolescents who work. The final three columns use log monthly income (2015 MXN) as the dependent variable and restrict the sample to adults aged 18+. Standard errors clustered at the household level are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels.

↩ Back to text

held fixed. The return to a year of schooling is not detectably different on the two banks, as one would expect in a single labor market (Aldeco et al., 2024). None of these measures captures wealth, liquidity, or the credit constraints of earlier generations, so those channels remain open.

Housing-market sorting. Cheaper housing on the east side does not account for the gap, although sorting on characteristics I do not observe remains possible. If housing prices capitalize local amenities or the stigma of the boundary, households with different resources or preferences may sort across it, as Baldomero-Quintana et al. (2025) document for Mexico City. Parcel-level cadastral data show a clear west–east gradient in land values (Figure B.4a), which correlate strongly with online housing listings (Figure B.4b). Re-estimating the schooling discontinuity at the block level while controlling for land values yields gaps from 0.80 to 0.97 years, close to the 0.79-year individual-level estimate for residents aged 6 and older (Tables B.7 and B.8 and Figure B.5).[18] Land values may themselves capitalize the boundary's history, and they may

[18]This exercise covers schooling only because the public block-level census suppresses blocks with few households, preventing a comparable attendance measure. Within the block-level specification, the standardized gap falls from

also reflect amenities and sorting that the census does not record, so conditioning on them can remove part of the effect or introduce bias in either direction. The estimates with land-value controls are a sensitivity check, and the gap survives it.

Residential turnover. The one episode in which the composition of part of the east side changed abruptly shows no negative selection on schooling into it. In April 1992, explosions along the Reforma collector destroyed about 1,400 houses along more than eight kilometers of streets, part of them in Analco, whose parish lost about 800 families (Reguillo, 1998). In 2000, adults in the rebuilt blocks had almost one more year of schooling than adults in nearby blocks of Analco, an advantage absent at the coarser geography of 1990 (Figure B.6). The advantage is gone by 2010, and in 2020 adult schooling in the rebuilt blocks is 0.34 years lower, a difference significant only at the 10 percent level. The age composition of the blocks changed over the same decade, and completed schooling cannot fall by a year among the same adults, so the reversal reflects a new turnover of residents. Educated households entered the east when new housing became available, and the census, which follows places and not households, cannot say on which characteristics later residents sorted.

School supply and quality. Households near the boundary face overlapping school choice sets, with broadly comparable measured provision and no residential catchments. Schools have no residential catchments and attendance is measured at the household, not at the school, so a gap in attendance need not show up as a gap in anything the schools report. The number of schools within a 30-minute walk, of every sector and shift, is not detectably different on the two sides, the few differences by level go in both directions, and students per teacher are higher on the east side, by 0.89 without weights ($q = 0.001$) and 0.45 with weights ($p = 0.112$) (Table B.9). Upper-secondary schools within reach, weighted by residents aged 15 to 17, show no detectable difference either (1.6 more on the east side, $p = 0.37$; 1.7 more without weights, $p = 0.085$), and the nearest one is 65 meters farther on the east side (78 meters with the same weights), under a minute on foot (Table B.9). Administrative records show lower approval rates, higher repetition, and larger classes on the eastern bank through about 2002, and differences detectable before the early 2000s are not detectable after (Figures B.7 to B.10). ENLACE scores are measured among students still enrolled, the operational indicators cover lower secondary, and neither links children to the schools they attend, so the evidence at the upper-secondary transition is thinner than the evidence of comparable provision and does not fully exclude differences in schooling constraints there. The 2020 attendance gap concerns cohorts schooled after these differences stopped being detectable, and the earlier quality differences are one of the historical channels bundled into the stock of schooling. Children from both sides can attend the same schools, although in the survey fewer than one in ten on either side cross the Calzada to do so, and most attend a school in their

−0.62 to between −0.40 and −0.48 standard deviations.

own *colonia* or very near it (Table B.10). These shares come from the 239 surveyed households and not from the census, and they describe where children go and not the quality of the schools they attend. Where children go differs between the sides at primary and lower-secondary ages, with 89 and 78 percent of east-side children attending a school in their *colonia* or very near it against 69 and 52 percent on the west side, and shows no detectable difference at upper secondary, where 66 and 64 percent do among 46 students, although the estimates by level are imprecise. The difference in where children attend lies at the ages where attendance shows no detectable difference, and it is not detectable at the transition where the gap opens. These findings weigh against an explanation based on fewer nearby schools or a systematic disadvantage in measured school conditions.

Crime exposure. Recorded crime per hectare shows no detectable discontinuity at the boundary in any category (Table B.11 and Figure B.11). This holds for homicide and vehicle theft, the categories that do not depend on victims reporting to the police.

Street infrastructure, urban form, and businesses. Street infrastructure is not worse east of the boundary and, where block fronts differ, the eastern side has more paving, sidewalks, and ramps (Table B.12). Residents and dwellings per block and parcel size show no detectable difference, while east-side blocks are larger, with more parcels, more persons per dwelling, and fewer residents per hectare (Table B.13), and establishments east of the boundary are smaller, with no detectable difference in sector mix (Tables B.14 and B.15).

What remains. Of the features that can divide a place, first-nature ones like the creek, second-nature ones like schools and infrastructure, and endogenous ones that depend on what residents do today (Lin and Rauch, 2022; Aliprantis and Lin, 2026), the first two no longer divide the two banks detectably, and what follows asks whether the third does. Parental human capital, household resources, the returns to schooling, and current school provision leave part of the attendance gap unexplained. They are not unimportant, since parental education accounts for up to a third of the gap and historical provision is embodied in today's stock of schooling. What they leave is a gap in a decision, among children whose parents have similar schooling and who face comparable measured school provision, and Section 4 turns to the parents who make it. Why that gap was much larger in 1930 and has not narrowed detectably since 2000 is a question the census cannot answer on its own, and Section 5 returns to it.

3.6 Robustness

The sign of the gaps is stable across polynomial order, kernel, variance estimator, and covariate set, and their precision varies, with p-values for attendance between 0.03 and 0.25 across specifications (Table B.1).

Migration. In 2020, 91.5 percent of Guadalajara residents lived in Jalisco five years earlier, and excluding inter-state migrants yields nearly identical estimates (Table B.1). The census follows places rather than households, so it cannot bound the selection of residents who left; Appendix B.13 reports a proxy for residential persistence and its limitations.

Donut estimates. Residential density jumps at the boundary, so the innermost observations come mostly from the east side. Dropping every resident within 50, 100, or 150 meters of the line leaves the schooling gap between −0.10 and −0.21 standard deviations and the attendance gap between −0.09 and −0.13 (Table B.17).

Spatial autocorrelation and placebo boundaries. Following Müller and Watson (2024), and using the package by Becker et al. (2025), I reject a spatial unit root and fail to reject spatial stationarity, supporting local BD estimators without spatial differencing. Randomly shifting and rotating the boundary 1,000 times within the municipality (Figure B.12) yields a placebo distribution centered at zero, with the true-boundary estimate in the left tail; following Borusyak and Hull (2023), a recentered specification makes the true-boundary discontinuity larger, not smaller (Figure B.13).

4 The Neighborhood and the Demand for Education: Survey and Experimental Evidence

4.1 Qualitative and Survey Evidence on the Neighborhood

Ethnographic evidence. The idea that neighbors could enter the decision to educate a child comes from the ethnography of Mexican cities. In a Mexico City settlement, Adler Lomnitz (1977) found that households survived recurrent periods without income through exchange networks among neighbors, sustained by trust and physical proximity, and that families whose incomes rose stopped asking for help and eventually moved away.

In working-class Guadalajara of the 1980s, household survival depended on pooling the work of several members, with children starting to work around age 15, which halted their schooling (González de la Rocha, 2001, pp. 79–80), and on support from relatives and neighbors (González de la Rocha, 1994). This support is costly to keep, since households must be available when others need help and have resources to reciprocate, and the poorest are often left out (González de la Rocha, 1999). On the east side of Guadalajara, residents describe the same arrangement and attribute to their neighbors and their *barrio* the difficulty of getting ahead (Jaramillo-Molina, 2022).

Trust in neighbors today. A representative survey of the city says the same in numbers. Residents east of the boundary trust their neighbors more and formal institutions less, and among them, trusting neighbors goes with less schooling. The 2022 Life Quality Survey interviews 289 residents in 23 census tracts across the municipality,[19] and I estimate discontinuities at the boundary with a separate linear distance gradient on each side (Table 2).

Residents east of the boundary are 37 percentage points more likely to report high trust in their neighbors, and show no detectable difference in trust in family, coworkers or classmates, or people on the street, so the difference I can detect is with neighbors. They also report lower trust in formal institutions, by 0.23 on a 0–1 index, consistent with exclusion shifting social life inward (Table C.1). Membership in neighborhood, school, political, religious, or assistance organizations shows no detectable difference, so the difference I can detect is in informal trust toward neighbors and not in formal participation (Table C.2). Life satisfaction, quality of life, and happiness show no detectable difference either (Table C.3). These patterns were available before I designed my own parent survey, and they suggested that social life on the east side is organized around neighbors in a way that could compete with educational investment.

Table 2: Trust, participation, and schooling at the boundary: Life Quality Survey (2022)

	Trust in				Institutional trust index	Member of					Well-being index	Years of schooling
	Neighbors	Family	Coworkers or classmates	People on the street		Neighborhood assoc.	School-related org.	Political org.	Religious group	NGO or assistance org.		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Panel A. Without schooling control												
$\mathbb{1}\{East_i\}$	0.372**	0.072	0.134	0.049	−0.235***	0.066	−0.039	0.023	−0.076	−0.062	0.033	−0.798
	(0.149)	(0.110)	(0.158)	(0.129)	(0.064)	(0.056)	(0.050)	(0.041)	(0.085)	(0.063)	(0.059)	(1.23)
Trust in neighbors												0.796
												(0.691)
Trust in neighbors × $\mathbb{1}\{East_i\}$												−1.74*
												(0.945)
Age-by-sex FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Distance to boundary	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Schooling control												
Survey weights	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
FDR q-value	0.035	0.390	0.329	0.481	0.004	0.268	0.347	0.399	0.329	0.328	1.000	
Observations	289	289	289	289	289	289	289	289	289	289	289	289
Dep. var. mean	0.655	0.912	0.585	0.217	0.229	0.041	0.018	0.036	0.065	0.018	0.810	11.33
R^2	0.089	0.051	0.020	0.084	0.093	0.033	0.034	0.029	0.068	0.050	0.041	0.234
Panel B. Controlling for years of schooling												
$\mathbb{1}\{East_i\}$	0.368**	0.084	0.165	0.035	−0.232***	0.080	−0.032	0.032	−0.064	−0.057	0.050	
	(0.151)	(0.111)	(0.156)	(0.129)	(0.065)	(0.057)	(0.050)	(0.044)	(0.087)	(0.063)	(0.058)	
Age-by-sex FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Distance to boundary	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Schooling control	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Survey weights	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
FDR q-value	0.041	0.404	0.295	0.480	0.006	0.200	0.404	0.404	0.404	0.361	1.000	
Observations	289	289	289	289	289	289	289	289	289	289	289	
Dep. var. mean	0.655	0.912	0.585	0.217	0.229	0.041	0.018	0.036	0.065	0.018	0.810	
R^2	0.089	0.057	0.036	0.089	0.094	0.054	0.046	0.037	0.079	0.054	0.080	

Notes: Estimates from the 2022 Life Quality Survey of *Jalisco Cómo Vamos*, representative of Guadalajara residents (289 respondents in 23 census tracts). The outcomes in columns (1)–(4) are indicators for reporting high trust in each group, and those in columns (6)–(10) are indicators for membership in each type of organization. The outcome in column (11) is the mean of three self-reported items, satisfaction with life, quality of life, and happiness, each on a five-point scale rescaled to the unit interval (Table C.3). Each column regresses the outcome on an indicator for residence east of the historical boundary and separate linear distance gradients on each side, with age–sex fixed effects and survey weights. Panel B adds a linear control for the respondent's years of schooling. Column (12) adds trust in neighbors and its interaction with the east-side indicator. The estimates extrapolate distance gradients toward the boundary from tracts at a median distance of 3.7 kilometers. Sharpened q-values (Anderson, 2008) adjust for the 26 trust and participation outcomes of the survey within each panel; the q-value in column (11) adjusts for the four well-being outcomes of Table C.3. Dep. var. mean is the survey-weighted mean of the outcome. Robust standard errors are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

↩ Back to text

[19] The survey is conducted by the NGO *Jalisco Cómo Vamos*.

4.2 A Survey of Parents Across the Boundary

The census does not record how much households rely on their neighbors or what parents expect from their children's education. To measure both, I collected new field data from parents and primary caregivers living within three kilometers of the historical boundary with at least one child aged 6–20.[20] The sample includes 239 households, 193 east of the boundary and 46 west, an allocation pre-specified to increase power on the historically excluded side. Sampling was stratified by side, and within each side I selected census blocks with residents aged 6–20 in the 2020 census, and enumerators screened dwellings in those blocks. The mean distance to the boundary is 1.5 km and the median 1.7 km. The sample reproduces the adult schooling gap of its own census blocks, since respondents on the east side have 0.77 fewer years of schooling, against 0.75 among residents aged 15 and older of the same blocks in the 2020 census.

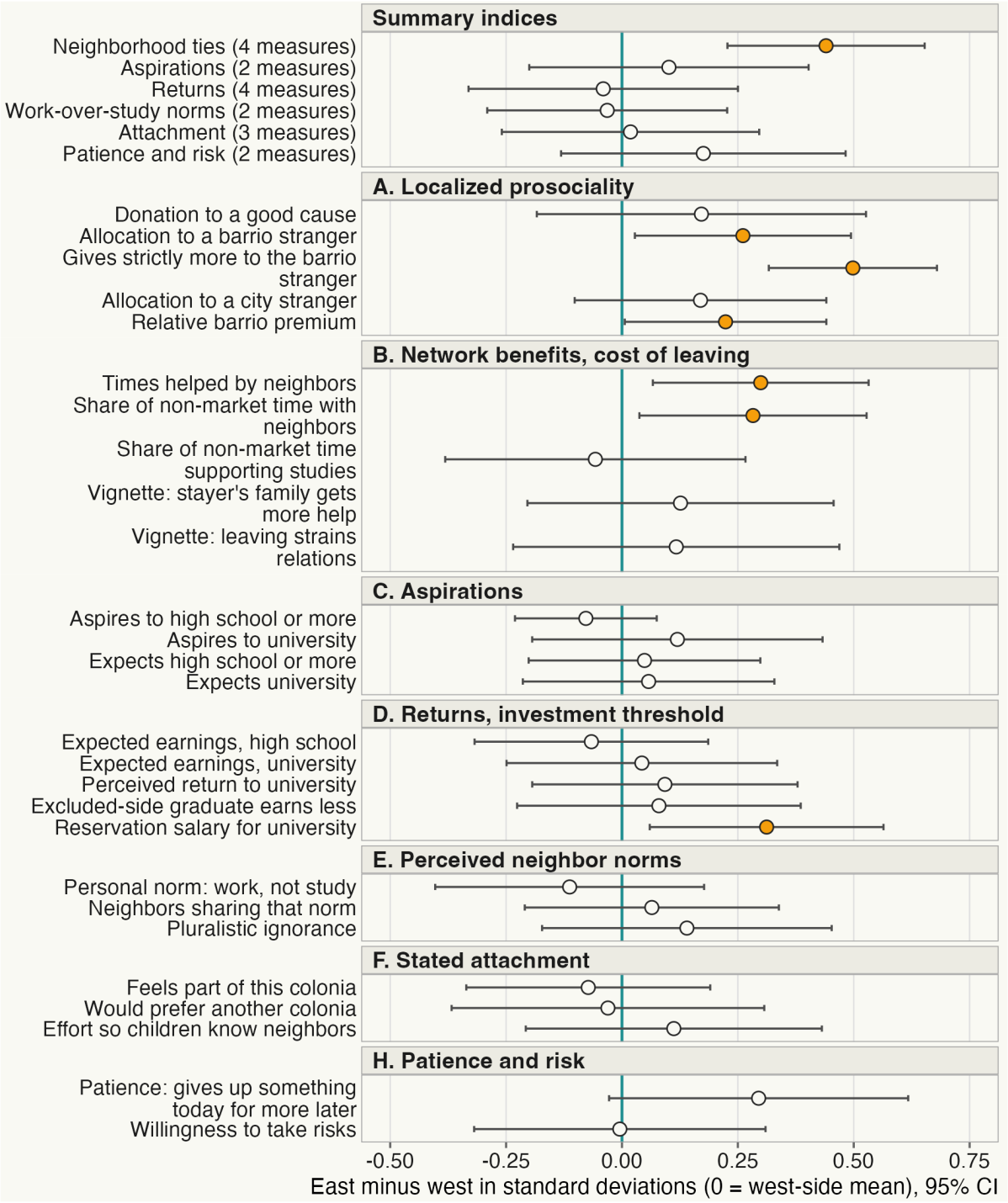
The survey has three parts, and the framework of Section 5 informed all three. An experimental block elicits demand for an educational investment while randomly varying whether the place where parents live has just been brought to mind (Section 4.4). Modules that follow it measure what the framework says should differ across the boundary if exchange with neighbors raises the opportunity cost of schooling, that is, how much households rely on their neighbors and the return they require to invest, and what should not differ, that is, how much parents value education. A second randomized exercise activates the history behind the place before respondents explain why young people stop studying (Section 4.7).

4.3 What Differs Between the Sides

The modules were measured after the experimental block, so the comparisons control for Map-First assignment, which does not change the measures of exchange with neighbors, and for respondent sex, age, schooling, and randomized question versions, and they are observational. Figure 7 reports each comparison in standard deviations, with the west-side mean as the reference, and Tables C.4 and C.5 report them in levels. The clearest differences between the sides are in exchange with neighbors. An index of neighborhood ties that combines help received from neighbors and time spent with them is 0.44 standard deviations higher on the east side. The two vignettes closest to the cost of leaving, that the stayer's family gets more help and that leaving for university strains relations, show no detectable difference across sides ($q = 0.63$), with 46 west-side households, and the substitution channel rests on the experiment rather than on them (Table C.4).

[20] Fieldwork began on May 9, 2026 and was carried out by Descifra Investigación Estratégica using SurveyToGo. The study was approved by the Paris School of Economics Institutional Review Board (approval 2026-015) and pre-registered in the AEA RCT Registry (AEARCTR-0018600) on May 14, 2026. Twenty-three households were interviewed before that date, and excluding them leaves the results unchanged (Appendix D.3). Every eligible household contacted participated, and all received MXN 100 regardless of their choices; the 17 households interviewed before the payment was introduced on May 13 were compensated retroactively.

Figure 7: Differences between the sides in the survey modules



Notes: Each row reports the coefficient on the east-side indicator for the outcome minus its west-side mean, divided by its standard deviation over both sides, so zero is the west-side mean. The specification is that of Table C.4: Map-First assignment, respondent sex, age and age squared, randomized question versions where applicable, and respondent-education fixed effects, with the focal child's sex, age, and school attendance for the aspiration rows. Outcomes that Table C.5 estimates by Poisson pseudo-maximum likelihood are estimated here by OLS in levels. Each summary index is the first principal component of its measures, standardized over the 239 households. Filled markers have sharpened false-discovery-rate q-values below 0.10 within their panel, and the bars are 95 percent confidence intervals with heteroskedasticity-robust standard errors. The modules were measured after the experimental block, and all comparisons are observational.

↩ Back to text

Reciprocal exchange and the cost of university. East-side respondents report receiving help from neighbors about twice as often during the previous year (0.30 standard deviations), and, conditional on work hours, time with neighbors accounts for a larger share of non-market time, about 7 percent against 4 percent on the west side (0.28 standard deviations).[21] Among the children who attend school, 78 percent on the east side and 59 percent on the west go to a school in their own *colonia* or very near it, and fewer than one in ten on either side crosses the Calzada to attend (Table B.10). At upper secondary the shares are similar, 66 and 64 percent, and schooling closer to home on the east side is what reliance on the neighborhood would predict for where a child is sent to school. Parents on the two sides report similar expected earnings and perceived university premia, but east-side respondents require an 18 percent higher salary at age 30, MXN 5,319 per month (0.31 standard deviations), before they consider university worthwhile (Figure C.1). The returns they expect are not detectably different, so the bar they set is higher for a reason that the expected returns do not capture, and the question does not say what it is. Conditioning the gap on the neighborhood-ties index lowers it by about a sixth, with an interval that includes zero, while the location and income measures the survey has leave it unchanged (Table C.6).

Giving within the neighborhood. In an allocation exercise adapted from Enke et al. (2023) and Cappelen et al. (2025), respondents divide MXN 500 between a stranger from their own *barrio* and an equally well-off stranger elsewhere in Guadalajara. East-side respondents give about MXN 20 more to the neighborhood stranger (0.26 standard deviations), and overall giving shows no detectable difference, so what differs detectably is the weight of the neighborhood and not generosity.[22]

What does not differ detectably. Parents on both sides report similarly high aspirations for upper-secondary school and university, similar perceived returns to education, similar norms about working rather than studying, and similar feelings of belonging. Indices that summarize each of these four groups of answers, built in the same way as the index of neighborhood ties, differ between the sides by 0.10 standard deviations or less (Figure 7). East-side respondents are equally willing to take risks and more patient, by 0.29 standard deviations, a difference that would raise investment on the east side, and an index of the two shows no detectable difference between the sides. Several measured household characteristics, including income, size, parental schooling, years of residence, and labor-market informality, show no statistically detectable difference between the sides after adjustment for multiple comparisons (Table C.7).

[21] Some east-side respondents define the neighborhood as the area "where people know me and I know the neighbors" or "as far as people in the barrio know each other"; most answers on both sides refer to streets, landmarks, and routes. Appendix C.2 reports the answers and shares by side.

[22] The equal split is modal on both sides; no west-side respondent gives the *barrio* stranger more than half, against 17 percent on the east side (0.50 standard deviations).

These non-rejections do not establish that the two samples are alike, since some estimates remain compatible with economically meaningful differences, and respondents' own schooling, measured in Section 4.2, is 0.77 years lower on the east side. Family in the neighborhood differs little, with 19.6 percent of close relatives living in the *colonia* on the east side against 14.1 percent on the west, and neither this nor the share of respondents whose parents lived there shows a detectable difference. The ties that differ detectably are with neighbors and not with kin, which separates the arrangement from the strong family ties of Banfield (1958) and Alesina and Giuliano (2014). Aspirations and expected returns show no detectable difference across sides, with intervals that admit differences of up to about 0.4 and 0.3 standard deviations and small point estimates (Table C.4), so these comparisons offer little support for accounts of the gap in Section 3 based on lower aspirations, different perceived returns, explicit norms, or detectable differences in socioeconomic composition, and they do not say which feature of the environment matters.

4.4 The Experiment: Prediction and Design

The prediction. Reciprocal exchange with neighbors is an economic arrangement. Neighbors look after each other's children, lend money and tools, share information about jobs, and step in during illness or emergencies, and they do so for households that remain available to do the same (Kranton, 1996; González de la Rocha, 1999).[23] Nothing in it requires altruism, a local identity, conformity, or a larger stock of social capital.

Schooling can compete with this arrangement. The competition grows in adolescence, when children begin to contribute to their households (Adler Lomnitz, 1977; González de la Rocha, 1999). The sign is not given in advance, because neighbors can also support schooling with information and norms (Coleman, 1988), and the framework of Section 5 treats education and exchange as substitutes or complements depending on which of the two dominates. In the Life Quality Survey, trust in neighbors goes with less schooling east of the boundary and not west of it (Table 2, column 12; Table C.8), which points to substitution on the east side.[24]

What schooling puts at risk. The ethnography of Section 4.1 points to what schooling puts at risk. Networks favor stable households with enough members to sustain their involvement, and they discriminate against mobile and newly arrived ones (González de la Rocha, 1999, pp. 21–22). Without enforcement, less exchange can be sustained with members likely to leave (Kranton, 1996; Ligon et al., 2002; Laczó, 2015). Young members also supplied the daily exchange with

[23] In Ahmedabad, households offered subsidized housing on the city's periphery lost access to informal insurance and borrowing, and many either did not move or later returned to central slums (Barnhardt et al., 2017).

[24] The closest precedent is Munshi and Rosenzweig (2006), where caste-based occupational networks affect schooling choices by raising the value of remaining connected to traditional occupations. Here the network is defined by residence rather than caste, its benefits arise from reciprocal exchange with neighbors rather than occupational referrals, and the experiment activates the place where the network lives rather than varying its value.

neighbors and relatives that households lost when they emigrated (González de la Rocha, 2001, p. 90), and parents resented the children who did better, moved to better *colonias*, and saw them less (González de la Rocha, 1999, p. 26). A child in school is less available, contributes later to what the household can reciprocate, and is more likely to move up and away, so schooling has an opportunity cost in exchange, borne by the household through its children.

The experiment tests this in the simplest way the framework allows. If educating a child puts some of the exchange with neighbors at risk, and parents do not always have the neighborhood in mind when they choose, then bringing it to mind should lower their demand for education where exchange is valuable and leave it unchanged where it is not. The first pre-registered hypothesis, which the registration calls neighborhood salience, is that bringing the neighborhood to mind lowers demand for tutoring. The second, that the response is concentrated on the historically excluded side, follows if exchange is worth more there, as the survey suggests. The experiment varies what parents have in mind, not the value of exchange itself, and Section 5 states the condition behind this prediction and what else it implies. An hour of tutoring is a marginal educational investment and not continued enrollment, so it need not carry the same neighborhood cost as schooling. What the experiment can show is whether bringing the neighborhood to mind lowers demand for an educational good where exchange is valuable, and the size of the cost that matters for schooling comes from the survey.

Activating the place. Work on activated history shows that a historical event can lie dormant in behavior until something brings it back to mind, and that its effect then appears where the event left a mark (Fouka and Voth, 2023; Ochsner and Roesel, 2024; Cantoni et al., 2026; Ghidoni and Schindler, 2026). The treatment activates a place instead, the neighborhood where the respondent lives, which on the east side is the place the colonial assignment marked and where Section 3 finds the schooling gap. It changes the order of two tasks. In the mapping exercise, respondents view a satellite map centered on their residence and trace the area they consider their neighborhood (*colonia* or *barrio*), following McCartan et al. (2024). A neighborhood, in Galster's sense, is the space over which what others do changes the well-being one gets from where one lives (Galster, 2019; Aliprantis and Lin, 2026), and the exchange with neighbors is one such externality. The map asks parents to draw the reach of that space as they perceive it. In the *Map-First* arm, they map immediately before the educational choice; in *Map-After*, immediately afterward. Assignment is randomized within side. The mapping exercise mentions neither education nor the historical boundary, so the treatment changes whether respondents have just thought about their neighborhood, holding information and material incentives fixed (Figure D.1).

Educational demand. Respondents make eleven choices between one hour of online mathematics tutoring for one of their children and an electronic gift card from MXN 0 to MXN 250 in MXN 25 increments, following the multiple-price-list design of Jack et al. (2022). The tutoring

session is the same for every participant on both sides of the boundary, so its quality is not a margin of the decision, nor does it depend on the quality of local schools, and the experiment is unaffected by what the school measures of Section 3.5 leave open. Row order and the side on which tutoring appears are randomized, one row is randomly selected to bind, and one in ten households receives the tutoring session or gift card implied by its choice.[25] The pre-registered outcomes are the share of choices in which tutoring is selected and latent willingness to pay for tutoring.

4.5 Activating the Place: Effects on Educational Demand

Activating the place lowers educational demand on the east side of the boundary, as Figure 8 shows. I estimate effects separately by side with enumerator and survey-day fixed effects, distance to the boundary with a separate slope on each side, controls for the randomized presentation of the choice list, and LASSO-selected controls, and I permute Map-First assignment within side for inference (Table D.1).[26]

On the east side, Map-First lowers the tutoring share by 12.3 percentage points, from a Map-After mean of 0.573 to 0.450, about one fifth. A model of the choices puts the fall in willingness to pay at MXN 75, 42 percent of its east-side Map-After mean, and most of that fall comes from what the model imputes beyond the price grid (Table D.2). Treating never-switchers as censored gives 32 percent. The effects on the share and on latent willingness to pay correspond to 0.33 and 0.30 standard deviations, and restricting the sample to the historical Analco area produces nearly identical estimates.[27] The difference with the west side has a randomization-inference p-value of 0.037 for the tutoring share and 0.051 for willingness to pay. Without the prime, demand shows no detectable difference across sides, and the comparison is imprecise, with 95 percent intervals from −42 to 44 percentage points for the tutoring share and from MXN −357 to 320 for willingness to pay (Table D.1, columns 2 and 5).

On the west side, both estimates are positive and imprecise; with 46 households, and with Map-First households there having lived in their neighborhood 10.6 fewer years than Map-After households, I do not interpret their sign.

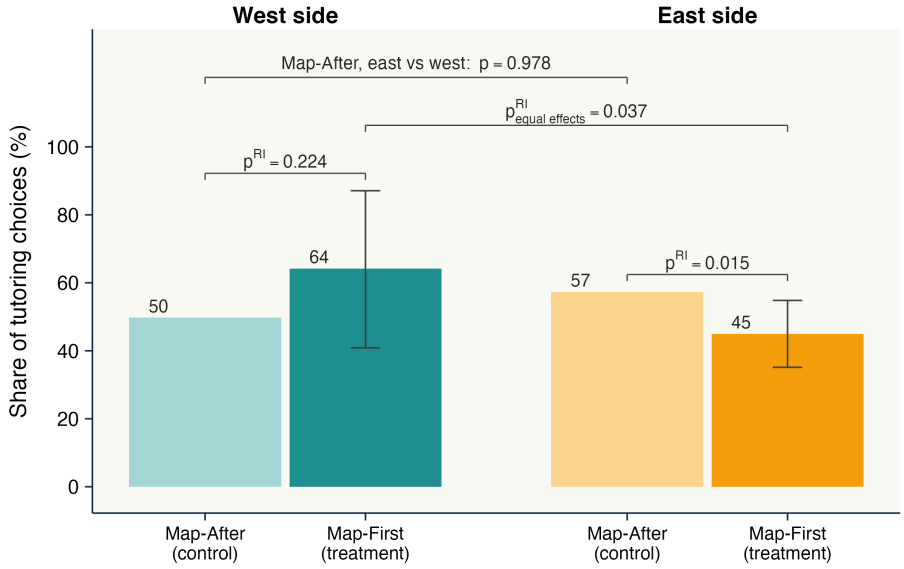
What the experiment establishes is that the place enters the decision on the east side when it is on parents' minds. It does not say why the response differs across sides or how much of the attendance gap in Section 3 it accounts for; Section 4.6 says which dimension of the neighborhood

[25] Before the task, respondents practice the same mechanism using a pen and quantities of marzipan candies and answer four comprehension questions; the mean score is 3.8 out of 4. Details are reported in Appendix D.

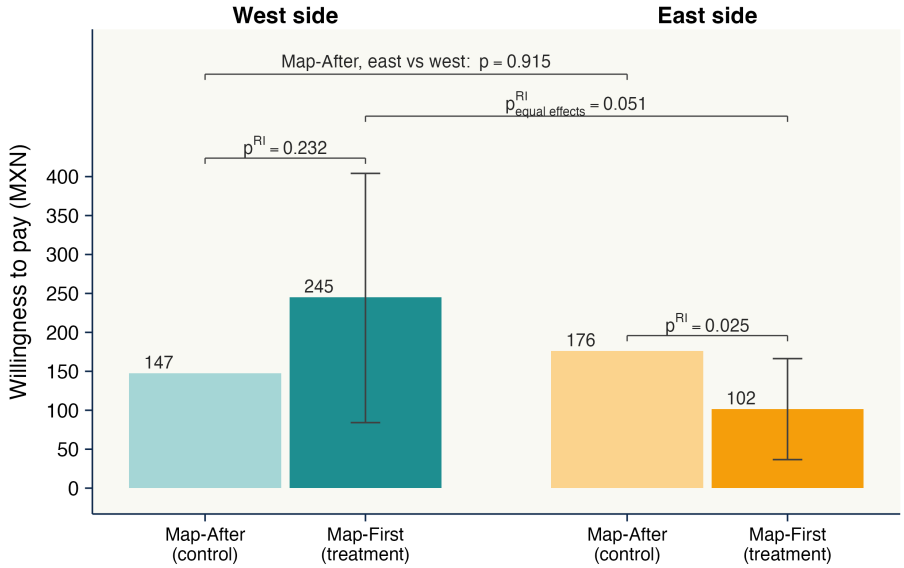
[26] Latent willingness to pay is recovered using the two-step estimator of Jack et al. (2022): a scaled random-effects probit of the ten binary choices with a positive gift-card value, followed by a regression of latent WTP on treatment assignment. Standard errors bootstrap both steps at the respondent level.

[27] These magnitudes are close to the average effect of primes on overt behavior in the meta-analysis of Dai et al. (2023), $d = 0.38$ across 867 effects, and above the estimates that correct for publication bias, between 0.20 and 0.28. Unlike most studies in that literature, the experiment was pre-registered and its outcome is an incentivized choice.

Figure 8: Effects of activating the place on educational demand



(a) Share of tutoring choices



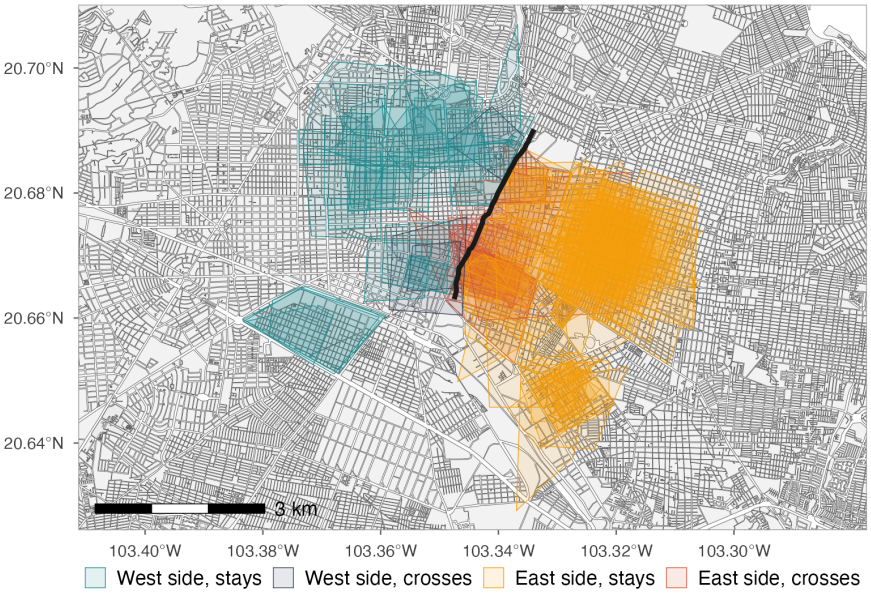
(b) Willingness to pay

Notes: Light bars report unadjusted Map-After means; dark bars add the regression-adjusted Map-First effect to those means and are therefore not raw Map-First means. Panel (a) reports the share of the eleven choices in which tutoring is selected, and Panel (b) latent willingness to pay in MXN. Whiskers are 95 percent intervals obtained by inverting the randomization-inference test. Brackets report randomization-inference p-values for the side-specific effects and for their difference (Appendix D.1), and the bracket over the Map-After bars reports the p-value of the coefficient on the east-side indicator in columns (2) and (5) of Table D.1, the regression-adjusted difference between the sides without the prime. Treatment labels are permuted 50,000 times within side, holding the number treated fixed. Regressions include enumerator and survey-day fixed effects, distance to the boundary, controls for choice-list presentation, and LASSO-selected controls.

↩ Back to text

the response goes with, and Section 5 uses the survey to put a size on the cost and the experiment to bound it from below. Assignment is randomized within each side and residence is not, so the contrast between sides identifies a difference in the map's effect between the two resident populations, and heterogeneity in that effect correlated with side, through characteristics the survey does not measure, cannot be ruled out. Tutoring is a marginal investment valued against gift cards, so activating the place could lower the value of tutoring or raise the value of the gift card, for instance if the card helps meet obligations with neighbors. Both readings make education more costly, and neither the experiment nor the reservation salary separates them (Section 5.5).

Figure 9: Neighborhoods traced by respondents, relative to the historical boundary



Notes: Each polygon represents the neighborhood delineated by one respondent in the interactive mapping exercise, drawn as the convex hull of the traced vertices. The dark line denotes the historical San Juan de Dios creek boundary, now Calzada Independencia. Polygons are colored by the respondent's own side of the historical boundary and by whether the traced neighborhood crosses it: teal for west-side respondents whose drawing stays on their side and navy for those whose drawing crosses, amber for east-side respondents whose drawing stays and brick for those whose drawing crosses. Polygons with obvious drawing errors, such as a misplaced distant vertex, are omitted.

↩ Back to text

Figure 9 shows the places participants drew. Most drew a compact area around their home, with a median of about half a square kilometer. The drawing contains the home for 88 percent of participants and crosses the boundary for 11 percent. Their descriptions refer mostly to the physical extent of the area and its everyday use (Tables D.3 and D.4). The place the map activates is the respondent's own side of the boundary, and the boundary itself is activated by a separate exercise (Section 4.7).

4.5.1 Robustness

The east-side result is not driven by imbalance, misunderstanding of the choice mechanism, a few respondents, or the measure of willingness to pay (Appendix D). The arms show no detectable difference in the 25 characteristics recorded before the randomized block, ten of them income brackets, after false-discovery-rate adjustment, comprehension is nearly identical across arms, the estimate is stable when any single household is omitted, and a bounded area-under-demand measure gives the same result (Tables D.2 and D.5 to D.7 and Figure D.2). On the west side, Map-First households have lived in their neighborhood 10.6 fewer years than Map-After households, so the pooled balance does not stand for balance within each side.

Thirty of 239 respondents inferred that the study linked neighborhoods to education, and Map-First raises this inference by 22 percentage points on the west side, an increase that is not significant, and by close to zero on the east side, where the behavioral effect appears. The east-side effect is stable under the post-treatment demand checks. The cross-side contrast, including its western component, moves by 4.2 percentage points in the tutoring share when guessed-theme controls are added and by 0.6 when respondents who guessed the mechanism are excluded, with randomization-inference intervals that include the headline contrast (Table D.8). These checks assess robustness to observed recognition of the study's theme, and they do not establish that the contrast is unaffected by experimenter demand, since guessing is measured after treatment and after the choices.

Within the east side, the response does not vary detectably with income or parental schooling, and if anything it is larger among households above the median income and with a parent who completed high school (Table D.9), with the limited power that Appendix D.5 reports. Taking part in exchange requires a base of income, and the poorest households are often left out of it (González de la Rocha, 1999, pp. 20–22). Demand falls only when the place has just been brought to mind, and not more among the poorest, which a stable trait acting on the level of demand would not produce.

4.6 Neighborhood Ties and the Demand for Education

If educating a child puts exchange with neighbors at risk where that exchange is valuable, households that rely more on their neighbors should demand less education on the east side and not on the west, as activating the place does. The measure is the index of neighborhood ties of Figure 7, the first principal component of the z-scores of the times the household received help from neighbors with child care, with loans, and with finding work, and of the time spent with neighbors relative to time at work. A nine-measure version that adds the two allocations, the two vignettes, and the reservation salary gives the same east-side slopes (Table D.10).

Table 3 reports the slope of the demand for tutoring on the index on each side, controlling for Map-First assignment and the controls of the main specification. On the east side, one

Table 3: Neighborhood ties and the demand for tutoring

	Share of tutoring choices (pp)										
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Neighborhood-ties index	−4.0**								−4.8**	7.0	11.0
	(1.9)								(2.0)	(13.6)	(12.8)
Aspirations index		−1.0							−2.4		1.9
		(2.6)							(2.9)		(9.4)
Returns index			1.3						1.5		16.7*
			(2.8)						(3.1)		(8.2)
Work-over-study norms index				0.7					0.5		4.2
				(2.7)					(2.7)		(8.4)
Attachment index					2.1				2.7		−8.9
					(2.4)				(2.5)		(8.6)
Patience and risk index						1.2			1.5		0.5
						(2.4)			(2.4)		(7.1)
Socioeconomic index							0.5		1.0		−21.4**
							(2.9)		(4.1)		(10.0)
Human capital index								−0.2	−0.3		14.9
								(2.8)	(3.7)		(8.8)
Sample	East side	East side	East side	East side	East side	East side	East side	East side	East side	West side	West side
Enumerator FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Day FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
LASSO controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Distance to boundary	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Map-First	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
East − West difference in the index (SD)	0.44***	0.10	−0.04	−0.03	0.02	0.18	−0.11	−0.18		0.44***	
	(0.11)	(0.15)	(0.15)	(0.13)	(0.14)	(0.16)	(0.16)	(0.15)		(0.11)	
p-value: east-side slope = west-side slope	0.070	0.661	0.137	0.795	0.847	0.715	0.898	0.705	0.027		
Observations	193	193	193	193	193	193	193	193	193	46	46
Dep. var. mean	53.8	53.8	53.8	53.8	53.8	53.8	53.8	53.8	53.8	59.3	59.3
R^2	0.403	0.393	0.393	0.393	0.395	0.393	0.392	0.392	0.412	0.554	0.666

Notes: The dependent variable is the share of the eleven price-list choices in which the respondent chooses tutoring, in percentage points, and each coefficient is per standard deviation of the index. Columns (1) to (9) use east-side households and columns (10) and (11) west-side households; columns (9) and (11) include the eight indices together. Every column includes enumerator and survey-day fixed effects, Map-First assignment, the signed distance to the boundary, and the LASSO-selected controls of Table D.1. The row East − West reports, for the index that enters the column alone, the difference between the sides with the specification of Table C.4, without the respondent-education fixed effects for the socioeconomic and human capital indices, which would absorb their schooling measures. The p-value row tests that the slope of the index in the column (in column (9), the neighborhood-ties index) is equal on both sides, in a model on both sides that interacts each index with the east-side indicator and lets the slope in distance to the boundary differ by side. Each index is the first principal component of its measures over the 239 households, standardized over the 239. The neighborhood-ties index combines the number of times the household received help from neighbors in the past twelve months with child care or care when sick, with loans, and with finding work, coded from the answer bins 1, 2, 3, 4, and 5 as 0, 1.5, 4, 8, and 12 times, and time with neighbors relative to work time. Table D.10 reports the slopes with the nine-measure index that adds the two allocations, the reservation salary, and the two vignettes, and with other variants. The placebo indices combine aspiring to and expecting university for the focal child; expected earnings with high school and with university, the perceived return, and the belief that an excluded-side graduate earns less; the personal and the perceived work-over-study norm; belonging, the preference for another *colonia*, and the effort to have children know the neighbors; and patience and the willingness to take risks. The socioeconomic index combines log household income, log income per household member, years of schooling of the more-schooled parent and of the more-schooled grandparent, a formal salary as main income, and rooms per household member, and the human capital index combines years of schooling of the more-schooled parent, of the respondent's father, and of the respondent's mother (Table C.7). Heteroskedasticity-robust standard errors are in parentheses and stars use them. All index measures were taken after the experimental block, and Map-First does not change the neighborhood-ties index. These analyses were not pre-registered. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

↩ Back to text

standard deviation more in the index goes with a tutoring share 4.0 percentage points lower and a willingness to pay MXN 27 lower (Table D.11), about 0.10 standard deviations of each outcome. On the west side both associations are positive and imprecise, and the difference between the sides is significant at the 10 percent level for both outcomes. Indices of aspirations, perceived returns, work-over-study norms, attachment to the neighborhood, and patience and risk, built in the same way, show no detectable difference across the boundary and do not predict demand on the east side (columns 2 to 6 and 9), and on the west side only the returns index does, at the 10 percent level (column 11). A socioeconomic index and a human capital index, built in the same way, do not predict demand on the east side either (columns 7 and 8), and the neighborhood-ties coefficient remains negative and statistically significant when the eight indices enter together (column 9).

Neighborhood ties are what differs most between the sides and what predicts lower demand within the east side, the side where activating the place lowers it. Measured school provision and recorded crime show no detectable discontinuity at the boundary, street infrastructure is no worse on the east side (Section 3), and aspirations, expected returns, norms, attachment, and patience and risk show no detectable difference either. I read the place the mapping task activates as the exchange with neighbors that the drawn area contains, and not the schools inside it or the dangers of the street.

Within the east side the two do not compound. The association between the index and demand is 7.1 percentage points per standard deviation among Map-After households and 1.3 among Map-First households, so the prime lowers demand most among the households with the fewest ties and brings them close to the level of those with the most.[28] This pattern is consistent with the neighborhood already being on the minds of those who rely on it most, although the interaction between Map-First and the index is not significant.

4.7 Activating History

The mapping task activates a place, and a second randomized exercise activates the history behind it. In the studies of activated history, the event is brought back to mind by a campaign, a news cycle, or a new wave of migration, and its effect is inferred from where the event happened (Fouka and Voth, 2023; Ochsner and Roesel, 2024; Cantoni et al., 2026; Ghidoni and Schindler, 2026). Here the activation is randomized. Before respondents explain why a young person from their neighborhood might stop studying, a random half hear the expression *de la Calzada para allá*, which residents use to refer to the east side and which names the historical divide without mentioning education. The explanations are a pre-registered secondary outcome. A language model classified them against a fixed codebook, as with the descriptions of the neighborhood, and an independent second classification agrees with it, with a mean Cohen's κ of 0.83.[29]

On the east side, the expression lowers by 15 percentage points the share of explanations that refer to money or the need to work, and references to the neighborhood and to the family rise, although not significantly; on the west side, no explanation changes (Table D.12). The treatment and the outcome differ from those of the tutoring experiment, so this exercise is not a replication.

[28] The interaction between Map-First and the index on the east side is 5.8 percentage points per standard deviation, with a randomization-inference p-value of 0.13. Neither individual measures of exchange with neighbors nor their averages among the other sampled households of the same *colonia*, or among those who live inside the neighborhood each respondent drew, detectably moderate the response to the map. Because these measures barely vary across *colonias* within the east side, the sample cannot test whether the response varies with how many neighbors take part.

[29] The registration specifies two independent coders, a scheme of six categories (economic necessity, family obligations, low perceived returns, peer influences, neighborhood-specific factors, and institutional barriers), and Cohen's κ. The answers were classified instead by `claude-opus-4-8` into five reasons (the local place, the surrounding social environment, the family, material constraints, and the young person's own choices), which Table D.12 groups into four categories, and the second classification was formed without access to the first.

Activating the history, like activating the place, moves east-side respondents away from money as the reason children leave school, and leaves the west side unchanged. In the framework, the earnings of adolescents matter through what they let the household give back to its neighbors, so explanations that move away from money as such are consistent with a cost that runs through the exchange.

5 A Framework of Neighborhood Ties and Reciprocal Exchange

5.1 The Decision to Educate

In neighborhood ℓ, a share q_ℓ of households take part in exchange: they look after each other's children, lend money and tools, pass on word of a job, and step in during an illness. A household that takes part obtains R when it needs help; with probability $1 - q_\ell$ no neighbor can help, and it falls back on outside channels, formal credit, public services, or the courts, which yield Z_ℓ (Kranton, 1996). The expected payoff from taking part is $q_\ell R + (1 - q_\ell) Z_\ell$, and its value over the outside alternative is

$$\Delta_\ell = q_\ell \, (R - Z_\ell), \tag{2}$$

which rises with the share of neighbors who take part and falls with the quality of outside channels, and is zero where no one takes part or where the bank, the clinic, and the police do as well as the neighbors. Z_ℓ is what residents obtain from formal institutions and not only what the state provides, since the same clinic is worth less to a resident who expects to be treated badly in it, and this is where the standing that exclusion denied enters. Household i relies on neighbors with intensity β_i and obtains $\beta_i \Delta_\ell$ from exchange.

Education yields a return ρ at a private cost C_i, lower where schools are closer and better, parents more educated, and households better off, the determinants of Section 3. It also puts a share ψ_n of the household's exchange at risk. Households stay in exchange by being available and able to reciprocate (González de la Rocha, 1999, pp. 16, 20), children begin to contribute to their households around age 15 (González de la Rocha, 2001, p. 80), young members supply part of the daily exchange (González de la Rocha, 2001, p. 90), and members who move up see their parents less (González de la Rocha, 1999, p. 26). A child in school is less available to the neighbors, delays the earnings with which the household reciprocates, and is more likely to leave when grown, so ψ_n rises with the age n of the child. The value of exchange is a linear version of the participation externality in Kranton (1996), the pull of school away from a network follows Munshi and Rosenzweig (2006), and the transmission of participation follows Bisin and Verdier (2001). Appendix E derives each step. Parents weigh everything that concerns the neighborhood by $a(s)$, below one when it is not on their minds and higher when it is (Gabaix, 2019).

Without the prime, demand shows no detectable difference across sides, with intervals wide enough to include large differences (Table D.1), which is consistent with low attention to the neighborhood when it is not on parents' minds, under comparable net tutoring values across sides. Decisions to leave school are taken when a neighbor needs help, a job appears, or the household falls short, so the neighborhood is on parents' minds when it matters, and the reservation salary for university was asked after the mapping task in both arms. Households that rely more on their neighbors may also have it on their minds without the prime, which is consistent with lower demand among them in Map-After and a smaller response to the prime (Section 4.6).

A household that educates keeps $\rho - C_i + a(s)(1-\psi_n)\beta_i\Delta_\ell$ and one that does not keeps $a(s)\beta_i\Delta_\ell$, so it educates if

$$\rho - C_i \;\geq\; a(s)\,\Pi_i, \qquad \Pi_i = \psi_n\,\beta_i\,\Delta_\ell, \tag{3}$$

where Π_i is the neighborhood cost of schooling, an opportunity cost that the household bears through its child, the part of the value of exchange that schooling puts at risk. It is zero outside exchange and where exchange is worth nothing, and largest for a household that relies on neighbors who rely on each other, with an adolescent child, when the neighborhood is on its mind. Neighbors can also support schooling with information and norms (Coleman, 1988), in which case Π_i is net of that support and can be negative; the sign is what the evidence has to reveal.

What the framework predicts. Four things follow from (3), and they are the four results of Section 4. The sides differ in reliance on neighbors and in nothing that enters the return. Bringing the neighborhood to mind raises $a(s)$ and lowers demand where Δ_ℓ is positive and not where it is close to zero. Within a side, demand falls with β_i only where Δ_ℓ is positive. Aspirations, expected returns, and norms enter ρ and C_i and not Π_i, so they neither differ across sides nor predict demand where Δ_ℓ is positive. The neighborhood cost also rises with the age of the child, which is why the gap opens in adolescence, and a larger Π_i raises the return a household requires, which is the 18 percent higher salary east-side parents demand.

With private costs distributed as F on both sides and a common return, attendance on side ℓ is

$$A_\ell = (1 - q_\ell)\,F(\rho) + q_\ell\,F\big(\rho - a(s)\Pi_\ell\big), \tag{4}$$

with Π_ℓ the neighborhood cost of a participating household (Figure E.1). Attendance is lower where more households take part and exchange is worth more, which is the gap among children with similarly educated parents. Even with a common private-cost distribution and return, differences in participation and in the value of exchange sustain an attendance gap. Better school provision lowers private costs without changing those neighborhood differences, and the gap's response depends on the mass of households near the schooling thresholds, so the framework permits a residual gap after provision converges and does not by itself derive the historical narrowing or its stalling.

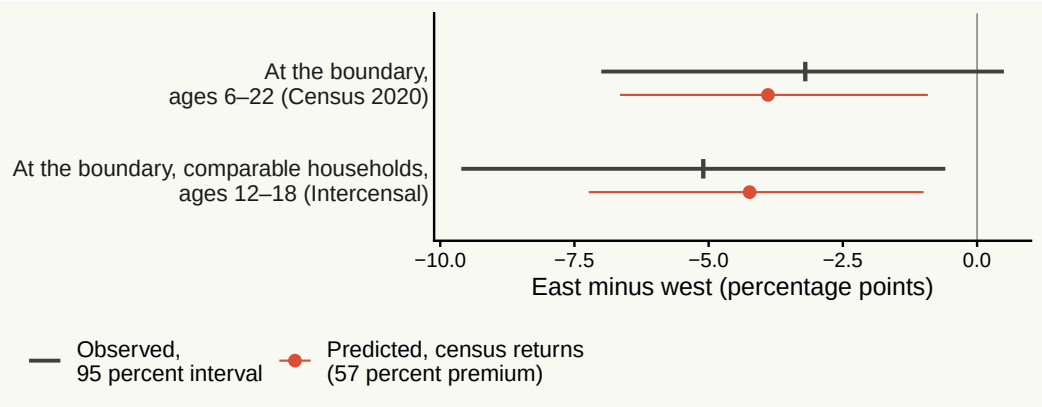
How large is the neighborhood cost? The survey measures it, and the experiment bounds it from below, in different units. Willingness to pay is proportional to $\rho - C_i - a(s)\Pi_i$ (Appendix E.1), so Map-First lowers the valuation of an hour of tutoring by a fifth on the choices within the grid and by two-fifths under the choice model, a reduction that is a lower bound on the neighborhood cost of tutoring under the assumptions below (Table D.2). The latent willingness to pay comes from a stochastic choice model that accommodates never-switching and multiple-switching responses. The observed choices and the bounded measure give the same sign on the east side, and the monetary magnitude and its percentage reduction depend on that model (Table D.2). The interval of that reduction reflects sampling uncertainty and not these modeling choices. The reservation salary for university was asked after the mapping task in both arms, which does not change it, so the east-west difference, MXN 5,319, puts it at 45 percent of the premium parents expect. The reservation-salary gap can reflect private schooling costs as well as the neighborhood cost. Conditioning on the income and location measures the survey has leaves it unchanged, and conditioning on reliance on neighbors lowers it by about a sixth, imprecisely (Table C.6). These comparisons do not identify the split. Under the assumptions that follow, I attribute the full gap to the neighborhood component for the calibration. Reading these two numbers this way rests on three assumptions, which Appendix E.2 derives. In the experiment, the private cost of an hour of free online tutoring is close to zero, so the net value of tutoring is approximately the return itself, and the prime raises the salience weight to close to one, so the drop in willingness to pay is at most the average neighborhood cost of tutoring and below it if some households already had the neighborhood in mind. The experimental bound also requires that the prime leaves the value of the gift card unchanged, so that the drop in willingness to pay reflects the value of tutoring rather than a higher value of cash for obligations with neighbors. In the survey, private costs are drawn from the same distribution on both sides, as in (4), and the west-side neighborhood cost is small, so the difference in reservation salaries identifies the east-side cost and not a difference between two costs. This is an assumption, and the balance tests of Section 4.3 do not establish it, although the response to the map does not vary detectably with income or parental schooling within the east side (Table D.9). The two exercises also refer to different investments, an hour of tutoring and a university degree, and the calibration uses the survey's share. Map-After demand without a detectable difference across sides and the absence of a detectable west-side response are consistent with low attention to the neighborhood without the prime and a small western neighborhood cost, under comparable private net tutoring values and salience changes across sides. They do not separately identify these quantities or establish that the map makes the neighborhood fully salient. I maintain the small western cost and the full salience of the prime as normalizations rather than infer them from the null contrasts, and the bound does not require low attention without the prime. Under them, the survey measures the cost as a share of the return parents expect, and the experiment lowers the valuation of a marginal educational good by a fifth to two-fifths, depending on how choices beyond the price grid are treated, a reduction the framework reads as a lower bound on its neighborhood cost (Figure E.2). Without them, the experiment gives the change in

net tutoring value that the prime produces, with a randomization-inference interval from 6 to 79 percent, and the survey gives the gap in the bar parents set for university, and their closeness is suggestive and not a joint measurement.

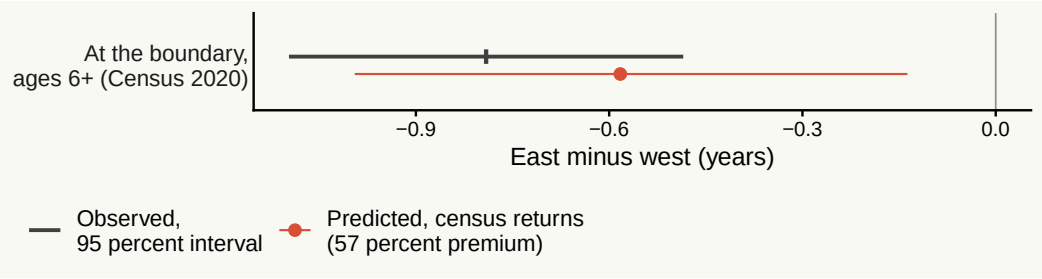
A back-of-the-envelope calculation asks whether a cost of that size can produce the census gap. Each parent reports what a university graduate earns and the salary at which university would be worth it; the difference measures how far the investment clears the bar. Parents who clear it by less than the neighborhood cost are those whose decision the cost can tip, the mass of F between the two thresholds in (4). I count that group in the survey after replacing stated returns with the 57 percent census premium, since parents overstate returns (Appendix C.2), and after subtracting the MXN 5,319 east-side gap in reservation salaries from east-side thresholds, so that the two sides' bars are measured from the same baseline. Correcting the earnings while keeping the reservation salaries assumes that the thresholds measure costs and would not move if beliefs were corrected. Across respondents, the reservation salary rises by 7 percent when the expected university salary rises by 10 percent (Appendix E.2), so the thresholds move with beliefs, less than one for one. Letting them move with the corrected earnings with that elasticity widens the implied attendance gap at the boundary from 3.9 to 7.6 percentage points. The cost I then impose on each household is 45 percent of the university premium that household states, the share measured in the survey, which averages MXN 5,267. A cost of that size tips 17 of 239 households, and applying that share to the age structure of each census estimate gives the gap implied by the framework. Relative to the premium the census pays, the same cost is larger than two-fifths, since parents overstate the premium, and Appendix E.2 reports what it represents. A fixed cost of MXN 5,319 for every household is a different schedule, and Table E.1 reports its predictions alongside.

Figure 10 compares those implied gaps with the census estimates from Section 3. Under the maintained assumptions, the calibration predicts an attendance gap of 3.9 percentage points at the boundary, and 7.6 points if the thresholds move with beliefs, against 3.2 points in the census among residents aged 6–22, and 4.2 against 5.1 points among comparable households aged 12–18 and 0.58 against 0.79 years of schooling among residents aged 6 and older. The predictions are statistically compatible with the observed gaps, and the observed attendance gap is imprecisely estimated, with a 95 percent interval from −7.0 to 0.5 points. The comparison establishes quantitative plausibility under the framework's assumptions, and it does not validate the mechanism or identify its contribution to the attendance gap. The attendance comparison asks whether a cost of the survey's size suffices to produce the gap in a current decision. The schooling comparison applies today's margins to every cohort as if in steady state, so the 0.58 years is a long-run benchmark and not an account of the existing stock, which carries earlier differences in provision, quality, and family transmission (Section 3). In the appendix, Figure E.2 shows that the Intercensal estimate requires a cost of at least 3 percent of each household's stated premium and the schooling estimate at least 21 percent, both below the measured costs. Using parents' stated returns instead makes the calculation overshoot by three to five times, and the

Figure 10: The neighborhood cost measured in the survey and the gaps measured in the census



(a) School attendance



(b) Years of schooling

Notes: Lines are observed east-minus-west estimates from Section 3 with 95 percent confidence intervals. Circles are the gaps implied by a neighborhood cost equal to 45 percent of each household's stated university premium, averaging MXN 5,267, applied to the margins of all 239 households computed with the 57 percent premium the census pays, after recentering east-side reservation salaries by the MXN 5,319 gap, passed through the age structure of each estimate; lines through them are 95 percent bootstrap intervals over the 239 households. Table E.1 reports equality tests, the southern boundary points, the predictions with the returns parents state, and those with a fixed cost of MXN 5,319 for every household; Appendix E.2 lists the assumptions.

↩ Back to text

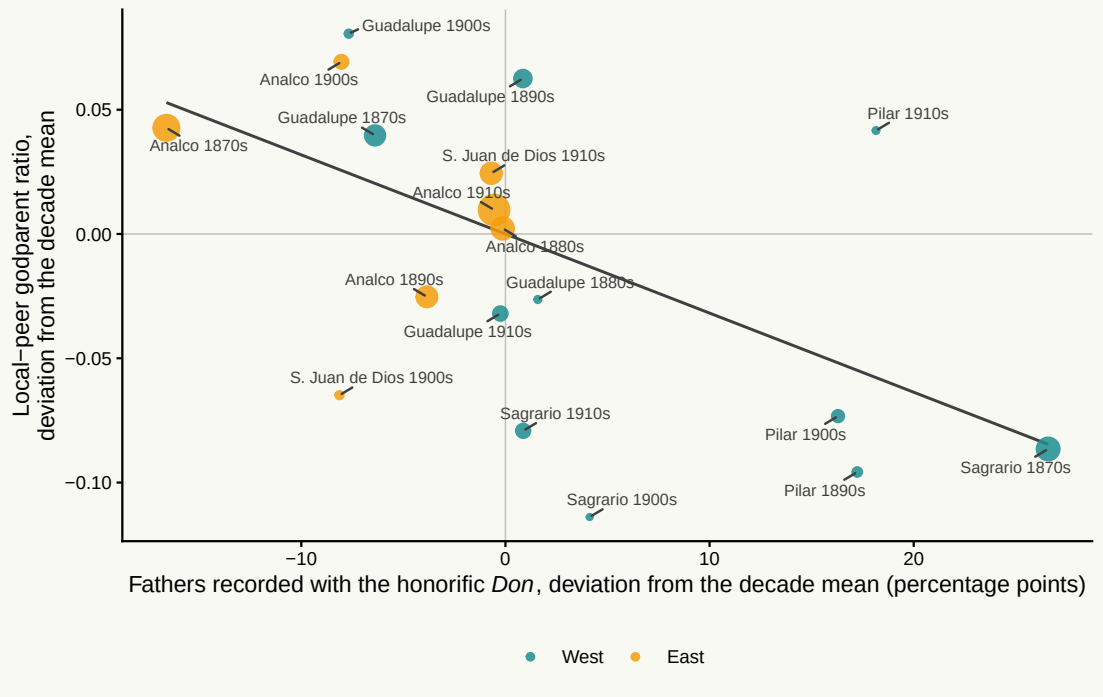
same exercise accounts for only about a third of the larger gaps in the colonial core, at the two southern boundary points (Table E.1 and Appendix E.2).

The calculation rests on three assumptions. The university margin stands in for the decision to stay in school at the transition to upper secondary, where the gap opens. The reservation salary asks about a young person from the respondent's neighborhood and not about the respondent's own child. The experimental bound is imprecise, with a randomization-inference interval from 6 to 79 percent, so its closeness to the survey's share does not pin the cost down. Appendix E.2 lists these three with the assumptions behind the experimental bound and the survey measure.

5.2 Why the Social Legacy Persists

The survey and the experiment measure neighborhood ties and their cost today. That these ties carried the gap across generations rests on an assumption the data cannot test, that participation in exchange on the east side has stayed high since well before the census period. The historical record documents the condition the framework says sustains participation, standing that was denied from 1791 onward and has not converged (Section 2.2). Figure 11 shows that in the baptism registers of 1870 to 1919, the closest measure of these ties, families chose godparents among couples of their own parish more often where fewer fathers carried the honorific *Don*, comparing parishes within the same decade. The relation separates the center of the city from its popular parishes on both banks, and it may reflect who lived in each parish as much as the parish itself.[30]

Figure 11: Local godparents and the honorific *Don* across parishes, within decades, 1870–1919



Notes: Each point is a parish and decade, sized by the number of baptisms in which the parents and both godparents can be read, with both measures expressed as deviations from the decade mean weighted by baptisms. The vertical axis is the local-peer godparent ratio and the horizontal axis the share of fathers whose entry in the register is preceded by the honorific *Don*, abbreviated "D." by the priests. The line is the weighted fit with decade fixed effects across all 18 cells, a slope of −0.032 per 10 percentage points (standard error 0.004); among the ten west-bank cells the slope is −0.040 (0.007). The parish of Guadalupe excludes the register of the hospital church of Belén. Source: FamilySearch, machine-read baptism registers.

↩ Back to text

[30] Ethnographies of Mexican cities describe the same arrangement from the 1970s to the 1990s (Adler Lomnitz, 1977; González de la Rocha, 1994; Reguillo, 1998; González de la Rocha, 2001). No source measures exchange with neighbors on both sides of the creek over time. The exercise uses five parishes and 18 parish-decades, the east is as local as the popular parish of Guadalupe on the west, and the registers record neither exchange with neighbors nor schooling (Appendix A.12).

Condition (3) takes q_ℓ as given, and the question is why it stays high on one side of a boundary that no longer separates schools. A child takes part in exchange when grown according to what the parents pass on and whom the child grows up among (Bisin and Verdier, 2001): a participating parent transmits the disposition with effort τ_1, a child who does not pick it up at home may pick it up from a neighbor, with probability q_ℓ, and a parent outside exchange transmits the opposite with effort τ_0. Parents try harder when more is at stake, $\lambda\Delta_\ell$ for a participating parent, with $\lambda > 0$, and $b(Z_\ell)$ for the other, and when their own disposition is rare. With a quadratic cost of effort scaled by k,

$$q_{\ell,t+1} - q_{\ell,t} = k\, q_{\ell,t}(1 - q_{\ell,t})\big[(1 - q_{\ell,t})\,\lambda\,\Delta_\ell - q_{\ell,t}\,b(Z_\ell)\big], \tag{5}$$

and Appendix E.3 derives it. The share settles where the bracket is zero at a positive share, higher where exchange is worth more and lower where outside channels are better, and moves there one generation at a time (Figure E.3a); exchange is worth more where more take part, by (2), and participating parents have more at stake where it is worth more, so the east side stays where it is under the condition that sustains participation (Appendix E.3).

This separates access from standing. In the framework, when provision converged, C_i fell at once, because it depends on the schools that exist today. Δ_ℓ need not fall with it, because q_ℓ adjusts one generation at a time and because Z_ℓ depends on standing as much as on provision. The historical record shows standing denied from 1791 into the twentieth century and a stigma that persists (Section 2.2), and residents east of the boundary today report less trust in formal institutions and more in their neighbors (Table 2), a contrast that extrapolates from tracts away from the boundary. These are consistent with a lower Z_ℓ on the east side, and none of them measures outside options or their path over time. The gap was much larger in 1930 and has not narrowed detectably since 2000, which is consistent with the framework if C_i fell as provision converged while Δ_ℓ did not, and the framework does not derive that path. Access no longer divides attendance, and the social legacy can. Residential turnover works through q_ℓ: entrants who take part less raise attendance, and a one-time change in a few blocks is followed by a return toward the previous share, since (5) pulls q_ℓ back when the rest point is stable, which holds when parents adjust slowly enough, a small k (Appendix E.3). The rebuilt blocks of Analco after the 1992 explosions followed this pattern, although the census cannot separate a return of q_ℓ from a simple re-sorting of residents (Section 3.5). Sorting works in the same direction. With strong local spillovers, residential sorting sustains large clusters that resist small shocks and change at their edges and in isolated locations (Aliprantis and Lin, 2026), so the transmission of participation within the east side and the sorting that keeps it a side reinforce each other.

5.3 Other Mechanisms

Other mechanisms can make social life a cost of education, and Appendix E.4 lists what each predicts and what the survey shows. A culture of poverty (Lewis, 1966) would make east-side parents value schooling less, and they do not, since their aspirations, expected returns, and views

on whether children should work or study match those of west-side parents.[31] Low aspirations or misinformed beliefs (Jensen, 2010; Genicot and Ray, 2020) lower the return parents expect and predict no response when the neighborhood is brought to mind, and the survey shows the opposite on both counts. Identity (Akerlof and Kranton, 2000; Benjamin et al., 2010) predicts the response to the map, and the three attachment items show no detectable difference across sides, which gives it little affirmative support and does not establish that schooling is equally compatible with neighborhood identity on both sides. Redistributive pressure predicts the opposite sign, since cash that neighbors can claim should be less attractive than tutoring they cannot (Jakiela and Ozier, 2016; Squires, 2024). A household's own obligations to its neighbors make cash more useful to it, while claims by others make cash less attractive because it can be taken.

Two other accounts share part of the mechanism and remain open. A penalty for getting ahead, as in the image of limited good (Foster, 1965), zero-sum thinking (Bergeron et al., 2026; Chinoy et al., 2026), or the tall poppy syndrome (Feather, 1989), shares its core, since a member who leaves or rises takes something from those who stay. These accounts locate the penalty in beliefs and norms, which the survey does not measure directly, while reciprocal exchange locates it in help received and time with neighbors, which differ across sides. Signaling requires that neighbors observe the choice (Austen-Smith and Fryer, 2005). Parents chose tutoring in private, so signaling through that choice cannot explain the response to the map, while tutoring or attendance that neighbors later observe can carry a social-image cost the experiment does not test. Reciprocal exchange is the reading the evidence favors, under which what differs between the sides, what predicts demand, and what the map brings to mind are the same, and all three are inert where exchange is worth little.

5.4 The Prediction Outside Guadalajara

Nothing in (2) is specific to Guadalajara. The opportunity cost that exchange with neighbors imposes on schooling exists wherever households rely on their neighbors because the institutions outside the neighborhood serve them badly, and it is absent where those institutions work. The same relation should then appear elsewhere, between how much people rely on neighbors rather than on anyone else and how much schooling they get, and it should be stronger where groups live apart and deal with institutions that were not built for them. Across countries, the closest proxy for that situation is ethnic segregation.

Wave 7 of the World Values Survey allows a descriptive check in 35 countries (the 35 of the 42 countries in Figure 1 that record ethnic group and parental education). I measure reliance

[31] Lewis (1952) found that migrants to Mexico City kept their ties to kin and community, and his later work on a "culture of poverty" shaped how scholars and policymakers saw the slum (de Antuñano, 2019). Adler Lomnitz (1977) shared his finding that these ties grew stronger in the city but rejected the culture of poverty, treating the networks of the poor as an active response to economic insecurity (de la Peña, 2019). The mechanism in this paper follows her reading.

on neighbors with trust concentrated in neighbors, the trust a respondent places in neighbors in excess of the trust placed in family and acquaintances, so that the measure captures where trust is concentrated and not how trusting a person is. Table 4 shows that trust concentrated in neighbors goes with less schooling, by 0.038 standard deviations per standard deviation in countries where ethnic segregation is high (Alesina and Zhuravskaya, 2011) and by 0.011 where it is low, controlling for parental education, while trust in close ties and trust in distant groups each go with more schooling (Tables F.1 and F.2). Table 4 shows the same pattern for dropping out before upper secondary school, with slopes of 0.027 standard deviations where ethnic segregation is high and 0.005 where it is low. The pattern is the one the framework predicts, and it is descriptive: trust concentrated in neighbors is not reciprocal exchange and segregation is not exogenous, and the survey and the experiment in Guadalajara test the mechanism directly.[32]

Table 4: Excess trust in neighbors, years of education, and dropout, World Values Survey Wave 7

	Years of education				Dropout before upper secondary			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Excess trust in neighbors	−0.018***	−0.025***	−0.022***		0.014***	0.019***	0.014***	
	(0.004)	(0.004)	(0.004)		(0.004)	(0.004)	(0.004)	
	[0.005]	[0.005]	[0.006]		[0.004]	[0.004]	[0.005]	
Trust in distant groups		0.051***	0.047***			−0.041***	−0.034***	
		(0.005)	(0.005)			(0.005)	(0.005)	
		[0.007]	[0.008]			[0.007]	[0.007]	
Trust in close ties			0.011**				−0.018***	
			(0.005)				(0.005)	
			[0.007]				[0.006]	
Excess trust in neighbors × low ethnic segregation				−0.011**				0.005
				(0.005)				(0.005)
				[0.006]				[0.006]
Excess trust in neighbors × high ethnic segregation				−0.038***				0.027***
				(0.007)				(0.008)
				[0.014]				[0.009]
Parental education FE	✓	✓	✓	✓	✓	✓	✓	✓
Region×year, region×settlement size FE	✓	✓	✓	✓	✓	✓	✓	✓
Sex×age, country of birth, religion, ethnic group FE	✓	✓	✓	✓	✓	✓	✓	✓
Survey weights	✓	✓	✓	✓	✓	✓	✓	✓
Observations	47,072	46,952	46,952	46,952	47,072	46,952	46,952	46,952
Dep. var. mean	11.68	11.68	11.68	11.68	0.302	0.302	0.302	0.302
R^2	0.5072	0.5092	0.5093	0.5095	0.4355	0.4369	0.4371	0.4372

Notes: OLS estimates from the World Values Survey / European Values Study integrated time series, wave 7 (2017–2023), for respondents in the 35 countries that contribute to the estimates (Table F.1). Both outcomes are standardized, and the unit of observation is the respondent. Columns (1)–(4) and (5)–(8) are columns (1)–(4) and (9)–(12) of Table F.3. Excess trust in neighbors (trust concentrated in neighbors in the text) is standardized trust in neighbors minus average trust in family and in people known personally, trust in close ties is trust in family and in people known personally, and trust in distant groups is trust in people met for the first time, of another religion, and of another nationality. A country has high ethnic segregation if its Alesina and Zhuravskaya (2011) index is at or above the cross-country mean. Columns (4) and (8) interact the three trust terms with the low- and high-segregation indicators and display the two slopes of excess trust in neighbors. Regions are subnational regions, so the region fixed effects absorb country effects. Heteroskedasticity-robust standard errors are in parentheses. Standard errors from a cluster bootstrap by country are in brackets, with 999 draws that each resample the 35 countries with replacement within each segregation regime, so that every draw keeps 12 high-segregation countries. Stars use the heteroskedasticity-robust standard errors. *, **, and *** denote significance at the 10%, 5%, and 1% levels. Table F.3 also reports upward educational mobility, and Table F.2 varies the fixed effects.

↩ Back to text

[32] A related exercise with the Facebook Social Connectedness Index finds lower schooling where more of a region's social ties are geographically local, a measure of network geography rather than of exchange (Figure F.1 and Table F.4).

5.5 What the Framework Does Not Explain

Three results need more than the framework's main condition. Without the prime, demand shows no detectable difference across sides in an imprecise comparison, and the framework does not say how little attention the neighborhood receives when it is not on parents' minds. Within the east side, the response to the map is largest among the households with the fewest ties, which is consistent with the framework if the neighborhood is already on the minds of those who rely on it most, so that attention without the prime rises with reliance on neighbors (Section 5.1). Under that reading the prime measures the increment in attention times the cost, and each of the 21 to 42 percent estimates is, under these assumptions, a lower bound on the average cost of tutoring rather than its level, which is how Appendix E.2 treats it. The survey does not test this reading, and the interaction between the map and the index of ties is not significant. The experiment cannot tell whether bringing the neighborhood to mind lowers the value of tutoring or raises the value of cash that can meet obligations with neighbors. The higher return east-side parents require is evidence beyond the gift-card choice, and it does not separate the two readings either, since the vignette includes educational expenses and forgone earnings. The framework is a possible mechanism of persistence and not an identified one.

6 Conclusion

Nearly five centuries after Guadalajara's colonial boundary was drawn, children on the historically excluded side are still less likely to attend school, and adults there have fewer years of schooling. The gap was much larger in 1930 and has not narrowed detectably since 2000, although the boundary design estimates today's attendance gap imprecisely. Measured school provision no longer differs detectably, standing still does, and the gap remains among children with similarly educated parents.

East-side parents report similar educational aspirations and expected returns to university, but they rely more on their neighbors and need a higher salary at age 30 to consider university worth it. Bringing their neighborhood to mind before an educational choice lowers their demand for tutoring, and the effect on the west side is imprecise. Exchange with neighbors is the dimension along which the historically excluded side differs and the one that predicts lower demand there, and the aspirations, expected returns, and norms measured in the survey do neither. Under the framework's assumptions, the survey puts the opportunity cost that these ties impose on schooling at about two-fifths of the return parents expect, and the experiment lowers the valuation of a marginal educational good by a fifth to two-fifths, depending on how choices beyond the price grid are treated, a reduction the framework reads as a lower bound on its neighborhood cost. A cost of that size could generate an attendance gap of the observed order at the boundary and, as a long-run benchmark, a schooling gap of that order, and it accounts for only about a third of the larger gaps in the colonial core. The experimental bound is an imprecise change in the net

value of tutoring, with an interval from 6 to 79 percent, and the calibration to the census rests on assumptions about how the cost and the returns measured in the survey carry over to it.

We still know little about what households give up when they invest in education, whether their children's time or the help they receive from neighbors, and how much it is worth to them. Nor do we observe these ties over time, so whether they carried the gap across generations remains the framework's hypothesis. The experiment shows that bringing the neighborhood to mind changes the decision, and the survey points to reliance on neighbors, although neither separates exchange with neighbors from identity or social image. The size it implies for the gap rests on a back-of-the-envelope calculation that assumes the cost of schooling and the distribution of returns measured in the survey hold for the population in the census. Future work should measure these costs and how they change as outside options improve.

Nothing in this paper says that east-side households are choosing badly. Residents on the two sides report similar life satisfaction, and the help of neighbors may be worth more to a household than an hour of tutoring. Each household does what is best for it, and all would do better if schooling did not put their ties at risk (Kranton, 1996). The room for policy lies in making the two compatible, by bringing to the neighborhood the services that neighbors now provide to each other, or by making it possible to study without leaving. Policies that ignore local ties may reach fewer of the households they target (Bau et al., 2025; Moscona et al., 2026), and take-up in relocation programs in the United States, India, and Brazil ranged from 48 to 60 percent (Chetty et al., 2016; Barnhardt et al., 2017; Belchior et al., 2026). Schooling can strain the same relationships without any move. Historical exclusion may persist through the ties that helped excluded households cope, and not only through the resources it denied them.

References

Aaronson, Daniel, Daniel A. Hartley, and Bhashkar Mazumder, "The Effects of the 1930s HOLC 'Redlining' Maps," *American Economic Journal: Economic Policy*, 2021, *13*, 355–92. (cited on p. 6).

Acemoglu, Daron, Simon Johnson, and James A. Robinson, "The Colonial Origins of Comparative Development: An Empirical Investigation," *American Economic Review*, 2001, *91* (5), 1369–1401. (cited on pp. 6 and 10).

_ , _ , and _ , "Institutions as a Fundamental Cause of Long-Run Growth," in Philippe Aghion and Steven N. Durlauf, eds., *Handbook of Economic Growth*, Vol. 1A, Amsterdam: Elsevier, 2005, chapter 6, pp. 385–472. (cited on p. 10).

Adler Lomnitz, Larissa, *Networks and Marginality: Life in a Mexican Shantytown*, New York: Academic Press, 1977. Translated by Cinna Lomnitz. (cited on pp. 1, 4, 24, 29, 42, and 44).

Ahlfeldt, Gabriel M., Stephen J. Redding, Daniel M. Sturm, and Nikolaus Wolf, "The Economics of Density: Evidence From the Berlin Wall," *Econometrica*, 2015, *83* (6), 2127–2189. (cited on p. 10).

Akerlof, George A. and Rachel E. Kranton, "Economics and Identity," *Quarterly Journal of Economics*, 2000, *115* (3), 715–753. (cited on pp. 7 and 44).

_ and _ , "Identity and Schooling: Some Lessons for the Economics of Education," *Journal of Economic Literature*, 2002, *40* (4), 1167–1201. (cited on p. 7).

Alberro, Solange, *Movilidad social y sociedades indígenas de Nueva España: Las élites, siglos XVI–XVIII*, Mexico City: El Colegio de México, 2019. (cited on p. 10).

Aldeco, Lucía, Mariana Calderón, Daniel Chiquiar, Gordon Hanson, Jorge Pérez Pérez, and Carlos Velázquez, "Local Labor Markets in Mexico: Definition, Databases, and Analysis," Working Paper, Banco de México, Mexico City 2024. (cited on pp. 8 and 21).

Alesina, Alberto and Ekaterina Zhuravskaya, "Segregation and the Quality of Government in a Cross Section of Countries," *American Economic Review*, 2011, *101* (5), 1872–1911. (cited on p. 45).

_ and Paola Giuliano, "Family Ties," in Philippe Aghion and Steven N. Durlauf, eds., *Handbook of Economic Growth*, Vol. 2A, Elsevier, 2014, chapter 4, pp. 177–215. (cited on pp. 7 and 29).

_ , Sebastian Hohmann, Stelios Michalopoulos, and Elias Papaioannou, "Religion and Educational Mobility in Africa," *Nature*, 2023, *618*, 134–143. (cited on p. 1).

_ , **Yann Algan, Pierre Cahuc, and Paola Giuliano**, "Family Values and the Regulation of Labor," *Journal of the European Economic Association*, 2015, *13* (4), 599–630. (cited on p. 7).

Aliprantis, Dionissi, "Making Our Neighborhoods, Making Our Selves: A Review Essay," *Journal of Economic Literature*, 2023, *61* (3), 1172–1187. (cited on p. 7).

_ **and Jeffrey Lin**, "Strong Spatial Spillovers Determine Neighborhood Shape and Neighborhood Change," in Rachel Meltzer and Vincent Reina, eds., *A Research Agenda for Neighborhoods*, Cheltenham: Edward Elgar, 2026, chapter 3, pp. 41–58. (cited on pp. 23, 30, and 43).

Alsan, Marcella and Marianne Wanamaker, "Tuskegee and the Health of Black Men," *The Quarterly Journal of Economics*, 2018, *133* (1), 407–455. (cited on p. 6).

Althoff, Lukas and Hugo Reichardt, "Jim Crow and Black Economic Progress after Slavery," *Quarterly Journal of Economics*, 2024, *139* (4), 2279–2330. (cited on p. 1).

Ambrus, Attila, Erica Field, and Robert Gonzalez, "Loss in the Time of Cholera: Long-Run Impact of a Disease Epidemic on the Urban Landscape," *American Economic Review*, 2020, *110* (2), 475–525. (cited on p. 6).

_ , **Markus Mobius, and Adam Szeidl**, "Consumption Risk-Sharing in Social Networks," *American Economic Review*, 2014, *104* (1), 149–182. (cited on p. 7).

_ , **Wayne Yuan Gao, and Pau Milán**, "Informal Risk Sharing with Local Information," *Review of Economic Studies*, 2022, *89* (5), 2329–2380. (cited on p. 7).

Ananat, Elizabeth Oltmans, "The Wrong Side(s) of the Tracks: The Causal Effects of Racial Segregation on Urban Poverty and Inequality," *American Economic Journal: Applied Economics*, 2011, *3* (2), 34–66. (cited on p. 11).

Anderson, Michael L., "Multiple Inference and Gender Differences in the Effects of Early Intervention: A Reevaluation of the Abecedarian, Perry Preschool, and Early Training Projects," *Journal of the American Statistical Association*, 2008, *103* (484), 1481–1495. (cited on pp. 18 and 25).

Anderson, Rodney D., "Race and Social Stratification: A Comparison of Working-Class Spaniards, Indians, and Castas in Guadalajara, Mexico in 1821," *The Hispanic American Historical Review*, 1988, *68* (2), 209–243. (cited on p. 9).

_ **and Tamara Spike**, "Making History Count: The Guadalajara Census Project (1791–1930)," *Hispanic American Historical Review*, 2007, *87* (2), 327–351. (cited on p. 12).

Arias, Luz Marina and Desha M. Dirod, "Indigenous Origins of Colonial Institutions," *Quarterly Journal of Political Science*, 2014, *9* (3), 371–406. (cited on p. 8).

Artiles, Miriam, "Ethnic Diversity, Historical Economic Exchange, and Development: Evidence from Andean Peru," *American Economic Journal: Applied Economics*, 2026, *18* (2), 229–265. (cited on pp. 6 and 10).

Asher, Sam, Paul Novosad, and Charlie Rafkin, "Intergenerational Mobility in India: New Measures and Estimates across Time and Social Groups," *American Economic Journal: Applied Economics*, 2024, *16* (2), 66–98. (cited on p. 1).

Austen-Smith, David and Roland G. Fryer, "An Economic Analysis of 'Acting White'," *The Quarterly Journal of Economics*, 2005, *120* (2), 551–583. (cited on pp. 7 and 44).

Bagagli, Sara, "The (Express)Way to Segregation: Evidence from Chicag," Technical Report 2025. (cited on p. 11).

Bailey, Michael, Rachel Cao, Theresa Kuchler, Johannes Stroebel, and Arlene Wong, "Social Connectedness: Measurement, Determinants, and Applications in Economic Research," *Journal of Economic Perspectives*, 2018, *32* (3), 259–280. (cited on p. 170).

Baldomero-Quintana, Luis, ⓡ Woo-Mora, L. Guillermo, and ⓡ De la Rosa-Ramos, Enrique, "Infrastructures of Race? Colonial Indigenous Segregation and Contemporary Land Values," *Regional Science and Urban Economics*, 2025, *110*, 104065. (cited on pp. 6 and 21).

Banerjee, Abhijit V. and Andrew F. Newman, "Information, the Dual Economy, and Development," *The Review of Economic Studies*, 1998, *65* (4), 631–653. (cited on p. 7).

Banfield, Edward C., *The Moral Basis of a Backward Society*, Glencoe, IL: Free Press, 1958. (cited on pp. 7 and 29).

Barnhardt, Sharon, Erica Field, and Rohini Pande, "Moving to Opportunity or Isolation? Network Effects of a Randomized Housing Lottery in Urban India," *American Economic Journal: Applied Economics*, 2017, *9* (1), 1–32. (cited on pp. 7, 29, and 47).

Bau, Natalie, Sara Lowes, and Eduardo Montero, "Culture, Policy, and Economic Development," 2025. Working paper, June 2025. (cited on p. 47).

Becker, Gary S. and Nigel Tomes, "Human Capital and the Rise and Fall of Families," *Journal of Labor Economics*, 1986, *4* (3, Part 2), S1–S39. (cited on p. 20).

Becker, Sascha O., P. David Boll, and Hans-Joachim Voth, "Spatial Unit Roots in Regressions: A Practitioner's Guide and a Stata Package," 2025. (cited on p. 24).

Belchior, Carlos Alberto, Gustavo Gonzaga, and Gabriel Ulyssea, "Who Benefits from Social Housing? Experimental Evidence from a Large-Scale Program in Brazil," *American Economic Review: Insights*, 2026. Forthcoming. (cited on pp. 7 and 47).

Benjamin, Daniel J., James J. Choi, and A. Joshua Strickland, "Social Identity and Preferences," *American Economic Review*, 2010, *100* (4), 1913–1928. (cited on p. 44).

Bergeron, Augustin, Jean-Paul Carvalho, Joseph Henrich, Nathan Nunn, and Jonathan L. Weigel, "Zero-Sum Environments, the Evolution of Effort-Suppressing Beliefs, and Economic Development," *Review of Economic Studies*, 2026. Forthcoming. (cited on p. 44).

Bergman, Peter, Raj Chetty, Stefanie DeLuca, Nathaniel Hendren, Lawrence F. Katz, and Christopher Palmer, "Creating Moves to Opportunity: Experimental Evidence on Barriers to Neighborhood Choice," *American Economic Review*, 2024, *114* (5), 1281–1337. (cited on p. 7).

Bezin, Emeline and Fabien Moizeau, "Cultural Dynamics, Social Mobility and Urban Segregation," *Journal of Urban Economics*, 2017, *99*, 173–187. (cited on p. 6).

Bisin, Alberto and Thierry Verdier, "'Beyond the Melting Pot': Cultural Transmission, Marriage, and the Evolution of Ethnic and Religious Traits," *Quarterly Journal of Economics*, 2000, *115* (3), 955–988. (cited on pp. 5 and 6).

_ **and** _ , "The Economics of Cultural Transmission and the Dynamics of Preferences," *Journal of Economic Theory*, 2001, *97* (2), 298–319. (cited on pp. 5, 6, 37, 43, and 158).

_ **and** _ , "Economic Models of Identity Formation and Cultural Transmission," 2026. Working paper, July 2026. (cited on p. 6).

Blanden, Jo, Matthias Doepke, and Jan Stuhler, "Educational Inequality," in Eric A. Hanushek, Stephen Machin, and Ludger Woessmann, eds., *Handbook of the Economics of Education*, Vol. 6, Elsevier, 2023, chapter 6, pp. 405–497. (cited on pp. 1, 6, and 20).

Bleakley, Hoyt and Jeffrey Lin, "Portage and Path Dependence," *The Quarterly Journal of Economics*, 2012, *127* (2), 587–644. (cited on pp. 6 and 10).

Boltz, Marie, Karine Marazyan, and Paola Villar, "Income Hiding and Informal Redistribution: A Lab-in-the-Field Experiment in Senegal," *Journal of Development Economics*, 2019, *137*, 78–92. (cited on p. 7).

Borusyak, Kirill and Peter Hull, "Nonrandom Exposure to Exogenous Shocks," *Econometrica*, 2023, *91* (6), 2155–2185. (cited on p. 24).

Bretón, Jorge Alberto Trujillo, "La ciudad del pecado: los bajos fondos y las drogas enervantes en Guadalajara, 1915–1946," in Jorge Alberto Trujillo Bretón, ed., *Por el mundo del delito y sus pormenores: historia, marginalidad y delito en América Latina*, Guadalajara, Mexico: Universidad de Guadalajara, 2018, pp. 307–336. (cited on pp. 2 and 13).

Bryan, Gharad, Simon Franklin, Tigabu Getahun, and Sarah Winton, "No Place Like Home? The Causal Effect of Forced Relocation from Central Addis Ababa," 2026. Working paper, January 2026. (cited on p. 7).

Buringh, E., "Urbanisation Hub: The Clio-Infra Database on Urban Settlement Sizes: 1500–2000," 2018. (cited on p. 10).

Bursztyn, Leonardo and Robert Jensen, "How Does Peer Pressure Affect Educational Investments?," *The Quarterly Journal of Economics*, 2015, *130* (3), 1329–1367. (cited on p. 7).

_ and _, "Social Image and Economic Behavior in the Field: Identifying, Understanding, and Shaping Social Pressure," *Annual Review of Economics*, 2017, *9*, 131–153. (cited on p. 7).

_, Georgy Egorov, and Robert Jensen, "Cool to Be Smart or Smart to Be Cool? Understanding Peer Pressure in Education," *The Review of Economic Studies*, 2019, *86* (4), 1487–1526. (cited on p. 7).

Calonico, Sebastian, Matias D. Cattaneo, and Rocío Titiunik, "Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs," *Econometrica*, 2014, *82* (6), 2295–2326. (cited on pp. 13 and 17).

Cantoni, Davide and Noam Yuchtman, "Historical Natural Experiments: Bridging Economics and Economic History," in Alberto Bisin and Giovanni Federico, eds., *The Handbook of Historical Economics*, London, UK: Academic Press, 2021, chapter 8, pp. 213–241. (cited on p. 10).

_, Felix Hagemeister, Leon Haziraj, and Mark Westcott, "Persistence and Activation of Right-Wing Political Ideology," 2026. Working paper, August 2026. (cited on pp. 4, 6, 30, and 36).

Cappelen, Alexander W., Benjamin Enke, and Bertil Tungodden, "Universalism: Global Evidence," *American Economic Review*, 2025, *115*, 43–76. (cited on p. 28).

Carranza, Eliana, Aletheia Donald, Florian Grosset-Touba, and Supreet Kaur, "The Social Tax: Redistributive Pressure and Labor Supply," *Econometrica*, 2025, *93*, 2273–2308. (cited on p. 7).

Carrillo, Bladimir, Carlos Charris, and Wilman Iglesias, "Moved to Poverty? A Legacy of the Apartheid Experiment in South Africa," *American Economic Journal: Economic Policy*, 2023, *15* (4), 183–221. (cited on p. 7).

Cattaneo, Matias D., Michael Jansson, and Xinwei Ma, "Manipulation Testing Based on Density Discontinuity," *The Stata Journal*, 2018, *18* (1), 234–261. (cited on p. 14).

_ , **Richard K. Crump, Max H. Farrell, and Yingjie Feng**, "On Binscatter," *American Economic Review*, 2024, *114* (5), 1488–1514. (cited on p. 2).

_ , **Rocío Titiunik, and Ruiqi Yu**, "Estimation and Inference in Boundary Discontinuity Designs: Location-Based Methods," *arXiv Working Paper 2505.05670*, 2025. (cited on pp. 14, 18, and 19).

_ , _ , **and** _ , "Boundary Discontinuity Designs: Theory and Practice," in R. Griffith, Y. Gorodnichenko, M. Kandori, and F. Molinari, eds., *Advances in Economics and Econometrics: Thirteenth World Congress*, Vol. 1, Cambridge University Press, 2026, chapter 2. To appear. (cited on pp. 1 and 13).

Chetty, Raj, Nathaniel Hendren, and Lawrence F. Katz, "The Effects of Exposure to Better Neighborhoods on Children: New Evidence from the Moving to Opportunity Experiment," *American Economic Review*, 2016, *106* (4), 855–902. (cited on pp. 7 and 47).

_ , **Rebecca Diamond, Thomas B. Foster, Lawrence F. Katz, Sonya R. Porter, Matthew Staiger, and Laura Tach**, "Creating High-Opportunity Neighborhoods: Evidence from the HOPE VI Program," Working Paper, Opportunity Insights 2026. (cited on p. 7).

Chinoy, Sahil, Nathan Nunn, Sandra Sequeira, and Stefanie Stantcheva, "Zero-Sum Thinking and the Roots of U.S. Political Differences," *American Economic Review*, 2026, *116* (3), 1052–1096. (cited on p. 44).

Chyn, Eric and Lawrence F. Katz, "Neighborhoods Matter: Assessing the Evidence for Place Effects," *Journal of Economic Perspectives*, 2021, *35* (4), 197–222. (cited on pp. 1 and 7).

_ , **Kareem Haggag, and Bryan A. Stuart**, "The Effects of Racial Segregation on Intergenerational Mobility: Evidence from Historical Railroad Placement," *American Economic Journal: Applied Economics*, 2026, *18*, 34–60. (cited on p. 11).

Coleman, James S., "Social Capital in the Creation of Human Capital," *American Journal of Sociology*, 1988, *94* (1), S95–S120. (cited on pp. 7, 29, and 38).

Cunha, Flavio and James J. Heckman, "The Technology of Skill Formation," *American Economic Review*, 2007, *97* (2), 31–47. (cited on p. 20).

Dai, Wenhao, Tianshu Yang, Benjamin X. White, Ryan Palmer, Emily K. Sanders, Jack A. McDonald, Melody Leung, and Dolores Albarracín, "Priming Behavior: A Meta-Analysis of the Effects of Behavioral and Nonbehavioral Primes on Overt Behavioral Outcomes," *Psychological Bulletin*, 2023, *149* (1–2), 67–98. (cited on p. 31).

Davis, Donald R. and David E. Weinstein, "Bones, Bombs, and Break Points: The Geography of Economic Activity," *American Economic Review*, 2002, *92* (5), 1269–1289. (cited on pp. 6 and 10).

de Antuñano, Emilio, "Mexico City as an Urban Laboratory: Oscar Lewis, the 'Culture of Poverty' and the Transnational History of the Slum," *Journal of Urban History*, 2019, *45* (4), 813–830. (cited on p. 44).

de Arregui, Domingo Lázaro, *Descripción de la Nueva Galicia*, Seville: Escuela de Estudios Hispano-Americanos, 1946. Edited by François Chevalier. (cited on pp. 9 and 10).

de la Peña, Guillermo, "Larissa Adler Lomnitz (1932–2019). Recordando su trayectoria y contribuciones a la antropología latinoamericana," *Antropología Americana*, 2019, *3* (6), 145–153. (cited on p. 44).

De la Torre, Renée, "Guadalajara vista desde la calzada: fronteras culturales e imaginarios urbanos," *Alteridades*, 1998, *8* (15), 45–55. (cited on p. 13).

Dell, Melissa, "The Persistent Effects of Peru's Mining Mita," *Econometrica*, 2010, *78* (6), 1863–1903. (cited on pp. 1, 6, and 13).

Durlauf, Steven N., "A Theory of Persistent Income Inequality," *Journal of Economic Growth*, 1996, *1* (1), 75–93. (cited on p. 1).

_ , "Neighborhood Effects," in J. Vernon Henderson and Jacques-François Thisse, eds., *Handbook of Regional and Urban Economics*, Vol. 4, Elsevier, 2004, chapter 50, pp. 2173–2242. (cited on p. 7).

Enke, Benjamin, Ricardo Rodríguez-Padilla, and Florian Zimmermann, "Moral Universalism and the Structure of Ideology," *The Review of Economic Studies*, 2023, *90*, 1934–1962. (cited on p. 28).

Feather, Norman T., "Attitudes towards the High Achiever: The Fall of the Tall Poppy," *Australian Journal of Psychology*, 1989, *41* (3), 239–267. (cited on p. 44).

Ferlenga, Francesco, "Symbols of Oppression: The Role of Confederate Monuments in the Great Migration," CAGE Working Paper 776, CAGE, University of Warwick 2025. (cited on pp. 4 and 6).

Foster, George M., "Peasant Society and the Image of Limited Good," *American Anthropologist*, 1965, *67* (2), 293–315. (cited on p. 44).

Fouka, Vasiliki and Hans-Joachim Voth, "Collective Remembrance and Private Choice: German–Greek Conflict and Behavior in Times of Crisis," *American Political Science Review*, 2023, *117* (3), 851–870. (cited on pp. 4, 6, 30, and 36).

Franklin, Simon, "Enabled to Work: The Impact of Government Housing on Slum Dwellers in South Africa," *Journal of Urban Economics*, 2020, *118*, 103265. (cited on p. 7).

_ , "The Demand for and Impacts of Government Housing: Evidence from Ethiopian Lotteries," 2025. Working paper, Queen Mary University of London, May 2025. (cited on p. 7).

Gabaix, Xavier, "Behavioral Inattention," in B. Douglas Bernheim, Stefano DellaVigna, and David Laibson, eds., *Handbook of Behavioral Economics: Applications and Foundations 2*, Vol. 2, North-Holland, 2019, pp. 261–343. (cited on pp. 37 and 151).

Galor, Oded, "Inequality, Human Capital Formation, and the Process of Development," in Eric A. Hanushek, Stephen Machin, and Ludger Woessmann, eds., *Handbook of the Economics of Education*, Vol. 4, Elsevier, 2011, chapter 5, pp. 441–493. (cited on p. 20).

_ **and Joseph Zeira**, "Income Distribution and Macroeconomics," *The Review of Economic Studies*, 1993, *60* (1), 35–52. (cited on p. 20).

Galster, George, *Making Our Neighborhoods, Making Our Selves*, Chicago: University of Chicago Press, 2019. (cited on p. 30).

Genicot, Garance and Debraj Ray, "Aspirations and Economic Behavior," *Annual Review of Economics*, 2020, *12*, 715–746. (cited on pp. 20 and 44).

Ghidoni, Riccardo and David Schindler, "Contagious Prejudice: The Marocchinate," 2026. Working paper, April 2026. (cited on pp. 4, 6, 30, and 36).

Giuliano, Paola and Nathan Nunn, "Understanding Cultural Persistence and Change," *Review of Economic Studies*, 2021, *88* (4), 1541–1581. (cited on p. 6).

Goldberg, Jessica, "The Effect of Social Pressure on Expenditures in Malawi," *Journal of Economic Behavior & Organization*, 2017, *143*, 173–185. (cited on p. 7).

Goldewijk, Kees Klein, Arthur Beusen, Jonathan Doelman, and Elke Stehfest, "Anthropogenic Land Use Estimates for the Holocene–HYDE 3.2," *Earth System Science Data*, 2017, *9* (2), 927–953. (cited on p. 10).

González de la Rocha, Mercedes, *The Resources of Poverty: Women and Survival in a Mexican City* Studies in Urban and Social Change, Oxford: Blackwell, 1994. (cited on pp. 1, 24, and 42).

_ , "La reciprocidad amenazada: un costo más de la pobreza urbana," in Rocío Enríquez Rosas, ed., *Hogar, pobreza y bienestar en México*, Guadalajara: ITESO, 1999, pp. 13–36. (cited on pp. 4, 5, 24, 29, 30, 34, and 37).

_ , "From the Resources of Poverty to the Poverty of Resources? The Erosion of a Survival Model," *Latin American Perspectives*, 2001, *28* (4), 72–100. (cited on pp. 4, 24, 30, 37, and 42).

Goraya, Sampreet Singh, Suanna Oh, and Yogita Shamdasani, "Absent but Not Idle: The Social Roots of Demand for Flexible Work," 2025. Working paper. (cited on p. 7).

Graham, Richard, *Independence in Latin America. Contrast and Comparisons*, University of Texas Press, 2013. (cited on p. 9).

Granovetter, Mark, "The Impact of Social Structure on Economic Outcomes," *Journal of Economic Perspectives*, 2005, *19*, 33–50. (cited on p. 6).

Groeger, André and Yanos Zylberberg, "Internal Labor Migration as a Shock Coping Strategy: Evidence from a Typhoon," *American Economic Journal: Applied Economics*, 2016, *8* (2), 123–153. (cited on p. 7).

Grosfeld, Irena and Ekaterina Zhuravskaya, "Cultural vs. Economic Legacies of Empires: Evidence from the Partition of Poland," *Journal of Comparative Economics*, 2015, *43*, 55–75. (cited on p. 6).

Gutiérrez, Felipe Castro, "El origen y conformación de los barrios de indios," in Felipe Castro Gutiérrez, ed., *Los indios y las ciudades de Nueva España*, 1ª ed., Mexico: UNAM, Instituto de Investigaciones Históricas, 2010, pp. 105–122. (cited on p. 9).

Haerpfer, C., R. Inglehart, A. Moreno, C. Welzel, K. Kizilova, J. Diez-Medrano, M. Lagos, P. Norris, E. Ponarin, B. Puranen, and others, "World Values Survey: Round Seven–Country-Pooled Datafile Version 5.0," 2022. (cited on p. 2).

Hanlon, W. Walker and Stephan Heblich, "History and Urban Economics," *Regional Science and Urban Economics*, 2022, *94*, 103751. (cited on p. 6).

Heblich, Stephan, Alex Trew, and Yanos Zylberberg, "East-Side Story: Historical Pollution and Persistent Neighborhood Sorting," *Journal of Political Economy*, 2021, *129* (5), 1508–1552. (cited on pp. 6 and 10).

Heckman, James J. and Paul A. LaFontaine, "The American High School Graduation Rate: Trends and Levels," *Review of Economics and Statistics*, 2010, *92* (2), 244–262. (cited on p. 1).

Hornbeck, Richard, Guy Michaels, and Ferdinand Rauch, "Identifying Agglomeration Shadows: Long-run Evidence from Ancient Ports," Technical Report 2025. (cited on p. 10).

Huillery, Elise, "History Matters: The Long-Term Impact of Colonial Public Investments in French West Africa," *American Economic Journal: Applied Economics*, 2009, *1*, 176–215. (cited on p. 6).

IMEPLAN, "Plan de Ordenamiento Territorial Metropolitano del Area Metropolitana de Guadalajara," Technical Report, Instituto Metropolitano de Planeación, Guadalajara 2016. (cited on p. 9).

Jack, B. Kelsey, Kathryn McDermott, and Anja Sautmann, "Multiple Price Lists for Willingness to Pay Elicitation," *Journal of Development Economics*, 2022, *159*, 102977. (cited on pp. 30 and 31).

Jakiela, Pamela and Owen Ozier, "Does Africa Need a Rotten Kin Theorem? Experimental Evidence from Village Economies," *Review of Economic Studies*, 2016, *83* (1), 231–268. (cited on pp. 7 and 44).

Jaramillo-Molina, Máximo Ernesto, "'De la Calzada para allá': desigualdad, segregación y estigmatización en el Área Metropolitana de Guadalajara, México," *Lavboratorio*, 2022, *32* (2), 11–55. (cited on pp. 2, 13, and 24).

_ **and Alejandra Saucedo**, "'De la Calzada para allá': Fronteras materiales y simbólicas de desigualdad, segregación y estigmatización en la ciudad de Guadalajara," in "Pobreza y exclusión: diagnósticos y estrategias," Vol. 6, COMECSO and CUCSH-UdeG, 2016, pp. 77–106. (cited on pp. 2 and 13).

Jensen, Robert, "The (Perceived) Returns to Education and the Demand for Schooling," *Quarterly Journal of Economics*, 2010, *125* (2), 515–548. (cited on p. 44).

Jiménez Pelayo, Águeda, Jaime Olveda, and Beatriz Núñez Miranda, *El crecimiento urbano de Guadalajara*, Zapopan: El Colegio de Jalisco, H. Ayuntamiento de Guadalajara and CONACYT, 1995. (cited on p. 11).

Karing, Anne, "Social Signaling and Childhood Immunization: A Field Experiment in Sierra Leone," *The Quarterly Journal of Economics*, 2024, *139*, 2083–2133. (cited on p. 7).

Katz, Lawrence F., "Presidential Address: Neighborhood Effects and Missing Markets for Opportunity," *American Economic Review*, 2026, *116* (9), 3197–3222. (cited on p. 7).

Keele, Luke and Rocío Titiunik, "Geographic Boundaries as Regression Discontinuities," *Political Analysis*, 2015, *23* (1), 127–155. (cited on p. 14).

_ **, Scott Lorch, Molly Passarella, Dylan Small, and Rocío Titiunik**, "An Overview of Geographically Discontinuous Treatment Assignments with an Application to Children's Health Insurance," in Matias D. Cattaneo and Juan Carlos Escanciano, eds., *Regression Discontinuity Designs: Theory and Applications*, Vol. 38 of *Advances in Econometrics*, Emerald Publishing Limited, 2017, pp. 147–194. (cited on p. 14).

Kling, Jeffrey R., Jeffrey B. Liebman, and Lawrence F. Katz, "Experimental Analysis of Neighborhood Effects," *Econometrica*, 2007, *75* (1), 83–119. (cited on p. 7).

Knight, Alan, *Mexico. The Colonial Era*, Cambridge University Press, 2002. (cited on p. 8).

Kranton, Rachel E., "Reciprocal Exchange: A Self-Sustaining System," *American Economic Review*, 1996, *86* (4), 830–851. (cited on pp. 5, 7, 29, 37, 47, and 151).

Laczó, Sarolta, "Risk Sharing with Limited Commitment and Preference Heterogeneity: Structural Estimation and Testing," *Journal of the European Economic Association*, 2015, *13* (2), 265–292. (cited on pp. 7 and 29).

Lee, David S. and Thomas Lemieux, "Regression Discontinuity Designs in Economics," *Journal of Economic Literature*, 2010, *48* (2), 281–355. (cited on p. 14).

Lee, Sanghoon and Jeffrey Lin, "Natural Amenities, Neighbourhood Dynamics, and Persistence in the Spatial Distribution of Income," *The Review of Economic Studies*, 2018, *85* (1), 663–694. (cited on p. 10).

Lewis, Oscar, "Urbanization without Breakdown: A Case Study," *The Scientific Monthly*, 1952, *75* (1), 31–41. (cited on p. 44).

_ , "The Culture of Poverty," *Scientific American*, 1966, *215* (4), 19–25. (cited on p. 43).

Ligon, Ethan, Jonathan P. Thomas, and Tim Worrall, "Informal Insurance Arrangements with Limited Commitment: Theory and Evidence from Village Economies," *Review of Economic Studies*, 2002, *69* (1), 209–244. (cited on pp. 7 and 29).

Lin, Jeffrey and Ferdinand Rauch, "What Future for History Dependence in Spatial Economics?," *Regional Science and Urban Economics*, 2022, *94*, 103628. (cited on pp. 6 and 23).

Lowes, Sara, "Culture in Historical Political Economy," in "The Oxford Handbook of Historical Political Economy," Oxford University Press, 2023, pp. 887–924. (cited on p. 6).

_ **and Eduardo Montero**, "The Legacy of Colonial Medicine in Central Africa," *American Economic Review*, 2021, *111* (4), 1284–1314. (cited on pp. 1 and 6).

_ **, Nathan Nunn, James A. Robinson, and Jonathan L. Weigel**, "The Evolution of Culture and Institutions: Evidence From the Kuba Kingdom," *Econometrica*, 2017, *85* (4), 1065–1091. (cited on p. 10).

Ludwig, Jens, Greg J. Duncan, Lisa A. Gennetian, Lawrence F. Katz, Ronald C. Kessler, Jeffrey R. Kling, and Lisa Sanbonmatsu, "Long-Term Neighborhood Effects on Low-Income Families: Evidence from Moving to Opportunity," *American Economic Review*, 2013, *103* (3), 226–31. (cited on p. 7).

Maloney, William F. and Felipe Valencia-Caicedo, "The Persistence of (Subnational) Fortune," *The Economic Journal*, 2016, *126* (598), 2363–2401. (cited on p. 10).

Matías, Alejandro Solís, *Analco*, Gobierno del Estado de Jalisco, 1986. (cited on p. 9).

_ , "San José de Analco en 1907," *Cuadernos de Estudios Jaliscienses*, 1992. (cited on pp. 9 and 10).

McCartan, Cory, Jacob R. Brown, and Kosuke Imai, "Measuring and Modeling Neighborhoods," *American Political Science Review*, 2024, *118* (4), 1966–1985. (cited on pp. 7 and 30).

McCrary, Justin, "Manipulation of the Running Variable in the Regression Discontinuity Design: A Density Test," *Journal of Econometrics*, 2008, *142* (2), 698–714. (cited on p. 14).

Michalopoulos, Stelios and Melanie Meng Xue, "Folklore," *The Quarterly Journal of Economics*, 2021, *136* (4), 1993–2046. (cited on p. 6).

Moscona, Jacob, Nathan Nunn, and James A. Robinson, "Searching for Fish in Trees? Economic Development when Context Matters," NBER Working Paper 34810, National Bureau of Economic Research 2026. (cited on p. 47).

Müller, Ulrich K. and Mark W. Watson, "Spatial Unit Roots and Spurious Regression," *Econometrica*, 2024, *92* (5), 1661–1695. (cited on p. 24).

Munshi, Kaivan, "Community Networks and the Process of Development," *Journal of Economic Perspectives*, 2014, *28* (4), 49–76. (cited on p. 7).

_ **and Mark Rosenzweig**, "Traditional Institutions Meet the Modern World: Caste, Gender, and Schooling Choice in a Globalizing Economy," *American Economic Review*, 2006, *96* (4), 1225–1252. (cited on pp. 7, 29, and 37).

_ **and** _ , "Networks and Misallocation: Insurance, Migration, and the Rural-Urban Wage Gap," *American Economic Review*, 2016, *106* (1), 46–98. (cited on p. 7).

Nunn, Nathan, "Culture and the Historical Process," *Economic History of Developing Regions*, 2012, *27* (sup1), S108–S126. (cited on p. 6).

_ , "Historical Development," in "Handbook of Economic Growth" 2014, pp. 347–402. (cited on p. 6).

_ , "The Historical Roots of Economic Development," *Science*, 2020, *367* (6485), eaaz9986. (cited on p. 6).

_ **and Leonard Wantchekon**, "The Slave Trade and the Origins of Mistrust in Africa," *American Economic Review*, 2011, *101* (7), 3221–52. (cited on pp. 1 and 6).

Ochsner, Christian and Felix Roesel, "Activated History: The Case of the Turkish Sieges of Vienna," *American Economic Journal: Applied Economics*, 2024, *16* (3), 76–112. (cited on pp. 4, 6, 30, and 36).

Oh, Suanna, "Does Identity Affect Labor Supply?," *American Economic Review*, 2023, *113* (8), 2055–2083. (cited on p. 7).

Portes, Alejandro, "Social Capital: Its Origins and Applications in Modern Sociology," *Annual Review of Sociology*, 1998, *24*, 1–24. (cited on p. 7).

Ramos-Toro, Diego, "Social Exclusion and Social Preferences: Evidence from Colombia's Leper Colony," *American Economic Review*, 2023, *113* (5), 1294–1333. (cited on pp. 1 and 6).

Reguillo, Rossana, "Semantizarás el territorio: los vecinos de Analco y las explosiones de 1992 en Guadalajara," *Alteridades*, 1998, *8* (15), 35–44. (cited on pp. 2, 13, 22, and 42).

Rojas-Ampuero, Fernanda and Felipe Carrera, "Sent Away: Displacement, Neighborhoods, and Children's Outcomes under Slum Clearance Policies," 2026. Working paper. (cited on p. 7).

Rosenthal, Stuart S. and Stephen L. Ross, "Change and Persistence in the Economic Status of Neighborhoods and Cities," in Gilles Duranton, J. Vernon Henderson, and William C. Strange, eds., *Handbook of Regional and Urban Economics*, Vol. 5B, Elsevier, 2015, chapter 16, pp. 1047–1120. (cited on p. 6).

Sacerdote, Bruce, "Peer Effects in Education: How Might They Work, How Big Are They and How Much Do We Know Thus Far?," in Eric A. Hanushek, Stephen Machin, and Ludger Woessmann, eds., *Handbook of the Economics of Education*, Vol. 3, Elsevier, 2011, pp. 249–277. (cited on p. 7).

Sampson, Robert J., Jeffrey D. Morenoff, and Thomas Gannon-Rowley, "Assessing 'Neighborhood Effects': Social Processes and New Directions in Research," *Annual Review of Sociology*, 2002, *28*, 443–478. (cited on p. 7).

Shayo, Moses, "Social Identity and Economic Policy," *Annual Review of Economics*, 2020, *12* (1), 355–389. (cited on p. 7).

Squires, Munir, "Kinship Taxation as an Impediment to Growth: Experimental Evidence from Kenyan Microenterprises," *Economic Journal*, 2024, *134*, 2558–2579. (cited on pp. 7 and 44).

Stack, Carol B., *All Our Kin: Strategies for Survival in a Black Community*, New York: Harper & Row, 1974. (cited on p. 1).

Tanck de Estrada, Dorothy, *Pueblos de indios y educación en el México colonial, 1750–1821*, Mexico City: El Colegio de México, Centro de Estudios Históricos, 1999. (cited on pp. 9 and 10).

_ , *Atlas ilustrado de los pueblos de indios de la Nueva España, 1800*, Fomento Cultural Banamex A C, 2005. (cited on p. 9).

Topa, Giorgio and Yves Zenou, "Neighborhood and Network Effects," in "Handbook of Regional and Urban Economics" 2015, pp. 561–624. (cited on p. 7).

Townsend, Robert M., "Risk and Insurance in Village India," *Econometrica*, 1994, *62* (3), 539–591. (cited on p. 7).

Valencia-Caicedo, Felipe, "The Mission: Human Capital Transmission, Economic Persistence, and Culture in South America," *The Quarterly Journal of Economics*, 2019, *134* (1), 507–556. (cited on pp. 1 and 6).

Valenzuela-Casasempere, Pablo, "Displacement and Infrastructure Provision: Evidence from the Interstate Highway System," Technical Report 2025. (cited on p. 11).

Voth, Hans-Joachim, "Persistence – Myth and Mystery," in Alberto Bisin and Giovanni Federico, eds., *The Handbook of Historical Economics*, Academic Press, 2021, pp. 243–267. (cited on p. 6).

Walton, John, "Guadalajara: Creating the Divided City," in Wayne A. Cornelius and Robert V. Kemper, eds., *Metropolitan Latin America: The Challenge and the Response*, Vol. 6 of *Latin American Urban Research*, Beverly Hills: Sage, 1978, pp. 25–50. (cited on pp. 8 and 11).

Wantchekon, Leonard, Marko Klašnja, and Natalija Novta, "Education and Human Capital Externalities: Evidence from Colonial Benin," *The Quarterly Journal of Economics*, 2015, *130*, 703–757. (cited on p. 6).

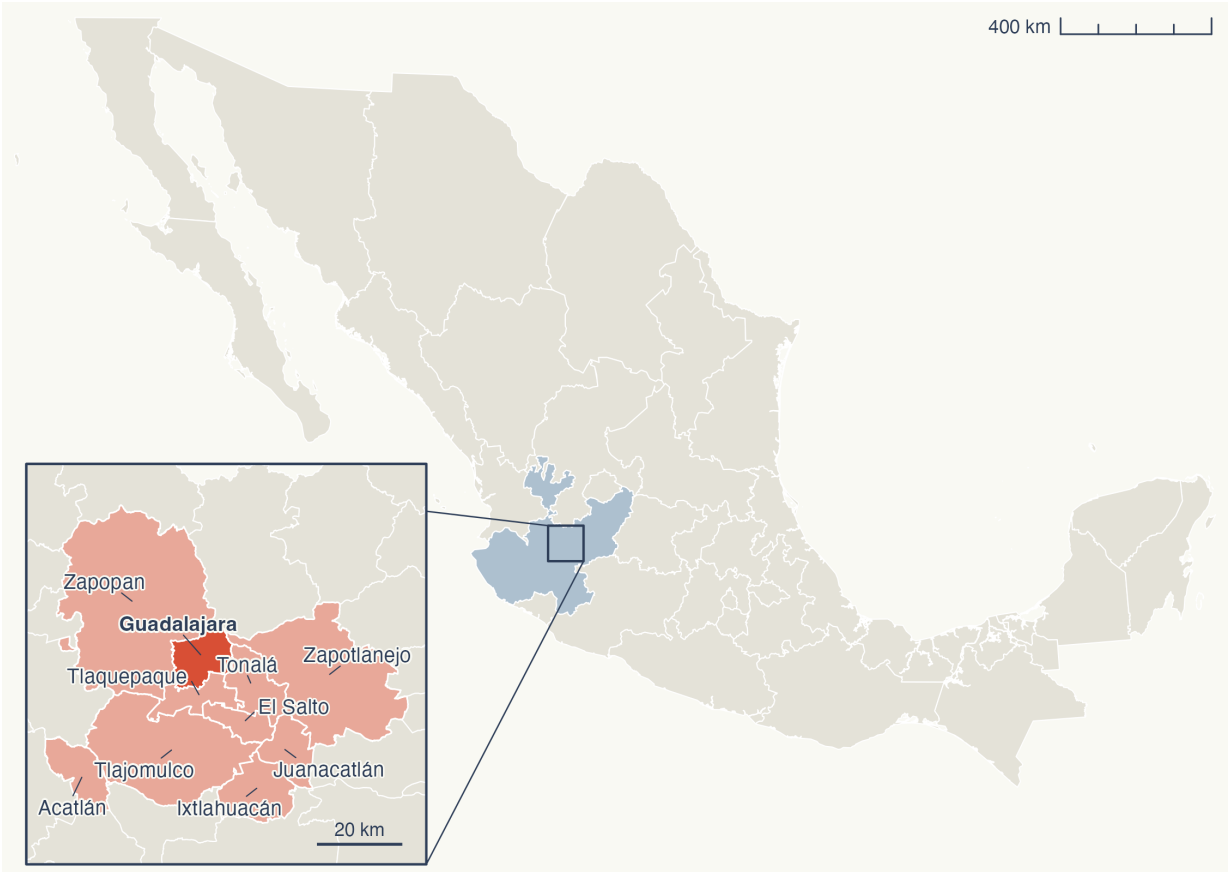
Woo-Mora, L. Guillermo, "Unveiling the Cosmic Race: Skin Tone and Intergenerational Economic Disparities in Latin America and the Caribbean," *Journal of Development Economics*, 2026, *179*, 103594. (cited on pp. 1 and 80).

A Context and Historical Natural Experiment

A.1 Guadalajara on the map and in the national distribution

Figure A.1 locates Guadalajara in Mexico and within its metropolitan area.

Figure A.1: Guadalajara in Mexico and in its metropolitan area



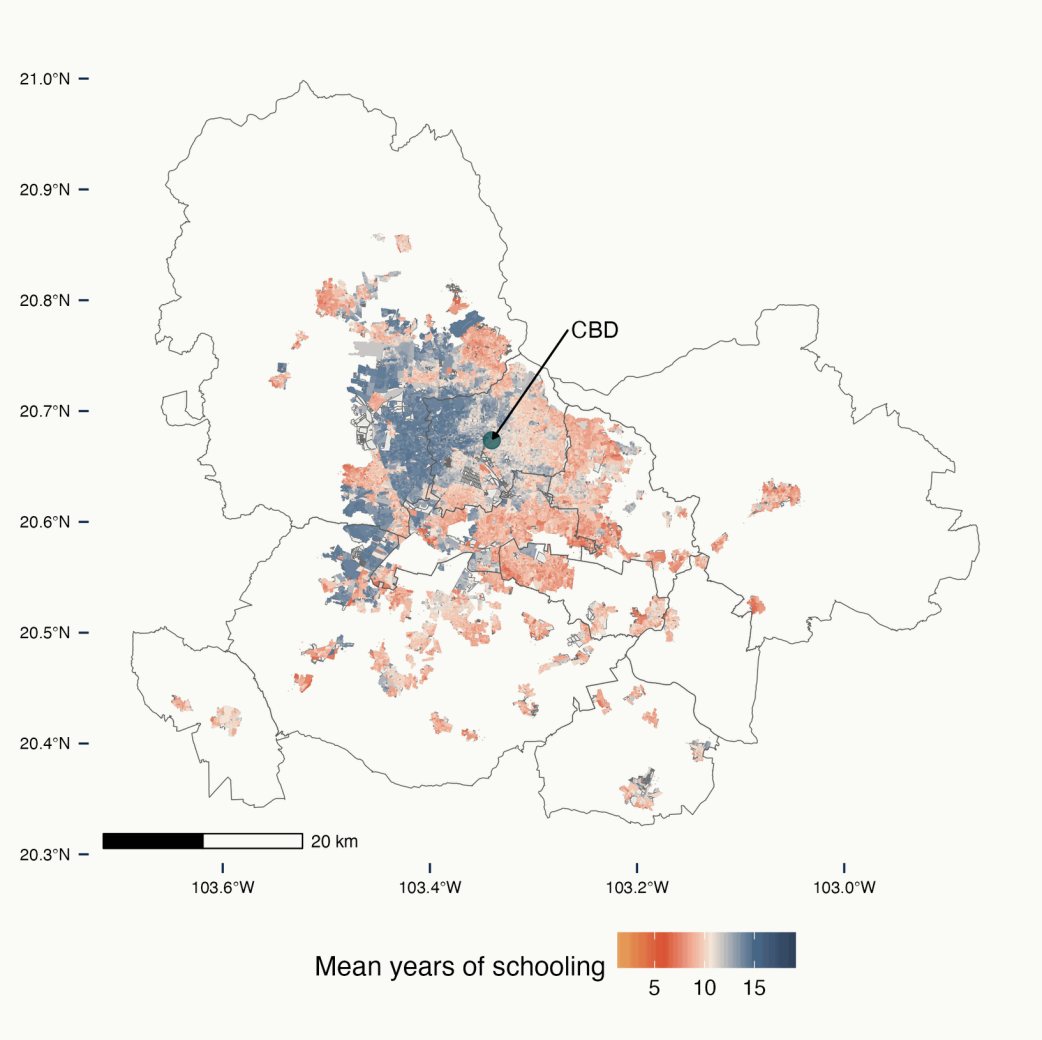
Notes: Mexico by state, with Jalisco in light blue. The box marks the Guadalajara metropolitan area, enlarged at the lower left. In the enlarged view, the ten municipalities of the metropolitan area are in light red, the municipality of Guadalajara in dark red, and other municipalities of Jalisco in gray. State and municipal boundaries are from INEGI's 2019 *Marco Geoestadístico*.

↩ Back to text

Figure A.2 extends the block-level map of Figure 2b to the ten municipalities of the Guadalajara metropolitan area. The east–west gradient does not stop at the municipal border: blocks west of the historical core, into Zapopan, hold the highest mean schooling, and blocks east of it, into Tonalá and San Pedro Tlaquepaque, the lowest.

Figure A.3 plots, for each claim in the context paragraph of Section 2, the distribution of the outcome across Mexico's urban municipalities, with Guadalajara and the national value marked. The municipality sits in the 98th percentile of household income and the 95th of mean schooling, and its school attendance exceeds the national rate at every level beyond primary.

Figure A.2: Mean years of schooling by census block, Guadalajara metropolitan area (2020)



Notes: Mean years of schooling at the census-block level for the ten municipalities of the Guadalajara metropolitan area, from the public block-level statistics of the 2020 *Censo de Población y Vivienda* (INEGI). Shading runs from red (lower mean schooling) to blue (higher mean schooling), on the same scale as Figure 2b; unshaded polygons are blocks with suppressed or zero values, and dark lines are municipal boundaries. The point marks the central business district. The east–west gradient visible at the historical boundary continues across municipal lines.

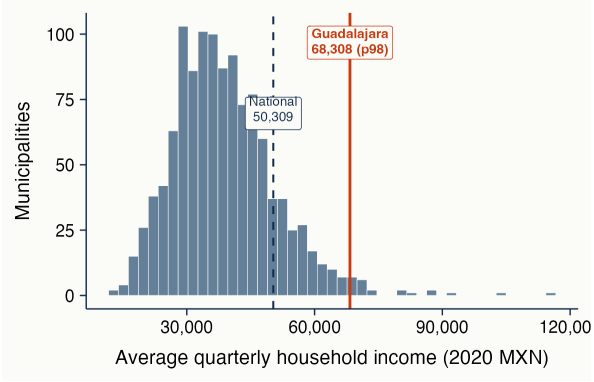
↩ Back to text

A.2 School attendance over time and where it varies

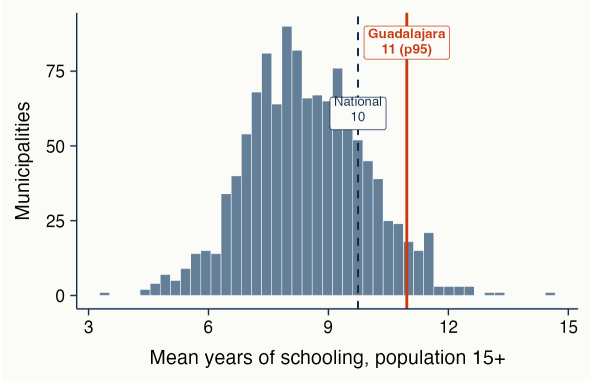
Figure A.4 extends the attendance comparison of Section 2 back to 1970 using census microdata. Guadalajara exceeds the national rate at ages 12 and above in every census, and the advantage is concentrated at the post-primary margins. At ages 6–11 the municipality led the national rate by six points in 1970, but the gap closed as attendance approached universal coverage nationwide, and in 2010 and 2020 Guadalajara sits marginally below it (93.7 against 94.5 percent in 2020). At ages 18–24 its lead instead widened, from 21.5 against 12.0 percent in 1970 to 42.7 against 32.4 percent in 2020.

Figure A.5 asks where that variation lives. Writing y for whether a young person aged 12–17

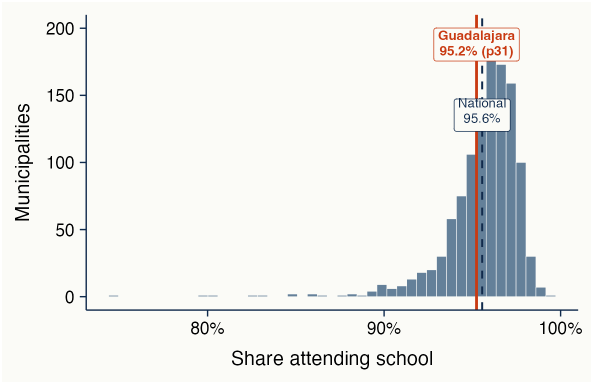
Figure A.3: Guadalajara in the national distribution of income, schooling, and attendance (2020)



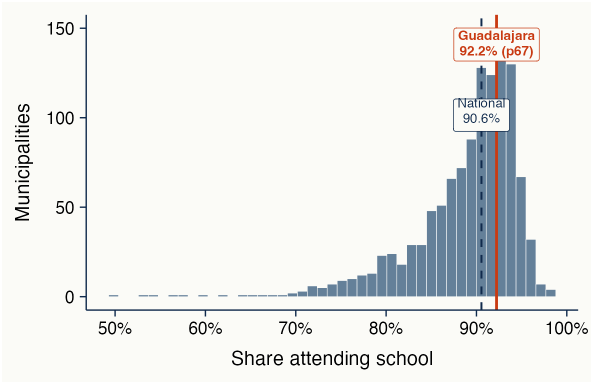
(a) Quarterly household income



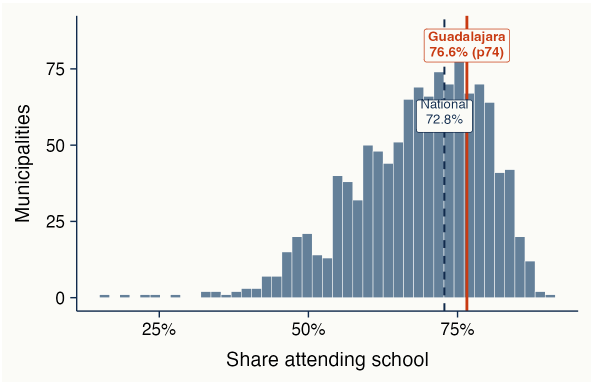
(b) Years of schooling, ages 15+



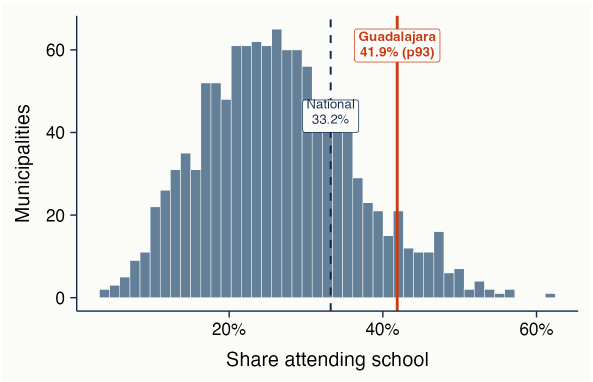
(c) School attendance, ages 6–11



(d) School attendance, ages 12–14



(e) School attendance, ages 15–17



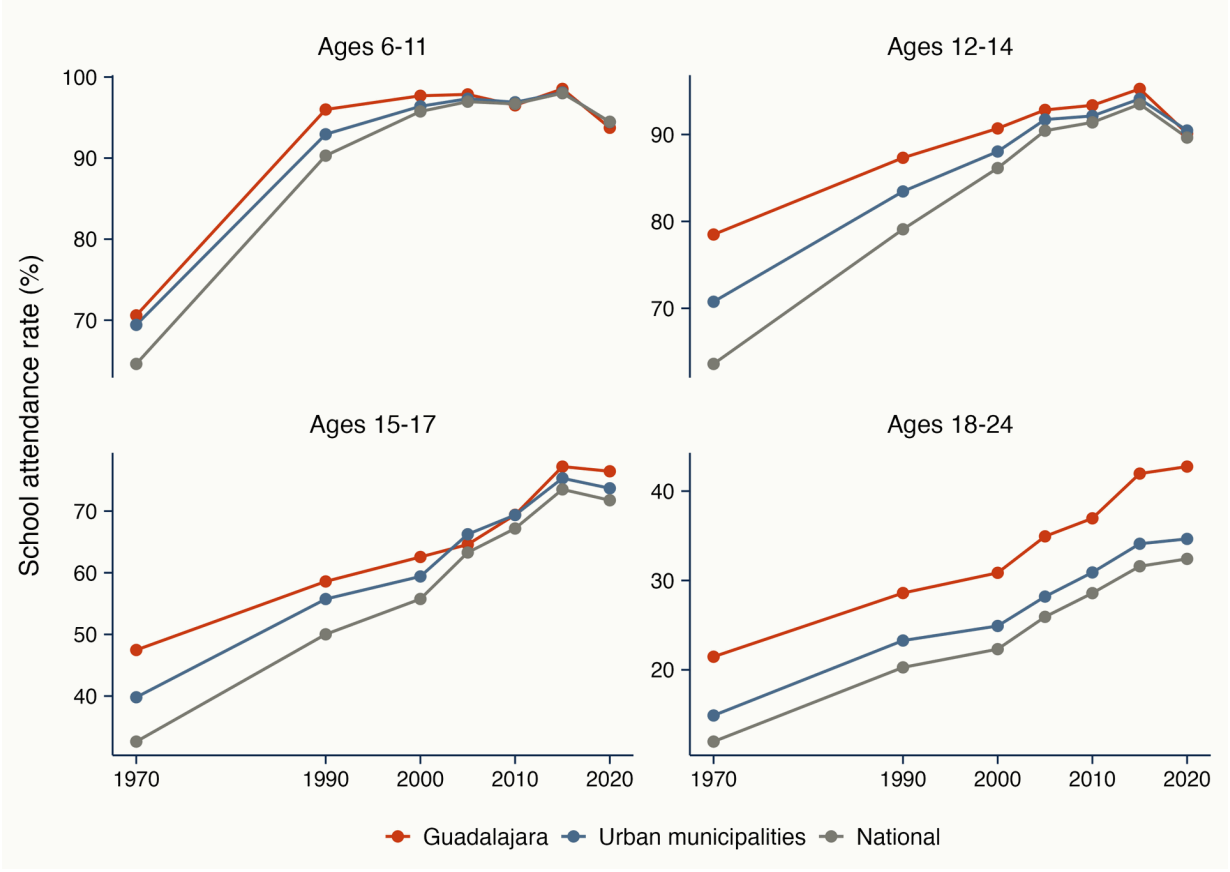
(f) School attendance, ages 18–24

Notes: Each panel plots the distribution of the outcome across the 1,169 urban municipalities of Mexico (municipalities with at least 15,000 residents in the 2020 census). The solid line marks Guadalajara, labeled with its value and its percentile within the urban distribution; the dashed line marks the national value. Household income is the 2020 average quarterly household income in pesos at the municipal level; schooling and attendance come from the municipal statistics of the 2020 *Censo de Población y Vivienda* (ITER). Attendance at ages 6–11 is near-universal everywhere, so Guadalajara's advantage appears only at the post-primary levels, as stated in the text.

↩ Back to text

attends school, the variance splits exactly into three nested parts: differences between municipalities, differences between households of the same municipality, and differences between siblings

Figure A.4: School attendance by age group, 1970–2020



Notes: Share of individuals in each age group reported as attending school, computed from the IPUMS International samples of the Mexican censuses of 1970, 1990, 2000, 2010 and 2020 and the intercensal counts of 2005 and 2015, weighted by person weights. "Urban municipalities" are the harmonized municipalities whose population in urban localities reaches at least 50 percent in the corresponding year, excluding Guadalajara; "National" pools all municipalities. The 1995 count is omitted because its sampling design and geographic coverage differ from those of the censuses. The 1960 sample does not record school attendance.

↩ Back to text

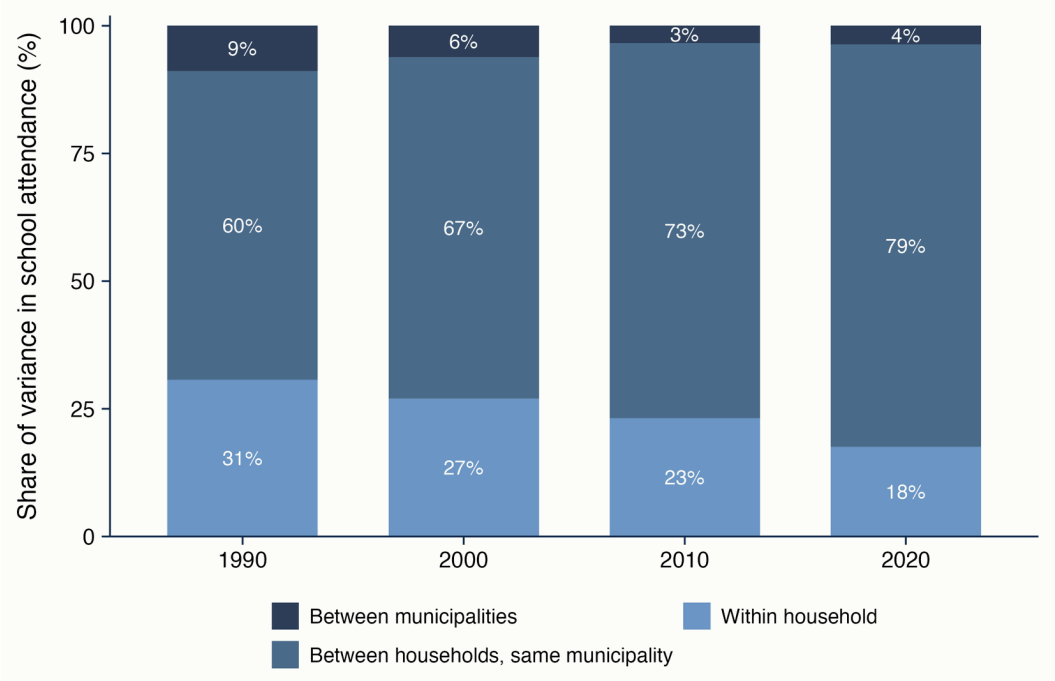
of the same household. The municipality accounts for very little and for steadily less: 8.8 percent of the variance in 1990 and 3.7 percent in 2020. Almost all of the remainder sits between households living in the same municipality, a component that rose from 60.5 to 78.7 percent over the same period. Educational inequality in Mexico thus moved away from differences across places and toward differences across households within a place.

Two further results follow. Observable household characteristics account for only part of the household component. Age, sex, the schooling of the household head or spouse, birth order and sibship size jointly explain 13.7 percent of the total variance in 2020, and net of them the municipality share falls to 1.6 percent while the household share rises to 80.2 percent. The modest explanatory share should not be read as parental schooling being unimportant, since it shifts attendance substantially: 57.5 percent of 12–17 year-olds attend school where the head or spouse has at most five years of schooling, against 92.4 percent where that figure reaches twelve

years. Parental schooling alone accounts for 8.3 percent of the variance because attendance is binary and most of its dispersion is idiosyncratic. And the household component is not merely the sum of individual differences. Among households with at least two children in the age range, where the split is separately identified, 59.0 percent of the variance lies between households and 36.2 percent between siblings within a household.

The exercise cannot go further than the municipality, because the census records no geography below it. The between-household component therefore pools whatever distinguishes households from whatever distinguishes the neighborhoods they live in. Separating the two requires variation in neighborhood below the municipality. The boundary in Section 3 supplies it, comparing households a few blocks apart within a single municipality, on either side of a line that other determinants of schooling vary smoothly across. That design compares different households rather than following one household across neighborhoods, so it does not by itself separate place effects from the composition of residents, a limitation Section 3 discusses.

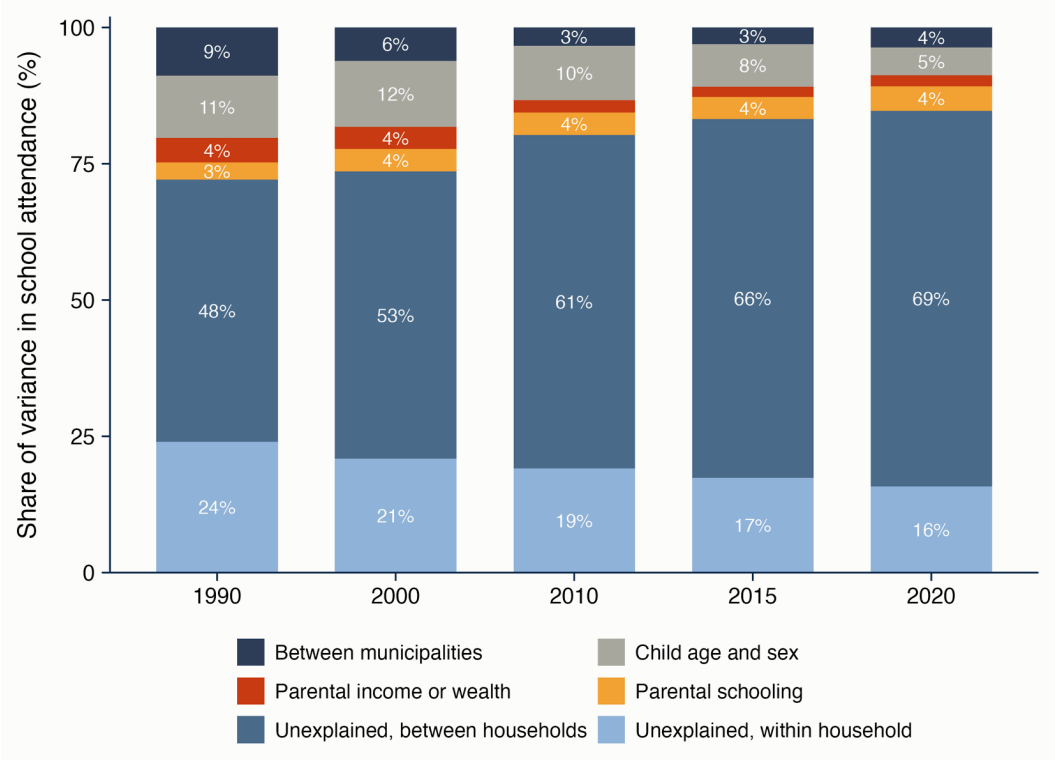
Figure A.5: Where the variance in school attendance lies, 1990–2020



Notes: Nested decomposition of the variance of an indicator for attending school among individuals aged 12–17 who are children of the household head, using the IPUMS International samples of the 1990, 2000, 2010 and 2020 Mexican censuses and weighting by person weights. The three components are $\mathrm{Var}_m(\bar{y}_m)$, $\mathbb{E}_m[\mathrm{Var}_h(\bar{y}_h)]$ and $\mathbb{E}_h[\mathrm{Var}(y)]$, where m indexes harmonized municipalities and h households nested within them; they sum to the total variance by construction. Households with a single child in the age range contribute no within-household variance, so their individual variation enters the household component; restricting to households with two or more such children, which identifies the split separately, yields 4.8, 59.0 and 36.2 percent in 2020. The census provides no geography below the municipality, so the household component contains both household and neighborhood differences.

↩ Back to text

Figure A.6: Parental resources and the variance in school attendance, 1990–2020



Notes: Decomposition of the variance of an indicator for attending school among individuals aged 12–17 who are children of the household head, using the IPUMS International samples of the 1990, 2000, 2010 and 2020 censuses and the 2015 Intercensal Survey, weighted by person weights. The between-municipality component is as in Figure A.5. The within-municipality variance is split with municipality fixed effects into the contributions of four blocks of indicators, allocated by Shapley values over all orderings: the child's age by sex; parental income or wealth (deciles of the head's and spouse's labor earnings per household member, and deciles of the first principal component of electricity, piped water, sewage, gas or electric cooking fuel, television, radio, home ownership, and rooms per person); and parental schooling (years of the more schooled of head and spouse). Labor earnings of other members are excluded because they include the earnings of working children. The unexplained remainder is split into its between-household and within-household parts. Components sum to the total variance.

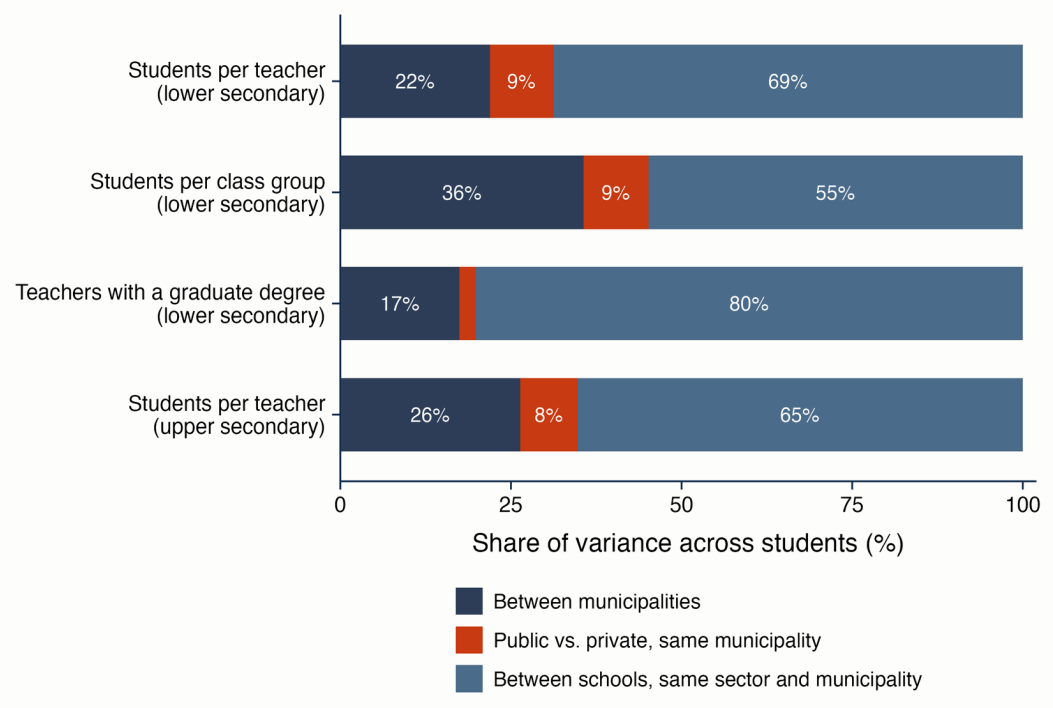
↩ Back to text

A.3 Pre-colonial population and locational fundamentals

The natural-experiment argument requires that nothing already distinguished the two banks of the SJdD creek when the settlements were founded in 1542 — that neither a pre-colonial population center nor an exogenous geographic feature dictated which bank each community occupied. This subsection presents the evidence behind that claim, first at the scale of the valley and then at the scale of the boundary itself.

Pre-colonial population. Figure A.10 shows pre-colonial population density across Mesoamerica circa 1500, drawing on the historical reconstruction of Goldewijk et al. (2017). The Anahuac Valley (Panel b) — the heartland of the Mexica state and home to Tenochtitlán — concentrated densely settled urban polities. The Atemajac Valley (Panel c), where Guadalajara would later be

Figure A.7: Where the variance in school supply lies



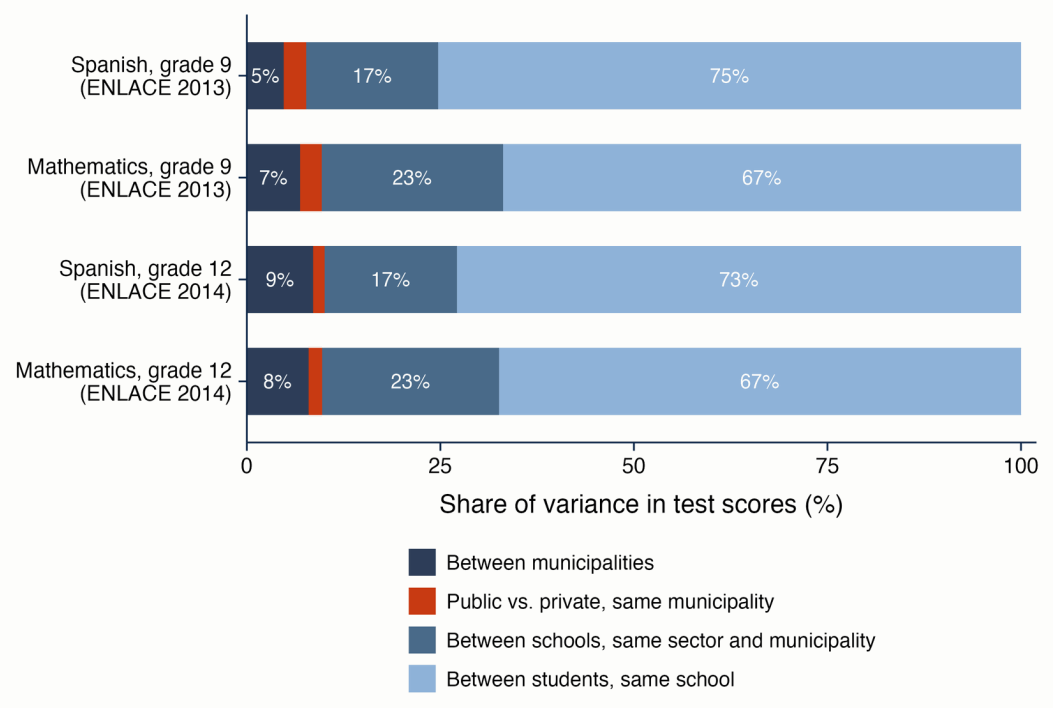
Notes: Nested decomposition of the national variance of school characteristics across students into differences between municipalities, between public and private schools of the same municipality, and between schools of the same sector and municipality. Each school is weighted by its enrollment. Private schools hold 9.4 percent of lower-secondary enrollment. Lower secondary: *Formato 911*, 2018–2019, 34,195 schools; upper secondary: *Formato 911* for general and technological high schools, 2018–2019. Student–teacher and student–group ratios above the 99.5th percentile are set to missing as likely capture errors. Attending a private school is a household choice, and differences between schools combine differences in supply with the sorting of students across schools.

↩ Back to text

founded, contained dispersed Indigenous populations but no major agglomerations. The valley therefore held no established urban center whose location could have dictated the 1542–1543 allocation of colonizers, allies, and conquered populations across the SJdD creek. That statement concerns the valley rather than the creek. The reconstruction is gridded at five arc-minutes, about nine kilometers at this latitude, so a single cell covers the entire estimation window and cannot speak to settlement or land use at the boundary itself. It also rests on the pre-colonial reconstruction alone. Figure A.11 reports population estimates from Buringh (2018) for Mexico City and Guadalajara from the 16th to 18th centuries, showing that Guadalajara grew slowly relative to Mexico City and from a low initial base, but that series begins after the foundation and documents how the asymmetry persisted rather than what preceded it. What the design requires is not an empty landscape but the absence of a discontinuity at the creek, which Table A.1 examines directly.

Locational fundamentals. A complementary concern is that, even absent pre-colonial agglomerations, geographic features could have determined which side of the SJdD creek was settled by

Figure A.8: Where the variance in test scores lies

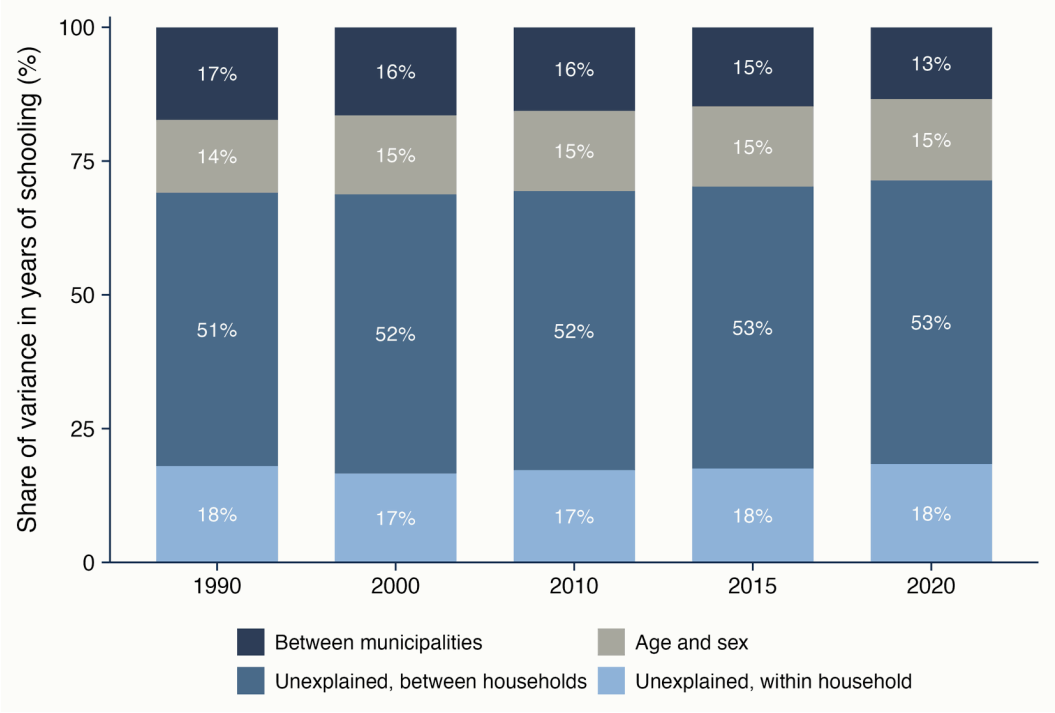


Notes: Nested decomposition of the national variance of individual ENLACE scores into differences between municipalities, between public and private schools of the same municipality, between schools of the same sector and municipality, and between students of the same school. Private schools are the *Particular* modality in grade 9 and the private *sostenimiento* in grade 12. Grade 9: ENLACE 2013, 1,589,219 students, excluding those flagged for copying. Grade 12: ENLACE 2014 for upper secondary, 972,272 students, excluding those who answered fewer than half the items in either subject, as in official averages; this file carries no copying flag.

↩ Back to text

colonizers and which by the conquered. Figure A.12 maps altitude at the grid-cell level, in cells of about 150 × 150 m, across the foundational sites, using INEGI's high-resolution raster data. The terrain is essentially uniform: there is no elevation differential between Spanish Guadalajara and Mexicaltzingo (west bank) and Analco (east bank), and no topographic feature that would have imposed a particular settlement pattern. Table A.1 formalizes this assessment. I implement a Boundary Discontinuity (BD) design on the grid cells covering the municipality of Guadalajara, with the cell's signed orthogonal distance to the SJdD creek as the running variable and MSE-optimal bandwidths, and test for differences across banks in five locational fundamentals: altitude, slope, mean temperature, precipitation, and humidity. Slope is the only estimate significant at the 10 percent level, with a p-value of 0.074 and a difference of 0.06 standard deviations, and no estimate survives the correction for multiple testing, with sharpened FDR-adjusted q-values (Anderson, 2008) of 0.585 or more. The two topographic tests are the informative ones, and they are precise. Altitude and slope come from INEGI's continuous elevation model at 15 m native resolution, a direct measurement that resolves the boundary, and their confidence intervals exclude differences larger than 0.02 and 0.12 standard deviations. The three climate layers are interpolated long-run normals for 1902–2011, so smoothness across the estimation window is in

Figure A.9: Where the variance in adult years of schooling lies, 1990–2020



Notes: Decomposition of the variance of years of schooling among individuals aged 25 and older, using the IPUMS International samples of the 1990, 2000, 2010 and 2020 censuses and the 2015 Intercensal Survey, weighted by person weights. The between-municipality component uses harmonized municipalities. The within-municipality variance is split with municipality fixed effects into the contribution of age-by-sex indicators and an unexplained remainder, which is divided into its between-household and within-household parts. Components sum to the total variance. Parental schooling and resources are not included because the parents of adults rarely live in the same household; household income and wealth are not included because for adults they depend on their own schooling.

↩ Back to text

part a property of the interpolation rather than an independent finding, and precipitation leaves an interval reaching about 0.30 standard deviations. No layer measures hydrology, and the one hydrological feature at the boundary is the creek itself, which defines the assignment and therefore cannot serve as a balance test. These estimates bound differences across banks rather than establish equivalence. What they show is no topographic feature large enough to have dictated which bank each community occupied, which is consistent with the documentary record that the allocation was administrative.

A.4 Persistence of the urban form: from creek to Calzada

Unlike many Mexican cities, where 19th- and 20th-century reorganizations erased earlier divisions, Guadalajara kept the initial 1542 structure as the basis of its long-run urban form.

From creek to avenue. Figure A.13 documents the transformation. Panel (a) shows the SJdD creek as a still-visible watercourse in the early 1900s; between 1903 and 1911 municipal authori-

Table A.1: Balance in locational fundamentals

Outcome	Mean within h		$\hat{\tau}$	SE	95% CI	p-value	FDR q-value	Poly	h (m)	b (m)	N_h	N_b
	West	East										
Altitude	1534.17	1538.68	0.001	0.012	[−0.022, 0.024]	0.929	1.000	1	576	1,281	222	622
Slope	1.18	1.56	−0.057	0.032	[−0.120, 0.006]	0.074	0.585	1	528	1,295	200	629
Temperature	20.39	20.37	0.007	0.023	[−0.037, 0.052]	0.751	1.000	1	603	1,285	105	278
Precipitation	934.74	928.33	−0.116	0.093	[−0.298, 0.066]	0.211	0.733	1	1,298	2,296	281	640
Humidity	45.97	45.73	−0.021	0.035	[−0.091, 0.048]	0.549	1.000	1	1,067	1,867	217	474

Notes: Estimates from a BD design at the grid-cell level, using INEGI raster data on locational fundamentals (cells of about 150 × 150 m for altitude and slope and of about 220 × 230 m for the climate normals) for the cells covering the municipality of Guadalajara, around the San Juan de Dios creek (today Calzada Independencia). Each row reports the estimated discontinuity for a different fundamental: altitude (m), slope (degrees), mean annual temperature (°C), annual precipitation (mm), and humidity (Lang aridity index, precipitation over temperature). All variables are standardized within the analysis window before estimation, so $\hat{\tau}$, its standard error, and its interval are in standard deviations. The running variable is the cell's signed orthogonal distance to the historical boundary, with positive values on the east side. Each estimate is a local linear regression with a triangular kernel, the MSE-optimal bandwidth h, a bias bandwidth b, the cell's longitude and latitude as covariates, and boundary-segment fixed effects. $\hat{\tau}$ is the bias-corrected estimate, and bandwidth selection and inference follow Calonico et al. (2014). The West and East columns are raw means within h of the boundary, in the units above. N_h and N_b are the numbers of cells within h and b of the boundary. Sharpened FDR-adjusted q-values (Anderson, 2008) are computed across the five fundamentals. Standard errors and 95% confidence intervals are robust and bias-corrected, with nearest-neighbor variance estimation. Slope is the only estimate significant at the 10 percent level, with a p-value of 0.074 and a difference of 0.06 standard deviations, and no estimate survives the correction for multiple testing, with q-values of 0.585 or more. The intervals bound the size of the differences across banks. They are tightest for altitude, which derives from a 15 m elevation model, and widest for precipitation, which comes from interpolated climate normals.

↩ Back to text

ties piped it underground and built a major north–south avenue along its course (panel b, Calzada Independencia). The transformation eliminated the creek as a physical barrier but preserved its position as the city's main east–west axis.

How often the Calzada is crossed. Figure A.14 compares the stretch of Calzada Independencia that runs along the boundary with stretches of similar length on two avenues in other Latin American cities, two urban rivers, and an urban expressway. Built blocks on the two sides of the Calzada stand 36 meters apart, and 22 streets, footbridges, and pedestrian crossings cross it, one every 151 meters. Built land stands 287 meters apart across the Seine in Paris and 314 meters across the Thames in London, with a crossing every 406 and 473 meters, and 253 meters apart across the Dan Ryan Expressway in Chicago, with a crossing every 549 meters. Avenida Caracas in Bogotá, with built land 43 meters apart and a crossing every 220 meters, and Bulevar 5 de Mayo in Puebla, with 37 meters and a crossing every 158 meters, are close to the Calzada.

A.5 Named institutions on each bank, 1800–1908

Sources. Four general maps of Guadalajara carry an index that names the city's institutions and keys each one to a block (Figure A.15). The 1800 sheet is a facsimile of a plan of the city "as it stood in the year 1800," copied from the original held at the *Archivo del Instituto de Geografía y Estadística*; its legend has 37 entries. The 1863 wave is a series of nine *cuartel* sheets held at the *Archivo Histórico de Jalisco*, surveyed by order of the state government for the use of its offices; the sheets for *cuarteles* 7 and 8 are signed by C. Vicente Ortigosa, 19 July 1863, and those for

cuarteles 1 and 2 by the *Dirección General de Rentas*, 5 April 1863. Each sheet numbers its blocks individually, labels its streets, and carries a legend of notable buildings. The 1890 wave is the *Plano General de la Ciudad de Guadalajara*, whose margins carry an index of establishments with *cuartel*, *manzana*, and street columns, divided into sections for notable buildings, markets, gardens, squares, arcades, hotels, baths, and former customs gates. The 1908 wave is the *Novísimo Plano General de la Ciudad de Guadalajara, con la nomenclatura últimamente aprobada por el Ayuntamiento*, by the engineer Regino Guzmán, published by Loreto y Ancira; it prints a red number on each block and lists the corresponding establishments in the margin under 105 entries.

Locating the entries. The 1863 and 1890 indices key their entries to a (*cuartel*, *manzana*) pair. I join them to the digitized 1890 plan, whose 713 polygons resolve into 672 numbered blocks once split pieces of the same block are merged, and whose numbering scheme the 1863 sheets share; the correspondence is validated against the 2010 national register of churches, restricted to within three kilometers of the creek before any name matching, with residuals between 0 and 75 meters. The 1800 legend has no block key: church entries are located through the same church register, and the remaining entries through name correspondence with the 1863 index. The 1908 plan requires reading the red numbers directly off the raster, which I do on magnified tiles; entries that survive in the earlier waves are additionally assigned by name correspondence with the georeferenced 1863 and 1890 sets. Every located entry is assigned to a bank by the sign of its position relative to the same boundary line used in the BD design.

Table A.2 counts 31 of the 37 entries in the 1800 legend, all 56 in the 1863 sheets, 83 of the 91 in the 1890 index, and 62 of the 105 in the 1908 index. The six 1800 entries that do not enter are of four kinds. One, *el río*, names the creek itself rather than an establishment. One, *las garitas*, names the customs gates at several points of the perimeter and belongs to no single bank. The *Casa de Enseñanza de Niñas* could not be found after a directed search. The remaining three, the *Casa de Recogidas*, the *Real Fábrica de Tabacos* and the *Puente de las Damas*, are named on the sheet but could not be tied to a block precisely enough to place them, and the count admits only entries with a location. The eight 1890 entries that do not enter are the *Escuela de Jurisprudencia*, the *Hospital de la Trinidad*, the *Teatro Novedades*, the *Hipódromo*, the *Estación del FCCM*, the *Penitenciaría*, the *Jardín de la Plaza de Armas* and the *Jardín de Aclimatación*, whose index keys do not match a digitized block; several of them stood outside the surveyed grid. Of the 62 entries assigned in 1908, 24 were located on the plan and 38 by name correspondence with the georeferenced 1863 and 1890 sets. The 24 divide further into seventeen read directly off the raster, four placed through the church register, one the *Seminario Conciliar*, whose number the plan prints on two blocks, one on each bank, resolved to the western occurrence because it matches its 1890 location, and two assigned by elimination after failing to appear in eighteen sampled tiles of the central grid. Elimination is a way of locating an entry on the plan rather than a third method beside the other two. The educational series is counted from a second and more intensive pass over the sheets, which admits entries read directly off the plan as well as georeferenced ones, and

therefore recovers entries the general count leaves out. Among the fifteen educational entries of the 1908 index, fourteen are western and one eastern; of the fourteen, twelve were located directly and two by elimination. In 1800 the same pass recovers two western educational entries, the *Colegio femenino* and the *Iglesia de Ex-Jesuitas y Universidad*.

Counts by category. Table A.2 reports the located entries by category, wave, and bank.

Table A.2: Located institutions by category, wave, and bank

	1800		1863		1890		1908	
Category	West	East	West	East	West	East	West	East
Church	15	3	18	3	0	0	21	3
Educational	4	0	3	0	10	0	14	1
Government	5	0	5	0	12	1	5	0
Square and garden	0	0	11	1	8	0	6	0
Commerce	0	0	0	0	20	0	0	0
Culture and recreation	1	0	4	0	7	1	2	1
Health and charity	1	0	1	1	5	2	3	2
Market	0	0	2	1	3	1	3	1
Baths	0	0	0	0	8	1	0	0
Infrastructure	2	0	1	0	3	0	0	0
Mill	0	0	3	0	0	0	0	0
Cemetery	0	0	1	1	0	1	0	0
Total	28	3	49	7	76	7	54	8

Notes: Counts of entries in each map's index that could be located on the urban grid and assigned to a bank of the SJdD creek. Categories are assigned by the author from the entry names and the sections of the original indices. Each map reflects its own editorial conventions about what merits an entry, so the composition of categories varies across waves. The 1890 index has a commercial section and no separate church section, while the 1908 index lists churches and no commercial establishments. Levels are comparable across banks within a wave, and across waves only within a category that appears in all four, as the educational category does.

↩ Back to text

In 1800, the only entries east of the creek are the three churches of Analco, San Sebastián, and San Juan de Dios. The eastern bank appears in the index of that year with places of worship and with no institution of any other kind.

Population on each bank. Two enumerations record residence by *cuartel* close enough to these waves to serve as a denominator. To assign *cuarteles* to banks I use the digitized blocks of the 1890 plan. *Cuartel* 9 lies entirely east of the creek and *cuarteles* 1 through 7 entirely west, with

no block on the far side of the line in either case. *Cuartel* 8 has 87 of its 91 blocks east. Those four blocks are the only departure from a partition that is otherwise exact at the level of the *cuartel*, and they are too few to make any other assignment of *cuartel* 8 sensible.

In the 1821 *padrón* of Anderson and Spike (2007), the two eastern *cuarteles* hold 6,873 residents against 10,346 in the seven western *cuarteles* of the central core, an eastern share of 39.9 percent. The comparison is restricted to that core, which is also the area the plans cover, and accounts for 49 percent of the 35,287 residents enumerated in 1821. Of the 24 populated *cuarteles* of the *padrón*, the remaining fifteen lie outside the core and cannot be assigned to a bank from what the enumeration records. That assignment rests on the *padrón*'s *cuartel* numbers denoting the same units as the plans, and the *padrón* does not allow me to verify it. Its scheme does not correspond to the one the maps use, and the natural check, matching street names, is unavailable for the unit that matters most, since the street field is empty in all 3,892 records of *cuartel* 8, the most populous unit in the enumeration and the one taken to be Analco. The 1821 share is therefore a benchmark conditional on that identification rather than an independently verified denominator. The 1930 census does not carry the same ambiguity. Its *cuartel* scheme runs from 1 to 11 and appears to continue the one the plans use, and there the two eastern *cuarteles* hold 33.6 percent of the population recorded for the city.

Limitations. Three limitations bear on how these counts should be read. First, the indices record the institutions their authors considered worth naming, and neighborhood primary instruction would not ordinarily appear in them; the counts describe where the city's endowed establishments stood rather than the number of school places available on each bank. Second, the share of located entries in 1908 is 59 percent, and the excluded entries are concentrated in categories that expanded during the Porfiriato, so the eastern share in that wave may be overstated. Third, the transcriptions of the indices have been coded once; a second independent reading of a subsample, with an agreement rate, has not yet been carried out.

A.6 Capitalization in the colonial core: 1882 land values

I introduce newly digitized microdata from Guadalajara's 1882 cadaster, a parcel-level record of assessed property values rare in its detail for nineteenth-century Latin America. Figure A.17, panel (a) reproduces an example façade entry from the original records; panel (b) shows the digitized distribution of land values across the central core. Historical maps were georeferenced to modern coordinates following the procedure detailed at the end of this subsection. Figure A.18 reports formal estimates of the discontinuity in 1882 land values at the SJdD creek using the Local Randomization RDD of Cattaneo et al. (2015). Across bandwidths from 150 to 500 meters, assessed land values are between 48 and 67 percent lower on the east bank. The narrowest window, 100 meters, gives a positive point estimate with an interval too wide to be informative. This divergence in 1882 — well before 20th-century industrialization, the federal

expansion of public schooling, or post-revolutionary demographic homogenization — captures the early capitalization of differential access to amenities, public goods, and social status into urban land markets, consistent with Baldomero-Quintana et al. (2025)'s evidence on land-value penalties from colonial Indigenous exclusion across Mexican cities.

Archival sources. The 1882 cadaster of Guadalajara is preserved at the *Archivo Municipal de Guadalajara* (AMG) in bound cadastral books organized by city block (*manzana*). Each *manzana* record contains, for every parcel, the property's unique identifier, street and house number, owner and renter names, structure type (*finca* versus *solar*), assessed land and structure values in pesos, and a free-text description often including a hand-drawn façade sketch and dimensions in *varas* (≈ 0.838 m). To my knowledge, this is the first time the 1882 cadaster has been systematically digitized and analyzed. The unit of analysis is the block rather than the parcel. For each *manzana* I add up the assessed land values of its *fincas* and divide the total by the surface of the block in square meters, which I obtain from the scale of the georeferenced cadastral map. The parcel dimensions recorded in *varas* do not enter this calculation, so the outcome does not depend on the legibility or completeness of the individual façade sketches, and no parcel is dropped for lacking a usable measurement. A block enters the estimation sample when its cadastral record and its polygon on the georeferenced map can both be identified.

Digitization. The cadastral books were photographed at the AMG and transcribed manually by a team of research assistants, with a first transcription pass and an independent verification pass; discrepancies were adjudicated against the original document, and illegible entries were flagged rather than imputed.

Georeferencing. Modern parcel-level geolocation is not directly available for the 1882 cadaster. I follow a three-step procedure. First, I obtain a high-resolution scan of the 1888 cadastral map published by the *Ayuntamiento de Guadalajara*, which uses the same *manzana* numbering. Second, I georeference the 1888 map to modern coordinates in QGIS using approximately 60 ground control points at locations unambiguously identifiable in both the historical map and modern satellite imagery (the Plaza de Armas, the Cathedral and surviving colonial churches, the *Mercado Corona*, and stable street intersections), using a thin-plate-spline transformation; the RMSE of the residuals is below 8 meters. Third, I match each 1882 parcel to its location in the georeferenced 1888 map by joining on the *manzana* number and within-manzana parcel ordering, prioritizing the 1882 record for substantive content. Parcels that cannot be unambiguously located account for less than 4% of the digitized parcels.

Validation. I cross-check parcel locations against (i) the modern *Catastro Municipal de Guadalajara*, (ii) Google Maps satellite imagery, and (iii) a sample of ≈ 200 parcels whose owner names can be matched to historical archives of notable families. Discrepancies arise primarily in zones

of major 20th-century transformation, the construction of Calzada Independencia between 1903 and 1911 and the 1950s market expansions. No block is excluded from the estimation sample on this basis; every located block within a given window enters the corresponding LR-RDD.

Final dataset. The resulting dataset contains block-level records for the central core of Guadalajara, geolocated to modern coordinates, with assessed 1882 land value per m^2, structure value, ownership and use information, and the block's signed orthogonal distance to the historical SJdD creek. The land-value variable used in Figure A.18 is the log of assessed land value per m^2 at the block level. Replication code and the digitized dataset will be deposited publicly upon publication, conditional on AMG authorization.

A.7 Marriage across the creek

The marriage evidence in Section 2.2 uses the hand-indexed FamilySearch marriage records of the parishes of Guadalajara, which name both spouses and their parents. I link each spouse to his or her own baptism, or to the baptism of a sibling with the same parents, in the machine-read baptism registers of the city's parishes. The link gives the parish of origin of both spouses for 1,062 couples, most of whom married between 1888 and 1941. Of these couples, 334, or 31 percent, joined spouses born on opposite banks of the creek, against 527, or 50 percent, had grooms and brides been paired at random.

Nearby parishes intermarry more, and parishes on opposite banks lie farther apart on average. Table A.3 reports, for each pair of parishes, the couples observed and those expected under random pairing within the decade of marriage, together with the distance between the parish churches on the 1863 plan. A Poisson regression of the couples in each pair on an indicator for opposite banks and the log of that distance, with the expected couples as exposure and pairs of two west-bank parishes as the reference, gives a ratio of 0.55 for opposite banks, with a 95 percent confidence interval from 0.44 to 0.69 from 1,000 resamples of couples within decade. The pair of Analco and San Juan de Dios is left out of the regression because many families of San Juan de Dios had earlier married and baptized in Analco. At about 600 meters, couples joining San Juan de Dios and the Sagrario occur at 0.55 times the random rate, and couples joining the Pilar and the Sagrario at 1.46 times.

The estimate rests on couples whose spouses were born in parishes with baptism books. When origins also come from the marriage of the parents, which adds the west-bank parish of Jesús, the ratio is 0.89, with a 95 percent confidence interval from 0.72 to 1.11. Among pairs between 1.1 and 1.3 kilometers apart alone, the comparison is too imprecise to separate the banks. FamilySearch holds almost no books from Mexicaltzingo, so the comparison is with the Spanish city and the popular parishes of the west bank.

Table A.3: Marriages between pairs of parishes, observed and expected at random

Pair of parishes	Distance (meters)	Couples	Expected at random	Ratio	95% CI
Opposite banks					
Sagrario–San Juan de Dios	653	13	23.7	0.55	[0.31, 0.79]
Pilar–San Juan de Dios	1,121	12	14.8	0.81	[0.45, 1.28]
Analco–Sagrario	1,220	120	198.6	0.60	[0.52, 0.69]
Guadalupe–San Juan de Dios	1,285	42	56.0	0.75	[0.58, 0.95]
Analco–Pilar	1,361	23	52.0	0.44	[0.30, 0.63]
Analco–Guadalupe	2,011	124	167.8	0.74	[0.64, 0.84]
Two west-bank parishes					
Pilar–Sagrario	602	35	23.9	1.46	[1.10, 1.84]
Guadalupe–Sagrario	837	87	72.4	1.20	[1.02, 1.41]
Guadalupe–Pilar	1,218	18	24.5	0.74	[0.44, 1.06]
Two east-bank parishes					
Analco–San Juan de Dios	796	102	83.3	1.22	[1.08, 1.38]

Notes: Couples in the hand-indexed FamilySearch marriage records whose two spouses have a parish of origin from their own baptism or a sibling's (1,062 couples). Couples counts couples joining a spouse born in each parish of the pair, in either direction, and expected at random counts those implied by pairing grooms and brides at random within the decade of marriage. Ratio is couples over expected, with 95 percent confidence intervals from 200 resamples of couples within decade. Distance is between the parish churches on the 1863 plan. Couples within a single parish are not shown.

↩ Back to text

A.8 Naming patterns: construction and additional results

Sources. The civil *padrones* of 1821–22 and 1824 compiled by Anderson and Spike (2007) record residents' given names, ages, and *cuarteles*. The national 10 percent sample of the 1930 census records the same fields. FamilySearch parish registers of baptisms and marriages span 1850–1949.

Distinct given names. For each census, I compute a distinctiveness index in the spirit of Fryer and Levitt (2004): the probability that a resident lives east of the creek given only their first name, estimated leave-one-out and benchmarked against a permutation null. Spellings are normalized before any comparison (Josef and José are treated as the same name written by different enumerators), and the index is estimated within census, so it never compares naming conventions across enumerators from different censuses.

Panel (a) of Figure A.19 shows a stable east–west gap for cohorts born between 1760 and

1839, with the 1821–22 and 1824 censuses agreeing cohort by cohort. The final bin of each enumeration is right-truncated by its census date, so the 1820–39 cohort is observed only through its earliest years in the two early censuses and the agreement across enumerations is exact only for the bins observed in full. In the 1930 census, the gap remains visible for the cohort born in 1860–79 but disappears for cohorts born after 1880, coinciding with the period in which Guadalajara's industrial expansion reorganized its social geography. The 1821 census records ethnicity (*calidad*), and estimating the comparison within *calidad* cells yields the same pattern.

A.9 Socio-demographic sorting in 1821

To document the persistence of the colonial division three centuries after the foundation, I rely on the 1821 *padrón* of Guadalajara, conducted in the immediate aftermath of independence and curated and digitized by the Guadalajara Census Project (Anderson and Spike, 2007). The census enumerated 35,287 residents. It numbers its units from 1 to 25 and records entries for 24 of them, with *cuartel* 16 absent, and gives names, age, sex, household membership, ethnicity (*calidad*), marital status, and detailed occupational information. To the best of my knowledge, the 1821 *padrón* of Guadalajara is the most detailed urban dataset available for nineteenth-century Latin America. Figure A.20 compares mean characteristics of residents on either side of the SJdD creek, restricting the sample to the eight *cuarteles* immediately adjacent to the boundary, the same spatial window used in the BD design of Section 3. Three patterns stand out. First, residents on the western bank were significantly more likely to hold noble titles (*Don*/*Doña*) and to identify as Spanish. Second, residents on the eastern bank were more likely to identify as Indigenous, to be in interracial marriages, and to hold journeyman or laborer occupations. Third, residential mobility — proxied by the share of migrants — was higher on the eastern bank. The differences survive controls for age and sex, household clustering, and FDR adjustment. The sign of this last pattern is the opposite of the long-term residence gap measured in 2020 (Table B.16). The two measures are two centuries apart and bracket the twentieth-century rural-to-urban migration that reshaped the city, so they describe different populations under different migration regimes rather than one series.

A.10 The Calzada as a present-day connector, not a barrier

Figure A.21 shows Calzada Independencia in 2020: a surface-level avenue integrated into the city grid, traversed continuously by pedestrians, vehicles, and a bus rapid transit line. It is not the kind of physical barrier studied in the U.S. literature on urban freeways and railroads (Ananat, 2011; Chyn et al., 2026; Bagagli, 2025; Valenzuela-Casasempere, 2025); if anything it functions as the city's main north–south arterial. For identification, this means the BD design exploits a historical line whose physical form has changed repeatedly, rather than a contemporary infrastructural divide imposing its own treatment effect.

Table A.4: 1821 socio-demographic gaps across the boundary, with and without a control for *calidad*

Panel A. Status and intermarriage

	Don/Doña			Interracial marriages		
	(1)	(2)	(3)	(4)	(5)	(6)
$\mathbb{1}\{East_i\}$	−0.374***	−0.337***	−0.176***	0.113***	0.113***	0.103***
	(0.016)	(0.021)	(0.022)	(0.018)	(0.018)	(0.018)
Sample	Full	*Calidad* recorded	*Calidad* recorded	Full	*Calidad* recorded	*Calidad* recorded
Sex and age indicators	✓	✓	✓	✓	✓	✓
Calidad indicators			✓			✓
Observations	12,872	6,650	6,650	2,869	2,869	2,869
Dep. var. mean	0.288	0.300	0.300	0.092	0.092	0.092
R^2	0.185	0.150	0.405	0.059	0.059	0.147

Panel B. Occupation

	Journeyman			Laborer		
	(7)	(8)	(9)	(10)	(11)	(12)
$\mathbb{1}\{East_i\}$	0.211***	0.195***	0.130***	0.171***	0.129***	0.089***
	(0.019)	(0.027)	(0.029)	(0.017)	(0.025)	(0.026)
Sample	Full	*Calidad* recorded	*Calidad* recorded	Full	*Calidad* recorded	*Calidad* recorded
Sex and age indicators	✓	✓	✓	✓	✓	✓
Calidad indicators			✓			✓
Observations	5,057	2,443	2,443	5,057	2,443	2,443
Dep. var. mean	0.396	0.373	0.373	0.247	0.245	0.245
R^2	0.075	0.130	0.157	0.071	0.085	0.099

Panel C. Household occupation and migration

	Similar occupations in household			Migrant		
	(13)	(14)	(15)	(16)	(17)	(18)
$\mathbb{1}\{East_i\}$	0.034	−0.054	−0.071*	0.240***	0.073***	0.088***
	(0.029)	(0.040)	(0.043)	(0.016)	(0.026)	(0.027)
Sample	Full	*Calidad* recorded	*Calidad* recorded	Full	*Calidad* recorded	*Calidad* recorded
Sex and age indicators	✓	✓	✓	✓	✓	✓
Calidad indicators			✓			✓
Observations	2,154	1,230	1,230	13,376	6,662	6,662
Dep. var. mean	0.501	0.455	0.455	0.262	0.227	0.227
R^2	0.075	0.100	0.118	0.103	0.038	0.051

Notes: Estimates from the individual-level 1821 *padrón* of Guadalajara (Anderson and Spike, 2007), on the *cuarteles* adjacent to the SJdD creek, 8 and 9 on the eastern bank (Analco) and 1, 2, 3, 10, 23, and 24 on the western bank. The outcomes are those of Figure A.20 that are not themselves measures of ethnicity, each an indicator, so the coefficients are east-minus-west differences in shares. All columns control for sex and age indicators. For each outcome, the first column uses the full sample and reproduces the difference annotated in Figure A.20. Self-reported *calidad* is missing for about half the records, so the second column re-estimates the first on the subsample in which it is recorded, and the third adds *calidad* indicators on that same subsample. The second and third columns of each outcome are the comparison that isolates the control from the change in sample. Standard errors clustered by household are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

↩ Back to text

A.11 Skin tone across the metropolitan area

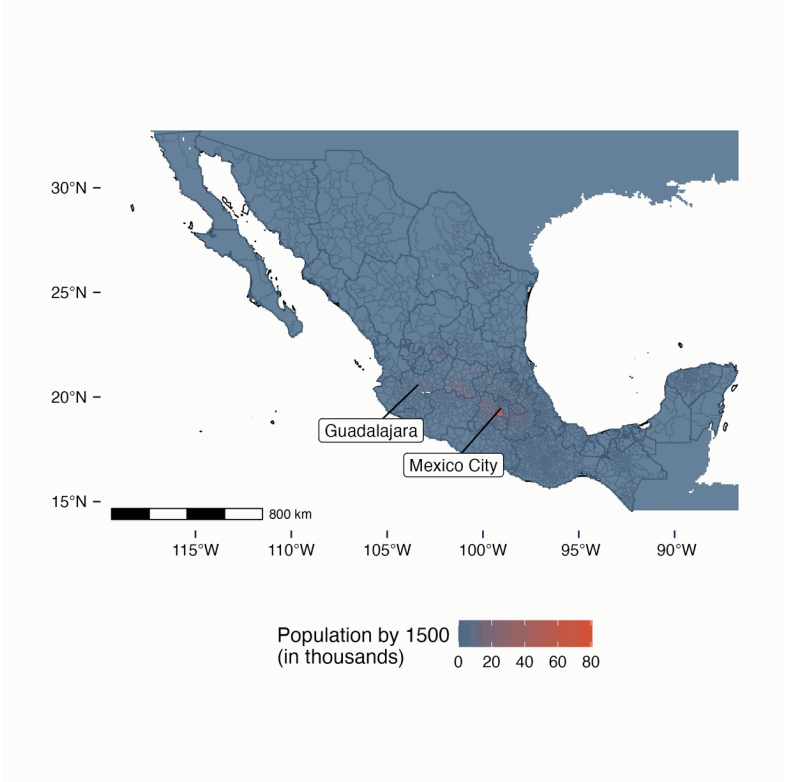
Mexican censuses do not record skin tone, so the ethnic composition of the two banks in Section 3 rests on Indigenous self-identification, household composition and language. Because skin tone rather than Indigenous identity may be the operative distinction (Woo-Mora, 2026), I turn to the 2017 ESRU Survey of Social Mobility (ESRU-EMOVI), which records interviewer-rated skin tone on the PERLA scale together with respondents' coordinates.

A.12 Godparents and the honorific across parishes, 1870–1919

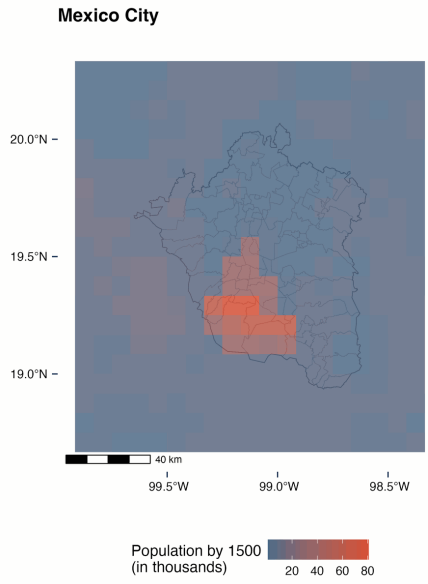
The godparent exercise of Section 5.2 uses the machine-read FamilySearch baptism registers of 1870 to 1919 in the parishes of the Sagrario, El Pilar, and Guadalupe on the west bank and Analco and San Juan de Dios on the east. For each baptism I ask whether the two godparents are a couple that baptizes its own children in the same parish within ten years. The local-peer ratio of a parish and decade is the share of baptisms with such godparents, net of the share expected when godmothers are reassigned at random within parish and five-year period, over the share of baptisms whose parents baptize another child in the same parish within ten years. Standing is the share of fathers named with the honorific in the same parish and decade. Each of the 18 parish-decade cells has at least 100 baptisms in which the parents and both godparents can be read.

Figure 11 plots both measures as deviations from their decade means. A weighted regression with decade fixed effects gives a local-peer ratio 0.032 lower per 10 percentage points more fathers with the title (standard error 0.004), and 0.040 lower (0.007) among the ten west-bank cells. Without decade fixed effects the estimate among west-bank cells is −0.017 (0.011), because the comparison then mixes parishes with the decline of the title, which almost disappears from every register by the 1910s, and within a parish over time both the title and local godparents fall. Without the Sagrario in the 1870s, the cell with the most fathers carrying the title, the estimate is −0.029 (0.016) with $p = 0.094$, and −0.047 (0.036) among west-bank cells. The honorific also marks where the city's elite lived, so the pattern does not separate standing from wealth.

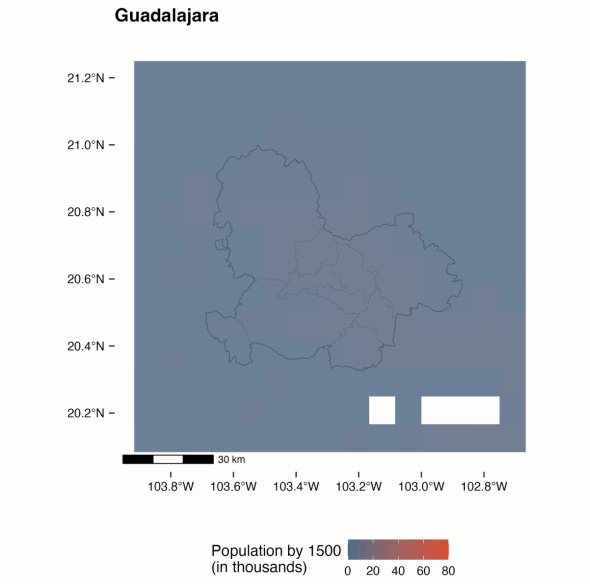
Figure A.10: Pre-colonial population density estimates in Mesoamerica



(a) Pre-colonial population density across Mexico



(b) Anahuac Valley (Central Mexico)

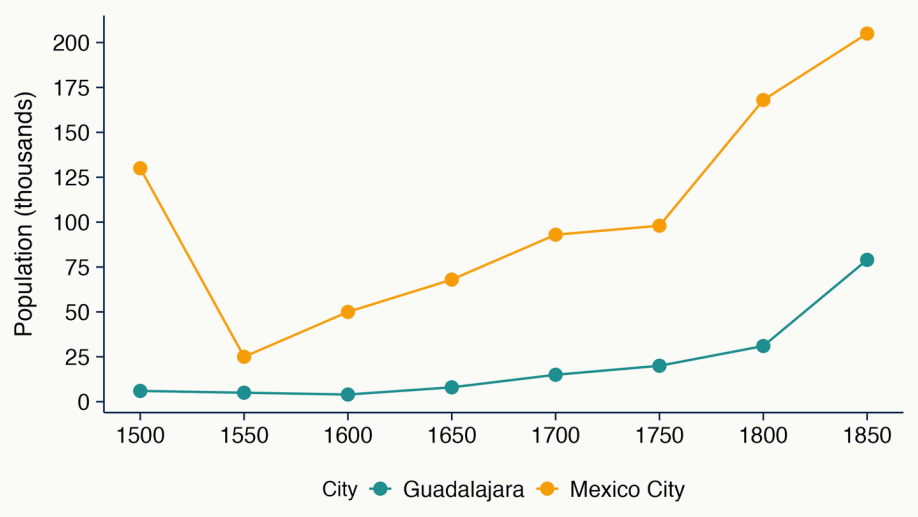


(c) Atemajac Valley (Guadalajara region)

Notes: Panel (a) shows pre-colonial population density across present-day Mexico circa 1500 CE, based on the historical land-use reconstruction of Goldewijk et al. (2017). Panels (b) and (c) zoom into the Anahuac Valley (the seat of Tenochtitlán and the Mexica state) and the Atemajac Valley (the future site of Guadalajara), respectively. The Anahuac Valley contained densely settled urban polities at the time of Spanish contact, while the Atemajac Valley hosted dispersed Indigenous populations but no major agglomerations within the area where Spanish Guadalajara, Mexicaltzingo, and Analco would later be founded. Local Indigenous groups including the Cazcanes, Cocas, and Tecuexes inhabited the broader region, but no urban center existed at the scale relevant for the 1542 foundation.

↩ Back to text

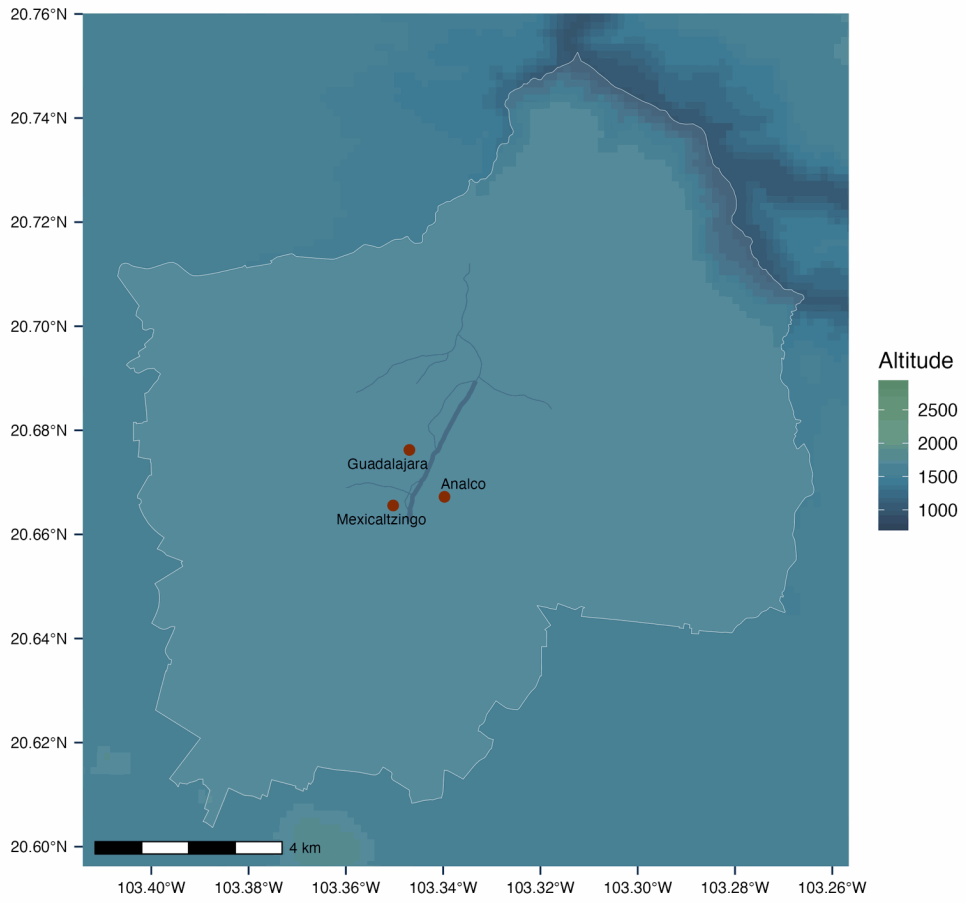
Figure A.11: Colonial population estimates for Mexico City and Guadalajara



Notes: Population estimates for Mexico City and Guadalajara from the 16th to 18th centuries, based on the historical urban-population reconstructions of Buringh (2018). The persistent population gap between the two cities — Mexico City consistently 5–10 times larger than Guadalajara through the colonial period — reflects the absence of pre-colonial urban agglomeration in the Atemajac Valley and the slower demographic growth of New Galicia relative to central New Spain. The figure weighs against the possibility that the 1542 foundation of Guadalajara built upon a substantial pre-existing urban base.

↩ Back to text

Figure A.12: Altitude across the historical boundary: 150 × 150 m grid-cell raster



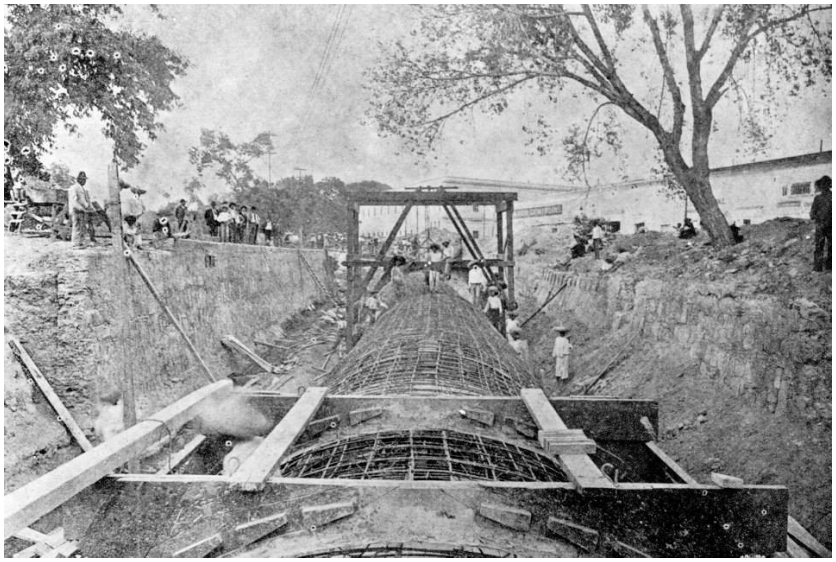
Notes: Altitude in meters above sea level at the 150 × 150 m grid-cell level, from INEGI's high-resolution digital elevation model. The figure overlays the historical boundary (San Juan de Dios creek, today Calzada Independencia) and the approximate locations of the three foundational settlements (Spanish Guadalajara, Mexicaltzingo, and Analco). Color shading indicates altitude, with no detectable differential across the SJdD creek. The visual impression is confirmed formally in Table A.1.

↩ Back to text

Figure A.13: Transformation of the SJdD creek into Calzada Independencia, 1903–1911



(a) Early 20th-century view of the SJdD creek

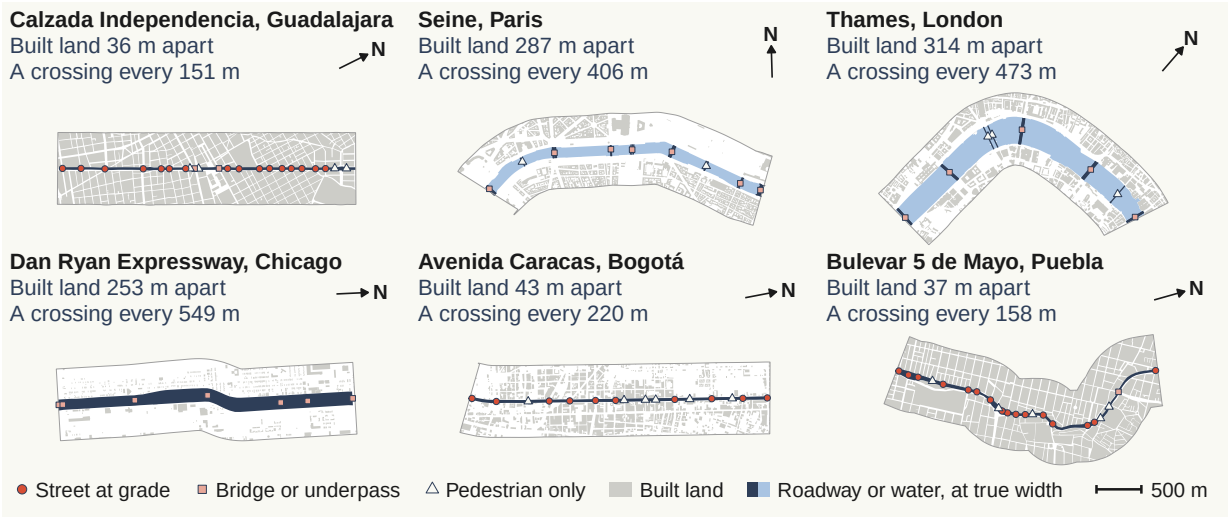


(b) Construction of Calzada Independencia along the creek's course

Notes: Panel (a) shows the SJdD creek in the early 20th century, immediately prior to its reorganization; described as a modest watercourse with multiple bridges in 17th-century accounts (Lázaro de Arregui, 1946), it retained its position as the dividing line through the late 19th century. Panel (b) documents the construction of Calzada Independencia along the same north–south axis, with the creek piped underground, between 1903 and 1911 as part of the Porfirian-era modernization of the city. Photographs from municipal historical archives.

↩ Back to text

Figure A.14: Calzada Independencia compared with other avenues, urban rivers, and an urban expressway



Notes: Each panel maps a stretch of about 3.3 kilometers at the same scale, with north marked by the arrow. Built land is the INEGI block polygons for Guadalajara and Puebla and OpenStreetMap building footprints elsewhere, and roadways and water are drawn at their true width. The distance between built land is the median, over points every 50 meters along the axis, of the distance between the nearest built polygons on the two sides. The spacing of crossings is the mean distance between consecutive crossings, and crossings closer than 30 meters along the axis count once. The stretch of the Calzada runs along the boundary of the BD design. The others run between named crossings, from Pont du Carrousel to Pont d'Iéna (Seine), Lambeth Bridge to Southwark Bridge (Thames), Pershing Road to Garfield Boulevard (Dan Ryan Expressway), Calle 32 to Calle 63 (Avenida Caracas), and Calle 32 Poniente to Avenida 27 Oriente (Bulevar 5 de Mayo). Sources: OpenStreetMap, extracted on October 4, 2026, and INEGI block polygons.

↩ Back to text

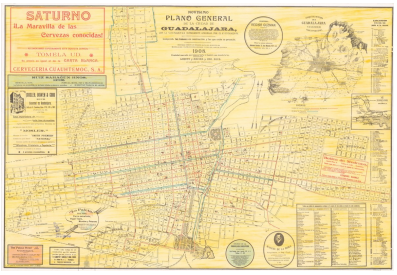
Figure A.15: Cartographic sources for the count of named institutions, 1800–1908



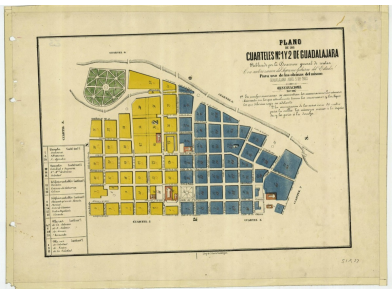
(a) 1800 facsimile plan



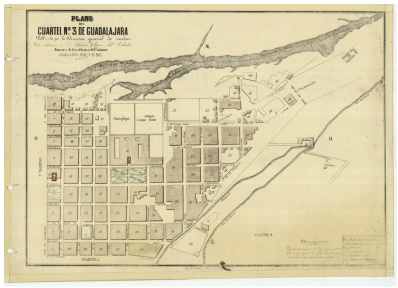
(b) 1890 *Plano General*



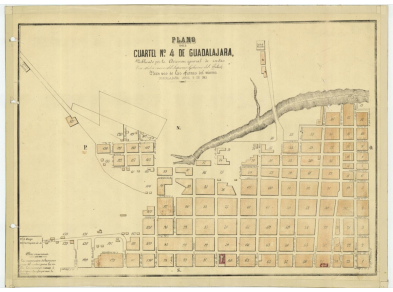
(c) 1908 *Novísimo Plano General*



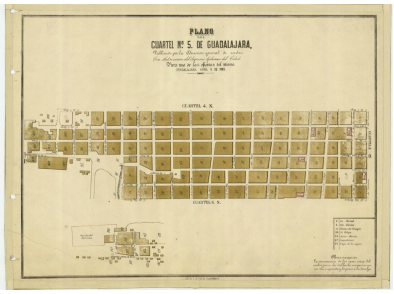
(d) 1863, *cuarteles* 1 and 2



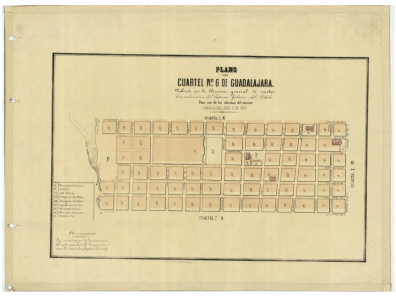
(e) 1863, *cuartel* 3



(f) 1863, *cuartel* 4



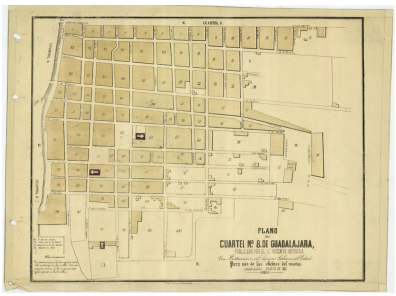
(g) 1863, *cuartel* 5



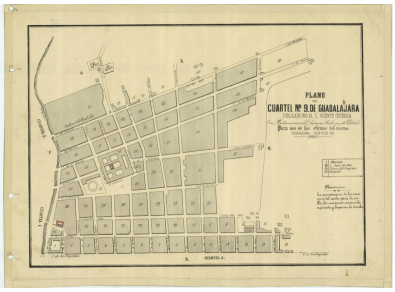
(h) 1863, *cuartel* 6



(i) 1863, *cuartel* 7



(j) 1863, *cuartel* 8 (Analco)

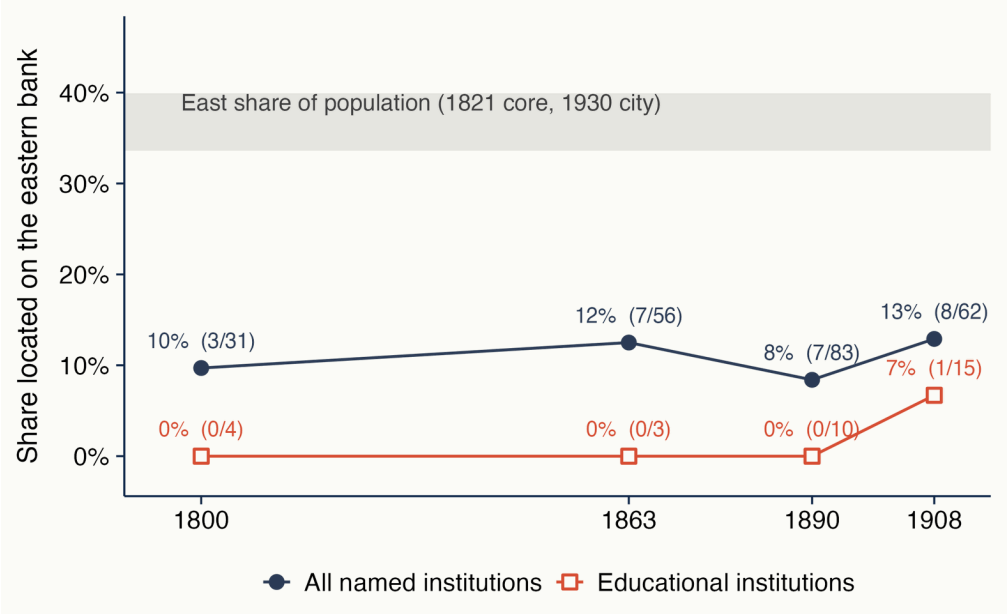


(k) 1863, *cuartel* 9

Notes: The four waves of Table A.2 and Figure A.16. Panels (a), (b) and (c) are the three general plans, each carrying an index in its margins that names the city's institutions and keys them to a block. Panels (d) through (k) are the 1863 *cuartel* survey held at the *Archivo Histórico de Jalisco*, eight sheets covering the city's nine *cuarteles*; the sheet in panel (d) covers *cuarteles* 1 and 2 together. Each 1863 sheet numbers its blocks individually, labels its streets, and carries its own legend of notable buildings, which is what allows an entry to be keyed to a (*cuartel*, *manzana*) pair. *Cuartel* 8, in panel (j), is Analco, on the eastern bank of the SJdD creek. Images are reproduced at reduced resolution; the counting was done on the full-resolution scans.

↩ Back to text

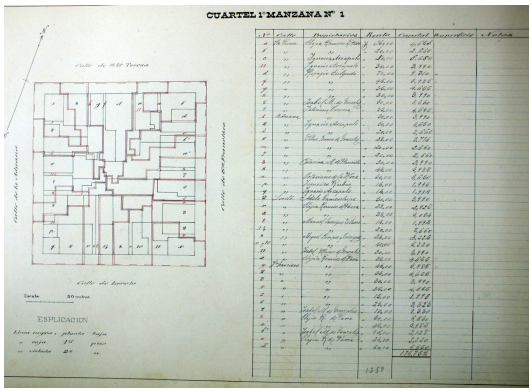
Figure A.16: Institutions located on each bank, 1800–1908



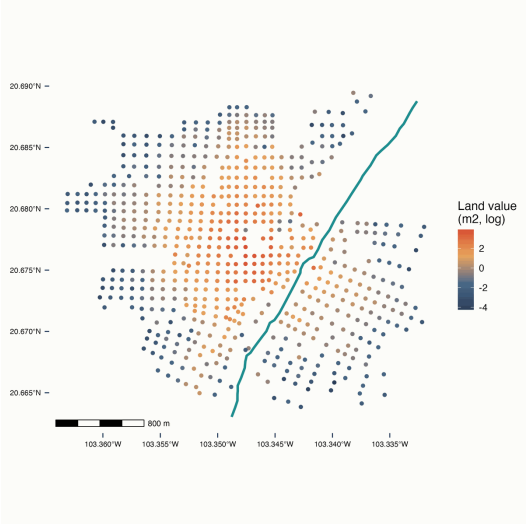
Notes: Each point is the share of the entries in a map's index that are located east of the SJdD creek, with the count of eastern entries over the total located in parentheses. The upper series covers all named institutions; the lower series covers educational establishments only. Each map reflects its own conventions about what merits an entry, so shares are comparable across banks within a wave, and across waves within a category present in all four, as the educational category is. The shaded band spans the eastern share of the population in the two enumerations that record residence by *cuartel*: 39.9 percent in the 1821 *padrón*, restricted to the central core for which sides can be assigned, and 33.6 percent in the 1930 census. Sides for the map entries follow the same boundary used in the BD design; sides for the *cuarteles* follow the digitized blocks of the 1890 plan, in which *cuarteles* 8 and 9 lie east of the creek and 1–7 lie entirely west. Of the 105 entries in the 1908 index, 62 could be assigned to a bank, 24 by direct location on the plan and 38 by name correspondence with an earlier georeferenced wave; the remaining 43 are excluded. Appendix A.5 reports the sources of the four maps, the location method for every entry, and the counts by category.

↩ Back to text

Figure A.17: Historical land values in 1882



(a) Property façade from the 1882 cadaster

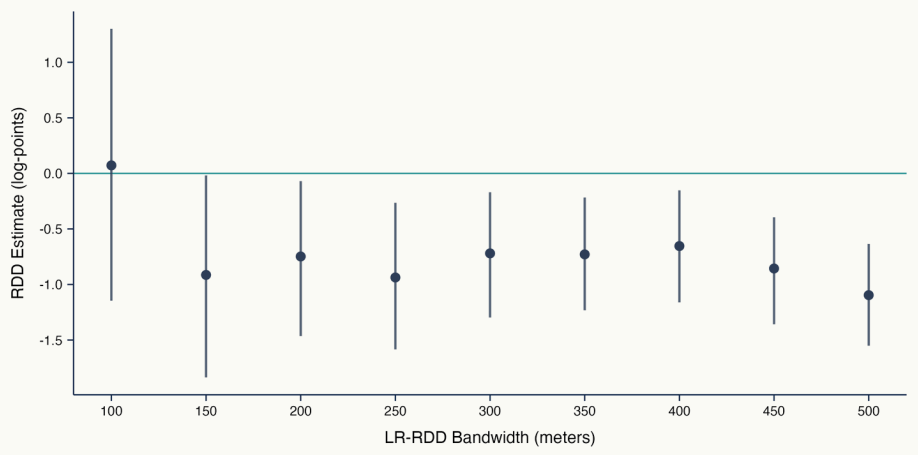


(b) Digitized land-value map (1882)

Notes: Panel (a) shows an example of a property façade entry as recorded in the 1882 cadaster of Guadalajara, alongside the formal assessed land value and an identifier locating the parcel within a city block (*manzana*). Panel (b) shows the digitized parcel-level land values across the historical core, with darker shades corresponding to higher assessed values per square meter. Historical cadastral maps were georeferenced to modern coordinates as described at the end of this subsection; the SJdD creek and the foundational settlements are overlaid. The clear west–east drop in assessed values across the creek is consistent with the formal RDD estimates in Figure A.18.

↩ Back to text

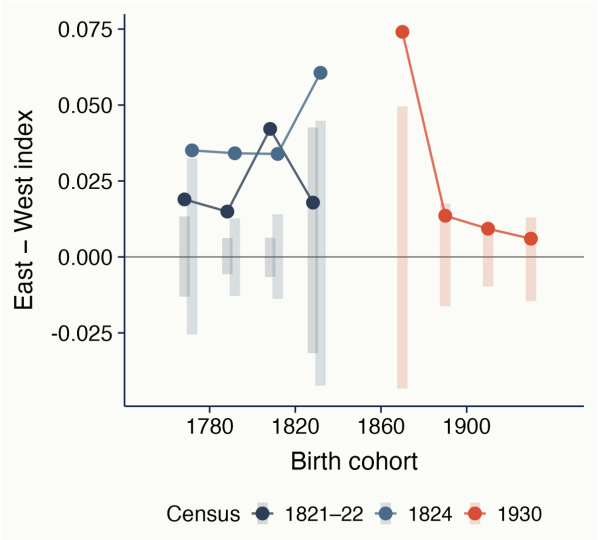
Figure A.18: Local-randomization RDD estimates of land-value discontinuities, 1882



Notes: Each point reports the estimated discontinuity in log land values per square meter across the SJdD creek in 1882, using the Local Randomization RDD (LR-RDD) of Cattaneo et al. (2015), which treats observations within a small window around the boundary as if assigned at random. Estimates are reported across bandwidths from 100 to 500 meters; vertical bars show 95% confidence intervals. Negative values indicate lower assessed land prices on the historically excluded east side. Estimates are negative and statistically significant at every window from 150 meters outward, with point estimates between -0.65 and -1.10 log points, which corresponds to land values 48 to 67 percent lower on the eastern bank. The narrowest window, 100 meters, gives a positive point estimate whose interval runs from -1.15 to 1.30 and is uninformative. The cadastral data and georeferencing procedure are documented at the end of this subsection.

↩ Back to text

Figure A.19: Given-name distinctiveness by birth cohort (1740–1939)

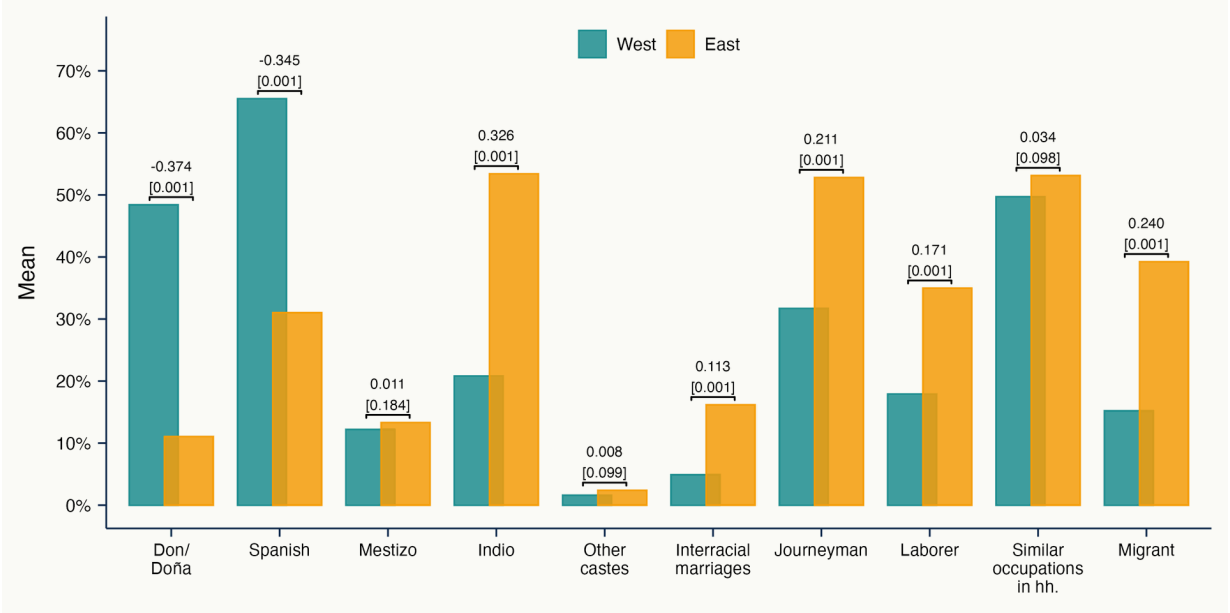


(a) Given names

Notes: Panel (a) plots, by twenty-year birth cohort, the east–west difference in the mean leave-one-out probability that a resident lives east of the creek given their first name, estimated separately within the censuses of 1821–22, 1824, and 1930. Shaded bars are 95 percent permutation bands under random assignment of side within cohort. The earliest cohort of each census is omitted (born 1740–59 in 1821–22, and 1840–59 in 1930): the cells are small and the permutation bands wide. Cohorts born in 1760–1819 are observed in full in two independent enumerations; the 1820–39 bin is right-truncated in each census by its enumeration date, so its year and age composition differs across censuses. Sides follow the *cuartel* recorded for each household, verified against the manuscripts' street lists; in 1930, *cuarteles* 8, 9, and 11 are assigned to the east and 1–7 to the west, where *cuarteles* 8 and 9 lie east of the creek in the digitized blocks of the 1890 plan and *cuartel* 11, which that plan does not cover, holds under one percent of the sample. Spellings are normalized within census before comparison. Levels are comparable within, not across, censuses.

↩ Back to text

Figure A.20: 1821 census: socio-demographic characteristics by bank of the San Juan de Dios creek



Notes: Mean values of selected socio-demographic characteristics in 1821, computed separately for residents of the six *cuarteles* immediately west of the SJdD creek (*cuarteles* 1, 2, 3, 23, and 24 in Spanish Guadalajara and 10 in Mexicaltzingo) and the two *cuarteles* immediately east (*cuarteles* 8 and 9, Analco). Data are from the individual-level 1821 *padrón* compiled by the Guadalajara Census Project (Anderson and Spike, 2007). Variables capture social status (share of household heads titled *Don/Doña*), self-reported ethnicity (share Spanish, share Indigenous), interracial marriage, occupational structure (journeymen and laborers), and residential mobility (share of migrants). Annotations above each pair of bars report the OLS estimate of the east-minus-west difference in means, controlling for age and sex, with standard errors clustered by household in parentheses. Sharpened FDR q-values (Anderson, 2008), adjusted across all variables, are in brackets. The sample is restricted to the eight *cuarteles* adjacent to the historical boundary for comparability with the BD sample. These regressions control for age and sex, not for ethnicity, and the western group includes Mexicaltzingo's Indigenous population, so bank and ethnicity are not the same partition. Table A.4 adds self-reported *calidad* as a control and the non-ethnic gaps remain, so the side of the creek predicts status, occupation, intermarriage and mobility beyond ethnic composition, consistent with variation in colonial status that is not reducible to ethnic composition.

↩ Back to text

Figure A.21: Calzada Independencia as a present-day surface-level avenue



(a) Westward view of Calzada Independencia

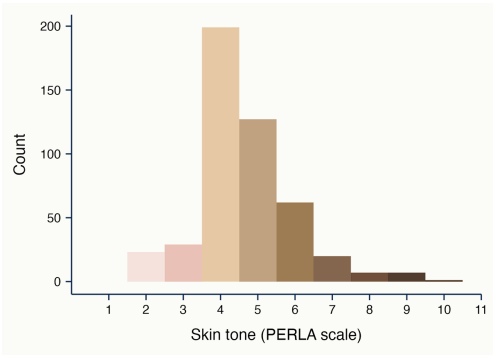


(b) Eastward view of Calzada Independencia

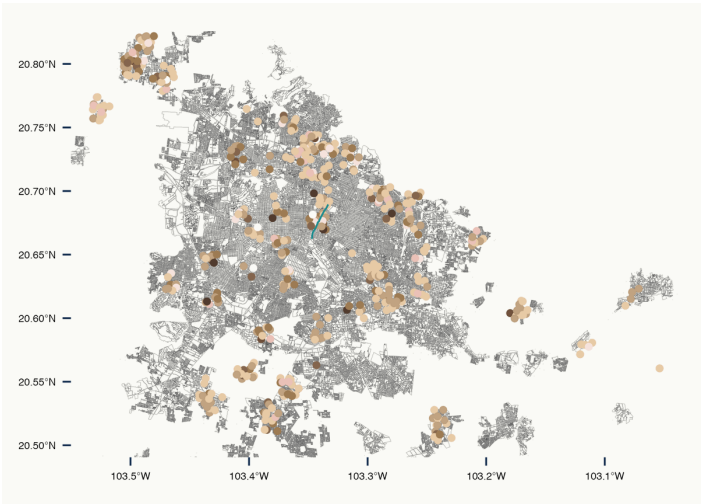
Notes: Photographs of Calzada Independencia in 2020. Panel (a) shows the view from the western (Spanish Guadalajara and Mexicaltzingo) side; panel (b) from the east side. The avenue is a surface-level urban arterial integrated into the city grid, with frequent pedestrian crossings, continuous commercial activity along both sides, and a bus rapid transit line (*Macrobús*) on its central median. Unlike the urban freeways studied in the U.S. literature (Ananat, 2011; Chyn et al., 2026; Bagagli, 2025; Valenzuela-Casasempere, 2025), Calzada Independencia connects rather than fragments the two banks. The persistence of the east–west divide that the BD design estimates therefore cannot be attributed to the Calzada functioning as a contemporary physical barrier.

↩ Back to text

Figure A.22: Skin tone across the Guadalajara metropolitan area



(a) Distribution of skin tone



(b) Location of respondents

Notes: Interviewer-rated skin tone on the PERLA scale for respondents of the 2017 ESRU Survey of Social Mobility residing in the Guadalajara metropolitan area. Panel (a) plots the distribution of ratings; panel (b) locates each respondent, coloured by rating, with the SJdD boundary overlaid in blue. Ratings concentrate between 4 and 6 and are dispersed across the metropolitan area rather than clustering on either bank. The survey is not dense enough near the boundary to support a discontinuity estimate, so the figure describes the metropolitan pattern rather than testing balance at the cutoff.

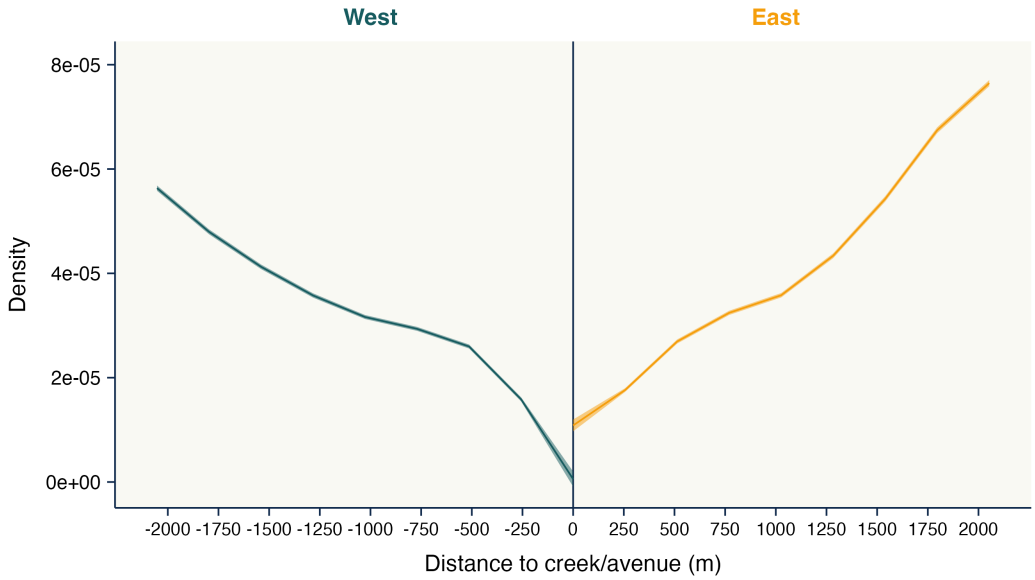
↩ Back to text

B Persistent Human Capital Gaps

B.1 Density at the boundary

A first concern in BD designs is whether the density of observations is itself discontinuous at the boundary (McCrary, 2008; Cattaneo et al., 2018). Figure B.1 plots the estimated density of individuals as a function of distance to the boundary, applying Cattaneo et al. (2018). The density declines toward zero on both sides, reflecting the lower residential population in the historical core. At the cutoff itself the estimated density is higher on the east side, and the two sides converge to within four percent of each other by 500 meters. Density rises monotonically with distance into the east side rather than piling up at the line, which is the pattern that manipulation would produce. It does not rule out the gradual sorting that Section 3 addresses with cadastral land values and measures of residential persistence, and its power is limited where the density is lowest.

Figure B.1: Density test at the historical boundary



Notes: Local-polynomial density estimate of individuals as a function of signed orthogonal distance to the historical boundary (Calzada Independencia, formerly the SJdD creek), following Cattaneo et al. (2018). The horizontal axis reports distance in meters, with positive values on the east side. Shaded bands report 95% confidence intervals. Density declines toward zero on both sides, reflecting the lower residential population in the historical core. The estimated density at the cutoff is higher on the eastern side, and the two sides converge to within four percent by 500 meters. The absence of a spike at the line is evidence against bunching; it is not evidence against gradual sorting across the two sides, and the test has limited power where the density approaches zero.

↩ Back to text

B.2 Main estimates and full robustness battery

Table B.1 reports the full set of BD estimates underlying Figure 4. The table is organized in two panels — years of schooling for individuals aged 6+ (Panel A) and school attendance for ages 6–22 (Panel B) — with, for each outcome, four main specifications followed by robustness checks.

The four main specifications progress from the unconditional levels estimate to the standardized estimate, add clean covariates, and finally add controls that may be partially affected by the treatment ("unclean" controls whose inclusion risks collider bias). Each robustness check holds the clean controls fixed and varies one design choice: the local-polynomial order, the kernel, the variance estimator, or the sample. Across the standardized specifications, the boundary effect on years of schooling lies between -0.08 and -0.17 SD, with -0.17 in the preferred specification without covariates and -0.14 once covariates are added. School-attendance estimates are noisier but consistently negative (-0.08 to -0.14 SD). The Calonico–Cattaneo–Titiunik nearest-neighbor (NN) variance estimator is used in the main specifications; the HC3 row reports the most conservative heteroskedasticity-consistent alternative.

Table B.1: BD estimates on human capital, 2020 Census

Specification	Level at the boundary		$\hat{\tau}$	SE	95% CI	p-value	FDR q-value	Poly	h (m)	b (m)	N_h	N_b	Kernel	VCE	Clean controls	Unclean controls
	West	East														
Panel A. Years of schooling (ages 6+)																
Main – levels	11.023	10.232	−0.791	0.156	[−1.097, −0.485]	< 0.001	< 0.001	1	571	1,489	25,011	107,104	Uniform	NN	No	No
Main – z-score	—	—	−0.165	0.033	[−0.229, −0.101]	< 0.001	< 0.001	1	571	1,489	25,011	107,104	Uniform	NN	No	No
Main – z-score + covariates	—	—	−0.141	0.048	[−0.235, −0.047]	0.003	0.007	1	395	770	13,833	42,070	Uniform	NN	✓	No
Main – z-score + covariates + unclean controls	—	—	−0.117	0.041	[−0.198, −0.036]	0.005	0.009	1	411	797	13,465	40,669	Uniform	NN	✓	✓
Rob. – Polynomial order (2)	—	—	−0.151	0.047	[−0.244, −0.058]	0.001	0.003	2	741	1,461	39,262	104,303	Uniform	NN	✓	No
Rob. – Kernel (triangular)	—	—	−0.103	0.047	[−0.194, −0.012]	0.027	0.034	1	478	897	18,525	51,626	Triangular	NN	✓	No
Rob. – Kernel (Epanechnikov)	—	—	−0.127	0.044	[−0.212, −0.042]	0.004	0.007	1	504	913	20,665	53,039	Epanechnikov	NN	✓	No
Rob. – VCE (HC3, most conservative)	—	—	−0.081	0.051	[−0.181, 0.018]	0.110	0.283	1	388	760	13,627	41,158	Uniform	HC3	✓	No
Rob. – Migrants (native-born only)	—	—	−0.122	0.043	[−0.205, −0.039]	0.004	—	1	407	796	12,717	38,784	Uniform	NN	✓	✓
Panel B. School attendance (ages 6–22)																
Main – levels	0.749	0.717	−0.032	0.019	[−0.070, 0.005]	0.094	0.049	1	1,200	2,059	18,265	45,016	Uniform	NN	No	No
Main – z-score	—	—	−0.075	0.045	[−0.163, 0.013]	0.094	0.049	1	1,200	2,059	18,265	45,016	Uniform	NN	No	No
Main – z-score + covariates	—	—	−0.104	0.072	[−0.245, 0.036]	0.146	0.079	1	571	1,082	6,101	15,789	Uniform	NN	✓	No
Main – z-score + covariates + unclean controls	—	—	−0.086	0.047	[−0.178, 0.006]	0.067	0.034	1	895	1,998	12,172	41,693	Uniform	NN	✓	✓
Rob. – Polynomial order (2)	—	—	−0.129	0.082	[−0.289, 0.032]	0.116	0.061	2	871	1,801	12,038	35,228	Uniform	NN	✓	No
Rob. – Kernel (triangular)	—	—	−0.136	0.063	[−0.260, −0.011]	0.032	0.034	1	703	1,619	8,847	29,378	Triangular	NN	✓	No
Rob. – Kernel (Epanechnikov)	—	—	−0.124	0.068	[−0.257, 0.009]	0.067	0.035	1	616	1,473	7,038	24,574	Epanechnikov	NN	✓	No
Rob. – VCE (HC3, most conservative)	—	—	−0.091	0.078	[−0.244, 0.063]	0.246	0.283	1	505	1,029	5,117	14,819	Uniform	HC3	✓	No

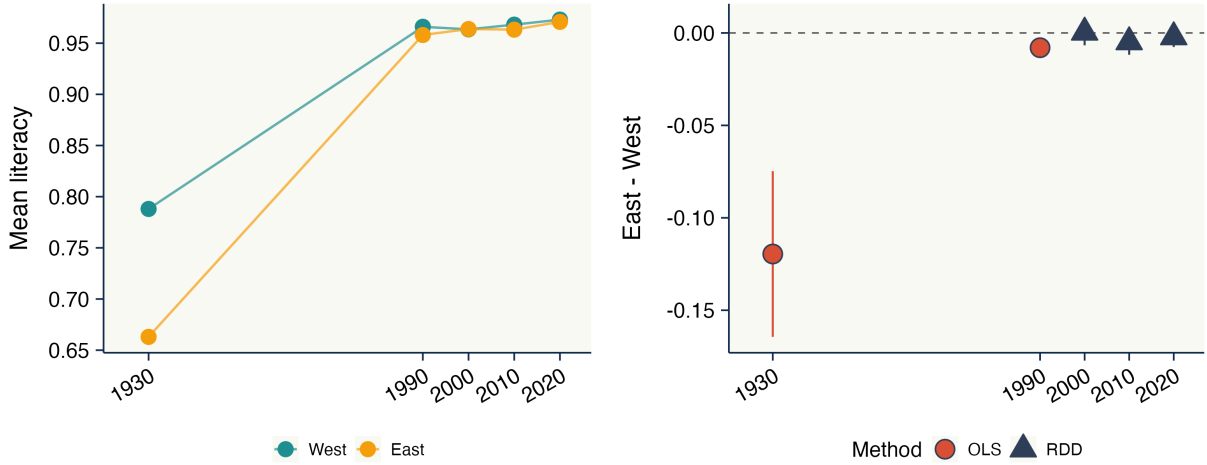
Notes: BD estimates of the boundary effect on human capital outcomes, using individual-level data from the complete 2020 *Censo de Población y Vivienda* for the municipality of Guadalajara, produced in the INEGI microdata laboratory. The running variable is the signed orthogonal distance from the resident's geolocated block to the historical SJdD creek (today Calzada Independencia), with positive values on the east side. Each row is one specification. "Main – levels" coefficients are in original units (years of schooling for Panel A); all others are in standardized (z-score) units. Inference follows Calonico et al. (2014): $\hat{\tau}$ is the bias-corrected treatment effect, and SE, the 95% CI, and the p-value are robust bias-corrected; h and b are the MSE-optimal main and bias-correction bandwidths in meters, and N_h and N_b the effective sample sizes. *Poly* is the local polynomial order, *Kernel* the weighting scheme, and *VCE* the variance estimator (NN the Calonico–Cattaneo–Titiunik nearest-neighbor estimator; HC0–HC3 the heteroskedasticity-consistent estimators). *Clean controls* include sex, age, longitude, latitude, and boundary-segment fixed effects. *Unclean controls* in Panel A include indicators for Afrodescendant or Indigenous self-identification, birth in Jalisco, residence in the municipality five years earlier, social security access, marital status, and dwelling type, all of which may themselves be affected by treatment. Unclean controls in Panel B include parental mean years of schooling, household size, and residence in the municipality five years earlier. The preferred estimate is the first row in Panel A, in levels and without covariates, and the fourth in Panel B. The two panels differ because the attendance comparison in Section 3 is between children of comparable households, and parental schooling and household size are the inherited human-capital stock that comparison holds fixed, at the cost of conditioning on variables measured after the assignment. The third row of Panel B, which omits them, gives −0.104 with a p-value of 0.146 against 0.067 for the fourth. For the specifications in levels, West and East report the estimated level of the outcome at the boundary on each side, whose difference is $\hat{\tau}$; the other rows are in standardized units, so these cells are marked with a dash. The p-value is unadjusted, and the q-value is the sharpened false-discovery-rate value of Anderson (2008), computed within each specification across the two outcomes; because the procedure estimates how many of the two null hypotheses are true, a q-value can fall below its p-value. The native-born row has no attendance counterpart, so no q-value is computed for it.

⟵ Back to text

B.3 Multi-wave evolution: additional outcomes

Figure B.2 traces literacy across the boundary from 1930 to 2020. Literacy among adults differed by 12.0 percentage points across the banks in 1930, and from 1990 on it is near-universal on both sides, with a gap under one percentage point in 1990 and none distinguishable from zero in 2000, 2010 and 2020.

Figure B.2: Literacy across the historical boundary, 1930–2020



Notes: As in Figure 5, for literacy. Literacy is measured among adults aged 15 or older in 1930 and among residents aged 6 or older from 1990 on. Panel (a) plots the mean literacy rate on each side of the boundary; panel (b) the east-minus-west gap, with OLS estimates for 1930 and 1990 (circles) and non-parametric BD estimates for 2000–2020 (triangles). In 1930 literacy among adults aged 15 or older stood at 78.8 percent on the western side against 66.3 on the eastern, a gap of 12.0 percentage points that is significantly different from zero. The series has no wave between 1930 and 1990, and by 1990 literacy is near-universal on both banks, close to 96 percent, with a gap under one percentage point. It remains indistinguishable from zero in 2000, 2010 and 2020.

↩ Back to text

B.4 Historical human capital gaps in 1930

The 1930 evidence highlighted in Figure 5 draws on the broader set of variables in Table B.2, restricting the sample to *cuarteles* adjacent to the boundary — cuarteles 1, 2, and 7 on the west (central Guadalajara and the historical Mexicaltzingo area) and cuarteles 8 and 9 on the east (Analco). Comparisons control flexibly for sex and age, with standard errors clustered at the household level and FDR adjustment. The pattern is consistent with the 2020 BD estimates. Human-capital outcomes divide the two banks sharply in 1930, with adult illiteracy 12.0 percentage points higher and school attendance among residents aged 6 to 18 15.3 points lower on the eastern side, and so do occupational composition and birthplace, with east-side residents more likely to be Jalisco-born. Property ownership shows no significant differential, but the two ownership indicators are coarse and the broader one is measured on a seventh of the sample, so that null does not establish that wealth in property was equal across the banks.

Table B.2: Socio-demographic differences across the historical boundary, 1930 Census

	Property		Birthplace	Occupation						Human capital	
	Some	Residence	Jalisco	Zapatero	Merchant	Laborer	Mason	Employed in firm	Maid	Housework	Illiterate	Studying
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
$\mathbb{1}\{East_i\}$	−0.013	−0.003	0.052***	0.015***	0.002	0.002	0.002	−0.005	−0.011***	0.033***	0.120***	−0.153***
	(0.024)	(0.003)	(0.013)	(0.003)	(0.006)	(0.005)	(0.003)	(0.003)	(0.003)	(0.009)	(0.023)	(0.026)
Sample	All ages	All ages	All ages	All ages	All ages	All ages	All ages	All ages	All ages	All ages	Ages 15+	Ages 6–18
Individual controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
FDR q-value	0.399	0.364	0.001	0.001	0.444	0.423	0.399	0.160	0.002	0.001	0.001	0.001
Observations	1,315	9,238	9,169	9,172	9,172	9,172	9,172	9,172	9,172	9,172	6,144	2,726
West mean	0.230	0.023	0.877	0.009	0.050	0.032	0.013	0.019	0.016	0.305	0.212	0.538
East mean	0.201	0.020	0.931	0.025	0.052	0.035	0.015	0.014	0.005	0.332	0.337	0.382
R^2	0.100	0.069	0.026	0.059	0.095	0.076	0.035	0.021	0.019	0.526	0.069	0.274

Notes: OLS estimates of the east-minus-west difference in mean characteristics among residents of the *cuarteles* adjacent to the historical boundary in the 1930 Population Census of Guadalajara (Anderson and Spike, 2007). The west-bank sample includes cuarteles 1, 2, and 7 (central Guadalajara and the historical Mexicaltzingo area); the east-bank sample includes cuarteles 8 and 9 (Analco), which $\mathbb{1}\{East_i\}$ identifies. Each dependent variable is an indicator: property ownership, birthplace (born in Jalisco), principal occupational categories, and two human-capital outcomes, illiteracy among adults aged 15 or older and current school attendance among individuals aged 6 to 18. Property answers coded 999 are treated as missing. *Individual controls* are sex and flexible age controls (age fixed effects). The FDR q-value row reports sharpened false-discovery-rate values (Anderson, 2008) across all twelve variables. West mean and East mean are the raw means on each bank. The east–west gaps in birthplace, occupational structure, illiteracy, and school attendance are large and significant after correction, while the property comparisons are not. The two property variables are indicators of whether a person reports owning any property or owning their residence, not values or any wider measure of wealth, and the first is recorded for 1,315 individuals against roughly 9,200 for occupation, so the null there is imprecise and silent about property values. The 1930 census records no ethnicity. The table describes where the measured differences lie at this date and does not identify the channels through which the boundary operated. Standard errors clustered by household, identified by cuartel and household number, are reported in parentheses. Stars refer to the unadjusted p-value of each estimate. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

⟷ Back to text

B.5 Limited urban expansion before the 20th century

The 1621 account of priest Domingo Lázaro de Arregui describes the SJdD creek as a modest watercourse crossed by multiple bridges (Lázaro de Arregui, 1946). Figure B.3, panel (a) reproduces a 1908 map: nearly four centuries after the foundation, the city's footprint remained largely contained within the colonial core, and the creek remained a legible east–west divide. Panel (b) reproduces a 1972 map: by the second half of the 20th century the metropolitan area had expanded dramatically following mid-century industrialization (Haber, 1987), but the colonial core retained its centrality and Calzada Independencia continued to demarcate the east–west structure.

B.6 Estimates along the boundary

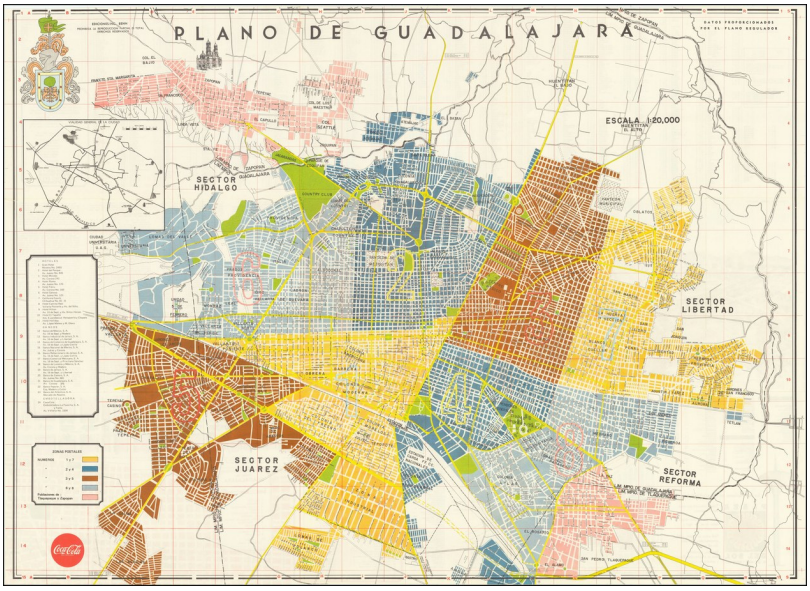
Because the historical boundary is irregular, Figure 6 estimates the discontinuity at each point along it. Table B.3 reports those estimates. The two southernmost points, which span the segment facing Mexicaltzingo, are the only ones whose uniform confidence band lies entirely below zero in both outcomes, and each differs significantly from every northern point. The two southern points show no detectable difference from each other, and the northern points show none among themselves.

Table B.4 estimates, at the same five points, the discontinuities in present-day conditions of the boundary and compares the southern points with the northern ones.

Figure B.3: Urban expansion of Guadalajara across the 20th century



(a) Limited urban expansion by the early 20th century (1908)



(b) Metropolitan expansion by the 1970s

Notes: Panel (a) reproduces a 1908 cartographic representation of Guadalajara: the footprint remained largely contained within the colonial core nearly four centuries after the 1542 foundation, and the SJdD creek divided Spanish Guadalajara and Mexicaltzingo on the western bank from Analco on the eastern bank. Panel (b) reproduces a 1972 map from the Instituto de Información Estadística y Geográfica (IIEG) at the height of Mexico's mid-century industrialization. The metropolitan footprint expanded by an order of magnitude, but the colonial core retained its centrality, and Calzada Independencia — which by 1972 had replaced the SJdD creek — continued to demarcate the east–west structure. Map sources: IMEPLAN (2016) and the IIEG historical map collection.

↩ Back to text

B.7 Robustness: supply- and demand-side determinants

Educational investment responds to what schools offer and to what households can and want to spend. The 2015 *Encuesta Intercensal* records household resources and housing that the full-

Table B.3: Location-based BD estimates by boundary point

Boundary point	Mean within h: West	Mean within h: East	$\hat{\tau}$	SE	95% CI	p-value	FDR q-value	Poly	h (m)	b (m)	N_h	N_b	95% uniform band
Panel A. Years of schooling													
Point 1 (north)	—	—	−0.427	0.457	[−1.324, 0.469]	0.350	—	—	597 × 542	—	9,253	—	[−1.60, 0.75]
Point 2 (north)	—	—	−0.724	0.264	[−1.242, −0.206]	0.006	—	—	781 × 709	—	17,159	—	[−1.40, −0.04]
Point 3 (north)	—	—	−0.894	0.522	[−1.917, 0.129]	0.087	—	—	637 × 578	—	8,860	—	[−2.23, 0.45]
Point 4 (south, Spanish Guadalajara)	—	—	−2.291	0.299	[−2.877, −1.704]	< 0.001	—	—	994 × 903	—	17,262	—	[−3.06, −1.52]
Point 5 (south, Mexicaltzingo)	—	—	−1.945	0.277	[−2.488, −1.401]	< 0.001	—	—	1,215 × 1,102	—	24,075	—	[−2.66, −1.23]
Panel B. School attendance													
Point 1 (north)	—	—	0.073	0.038	[−0.003, 0.148]	0.059	—	—	970 × 912	—	7,230	—	[−0.02, 0.17]
Point 2 (north)	—	—	−0.033	0.034	[−0.100, 0.034]	0.329	—	—	1,055 × 991	—	7,653	—	[−0.12, 0.05]
Point 3 (north)	—	—	0.013	0.020	[−0.025, 0.052]	0.497	—	—	2,012 × 1,892	—	26,679	—	[−0.04, 0.06]
Point 4 (south, Spanish Guadalajara)	—	—	−0.126	0.030	[−0.184, −0.068]	< 0.001	—	—	1,678 × 1,577	—	16,206	—	[−0.20, −0.05]
Point 5 (south, Mexicaltzingo)	—	—	−0.118	0.036	[−0.188, −0.048]	< 0.001	—	—	1,637 × 1,539	—	11,708	—	[−0.21, −0.03]
Panel C. Years of schooling, southern point minus northern point													
Point 4 − point 1			−1.863	0.547		< 0.001	0.002						
Point 4 − point 2			−1.567	0.399		< 0.001	< 0.001						
Point 4 − point 3			−1.397	0.602		0.020	0.008						
Point 5 − point 1			−1.517	0.535		0.005	0.004						
Point 5 − point 2			−1.221	0.383		0.001	0.002						
Point 5 − point 3			−1.051	0.591		0.075	0.025						
Panel D. School attendance, southern point minus northern point													
Point 4 − point 1			−0.198	0.048		< 0.001	< 0.001						
Point 4 − point 2			−0.092	0.045		0.041	0.017						
Point 4 − point 3			−0.139	0.035		< 0.001	< 0.001						
Point 5 − point 1			−0.190	0.052		< 0.001	< 0.001						
Point 5 − point 2			−0.084	0.049		0.088	0.030						
Point 5 − point 3			−0.131	0.041		0.001	0.001						

Notes: Location-based BD estimates following Cattaneo et al. (2025), using individual-level data from the 2020 *Censo de Población y Vivienda* geolocated at the block level and produced in the INEGI microdata laboratory. Boundary points are ordered from north to south; points 4 and 5 span the southern segment, the one facing the historical *pueblo* of Mexicaltzingo. Outcomes are in levels: years of schooling in years and school attendance as a share. Panels A and B report, for each point, the robust bias-corrected estimate with its standard error, 95% confidence interval, and p-value, and the 95% uniform confidence band, which is centred on that estimate and is simultaneous over the five points reported. These are the estimates plotted in Figure 6, so the table and the figure report the same quantities. `rd2d` selects one bandwidth per coordinate, common to both sides; h reports the east–west and north–south half-widths of the estimation window, converted from the exported bandwidths in degrees at the latitude of each point, and N_h the observations inside it on both sides. The export records neither the polynomial order nor a separate bias-correction bandwidth, so the Poly, b, and N_b cells are marked with a dash. The export does not include side means, so the West and East cells are marked with a dash. The uniform band covers the five points jointly, so no false-discovery-rate q-values are computed for Panels A and B. Panels C and D report, in the $\hat{\tau}$, SE, p-value, and q-value columns, the difference between each southern point and each northern point, computed from the bias-corrected estimates. Because `rd2d` does not export the covariance between boundary points, and the version available when these estimates were produced offered no joint test of equality across points, the standard error of the difference treats the two estimates as independent. That overstates the standard error when the covariance between the two estimates is positive, which overlapping local neighbourhoods would ordinarily produce, but bias-corrected local-polynomial estimates carry influence weights that can change sign, so the direction is not guaranteed. These are pointwise contrasts rather than a simultaneous test over the family of comparisons. The p-value of each contrast is unadjusted, and the q-value is the sharpened false-discovery-rate value of Anderson (2008), computed across the six comparisons within each outcome. A sixth boundary point is estimated but returns no bias-corrected estimate or band, so neither the band nor the contrasts cover it.

↩ Back to text

count census does not. Table B.5 reports BD estimates at the boundary for each of these measures. They cover several of the determinants discussed in Section 3.5 — household resources, credit access, and parental human capital — but not all of them: wealth, liquidity, collateral, and the concentration of landownership are not measured here (Galor, 2011).

Household income. If eastern households were poorer today, the schooling gap could reflect current budgets rather than anything the assignment set in motion. Household labor income per capita shows no detectable discontinuity at the boundary, although its interval admits an east-side shortfall of up to about half without the additional controls and two-fifths with them, and government transfers, pensions, and remittances show no detectable discontinuity either (Table B.5). The boundary estimate for years of schooling falls by about a third, from 0.31 to 0.21 standard deviations, when household income per capita is added, in a smaller sample, and current income is itself partly a product of schooling. Household income includes the earnings of members aged 12 to 17 who work, so conditioning on it conditions partly on attendance. In

Table B.4: Present-day conditions at the boundary, north and south

	Boundary estimate (SD)		South − North					
Outcome	North (1–2)	South (4–5)	Diff.	SE	95% CI	p-value	FDR q-value	Blocks
Land								
Log land value per m^2 (2017 cadastre)	−0.72	−0.62	0.10	0.31	[−0.51, 0.72]	0.744	0.368	11,401
Schools								
Schools within 30 minutes' walk	−0.15	0.00	0.16	0.09	[−0.01, 0.32]	0.068	0.074	12,077
High schools within 30 minutes' walk	0.01	0.06	0.04	0.08	[−0.11, 0.20]	0.599	0.368	12,077
Crime								
All crimes, incidents per hectare (2017–2024)	−0.17	−0.50	−0.33	0.25	[−0.82, 0.15]	0.179	0.140	12,077
Street infrastructure								
Fronts paved (share)	0.09	1.37	1.28	0.35	[0.60, 1.96]	< 0.001	0.002	3,827
Fronts with sidewalk (share)	0.31	0.53	0.22	0.34	[−0.45, 0.90]	0.513	0.368	3,827
Fronts with ramps (share)	−0.53	0.95	1.47	0.46	[0.57, 2.37]	0.001	0.006	3,827
Urban form								
Residents per block	0.19	−0.19	−0.38	0.16	[−0.68, −0.08]	0.014	0.035	12,077
Occupied dwellings per block	0.14	−0.26	−0.40	0.18	[−0.76, −0.05]	0.026	0.045	12,029

Notes: Location-based BD estimates following Cattaneo et al. (2025) at the five boundary points of Figure 6 and Table B.3, ordered from north to south, using the census blocks of Guadalajara. Each outcome is standardized over blocks, so estimates are in standard deviations, and each point estimate is the robust bias-corrected estimate, as in Table B.3. North is the mean of the estimates at points 1 and 2, along the stretch urbanized in the twentieth century, and South the mean at points 4 and 5, where the colonial settlements faced each other; point 3 is left out, as in the text. Land values come from the 2017 cadastre, schools within 30 minutes' walk from Table B.9, crime is annual incidents per hectare from the data of Table B.11, street fronts from the 2020 national housing inventory of Table B.12, and residents and occupied dwellings from the 2020 Census. Because `rd2d` does not report the covariance between boundary points, the standard error of the difference treats the four estimates as independent, as in Table B.3. The p-value is unadjusted, and the q-value is the sharpened false-discovery-rate value of Anderson (2008) across the outcomes of the table. Blocks is the number of blocks with the outcome.

↩ Back to text

the municipality-wide comparison of Table 1, adding household income leaves the adolescent attendance gap essentially unchanged, in a slightly smaller sample.

Parental human capital. Parental education raises the productivity of investment in children (Cunha and Heckman, 2007), so an initial difference in human capital can reproduce itself across generations even with equal schools and perfect capital markets (Galor and Tsiddon, 1997). Parental schooling absorbs about a third of the adolescent attendance gap, which falls from 8.1 to 5.4 percentage points in Table 1. That attenuation is not a confound removed: a parent's own schooling was itself shaped by the provision asymmetry documented in Section 2.2, so it measures how much of the gap runs through the previous generation. In the experiment, the Map-First effect is no smaller among households whose parents completed high school (Table D.9).

Credit access. With imperfect credit markets, lower initial wealth can sustain lower investment in children across generations even after provision converges (Galor and Zeira, 1993; Galor, 2011). The survey records how the dwelling was acquired and financed, whether the household is still paying for it, and whether employment carries housing credit and retirement savings. With the table's additional controls, east-side households are more likely to own, to have bought the dwelling ready-built, and to have paid for it with their own resources or with bank credit, and less likely to rent or to have inherited it, and without those controls east-side credit users are

Table B.5: Household resources, housing, and commuting at the boundary (2015 Intercensal Survey)

	Mean within h												
Outcome	West	East	$\hat{\tau}$	SE	95% CI	p-value	FDR q-value	Poly	h (m)	b (m)	N_h	N_b	Unclean controls
Panel A. Human capital													
Years of schooling	—	—	−0.305	0.097	[−0.496, −0.115]	0.002	—	1	835	1,996	3,077	10,876	No
	—	—	−0.213	0.101	[−0.411, −0.016]	0.034	—	1	835	1,996	2,481	9,106	✓
Attends school in Guadalajara	—	—	−0.037	0.032	[−0.100, 0.025]	0.242	—	1	1,258	2,473	1,038	3,236	No
Panel B. Household resources													
Household income (log)	—	—	−0.281	0.194	[−0.661, 0.098]	0.146	—	1	867	1,623	832	1,903	No
	—	—	−0.102	0.201	[−0.496, 0.292]	0.613	—	1	664	1,512	592	1,608	✓
Government transfers	—	—	−0.013	0.073	[−0.155, 0.130]	0.859	—	1	827	1,490	867	2,094	No
	—	—	0.009	0.063	[−0.115, 0.133]	0.882	—	1	852	1,548	1,092	2,159	✓
Pension income	—	—	−0.067	0.092	[−0.248, 0.114]	0.470	—	1	767	1,693	832	2,653	No
	—	—	0.001	0.057	[−0.110, 0.113]	0.980	—	1	1,199	2,219	1,632	4,306	✓
International remittances	—	—	0.002	0.027	[−0.050, 0.055]	0.933	—	1	1,778	3,137	2,745	7,449	No
	—	—	−0.007	0.035	[−0.075, 0.061]	0.838	—	1	1,353	2,408	1,820	4,911	✓
Domestic remittances	—	—	−0.005	0.051	[−0.106, 0.095]	0.917	—	1	1,102	2,400	1,551	4,923	No
	—	—	−0.016	0.046	[−0.106, 0.074]	0.727	—	1	1,465	2,667	2,029	5,685	✓
Panel C. Housing tenure													
Owner-occupied	—	—	0.141	0.102	[−0.060, 0.342]	0.169	—	1	768	1,511	837	2,099	No
	—	—	0.247	0.068	[0.114, 0.380]	< 0.001	—	1	1,298	2,390	1,345	3,848	✓
Rented	—	—	−0.170	0.116	[−0.398, 0.059]	0.145	—	1	485	1,380	561	1,825	No
	—	—	−0.304	0.061	[−0.423, −0.185]	< 0.001	—	1	1,769	3,183	2,129	6,027	✓
Owned by relatives, or lent	—	—	−0.098	0.067	[−0.229, 0.033]	0.142	—	1	682	1,547	786	2,168	No
	—	—	−0.056	0.055	[−0.165, 0.052]	0.308	—	1	1,226	2,939	1,319	5,337	✓
Title in the respondent's name	—	—	0.002	0.060	[−0.116, 0.120]	0.974	—	1	2,417	4,670	2,036	6,210	No
	—	—	0.158	0.124	[−0.085, 0.401]	0.201	—	1	1,387	2,812	512	1,901	✓
Panel D. How the dwelling was acquired													
Bought ready-built	—	—	0.271	0.148	[−0.020, 0.562]	0.068	—	1	897	2,081	423	1,578	No
	—	—	0.503	0.187	[0.136, 0.871]	0.007	—	1	1,042	1,921	423	959	✓
Had it built	—	—	−0.044	0.085	[−0.210, 0.123]	0.607	—	1	1,016	2,192	517	1,720	No
	—	—	−0.040	0.069	[−0.175, 0.096]	0.566	—	1	1,551	3,293	603	2,657	✓
Built by the household	—	—	−0.035	0.077	[−0.185, 0.116]	0.653	—	1	985	1,654	470	1,035	No
	—	—	−0.039	0.093	[−0.222, 0.144]	0.675	—	1	963	1,734	342	793	✓
Inherited	—	—	−0.253	0.134	[−0.516, 0.010]	0.059	—	1	1,118	2,335	602	2,008	No
	—	—	−0.401	0.157	[−0.709, −0.093]	0.011	—	1	1,091	2,763	427	1,850	✓
Panel E. How the dwelling was financed													
Own resources	—	—	0.143	0.082	[−0.018, 0.304]	0.081	—	1	627	1,716	686	2,702	No
	—	—	0.169	0.050	[0.071, 0.267]	< 0.001	—	1	1,759	3,138	2,136	5,917	✓
Loan from family or friends	—	—	0.055	0.033	[−0.010, 0.120]	0.096	—	1	1,006	1,889	1,359	3,077	No
	—	—	0.021	0.019	[−0.016, 0.059]	0.263	—	1	1,849	3,030	2,365	5,628	✓
Bank or other institutional credit	—	—	−0.014	0.027	[−0.066, 0.038]	0.596	—	1	347	1,125	354	1,577	No
	—	—	0.086	0.023	[0.041, 0.131]	< 0.001	—	1	1,069	2,259	1,151	3,466	✓
Public housing credit	—	—	0.007	0.025	[−0.043, 0.057]	0.783	—	1	483	1,219	533	1,652	No
	—	—	−0.011	0.025	[−0.060, 0.038]	0.658	—	1	1,095	2,348	1,161	3,785	✓
Fully paid	—	—	0.760	0.331	[0.112, 1.408]	0.022	—	1	895	1,979	111	329	No
	—	—	0.257	0.197	[−0.130, 0.643]	0.193	—	1	1,466	3,293	154	706	✓
Still paying	—	—	−0.760	0.331	[−1.408, −0.112]	0.021	—	1	891	1,983	111	330	No
	—	—	−0.258	0.197	[−0.644, 0.129]	0.192	—	1	1,461	3,301	154	706	✓
Panel F. Commute to school													
Up to 15 minutes	—	—	−0.227	0.126	[−0.472, 0.019]	0.071	—	1	581	1,484	591	1,704	No
16 to 30 minutes	—	—	0.083	0.068	[−0.051, 0.217]	0.224	—	1	1,103	2,493	1,229	4,457	No
31 to 60 minutes	—	—	0.059	0.060	[−0.059, 0.178]	0.325	—	1	553	1,894	536	2,600	No
More than 60 minutes	—	—	0.003	0.016	[−0.029, 0.034]	0.872	—	1	804	1,518	734	1,711	No
Public transport	—	—	0.174	0.081	[0.016, 0.333]	0.031	—	1	1,034	2,026	1,186	2,982	No
Private transport	—	—	0.104	0.100	[−0.093, 0.301]	0.300	—	1	341	898	255	910	No

Notes: BD estimates at the historical boundary using the 2015 *Encuesta Intercensal* for the municipality of Guadalajara, produced in the INEGI microdata laboratory. Years of schooling are in standard deviations and household income in log points; all other outcomes are indicators, so their estimates are differences in shares. Household income is INEGI's labor income of the household (*ingtrhog*), the earnings of all members aged 12 and older, divided by the number of members, so it includes the earnings of adolescents who work. In Panel E the last two rows are asked only of households that used credit. Panel F covers sons and daughters aged 24 or less. Each row is a separate estimate with a local linear polynomial, a uniform kernel, nearest-neighbor variance, survey weights, and an MSE-optimal bandwidth selected in each row (Calonico et al., 2014). All rows include the clean controls: age, sex, longitude, latitude, and boundary-segment fixed effects for person-level outcomes, and longitude, latitude, and boundary-segment fixed effects for household-level outcomes. *Unclean controls* add log household income per capita for person-level outcomes and, for household-level outcomes, log distance to the central business district, the sex and age of the household head, parental schooling, and log household income per capita; income is omitted from the controls in the row where it is the dependent variable. $\hat{\tau}$ is the bias-corrected estimate, and SE, the 95% CI, and the p-value are robust bias-corrected; h and b are the MSE-optimal main and bias-correction bandwidths in meters, and N_h and N_b the effective sample sizes. The laboratory export does not include the side means within h, so the West and East cells are marked with a dash, and no false-discovery-rate q-values were computed for these outcomes. Employment benefits and the commute to work come from a municipality-wide comparison, not a BD design, and are reported in Table B.6.

↩ Back to text

Table B.6: Employment benefits and commute to work, east and west of the boundary (2015 Intercensal Survey)

	Employment benefits		Commute time				Commute mode			Works in Guadalajara
	Housing credit	Retirement savings	Up to 15 min	16–30 min	31–60 min	More than 60 min	Walking	Public transport	Private transport	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
$\mathbb{1}\{East_i\}$	−0.023	−0.008	−0.026	−0.037**	0.037**	0.058***	−0.017	0.079***	−0.059***	0.051***
	(0.021)	(0.020)	(0.017)	(0.016)	(0.015)	(0.009)	(0.014)	(0.018)	(0.018)	(0.012)
Sample	Workers	Workers	Workers	Workers	Workers	Workers	Workers	Workers	Workers	Workers
Individual controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Geographic controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Boundary-segment FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observations	26,202	26,197	36,039	36,039	36,039	36,039	31,985	36,039	36,039	35,673
Dep. var. mean	—	—	—	—	—	—	—	—	—	—
R^2	0.070	0.072	0.028	0.009	0.018	0.016	0.036	0.040	0.051	0.011

Notes: OLS estimates from individual-level data on workers in the 2015 *Encuesta Intercensal* for the municipality of Guadalajara, produced in the INEGI microdata laboratory. Each dependent variable is an indicator: the job provides access to housing credit or to a retirement savings account (SAR or AFORE), the travel time to work falls in the stated band, the worker reaches work by the stated mode, or the workplace is in the municipality of Guadalajara. The benefit outcomes are recorded for fewer workers than the commuting outcomes. This is a municipality-wide comparison and not a BD estimate: the coefficient is that of the east-side indicator $\mathbb{1}\{East_i\}$, with side-specific linear trends in distance to the boundary. *Individual controls* are age and sex fixed effects; *Geographic controls* are longitude, latitude, log distance to the central business district, distance to the boundary, and its interaction with the east-side indicator. Estimates use the survey weights. The laboratory export does not include the dependent-variable means, so that row is marked with a dash. Standard errors clustered by dwelling are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

↩ Back to text

more likely to have paid off their credit, and none of these is the pattern of a more constrained side (Table B.5). In a municipality-wide comparison, the shares of workers with housing credit or retirement savings through their employment show no detectable difference between the two sides (Table B.6). These measures capture the use of housing finance and formal access through employment, not liquidity or collateral, and they describe households today. They do not speak to whether credit constraints operated when the schooling stock was formed.

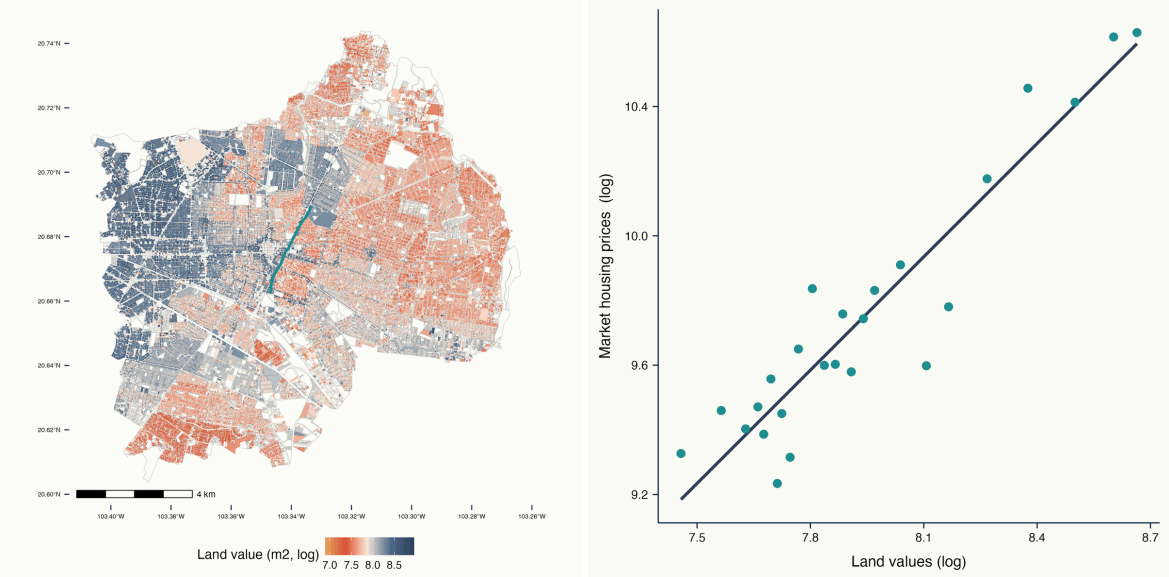
Access to schools and to jobs. Schools have no residential catchments here, and short commutes to school show no detectable difference across the boundary, with an interval that admits a shortfall of up to about 47 percentage points on the east side, while public transport is used for school 17 points more often there, and these estimates do not establish longer journeys to school (Table B.5). The commute to work is another matter. In the municipality-wide comparison, eastern workers are more likely to travel more than an hour and to use public transport (Table B.6). Access to jobs differs more clearly across the two sides than access to schools.

B.8 Housing-market sorting and capitalization

If housing prices capitalize amenities or the boundary's stigma, socioeconomic groups may sort across space, and gaps could reflect selection rather than the legacy of exclusion.

The west–east gradient in land values. Figure B.4a maps parcel-level cadastral land values, documenting a clear west–east gradient. Figure B.4b validates the cadastral measure against modern market housing prices at the neighborhood (*colonia*) level (IIEG web-scraped listings); the two are tightly correlated, so the cadastral measure is a credible proxy for the price-relevant component of land value.

Figure B.4: Land values and housing-market capitalization in Guadalajara



(a) Spatial distribution of land values (log) across Guadalajara

(b) Cadastral land values vs. market housing prices (log)

Notes: Panel (a) maps parcel-level cadastral land values (log pesos per m^2) from the Guadalajara Municipal Cadaster (2017). Each polygon is a parcel; shading runs from red (lower) to blue (higher). The thick green line marks Calzada Independencia. Parcels east of the boundary are systematically less valuable per m^2. Panel (b) validates the cadastral measure against modern market housing prices. Both variables are aggregated to the neighborhood (*colonia*): the horizontal axis is the log of the neighborhood mean of cadastral land value per m^2 and the vertical axis the log of the neighborhood mean of market housing price per m^2 from web-scraped listings, so the two panels are on the same logarithmic scale. Each point is one of 25 bins; the solid line is a linear fit and the dashed line is the 45-degree line. Sources: Guadalajara Municipal Cadaster (2017); IIEG web-scraped housing listings.

↩ Back to text

Land-value differences by land use. Table B.8 estimates the discontinuity in parcel-level land values separately by the land use of the parcel's block. Blocks are classified with the 2020 census and the national business directory (DENUE). An establishment serves the neighborhood when it employs at most ten people, is not located in a market, shopping center, or commercial building, and belongs to a listed set of everyday activities such as grocery stores, tortillerías, pharmacies, small restaurants, beauty salons, and repair shops. A block is residential if it has inhabited dwellings and only neighborhood-serving establishments, non-residential if it has no inhabited dwellings or more than half of its establishments do not serve the neighborhood, and mixed otherwise. Land values are lower east of the boundary for non-residential parcels (−0.30 log points) and mixed parcels (−0.25), while the estimate for residential parcels (−0.21) is imprecise

and not statistically significant. The differences between classes are not statistically significant.

Table B.7: BD estimates controlling for land values (block level, 2020 Census)

	Mean within h											
Specification	West	East	$\hat{\tau}$	SE	95% CI	p-value	FDR q-value	Poly	h (m)	b (m)	N_h	N_b
Main (unweighted)	12.10	11.26	−0.622	0.189	[−0.992, −0.253]	< 0.001	—	1	821	1,651	593	1,541
Main (Poly 2)	12.61	11.30	−0.609	0.175	[−0.952, −0.267]	< 0.001	—	2	1,821	2,859	1,797	3,719
With clean controls	12.07	11.10	−0.643	0.164	[−0.965, −0.321]	< 0.001	—	1	691	1,646	463	1,535
With clean controls (Poly 2)	12.55	11.30	−0.733	0.167	[−1.060, −0.406]	< 0.001	—	2	1,542	2,768	1,397	3,536
+ Land value control	12.10	11.26	−0.402	0.143	[−0.682, −0.122]	0.005	—	1	805	1,724	570	1,606
+ Land value control (Poly 2)	12.59	11.27	−0.483	0.145	[−0.766, −0.199]	< 0.001	—	2	1,780	2,866	1,687	3,556

Notes: BD estimates of the boundary effect on mean years of schooling at the census-block level, using aggregated 2020 census data for Guadalajara. $\hat{\tau}$, its standard error, and its confidence interval are in standardized (z-score) units; block-level mean years of schooling is 11.22 (SD 2.00). "Main (unweighted)" is the baseline block-level BD estimate without land-value controls. "Clean controls" include block longitude, latitude, and boundary-segment dummies. "Land value control" adds the log of cadastral land values (pesos per m^2). All specifications are unweighted, so that the census block is the unit of analysis; the person-weighted estimand is the individual-level estimate reported in Table B.1. "+ Land value control" adds land values to the clean controls and is a sensitivity specification, and the main estimate of the gap is the individual-level one of Table B.1. Land-value controls leave the effect economically and statistically significant, and equal to between 0.80 and 0.97 years once block-level standardized units are converted at the block standard deviation of 2.00. The West and East columns are mean years of schooling within h of the boundary. $\hat{\tau}$ is the bias-corrected estimate. h and b are the MSE-optimal main and bias-correction bandwidths, and N_h and N_b are the numbers of blocks within them. The rows are alternative specifications for one outcome, so no false-discovery-rate q-values are computed. Standard errors, confidence intervals, and p-values are robust bias-corrected (Calonico et al., 2014).

↩ Back to text

Robustness of the boundary effect to land-value controls. Table B.7 reports BD estimates after adding flexible controls for block-level land values. The sensitivity specification adds land values to the clean controls, giving −0.40 standardized block-level units, or about 0.80 years of schooling, and the main estimate of the gap remains the individual-level one of 0.79 years (Table B.1). Across the land-value specifications the estimate ranges from −0.40 to −0.48 in block-level standard deviations, that is from 0.80 to 0.97 years, close to the estimate obtained on individual data; land values absorb part of the cross-boundary differential without changing the magnitude the individual-level design reports.

Table B.8: BD estimates on parcel land values by block land use

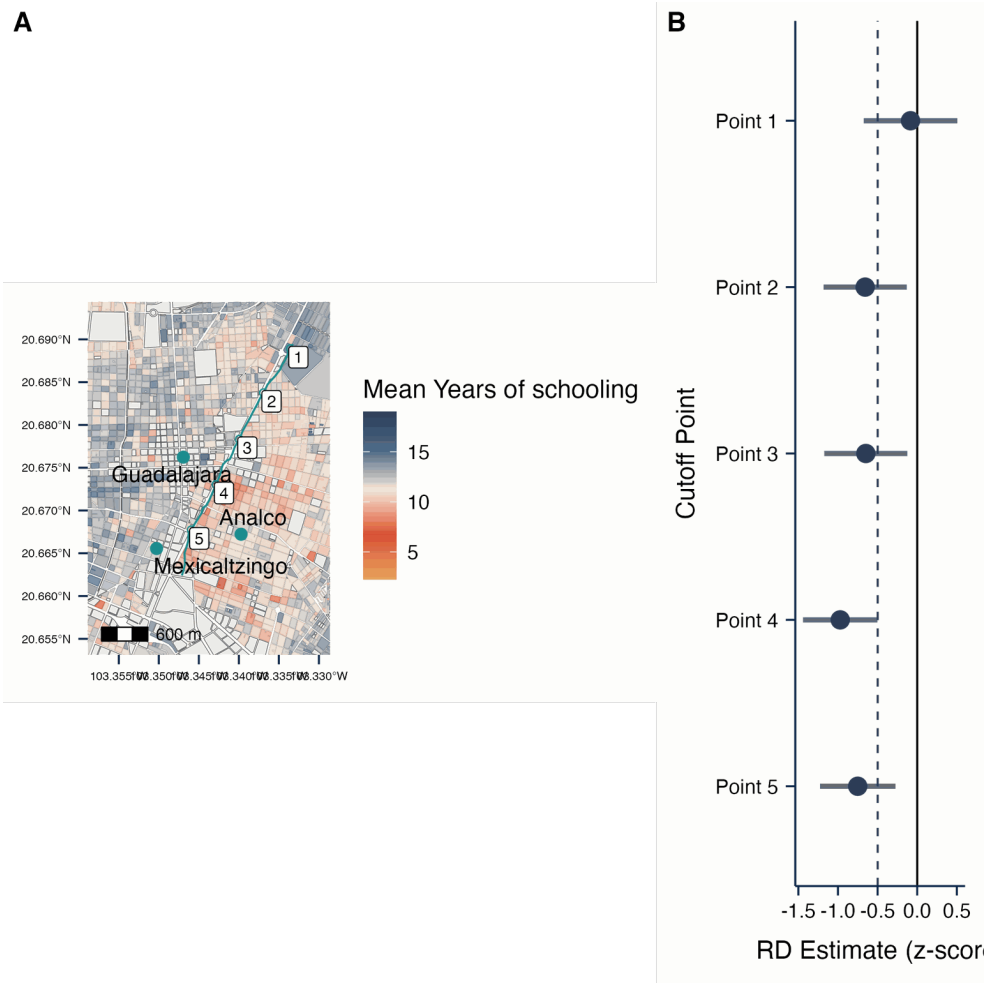
	Mean within h											
Block class	West	East	$\hat{\tau}$	SE	95% CI	p-value	FDR q-value	Poly	h (m)	b (m)	N_h	N_b
Residential	7.889	7.926	−0.208	0.541	[−0.883, 1.238]	0.743	—	1	488	488	495	495
Mixed	7.816	7.877	−0.249	0.207	[−0.887, −0.075]	0.020	—	1	695	695	3,471	3,471
Non-residential	8.034	7.882	−0.302	0.065	[−0.487, −0.232]	< 0.001	—	1	812	812	7,702	7,702
Differences												
Mixed − Residential			−0.040	0.579	[−1.794, 0.477]	0.256						
Non-residential − Residential			−0.094	0.545	[−1.605, 0.531]	0.325						
Non-residential − Mixed			−0.054	0.217	[−0.304, 0.547]	0.575						

Notes: BD estimates of the boundary effect on the log of cadastral land value per m^2 at the parcel level, estimated jointly by block land-use class with class-specific MSE-optimal bandwidths (Calonico et al., 2014). Each parcel takes the class of the census block that contains its centroid. Covariates for efficiency: longitude, latitude, and boundary-segment dummies. The West and East columns are mean log land values per m^2 within h of the boundary. $\hat{\tau}$ is the conventional estimate. h is the main bandwidth (meters) and N_h the number of parcels within it; the estimator fits the bias-correcting quadratic within h, so b equals h and N_b equals N_h. No false-discovery-rate q-values are computed for this table. Differences between classes use the bias-corrected estimates and treat the classes as independent subsamples. Standard errors, confidence intervals, and p-values are robust bias-corrected, clustered by census block.

↩ Back to text

Location-based estimates with land-value controls. Figure B.5 extends the location-based BD estimator of Cattaneo et al. (2025) to the land-value-controlled specification. The pointwise estimates remain negative and concentrated near the historical core.

Figure B.5: Location-based BD estimates with block-level land-value controls



Notes: Location-based BD estimates and 95% uniform confidence bands following Cattaneo et al. (2025), using block-level 2020 census data for Guadalajara and including block-level controls for log cadastral land values. The horizontal axis indexes points along the boundary. The shaded region is the 95% uniform confidence band; the dashed line marks the pooled (AATE) estimate. The vertical axis is in standardized units of mean years of schooling (sample mean 11.22, SD 2.00). Pointwise estimates remain negative along the boundary, consistent with the univariate land-value-controlled estimate in Table B.7.

↩ Back to text

B.9 The 1992 explosions: design details

A block is directly affected if it intersects an 80-meter buffer around the collector's path, digitized from a georeferenced photomosaic of the damaged corridor. First-degree neighbors of the affected blocks are excluded from the analysis as a buffer zone; second- through fifth-degree neighbors form the comparison group. The sample is restricted to the east side, within two kilometers of the historical boundary.

The 2000–2020 estimates use individual-level census microdata at the block level, accessed

through INEGI's microdata laboratory. Each wave-specific coefficient compares residents of affected and comparison blocks, with age and sex fixed effects and standard errors clustered by block; the estimates are unweighted. Years of schooling are measured among adults aged 18 and older. School attendance is measured among children aged 6–18, conditional on parental years of schooling. The 1990 census is geolocated at the AGEB level, so the pre-shock placebo compares the AGEBs intersecting the affected corridor with the remaining east-side AGEBs under the same specification, clustering by AGEB.

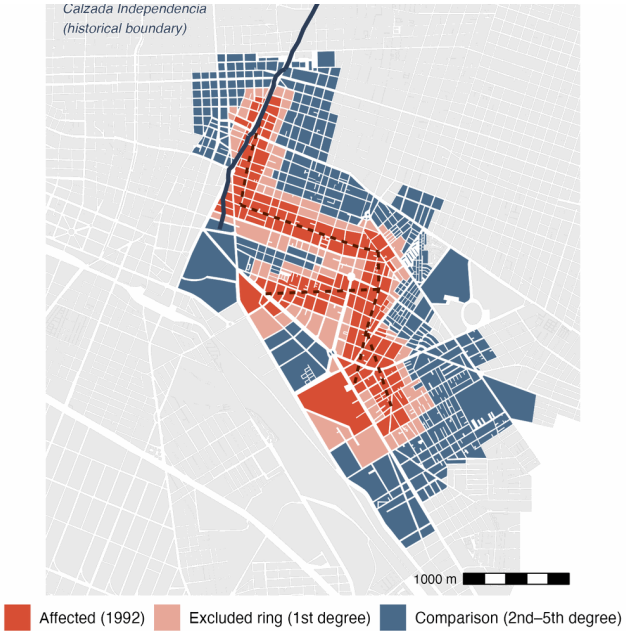
The same comparison for the composition of adults gives a difference in age of 0.88 years in 1990 (standard error 0.39), 1.35 in 2000 (0.64), −0.86 in 2010 (0.43), and 0.02 in 2020 (0.63). The change between 2000 and 2010 is −2.2 years (0.77), against a change of −1.07 years (0.43) in adult schooling. Sex, Indigenous language, and household size show no detectable difference in any wave. In the public block tabulations, affected blocks have fewer residents aged 65 and older and fewer female-headed households than comparison blocks in 2010 and 2020, and more children per household in 2010 ($p = 0.063$) but not in 2020 ($p = 0.19$), while mean schooling shows no detectable difference in either year. In 2020, the seven composition indicators differ jointly between affected and comparison blocks ($F = 4.44$, $p = 0.0001$), and 1.5 percent of residents of affected blocks aged five and older lived in another state five years earlier, against 3.9 percent in comparison blocks.

B.10 School supply and quality

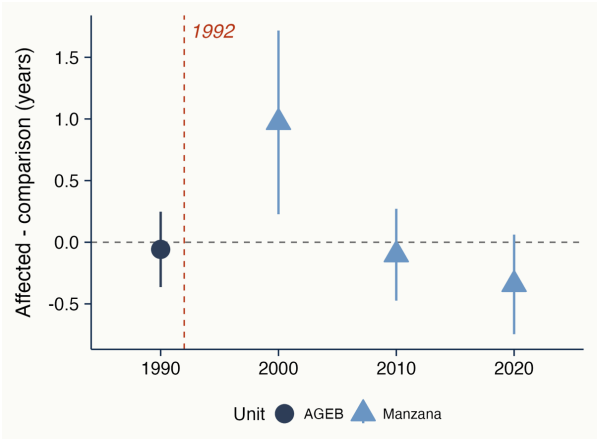
School supply. Table B.9 reports BD estimates on the schools within 30 minutes' walk of each block and on the walking distance to the nearest school, from 2018–2019 administrative records, without weights and weighting blocks by their share of school-age residents. Within the bandwidth, households on both sides reach about 130 schools, the numbers of shifts, public and private schools, and primary schools and the public share are not detectably different, and students per teacher are higher on the east side, by 0.89 without weights ($q = 0.001$) and 0.45 with weights ($p = 0.112$). Without weights, the nearest primary school is 52 meters closer on the east side and the nearest high school 65 meters farther, each under a minute on foot at 80 meters per minute, and with weights only the high-school difference remains, at 78 meters ($p = 0.058$).

School quality. I assess quality via *Formato 911* (approval, repetition, dropout, class size, student–teacher ratios; 1998–2015) and ENLACE standardized tests (Spanish and Math; 2006–2014). These indicators reflect the students each school enrolls as well as the school, and the operational ones cover mainly lower secondary, while ENLACE scores at the upper-secondary level come from students who reach the final grade. For each source I run a municipality-wide OLS comparison (Figures B.7 and B.9) and a BD design within 1.5 km of the boundary (Figures B.8 and B.10). Approval, repetition, and class size favor western schools at the start of the series, and differences detectable before the early 2000s are not detectable after. Measured

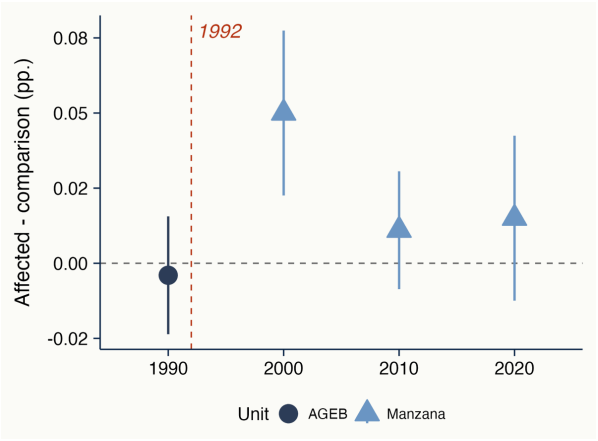
Figure B.6: The 1992 Reforma explosions: affected blocks, selection, and school attendance



(a) Affected and comparison blocks



(b) Selection: years of schooling



(c) School attendance of children

Notes: Panel (a) maps blocks directly affected by the April 1992 explosions (red), defined as intersecting an 80-meter buffer around the collector's path (dashed line); immediate first-degree neighbors, excluded from the comparison (yellow); and second- through fifth-degree neighboring blocks used as the comparison group (teal). The dark line is Calzada Independencia. Panels (b) and (c) plot differences between affected and comparison areas by census wave. Panel (b) reports years of schooling among adults aged 18 and older; panel (c) reports school attendance among children aged 6–18 conditional on parental schooling. The 1990 estimate is a pre-shock comparison at the AGEB level; the 2000–2020 estimates use census blocks. Vertical bars show 95 percent confidence intervals. All comparisons are within Analco, between directly affected blocks and their second- to fifth-degree neighbors. Appendix B.9 reports the construction of the affected area, comparison group, and specifications.

↩ Back to text

Table B.9: BD estimates on walking access to schools (block level, 2018–2019)

Outcome	Mean within h		$\hat{\tau}$	SE	95% CI	p-value	FDR q-value	Poly	h (m)	b (m)	N_h	N_b
	West	East										
Panel A. Schools within 30 minutes' walk, no weights												
Total schools	138.4	128.0	0.35	1.59	[−2.37, 3.86]	0.641	1.000	1	802	1,607	776	1,871
Total schools × shifts	165.3	153.9	1.99	2.08	[−2.48, 5.69]	0.442	1.000	1	853	1,334	840	1,487
Public schools	57.6	58.2	0.27	0.75	[−1.00, 1.93]	0.536	1.000	1	1,034	1,676	1,077	1,980
Private schools	79.8	69.8	0.09	1.12	[−1.83, 2.57]	0.742	1.000	1	761	1,586	720	1,839
Primary schools	62.7	60.7	−0.46	0.70	[−1.73, 1.02]	0.614	1.000	1	894	1,669	880	1,967
Secondary schools	16.6	14.9	−0.37	0.24	[−0.77, 0.18]	0.220	1.000	1	626	1,576	568	1,824
High schools	57.8	49.8	1.73	1.00	[−0.24, 3.67]	0.085	0.518	1	945	1,484	941	1,696
Public share	0.437	0.480	−0.002	0.004	[−0.010, 0.007]	0.785	1.000	1	914	1,467	912	1,672
Students per teacher	16.9	17.0	0.89	0.16	[0.54, 1.17]	< 0.001	0.001	1	985	1,641	1,004	1,922
Panel B. Schools within 30 minutes' walk, weighted by the share of block residents of school age												
Total schools	130.0	128.4	−1.20	2.59	[−6.69, 3.46]	0.533	0.991	1	870	1,432	608	1,206
Total schools × shifts	156.3	157.0	−0.03	2.99	[−6.67, 5.05]	0.786	1.000	1	838	1,421	579	1,199
Public schools	56.4	58.4	−0.79	0.96	[−2.41, 1.34]	0.578	0.991	1	1,195	1,975	944	1,992
Private schools	72.1	68.6	−0.17	1.85	[−3.99, 3.26]	0.843	1.000	1	864	1,421	601	1,199
Primary schools	60.0	60.1	−1.30	1.20	[−4.02, 0.69]	0.166	0.794	1	1,143	2,213	821	2,245
Secondary schools	15.2	13.3	−1.20	0.48	[−2.09, −0.20]	0.018	0.189	1	1,263	2,261	847	2,102
High schools	53.7	53.0	1.65	1.59	[−1.67, 4.55]	0.365	0.991	1	914	1,476	562	1,080
Public share	0.458	0.474	−0.005	0.006	[−0.016, 0.006]	0.331	0.991	1	912	1,563	643	1,363
Students per teacher	16.8	17.0	0.45	0.24	[−0.09, 0.85]	0.112	0.794	1	896	1,648	626	1,473
Panel C. Walking distance to the nearest school (m), no weights												
Nearest school	342	378	14.6	20.9	[−18.2, 63.9]	0.276	0.226	1	1,071	1,737	1,124	2,078
Nearest primary school	414	441	−51.9	21.7	[−96.8, −11.7]	0.012	0.053	1	1,166	1,916	1,248	2,425
Nearest secondary school	680	747	−7.0	33.5	[−85.2, 46.0]	0.558	0.388	1	1,050	1,673	1,094	1,975
Nearest high school	445	629	65.4	29.4	[2.1, 117.5]	0.042	0.068	1	911	2,032	908	2,665
Panel D. Walking distance to the nearest school (m), weighted as in Panel B												
Nearest school	360	378	24.4	24.2	[−12.2, 82.5]	0.146	0.303	1	1,016	2,844	755	3,555
Nearest primary school	393	400	0.8	35.2	[−45.6, 92.3]	0.507	0.455	1	846	1,757	545	1,520
Nearest secondary school	711	720	−20.0	70.8	[−173.2, 104.1]	0.625	0.455	1	1,194	1,986	785	1,685
Nearest high school	498	630	78.2	40.4	[−2.6, 155.6]	0.058	0.303	1	1,121	2,106	745	1,915

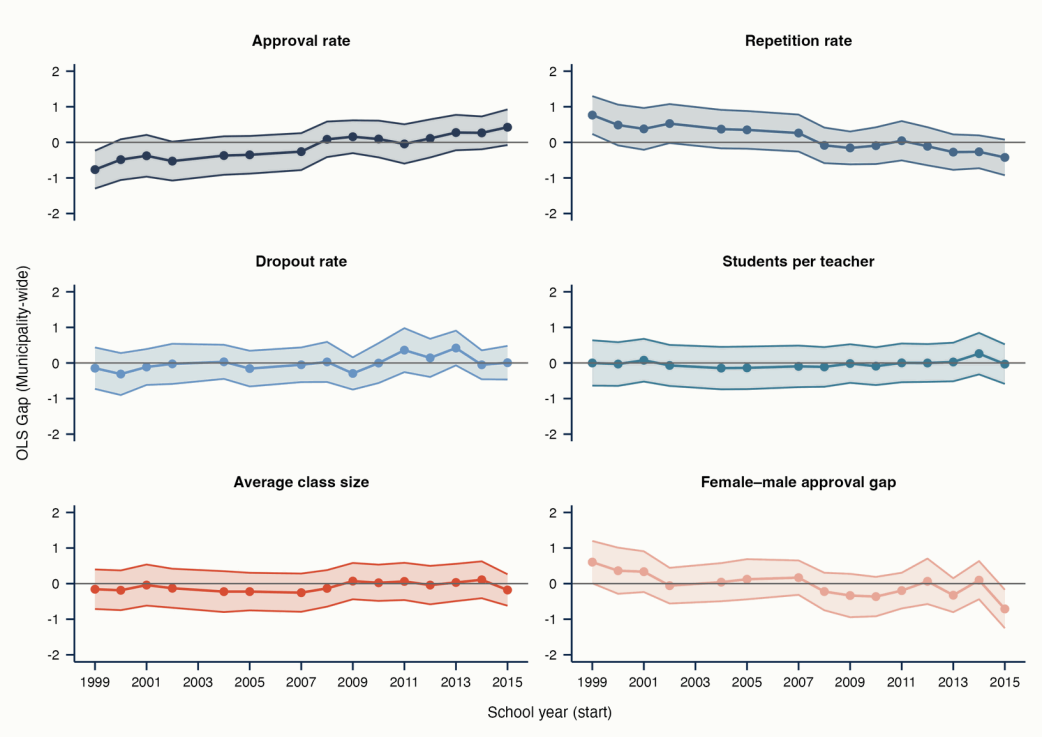
Notes: BD estimates of the boundary effect on walking access to schools from each block, using administrative school records for 2018–2019 and the 13,791 census blocks of Guadalajara. A school is within reach if the walking distance along the INEGI street network from the block centroid is at most 2,400 meters, 30 minutes at 4.8 km per hour; schools in neighboring municipalities count. Schools are primary, secondary, and high schools (including technical upper secondary); schools of public autonomous universities count as public. Dependent variables: the number of schools within reach, of school shifts (*turnos*), and of public, private, primary, secondary, and high schools; the public share of the schools within reach; and their students per teacher. Panels C and D report the walking distance in meters along the same network from the block centroid to the nearest school of any level and to the nearest primary, secondary, and high school. Specifications include boundary-segment fixed effects and geographic coordinates. Panels B and D weight each block by the share of its residents of the matching school age (6–17, or 6–11 for primary, 12–14 for secondary, and 15–17 for high schools); the 2020 census hides counts of one and two, which are set to 1.5. The West and East columns are means within h of the boundary, weighted in Panels B and D. $\hat{\tau}$ is the conventional estimate. b is the bias-correction bandwidth, and N_h and N_b are the numbers of blocks within h and b. q-values are sharpened false-discovery-rate values across the nine outcomes of each of Panels A and B and the four outcomes of each of Panels C and D (Anderson, 2008). Standard errors, confidence intervals, and p-values are robust bias-corrected (Calonico et al., 2014).

↩ Back to text

school provision is broadly comparable on the two sides, which weighs against school access and measured quality as the channel.

Where surveyed children attend school. The parent survey asks, for each child aged 6 to 20 who attends school, in what part of the city the school is. Among these children, 78 percent on the east side and 59 percent on the west attend a school in their own *colonia* or very near it, and the difference of 17 percentage points after controls mirrors the larger share of west-side children

Figure B.7: Municipality-wide OLS estimates of school quality (Formato 911, secondary)



Notes: East-minus-west gaps in school-level operational indicators for secondary schools across Guadalajara, using *Formato 911* data (1998–2015). Each panel reports a separate outcome (approval, repetition, dropout, class size, student–teacher ratio, female-minus-male approval gap), standardized within year. Specifications include school-location controls, a public/private indicator, and year fixed effects. Confidence bands are 95% and are pointwise, without adjustment for the six outcomes or the seventeen years. The approval and repetition gaps favour the western schools at the start of the series, with bands away from zero in the first years and point estimates of about 0.3 standard deviations in absolute value through 2007. Differences detectable before the early 2000s are not detectable after. The remaining indicators show no detectable difference in most years.

↩ Back to text

who attend a school elsewhere on their own side (Table B.10). Fewer than one in ten children on either side attends a school on the other side of the boundary, and among the 46 upper-secondary students the shares do not differ detectably. The comparison describes where children go and not the quality of the schools they attend.

B.11 Crime exposure

Recorded crime per hectare shows no detectable discontinuity at the boundary in any category (Table B.11 and Figure B.11). Reporting depends on the victim going to the prosecutor's office (*Fiscalía*), and at the boundary east-side residents report less trust in the police (−0.26 on a 0–1 scale, Table C.1), so recorded crime could understate east-side crime in the categories that depend on reporting. Homicide enters the record from the body, and vehicle theft is reported for insurance and registration, so neither depends on trust in the police. Neither shows a detectable discontinuity, so the absence of a detectable difference in recorded crime also holds in the categories where reporting does not depend on trust in the police.

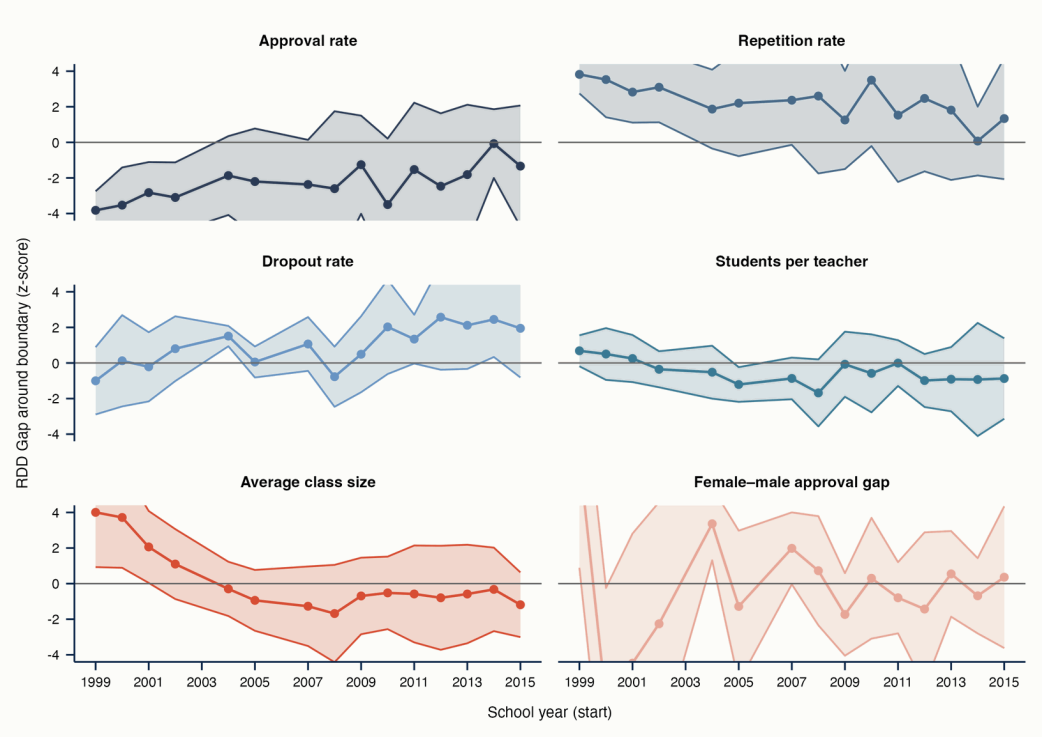
Table B.10: Where surveyed children attend school

	School the child attends (percent of children who attend school)			
	Own *colonia* or very near	Own side, other *colonia*	Other side of the boundary	Elsewhere in the city or outside it
	(1)	(2)	(3)	(4)
Panel A. All levels				
East side	16.7**	−19.1***	3.5	−1.0
	(7.3)	(6.9)	(3.9)	(3.7)
FDR q-value	0.038	0.025	0.330	0.642
West-side mean	58.7	26.7	6.7	8.0
East-side mean	77.5	8.3	9.1	5.1
Observations	350	350	350	350
R^2	0.273	0.104	0.100	0.230
Panel B. Primary				
East side	18.1*	−24.0***	2.4	3.4
	(9.2)	(8.8)	(4.3)	(3.1)
FDR q-value	0.083	0.031	0.401	0.216
West-side mean	69.4	25.0	5.6	0.0
East-side mean	89.1	3.4	6.1	1.4
Observations	183	183	183	183
R^2	0.177	0.199	0.111	0.253
Panel C. Lower secondary				
East side	19.3	−20.4	−2.4	3.5
	(15.5)	(13.4)	(8.1)	(3.4)
FDR q-value	0.677	0.677	0.677	0.677
West-side mean	52.4	33.3	9.5	4.8
East-side mean	77.8	12.5	8.3	1.4
Observations	91	91	91	91
R^2	0.138	0.189	0.208	0.520
Panel D. Upper secondary				
East side	10.2	1.6	9.6	−21.4
	(21.3)	(19.0)	(8.2)	(15.3)
FDR q-value	1.000	1.000	1.000	1.000
West-side mean	63.6	9.1	0.0	27.3
East-side mean	65.7	17.1	8.6	8.6
Observations	45	45	45	45
R^2	0.201	0.246	0.374	0.278
Panel E. University				
East side	16.1	−26.2	18.3	−8.2
	(16.5)	(23.1)	(21.0)	(25.6)
FDR q-value	1.000	1.000	1.000	1.000
West-side mean	0.0	50.0	16.7	33.3
East-side mean	18.2	13.6	31.8	36.4
Observations	25	25	25	25
R^2	0.285	0.450	0.290	0.271
Map-First	✓	✓	✓	✓
Respondent sex, age, and age squared	✓	✓	✓	✓
Respondent-education FE	✓	✓	✓	✓
Child sex and age	✓	✓	✓	✓
School-level FE	Panel A	Panel A	Panel A	Panel A

Notes: The sample is the 351 children aged 6 to 20 who attend school in the households of the parent survey, 276 on the east side and 75 on the west, from 222 of the 239 surveyed households. The question was asked for each child in the household roster, children who do not attend school (44) are excluded because it was not asked about them, and no child attends online. The dependent variable is an indicator, times 100, for the respondent's answer to "In what part of the city is the school that the child attends?". The categories are the same *colonia* or very near (column 1), the east or the west of Guadalajara, coded as the household's own side of the boundary (column 2) or the other side (column 3), and another part or municipality of the city or outside Guadalajara (column 4), so the four columns sum to 100. Column 4 pools other parts of Guadalajara, other municipalities of the city, and places outside it, since the questionnaire asks about other parts of the city and other municipalities in one category. Each coefficient is on the east-side indicator with the specification of Table C.4, Map-First assignment, respondent sex, age and age squared, and respondent-education fixed effects, plus the child's sex and age. Panel A adds fixed effects for the school level, and Panels B to E split the sample by level, with upper secondary being *preparatoria*. The one child in a technical program enters only Panel A, and observations exclude fixed-effect singletons. The West-side and East-side means are unadjusted shares. The FDR q-value row reports sharpened false-discovery-rate values (Anderson, 2008) across the four columns of each panel. Standard errors clustered by household are in parentheses. The survey records where children go to school and not the quality of the school, and all estimates are observational cross-boundary comparisons. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

↩ Back to text

Figure B.8: BD estimates of school quality (Formato 911, secondary)



Notes: BD estimates of the boundary effect on school-level operational indicators for secondary schools within 1.5 km of the boundary, *Formato 911* (1998–2015), standardized within year, following Calonico et al. (2014). Confidence bands are 95% and are pointwise, without adjustment for the six outcomes or the seventeen years. The approval, repetition and class-size gaps favour the western schools at the start of the series, with bands away from zero through about 2002. From the mid-2000s the bands widen to include zero, while the approval and repetition point estimates stay away from zero, and dropout estimates are positive after 2010. Differences detectable before the early 2000s are not detectable after.

↩ Back to text

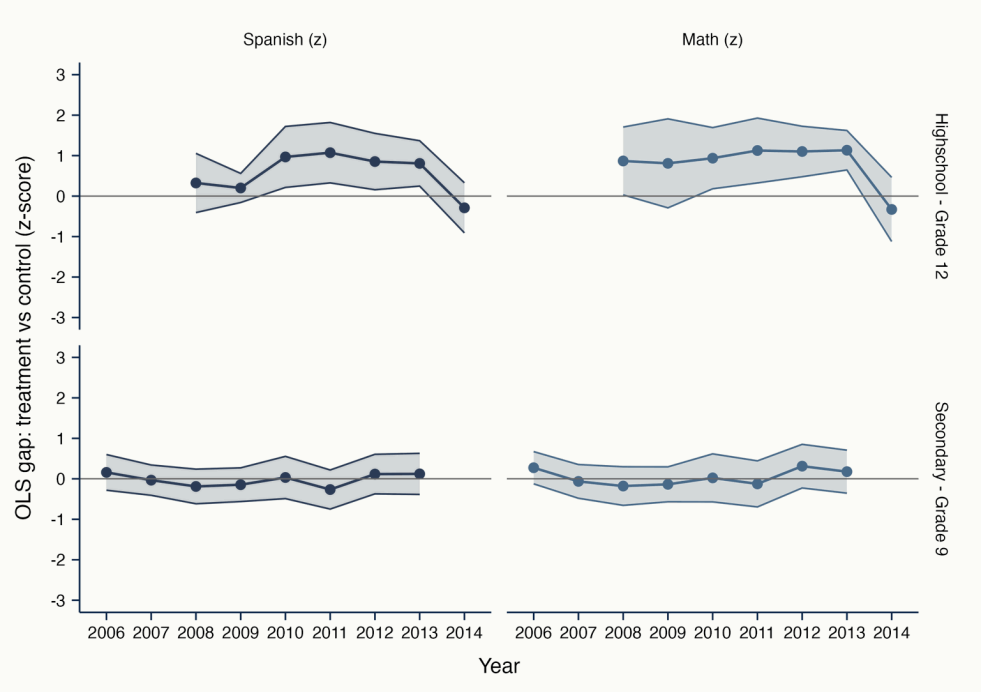
Table B.11: BD estimates on recorded crime (block level, 2017–2024)

	Mean within h											
Outcome	West	East	$\hat{\tau}$	SE	95% CI	p-value	FDR q-value	Poly	h (m)	b (m)	N_h	N_b
Total crime	3.888	2.208	−2.422	4.194	[−10.643, 5.799]	0.564	1.000	1	1,914	2,601	2,418	3,824
Homicide	0.031	0.034	0.033	0.031	[−0.028, 0.093]	0.290	1.000	1	1,530	2,357	1,764	3,336
Vehicle theft	0.683	0.447	0.237	0.292	[−0.336, 0.809]	0.417	1.000	1	919	1,568	915	1,808
Robbery from persons	1.312	0.601	−1.500	3.232	[−7.835, 4.835]	0.643	1.000	1	1,960	2,626	2,518	3,869
Burglary and business robbery	0.934	0.474	−0.556	0.494	[−1.524, 0.411]	0.260	1.000	1	1,460	2,282	1,664	3,189
Intentional injuries	0.235	0.165	−0.166	0.236	[−0.627, 0.296]	0.482	1.000	1	1,899	2,586	2,381	3,792
Violence against women	0.329	0.265	−0.020	0.192	[−0.397, 0.357]	0.916	1.000	1	1,322	2,267	1,461	3,145
Other	0.473	0.221	−0.054	0.245	[−0.535, 0.427]	0.825	1.000	1	1,748	2,503	2,096	3,639

Notes: BD estimates of the boundary effect on recorded crime, from the investigation files of the Jalisco Attorney General's Office for the municipality of Guadalajara, 2017–2024, as georeferenced by the IIEG (187,389 incidents). Each incident is assigned to the nearest census block within 50 meters, since incidents are geocoded on the street, and 187,182 incidents are assigned. Outcomes are annual incidents per hectare of block area, averaged over the eight years. They are in levels, since with many blocks at zero a discontinuity in $\log(1+x)$ depends on the units of x (Chen and Roth, 2024). Each block is weighted by its area. Homicide is intentional homicide and feminicide, which accounts for 14 of the 440 homicides within 2 km of the boundary. Vehicle theft is theft of private cars and motorcycles, robbery from persons includes robbery of bank customers, and burglary and business robbery joins residential burglary and robbery of businesses. Intentional injuries are *lesiones dolosas*, and violence against women is domestic violence, rape, and child sexual abuse. "Other" is auto-parts theft, theft from the interior of vehicles, and the offences with fewer than 200 incidents within 2 km of the boundary, robbery of heavy cargo (123) and bank robbery (12). Specifications use the blocks within 3.5 km of the boundary and include boundary-segment fixed effects and geographic coordinates. The West and East columns are area-weighted means of the outcome within h of the boundary. $\hat{\tau}$ is the bias-corrected estimate. h and b are the MSE-optimal main and bias-correction bandwidths, and N_h and N_b are the numbers of blocks within them. q-values are sharpened false-discovery-rate values across the eight outcomes (Anderson, 2008). Standard errors, confidence intervals, and p-values are robust bias-corrected (Calonico et al., 2014).

↩ Back to text

Figure B.9: Municipality-wide OLS estimates of standardized test performance (ENLACE)



Notes: East-minus-west gaps in mean ENLACE scores in Spanish and Math at the last grade of secondary and high school across Guadalajara (2006–2014). Specifications include school-location controls, a public/private indicator, and year fixed effects. Confidence bands are 95%. These scores begin in 2006, after the years in which Figure B.7 shows detectable differences. Grade-9 gaps show no detectable difference, and grade-12 estimates are positive and away from zero from 2010 to 2013. These are municipality-wide comparisons by school location.

↩ Back to text

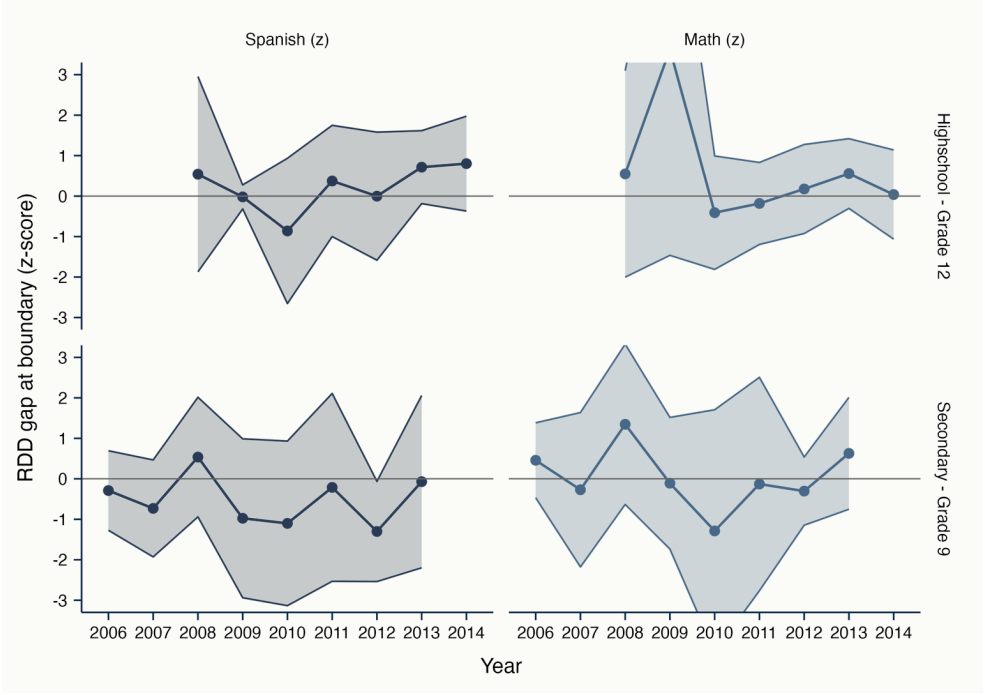
B.12 Street infrastructure, urban form, and businesses

Table B.12 reports BD estimates for eleven features of the street that the 2020 National Housing Inventory records on each block front. Paved streets, sidewalks, wheelchair ramps, and mobile street stalls are more frequent on the eastern side, and no feature is less frequent there. Table B.13 reports block-level measures of urban form and the area of parcels. Residents and occupied dwellings per block and parcel area show no detectable difference at the boundary, and east-side blocks are larger, with more parcels, more persons per occupied dwelling, and fewer residents per hectare, each with a sharpened q-value of 0.098. Tables B.14 and B.15 use the establishments of the December 2019 business directory. Establishments east of the boundary are smaller, with fewer establishments of more than thirty employees and fewer establishments per resident, and the sector shares show no detectable difference after the false-discovery-rate adjustment.

B.13 Residential mobility and selective in-migration

According to the 2020 census, 91.5% of Guadalajara residents lived in Jalisco five years earlier, and re-estimating the BD design excluding inter-state migrants leaves the results essentially unchanged (last row of Panel A, Table B.1). To approximate within-city mobility, I match adults aged 34+ in 2020 to plausibly the same individuals in the same block in 2010, using nearest-

Figure B.10: BD estimates of standardized test performance (ENLACE)



Notes: BD estimates of the boundary effect on mean ENLACE scores in Spanish and Math at secondary and high school within 1.5 km of the boundary (2006–2014), following Calonico et al. (2014). Confidence bands are 95%. These scores begin in 2006, after the years in which Figure B.8 shows detectable differences. The estimates show no detectable discontinuity in most years, with bands that admit gaps larger than one standard deviation in either direction, and grade-9 Spanish estimates are mostly negative.

↩ Back to text

Table B.12: BD estimates on street infrastructure (block fronts, 2020)

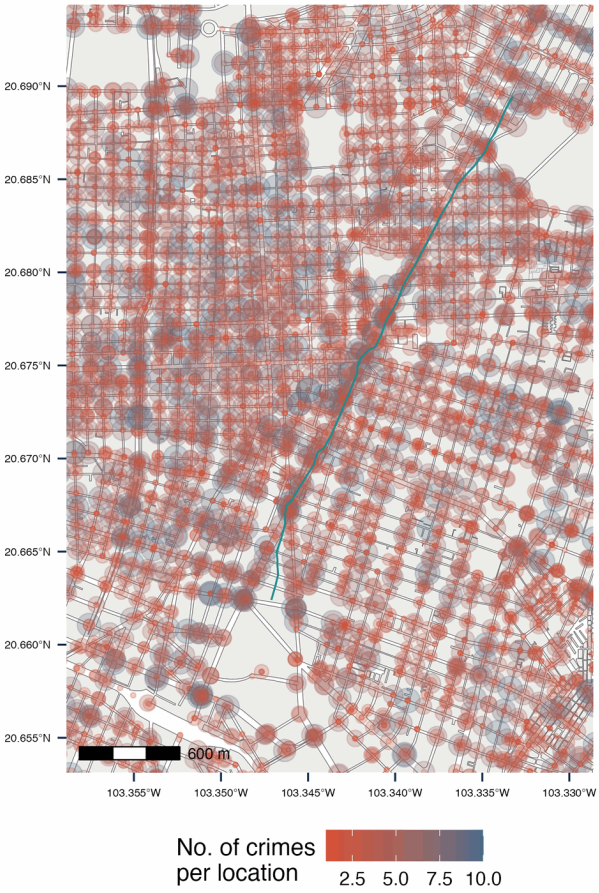
	Mean within h											
Outcome	West	East	$\hat{\tau}$	SE	95% CI	p-value	FDR q-value	Poly	h (m)	b (m)	N_h	N_b
Paved street (asphalt or concrete)	0.827	0.950	0.474	0.147	[0.166, 0.743]	0.002	0.012	1	468	872	1,713	3,619
Sidewalk	0.905	0.917	0.210	0.075	[0.074, 0.368]	0.003	0.012	1	816	1,261	3,339	5,878
Wheelchair ramp	0.588	0.784	0.361	0.110	[0.141, 0.572]	0.001	0.012	1	584	871	2,174	3,616
Storm drain	0.230	0.306	0.205	0.109	[−0.014, 0.412]	0.067	0.104	1	702	1,030	2,759	4,516
Street lighting	0.552	0.557	0.138	0.096	[−0.028, 0.350]	0.095	0.125	1	760	1,167	3,037	5,337
Trees	0.716	0.658	0.029	0.108	[−0.177, 0.246]	0.749	0.599	1	893	1,321	3,730	6,245
Pedestrian crossing	0.841	0.845	−0.023	0.078	[−0.181, 0.124]	0.714	0.599	1	421	666	1,525	2,575
Pedestrian traffic light	0.115	0.063	0.023	0.162	[−0.312, 0.322]	0.976	0.602	1	660	977	2,552	4,201
Public transport stop	0.071	0.087	−0.082	0.111	[−0.309, 0.127]	0.412	0.347	1	919	1,342	3,869	6,372
Street stall (semi-fixed)	0.063	0.062	0.123	0.146	[−0.114, 0.457]	0.239	0.212	1	726	1,245	2,881	5,777
Street stall (mobile)	0.061	0.078	0.279	0.129	[0.074, 0.580]	0.011	0.024	1	705	1,231	2,777	5,699

Notes: BD estimates of the boundary effect on street infrastructure, using the block fronts of the 2020 National Housing Inventory (*Inventario Nacional de Viviendas*) within 3 km of the former creek. Each outcome equals one if the inventory records the feature on the front and is missing when it does not apply or is not specified; paved streets are asphalt or concrete. $\hat{\tau}$, its standard error, and its confidence interval are in standard deviations of the outcome. Each row is a separate estimate, east minus west, from a local linear regression with an MSE-optimal bandwidth h, separate linear fits on each side, and longitude, latitude, and boundary-segment fixed effects as covariates. The side and the boundary segment are those of the front's census block. The West and East columns are raw shares within h of the boundary. $\hat{\tau}$ is the conventional estimate. b is the bias-correction bandwidth, and N_h and N_b are the numbers of fronts within h and b. q-values are sharpened false-discovery-rate values across the eleven outcomes (Anderson, 2008). Standard errors, confidence intervals, and p-values are robust bias-corrected (Calonico et al., 2014).

↩ Back to text

neighbor distance on age, sex, years of schooling, and birthplace, and classify an individual as a long-term resident if the match distance falls below the median. Table B.16 reports the boundary effect on that indicator, which is positive and significant once the clean controls are added. Two

Figure B.11: Spatial distribution of reported crimes in central Guadalajara, 2017–2024



Notes: Each circle represents the count of reported crimes (*carpetas de investigación*) at a block-level coordinate, darker shades indicating higher incidence. Data include all crime categories reported to the Jalisco Attorney General's Office (2017–2024), georeferenced by IIEG. The thick green line marks the boundary. Recorded crime is densely clustered on both sides of the boundary. The map shows counts, and Table B.11 normalizes them by block area.

↩ Back to text

features of the construction limit what the indicator can carry. The census records are anonymous, so the match links a 2020 resident to a demographically similar 2010 resident of the same block rather than to the same person, and a block whose households turned over completely would still return close matches if the arrivals resembled the departures. The nearest-neighbour distance also falls mechanically with the number of 2010 records available in the block, and residential density at the boundary is higher on the eastern side (Figure B.1), so eastern blocks offer larger candidate pools and shorter distances for that reason alone. That bias runs in the same direction as the estimate. I therefore do not read the coefficient as a measured difference in residential turnover, and the argument below rests on the migrant-exclusion row rather than on it. What the two together weigh against is one specific account of the gap, in which the eastern side looks worse because it has absorbed recent and less-educated arrivals. Dropping inter-state migrants moves the estimates hardly at all, and the mobility evidence does not show the excluded side to be the more transient one. The framework does not derive that pattern either. Residential turnover enters it as an exogenous parameter whose effect on local composition the framework traces, and not the

Table B.13: BD estimates on urban form (blocks and parcels)

Outcome	Mean within h		$\hat{\tau}$	SE	95% CI	p-value	FDR q-value	Poly	h (m)	b (m)	N_h	N_b
	West	East										
Panel A. Blocks (2020 Census, 2017 cadastre)												
Residents per block	75.3	92.1	2.273	8.911	[−17.713, 17.218]	0.978	0.432	1	969	1,920	735	1,978
Occupied dwellings per block	23.2	27.2	−0.740	2.966	[−7.364, 4.262]	0.601	0.430	1	922	1,794	682	1,731
Parcels per block	19.1	25.0	5.555	1.842	[1.211, 8.431]	0.009	0.098	1	1,020	1,965	799	2,065
Log block area (ha)	−0.726	−0.407	0.322	0.132	[0.018, 0.537]	0.036	0.098	1	1,173	1,932	969	1,991
Residents per hectare	193	155	−66.104	40.747	[−161.446, −1.720]	0.045	0.098	1	1,445	2,682	1,277	3,358
Occupied dwellings per hectare	66.3	48.0	−12.878	13.145	[−42.207, 9.322]	0.211	0.178	1	1,620	2,835	1,487	3,653
Parcels per hectare	36.6	34.8	5.115	2.868	[1.007, 12.250]	0.021	0.098	1	1,128	2,524	917	3,081
Persons per occupied dwelling	2.983	3.303	0.205	0.105	[0.012, 0.423]	0.038	0.098	1	1,811	2,710	1,774	3,413
Log median parcel area (m^2)	5.170	5.126	0.010	0.095	[−0.208, 0.164]	0.818	0.432	1	1,483	2,333	1,272	2,555
Share of parcels in condominium	0.009	0.007	−0.015	0.015	[−0.048, 0.009]	0.179	0.176	1	2,001	3,101	2,035	3,929
Panel B. Parcels (2017 cadastre; SEs clustered by block)												
Log parcel area (m^2), parcel level	5.165	5.130	0.007	0.061	[−0.124, 0.113]	0.927	—	1	1,456	2,475	29,102	73,007

Notes: BD estimates of the boundary effect on urban form. Panel A uses the 12,077 populated census blocks of Guadalajara, with the 2020 Census and the 2017 cadastre, whose 289,935 parcels are assigned to blocks by their centroid. Panel B uses parcels. Each row is a separate estimate, east minus west, from a local linear regression with an MSE-optimal bandwidth h, separate linear fits on each side, and longitude, latitude, and boundary-segment fixed effects as covariates. The West and East columns are means within h of the boundary. $\hat{\tau}$ is the conventional estimate. b is the bias-correction bandwidth, and N_h and N_b are the numbers of observations within h and b. q-values are sharpened false-discovery-rate values across the ten block-level outcomes of Panel A (Anderson, 2008); the parcel-level estimate of Panel B is not part of that family. Standard errors, confidence intervals, and p-values are robust bias-corrected (Calonico et al., 2014). In Panel B, standard errors are clustered by block.

↩ Back to text

Table B.14: BD estimates on business structure (establishments, 2019)

Outcome	Mean within h		$\hat{\tau}$	SE	95% CI	p-value	FDR q-value	Poly	h (m)	b (m)	N_h	N_b
	West	East										
Panel A. Establishment characteristics (establishment level)												
0-5 employees	0.788	0.844	0.053	0.028	[0.004, 0.113]	0.035	0.132	1	1,434	2,485	16,083	29,410
6-10 employees	0.114	0.087	−0.022	0.014	[−0.051, 0.004]	0.098	0.172	1	1,746	2,778	19,916	34,358
11-30 employees	0.066	0.050	−0.007	0.010	[−0.028, 0.011]	0.408	0.374	1	1,423	2,303	15,985	26,811
31+ employees	0.032	0.020	−0.024	0.010	[−0.045, −0.008]	0.006	0.072	1	1,232	2,190	14,077	25,404
Log employees (bracket midpoint)	1.444	1.344	−0.114	0.054	[−0.232, −0.022]	0.018	0.108	1	1,368	2,421	15,319	28,413
Registered legal name (formality proxy)	0.293	0.203	−0.040	0.030	[−0.090, 0.026]	0.282	0.320	1	2,075	2,983	23,932	38,297
Lists a phone	0.404	0.402	0.033	0.049	[−0.062, 0.129]	0.490	0.417	1	1,426	2,065	16,008	23,804
Lists an e-mail	0.256	0.179	0.048	0.035	[−0.003, 0.133]	0.060	0.158	1	897	1,928	10,032	22,067
Lists a website	0.122	0.068	−0.038	0.019	[−0.072, 0.003]	0.068	0.158	1	1,834	2,699	20,913	33,012
Inside market / plaza / commercial building	0.091	0.091	0.050	0.146	[−0.216, 0.358]	0.629	0.523	1	2,126	2,900	24,565	36,675
Semi-fixed unit	0.017	0.015	0.011	0.012	[−0.012, 0.035]	0.336	0.337	1	1,796	2,356	20,463	27,565
Neighborhood-serving activity	0.334	0.307	0.000	0.044	[−0.075, 0.097]	0.805	0.675	1	1,593	2,705	17,809	33,077
Panel B. Density (block level)												
Block has any establishment	0.942	0.901	0.062	0.037	[−0.005, 0.141]	0.066	0.071	1	1,071	2,094	851	2,289
Log(1 + establishments in block)	2.089	2.050	−0.175	0.166	[−0.530, 0.119]	0.214	0.111	1	1,510	2,318	1,356	2,703
Establishments per 100 residents (populated blocks)	37.8	25.0	−47.301	23.581	[−101.868, −9.432]	0.018	0.059	1	1,745	2,687	1,677	3,364

Notes: BD estimates of the boundary effect on business structure, using the 89,893 establishments of the December 2019 *Directorio Estadístico Nacional de Unidades Económicas* (DENUE), each placed in the nearest 2020 census block within 50 meters. Panel A has one observation per establishment and Panel B one per block. The running variable is the signed distance of the block to the boundary. Each row is a separate estimate, east minus west, from a local linear regression with an MSE-optimal bandwidth h, separate linear fits on each side, and longitude, latitude, and boundary-segment fixed effects as covariates. The West and East columns are means within h of the boundary. $\hat{\tau}$ is the conventional estimate. b is the bias-correction bandwidth, and N_h and N_b are the numbers of observations within h and b. q-values are sharpened false-discovery-rate values within each panel (Anderson, 2008). Standard errors, confidence intervals, and p-values are robust bias-corrected (Calonico et al., 2014). Standard errors are clustered by block.

↩ Back to text

reverse. It does not rule out selective migration more generally. What matters for the schooling gap is the educational composition of those who arrive and those who leave, not the amount of turnover, so a small but strongly selected flow could still open a compositional difference. The census records nothing about former residents, so the selection of leavers is unobserved here and this exercise bounds neither margin. The measure also refers to one period. It compares residents in 2020 with residents of the same block in 2010, so it describes turnover over the last decade. Guadalajara's population was reshaped earlier, by the rural-to-urban migration of the twentieth

Table B.15: BD estimates on the sector mix of establishments (2019)

	Mean within h											
Outcome	West	East	$\hat{\tau}$	SE	95% CI	p-value	FDR q-value	Poly	h (m)	b (m)	N_h	N_b
SCIAN 2-digit sector shares (establishment level)												
Manufacturing	0.097	0.076	−0.018	0.024	[−0.057, 0.036]	0.655	0.859	1	1,556	2,583	17,441	31,014
Wholesale trade	0.060	0.062	0.022	0.023	[−0.018, 0.071]	0.237	0.668	1	1,070	1,769	12,266	20,108
Retail trade	0.340	0.430	0.100	0.049	[−0.003, 0.189]	0.057	0.667	1	2,440	3,546	28,629	47,272
Information	0.003	0.001	0.001	0.002	[−0.002, 0.004]	0.545	0.859	1	1,632	2,508	18,265	29,783
Finance and insurance	0.017	0.010	−0.013	0.012	[−0.041, 0.005]	0.129	0.667	1	2,276	3,261	26,483	42,724
Real estate and rental	0.012	0.008	−0.003	0.005	[−0.011, 0.008]	0.732	0.859	1	2,305	3,180	26,811	41,069
Professional services	0.041	0.010	−0.009	0.010	[−0.024, 0.015]	0.644	0.859	1	1,245	2,096	14,141	24,279
Business support services	0.019	0.011	0.010	0.006	[0.000, 0.023]	0.048	0.667	1	1,687	3,885	19,077	53,387
Education	0.025	0.016	−0.007	0.006	[−0.019, 0.003]	0.143	0.667	1	1,500	2,669	16,776	32,512
Health and social assistance	0.037	0.026	−0.002	0.007	[−0.016, 0.011]	0.719	0.859	1	1,087	1,953	12,469	22,359
Recreation	0.010	0.009	−0.001	0.003	[−0.008, 0.005]	0.665	0.859	1	2,554	3,718	30,482	50,420
Food and lodging	0.138	0.104	−0.006	0.028	[−0.057, 0.055]	0.967	0.936	1	1,741	2,795	19,801	34,718
Other services (repair, personal)	0.131	0.151	−0.028	0.026	[−0.085, 0.018]	0.198	0.667	1	1,382	1,976	15,493	22,698
Government	0.015	0.004	−0.013	0.009	[−0.033, 0.002]	0.092	0.667	1	2,244	3,075	26,040	39,623

Notes: BD estimates of the boundary effect on the share of establishments in each two-digit SCIAN sector, with the sample and the specification of Table B.14. There is one observation per establishment. The West and East columns are sector shares within h of the boundary. $\hat{\tau}$ is the conventional estimate. b is the bias-correction bandwidth, and N_h and N_b are the numbers of establishments within h and b. q-values are sharpened false-discovery-rate values across the fourteen sectors (Anderson, 2008). Standard errors, confidence intervals, and p-values are robust bias-corrected (Calonico et al., 2014). Standard errors are clustered by block.

↩ Back to text

century, and that episode is the kind of sustained inflow of entrants that, in the framework, can lower the rest point of participation (Appendix E.3). Neither this measure nor the 1821 migrant shares of Appendix A.9 observes it, and the two sit on either side of it.

Table B.16: BD estimates on probability of long-term residence (individual level, 2020 Census)

	Mean within h												
Outcome	West	East	$\hat{\tau}$	SE	95% CI	p-value	FDR q-value	Poly	h (m)	b (m)	N_h	N_b	Clean controls
Probability to stay in place	—	—	0.144	0.075	[−0.003, 0.291]	0.054	—	1	132	255	1,315	3,672	No
Probability to stay in place	—	—	0.178	0.080	[0.022, 0.334]	0.026	—	1	116	209	1,121	2,605	✓

Notes: BD estimates of the boundary effect on the probability of long-term residence, individual-level 2020 census data for Guadalajara produced in the INEGI microdata laboratory, restricted to adults aged 34+. The outcome is constructed via nearest-neighbor matching of each 2020 resident to a 2010 resident of the same block on age, sex, years of schooling, and birthplace; an individual is a long-term resident if the match distance is below the median. Inference follows Calonico et al. (2014), with a local linear polynomial, a uniform kernel, and the NN variance estimator: $\hat{\tau}$ is the bias-corrected estimate, and SE, the 95% CI, and the p-value are robust bias-corrected; h and b are the MSE-optimal main and bias-correction bandwidths in meters, and N_h and N_b the effective sample sizes. *Clean controls* add sex, age, longitude, latitude, and boundary-segment fixed effects. Once the controls are added, the effect is positive and significant, and residents east of the boundary are about 18 percentage points more likely to be long-term residents. Because the census does not follow former residents and the indicator is not a measured turnover rate, this does not rule out selective residential sorting or selection among unobserved leavers. The laboratory export does not include the side means within h, so the West and East cells are marked with a dash. The two rows are specifications of the same outcome, so no false-discovery-rate q-values are computed.

↩ Back to text

B.14 Donut estimates

B.15 Placebo boundaries and recentered specification

I randomly shift and rotate the boundary 1,000 times within the municipality and re-estimate the BD design at each placebo location. Three examples are shown in Figure B.12; each preserves the length and roughly the orientation of the original but is otherwise drawn at random. The draws are generated as follows. Each placebo takes the original boundary and applies a rigid translation

Table B.17: Donut BD estimates on human capital, 2020 Census

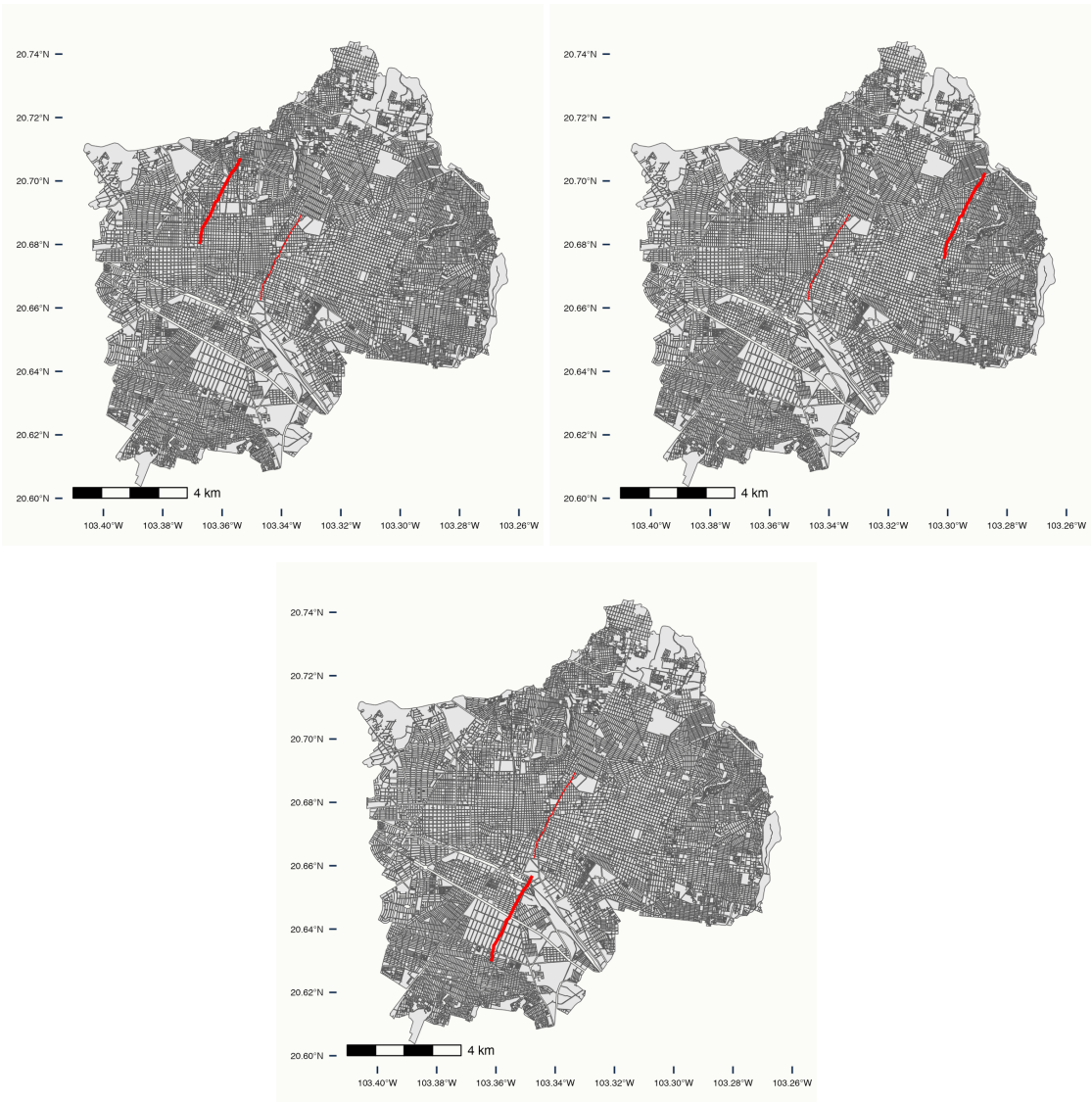
Donut radius (m)	Mean within h West	Mean within h East	$\hat{\tau}$	SE	95% CI	p-value	FDR q-value	Poly	h (m)	b (m)	N_h	N_b
Panel A. Years of schooling (ages 6+)												
0	—	—	−0.165	0.033	[−0.229, −0.101]	< 0.001	—	1	571	1,489	25,011	107,104
50	—	—	−0.101	0.026	[−0.151, −0.050]	< 0.001	—	1	976	1,702	57,387	135,906
100	—	—	−0.211	0.031	[−0.272, −0.149]	< 0.001	—	1	667	2,146	32,464	202,382
150	—	—	−0.151	0.031	[−0.212, −0.091]	< 0.001	—	1	917	1,763	49,857	141,296
Panel B. School attendance (ages 6–22)												
0	—	—	−0.075	0.045	[−0.163, 0.013]	0.094	—	1	1,200	2,059	18,265	45,016
50	—	—	−0.086	0.047	[−0.178, 0.007]	0.069	—	1	1,023	2,294	14,606	55,266
100	—	—	−0.100	0.048	[−0.194, −0.006]	0.037	—	1	1,098	2,266	15,686	53,590
150	—	—	−0.127	0.056	[−0.237, −0.016]	0.025	—	1	955	2,413	12,429	59,950

Notes: BD estimates excluding all residents within the stated distance of the historical boundary, in standardized units, following Calonico et al. (2014) with a uniform kernel, local linear polynomial and the NN variance estimator. Data are the individual-level 2020 *Censo de Población y Vivienda* for Guadalajara, produced in the INEGI microdata laboratory. Every row, including the zero-donut row, is estimated without covariates, since covariate coverage is thinnest in the blocks nearest the boundary; the zero-donut rows are the no-covariate specifications of Table B.1. $\hat{\tau}$ is the bias-corrected estimate, and SE, the 95% CI, and the p-value are robust bias-corrected; h and b are the MSE-optimal main and bias-correction bandwidths in meters, and N_h and N_b the effective sample sizes. The bandwidth is re-selected for each radius and moves substantially across rows, so the variation in the estimates reflects both the exclusion and the width of the estimation window. The table is informative about the stability of the sign rather than a controlled comparison isolating the innermost observations. Residential density at the cutoff is far lower on the western side, which is Guadalajara's historic commercial core, so the innermost observations come disproportionately from one side. The two densities converge only at about 500 meters (Figure B.1), so all three radii fall inside the range over which they differ. The donut removes the observations closest to the line without reaching the distance at which the two sides are comparably populated. The laboratory exports do not include the side means within h, so the West and East cells are marked with a dash. The rows vary the excluded radius for the same two outcomes, so no false-discovery-rate q-values are computed.

↩ Back to text

and a rotation about its centroid, leaving its length and shape unchanged. The longitude shift is drawn uniformly from $\pm[0.01, 0.05]$ degrees and the latitude shift from $\pm[0.01, 0.035]$ degrees, with the two signs equally likely and the interval around zero excluded, so that no placebo lies within roughly one kilometre of the true line; at this latitude the shifts span 1.0 to 5.2 km east or west and 1.1 to 3.9 km north or south. The rotation is drawn uniformly from $\{0°, 45°, 90°, 135°\}$. A draw is admissible if the shifted boundary still intersects the municipality of Guadalajara; 984 of the 1,000 draws satisfy that rule and 16 are dropped, all of them at 45° or 90°, and a draw is dropped as well if the estimator fails to return at that location. Sides are assigned by closing the shifted line into a polygon and labelling blocks by whether their centroid falls inside it, with the closing orientation set by the rotation, so that the side coding rotates with the boundary rather than being fixed to east and west. As at the true boundary, the estimator is evaluated at five regularly spaced points along the shifted line and the placebo statistic is the equally weighted average of the five pointwise effects, which keeps the support and the number of evaluation points the same across draws. Recentering follows Borusyak and Hull (2023). For each census block I compute its expected exposure across the admissible draws, meaning the share of placebo boundaries that place it on the treated side and its average signed distance to them, regress block schooling on those two variables and their interaction, and re-estimate the BD design on the residuals. The exposure measures are therefore block-level and are built from the same draws that generate the placebo distribution. Figure B.13 reports the distribution of placebo coefficients: approximately symmetric and centered near zero, with the true-boundary estimate in the left tail. Following

Figure B.12: Examples of placebo boundaries for the spatial falsification test

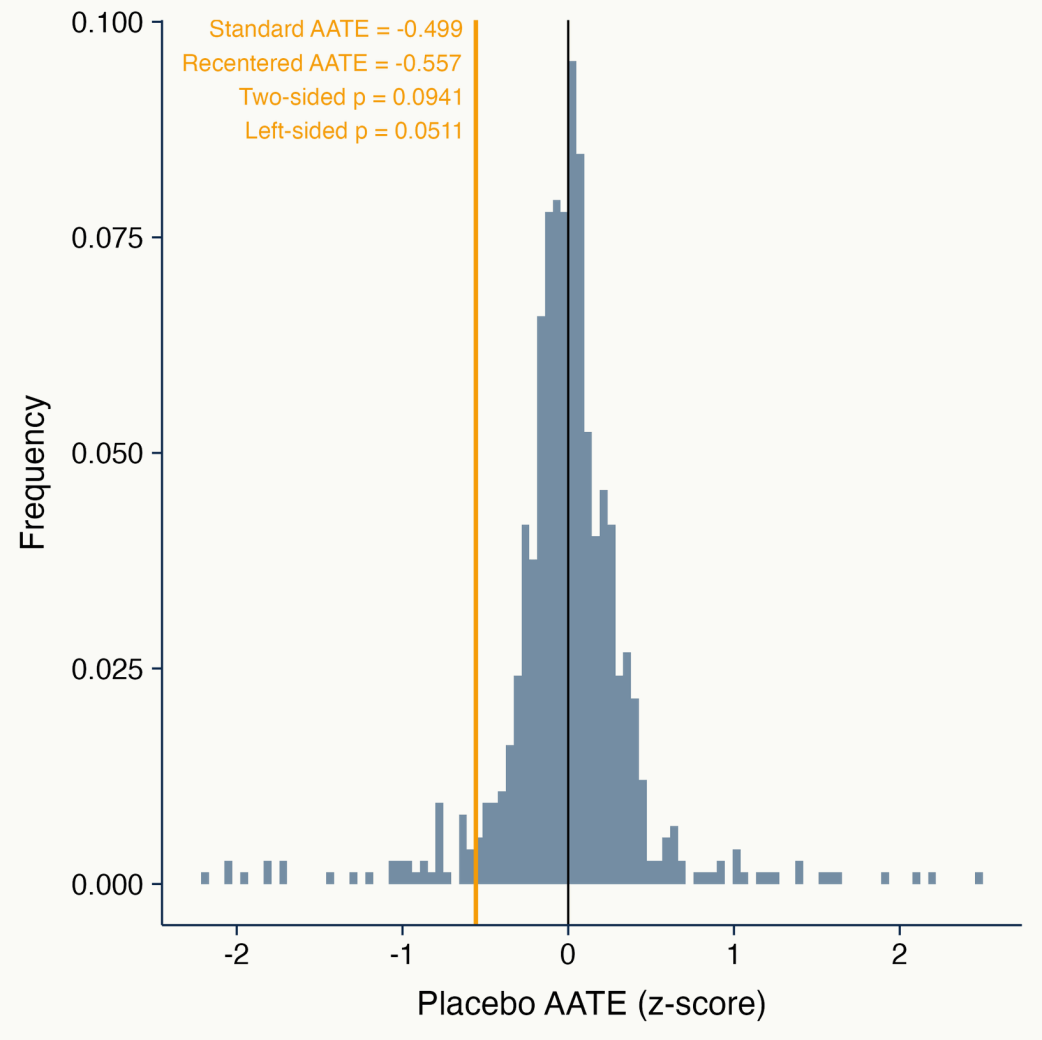


Notes: Three illustrative examples (out of 1,000) of placebo boundaries used in the spatial falsification test of Figure B.13. Each panel shows one placebo realization (thick red line) generated by randomly shifting and rotating the historical SJdD creek boundary (thin red line) within Guadalajara. Placebos preserve the length and approximate orientation of the original but are otherwise random. Each placebo is used to estimate a BD design on census-block mean years of schooling; the resulting distribution is in Figure B.13.

↩ Back to text

Borusyak and Hull (2023), a recentered specification that adjusts each block's outcome for its average exposure to placebo boundaries gives a true-boundary estimate of −0.557 SD — *larger* than the unadjusted −0.499. The two-sided randomization-inference p-value is 0.094 and the left-sided p-value 0.051.

Figure B.13: Distribution of placebo-boundary treatment effects with recentering



Notes: Distribution of standardized BD estimates from 1,000 placebo boundaries (Figure B.12), applied to census-block mean years of schooling in the 2020 census for Guadalajara. The horizontal axis is the estimated effect in standardized units. The black vertical line at zero marks the null. The unadjusted true-boundary estimate is −0.499 SD and lies in the left tail. Following Borusyak and Hull (2023), the orange line marks the recentered estimate (−0.557 SD), which adjusts each block's outcome for its average exposure to the placebo boundaries, removing geographic-exposure bias. The two-sided randomization-inference p-value is 0.094 and the left-sided p-value 0.051. The exercise weighs against boundary-magnitude discontinuities of this size being common at arbitrary locations.

↩ Back to text

C Neighborhood Social Differences: Supporting Material

C.1 Additional Life Quality Survey results

C.1.1 Trust and participation across the boundary

Table 2 reports the comparisons of schooling and interpersonal trust summarized in Section 4.1, including the relation between trust in neighbors and schooling; institutional trust and association membership are reported below. These are descriptive patterns. The survey interviews 289 Guadalajara residents in 23 census tracts, 26 of them within 1.5 kilometers of the boundary, and unconditional means are similar across sides, with 67 percent of western and 64 percent of eastern respondents reporting that they trust their neighbors and an institutional-trust index that averages 0.23 on both sides. The coefficient on the east-side indicator is the discontinuity at the boundary implied by a separate linear distance gradient on each side. All estimates use the survey's expansion factors. Panel A includes no schooling control and Panel B controls linearly for the respondent's years of schooling; the two panels give similar estimates. These estimates depend on the extrapolation.

C.1.2 Institutional trust

Table C.1 reports the institutional-trust index (Col. 1) and disaggregates it into its ten items (Cols. 2–11). The point estimate is negative for every institution in both panels. Controlling for schooling, six of the ten items are significant at the 5 percent level after the multiple-outcome adjustment, and two more, the local congress ($q = 0.056$) and the electoral institute ($q = 0.058$), at the 10 percent level; the exceptions are churches ($q = 0.230$) and the prosecutor's office ($q = 0.249$). The largest gaps are for the Navy (-0.47) and the Army (-0.37), on the 0–1 scale of each item.

C.1.3 Social participation

Table C.2 disaggregates participation. The estimated discontinuities in belonging to any organization (Col. 1) and in the number of memberships (Col. 2) are small and imprecise. Controlling for schooling, the point estimate for neighborhood or residents' associations is 8.0 percentage points (Col. 3), about twice the sample mean of 4.1 percent, but it is imprecisely estimated ($q = 0.200$), and no other type of membership differs (Cols. 4–7).

C.1.4 Subjective well-being

Table C.3 reports the well-being index of Table 2 (Col. 1) and its three items, satisfaction with life, quality of life, and happiness (Cols. 2–4). Each item is a five-point scale rescaled to the unit

Table C.1: Trust in institutions across the boundary

	Institutional trust index	Trust in									
		Churches	Local congress	Electoral institute	Political parties	Government	Judiciary	Police	Army	Navy	Prosecutor's office
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Panel A. Without schooling control											
$\mathbb{1}\{East_i\}$	−0.235***	−0.111	−0.182**	−0.234**	−0.178***	−0.259***	−0.195**	−0.259***	−0.380***	−0.480***	−0.111
	(0.064)	(0.116)	(0.081)	(0.103)	(0.063)	(0.075)	(0.077)	(0.088)	(0.111)	(0.112)	(0.085)
Age-by-sex FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Distance to boundary	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Schooling control											
Survey weights	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
FDR q-value	0.004	0.328	0.049	0.049	0.018	0.005	0.035	0.017	0.005	0.001	0.221
Observations	289	289	289	289	289	289	289	289	289	289	289
Dep. var. mean	0.229	0.472	0.164	0.265	0.142	0.178	0.199	0.203	0.349	0.373	0.204
R^2	0.093	0.105	0.048	0.061	0.059	0.062	0.070	0.050	0.122	0.130	0.055
Panel B. Controlling for years of schooling											
$\mathbb{1}\{East_i\}$	−0.232***	−0.147	−0.182**	−0.225**	−0.178***	−0.258***	−0.187**	−0.252***	−0.370***	−0.466***	−0.105
	(0.065)	(0.114)	(0.083)	(0.105)	(0.062)	(0.075)	(0.077)	(0.089)	(0.112)	(0.113)	(0.087)
Age-by-sex FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Distance to boundary	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Schooling control	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Survey weights	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
FDR q-value	0.006	0.230	0.056	0.058	0.019	0.006	0.041	0.019	0.007	0.002	0.249
Observations	289	289	289	289	289	289	289	289	289	289	289
Dep. var. mean	0.229	0.472	0.164	0.265	0.142	0.178	0.199	0.203	0.349	0.373	0.204
R^2	0.094	0.139	0.048	0.064	0.059	0.062	0.073	0.053	0.125	0.136	0.057

Notes: OLS estimates from the 2022 wave of the local Life Quality Survey (*Jalisco Cómo Vamos*) for the municipality of Guadalajara. Each column is a separate 0–1 institutional-confidence outcome, and column (1) is the institutional-trust index (first principal component of the ten items). The coefficient on the east-side indicator is the discontinuity at the boundary implied by a separate linear distance gradient on each side; unconditional means are similar across sides. All specifications include age-by-sex fixed effects and side-specific linear distance gradients, and are weighted by the survey's expansion factors. Panel A includes no schooling control, and Panel B controls linearly for the respondent's years of schooling. Sharpened FDR q-values (Anderson, 2008) adjust across all 26 social-capital outcomes within each panel. Dep. var. mean is the survey-weighted mean of the outcome. Heteroskedasticity-robust standard errors are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

↩ Back to text

interval, and the index is their mean. The point estimates are positive and small in both panels, between 0.003 and 0.096, and none is distinguishable from zero.

C.2 Survey modules: additional detail

Types of help from neighbors. The help index in Panel B of Table C.4 sums three items asking how often, over the previous twelve months, the respondent received each type of help from neighbors, with frequency bins converted to counts. The most common type combines childcare, support during an illness, and household emergencies: 31% of east-side respondents and 20% of west-side respondents report receiving it at least once, and the difference in the number of times is imprecise. Loans of money or goods are reported by 13% and 11%. Help finding work shows the sharpest difference, reported by 17% of east-side respondents against 2% on the west side. These are unadjusted comparisons, and the first item does not separate childcare from the other two forms of help it groups.

Table C.2: Organizational membership across the boundary

	Member of any organization	Number of memberships	Member of				
			Neighborhood assoc.	School-related org.	Political org.	Religious group	NGO or assistance org.
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A. Without schooling control							
$\mathbb{1}\{East_i\}$	−0.086	−0.088	0.066	−0.039	0.023	−0.076	−0.062
	(0.100)	(0.188)	(0.056)	(0.050)	(0.041)	(0.085)	(0.063)
Age-by-sex FE	✓	✓	✓	✓	✓	✓	✓
Distance to boundary	✓	✓	✓	✓	✓	✓	✓
Schooling control							
Survey weights	✓	✓	✓	✓	✓	✓	✓
FDR q-value	0.329	0.443	0.268	0.347	0.399	0.329	0.328
Observations	289	289	289	289	289	289	289
Dep. var. mean	0.114	0.178	0.041	0.018	0.036	0.065	0.018
R^2	0.068	0.046	0.033	0.034	0.029	0.068	0.050
Panel B. Controlling for years of schooling							
$\mathbb{1}\{East_i\}$	−0.058	−0.041	0.080	−0.032	0.032	−0.064	−0.057
	(0.101)	(0.188)	(0.057)	(0.050)	(0.044)	(0.087)	(0.063)
Age-by-sex FE	✓	✓	✓	✓	✓	✓	✓
Distance to boundary	✓	✓	✓	✓	✓	✓	✓
Schooling control	✓	✓	✓	✓	✓	✓	✓
Survey weights	✓	✓	✓	✓	✓	✓	✓
FDR q-value	0.404	0.480	0.200	0.404	0.404	0.404	0.361
Observations	289	289	289	289	289	289	289
Dep. var. mean	0.114	0.178	0.041	0.018	0.036	0.065	0.018
R^2	0.098	0.073	0.054	0.046	0.037	0.079	0.054

Notes: OLS estimates from the 2022 wave of the local Life Quality Survey (*Jalisco Cómo Vamos*) for the municipality of Guadalajara. Columns (1) and (2) report any membership and the count of memberships, and columns (3)–(7) report membership in specific organization types. The coefficient on the east-side indicator is the discontinuity at the boundary implied by a separate linear distance gradient on each side; unconditional means are similar across sides. All specifications include age-by-sex fixed effects and side-specific linear distance gradients, and are weighted by the survey's expansion factors. Panel A includes no schooling control, and Panel B controls linearly for the respondent's years of schooling. Sharpened FDR q-values (Anderson, 2008) adjust across all 26 social-capital outcomes within each panel. Dep. var. mean is the survey-weighted mean of the outcome. Heteroskedasticity-robust standard errors are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

↩ Back to text

Returns and the reservation salary. Parents on both sides expect returns well above the realized premium: the median respondent expects a university graduate to earn 100% more than a high-school graduate at age 30, compared with a 57% premium in 2020 census microdata for full-time Guadalajara workers aged 25–35 (Figure C.1). The reservation-salary question describes a local young person who has completed high school and received a job offer but could instead attend university. The prompt states that university entails tuition, materials, transportation, and forgone earnings, and respondents are asked how much the young person would need to earn per month at age 30 for university to have been worthwhile. The MXN 5,319 difference is roughly 45% of the university wage premium respondents expect.

Table C.3: Subjective well-being across the boundary

	Well-being index	Life satisfaction	Quality of life	Happiness
	(1)	(2)	(3)	(4)
Panel A. Without schooling control				
$\mathbb{1}\{East_i\}$	0.033	0.003	0.024	0.074
	(0.059)	(0.072)	(0.072)	(0.076)
Age-by-sex FE	✓	✓	✓	✓
Distance to boundary	✓	✓	✓	✓
Schooling control				
Survey weights	✓	✓	✓	✓
FDR q-value	1.000	1.000	1.000	1.000
Observations	289	289	289	289
Dep. var. mean	0.810	0.857	0.731	0.843
R^2	0.041	0.052	0.045	0.039
Panel B. Controlling for years of schooling				
$\mathbb{1}\{East_i\}$	0.050	0.018	0.037	0.096
	(0.058)	(0.072)	(0.072)	(0.074)
Age-by-sex FE	✓	✓	✓	✓
Distance to boundary	✓	✓	✓	✓
Schooling control	✓	✓	✓	✓
Survey weights	✓	✓	✓	✓
FDR q-value	1.000	1.000	1.000	1.000
Observations	289	289	289	289
Dep. var. mean	0.810	0.857	0.731	0.843
R^2	0.080	0.071	0.060	0.079

Notes: OLS estimates from the 2022 Life Quality Survey (*Jalisco Cómo Vamos*) for the municipality of Guadalajara. Columns (2)–(4) are self-reported satisfaction with life, quality of life, and happiness, each on a five-point scale rescaled to the unit interval, and column (1) is the mean of the three items. The coefficient on the east-side indicator is the discontinuity at the boundary implied by a separate linear distance gradient on each side; unconditional means are similar across sides. All specifications include age-by-sex fixed effects and side-specific linear distance gradients, and are weighted by the survey's expansion factors. Panel A includes no schooling control, and Panel B controls linearly for the respondent's years of schooling. Sharpened FDR q-values (Anderson, 2008) adjust across the four well-being outcomes of this table within each panel. Dep. var. mean is the survey-weighted mean of the outcome. Heteroskedasticity-robust standard errors are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

↩ Back to text

Table C.4: Neighborhood ties and alternative mechanisms

Summary indices (first principal component, standard deviations)	Neighborhood ties (4 measures)	Aspirations (2 measures)	Returns (4 measures)	Work-over-study norms (2 measures)	Attachment (3 measures)	Patience and risk (2 measures)
	(1)	(2)	(3)	(4)	(5)	(6)
$\mathbb{1}\{East_i\}$	0.44***	0.10	−0.04	−0.03	0.02	0.18
	(0.11)	(0.15)	(0.15)	(0.13)	(0.14)	(0.16)
Question-version controls			✓			
Focal-child controls		✓				
FDR q-value	0.00	1.00	1.00	1.00	1.00	1.00
Observations	238	238	238	238	238	238
Control mean (west side)	−0.33	0.00	0.03	−0.02	−0.06	−0.18
East-side mean	0.08	0.00	−0.01	0.00	0.02	0.04
R^2	0.065	0.236	0.079	0.115	0.083	0.114

Panel A. Localized prosociality (allocation exercises)	Donation to a good cause	Allocation to a *barrio* stranger	Gives strictly more to the *barrio* stranger	Allocation to a city stranger	Relative *barrio* premium
	(7)	(8)	(9)	(10)	(11)
$\mathbb{1}\{East_i\}$	23.0	19.7**	0.170***	14.6	0.088**
	(24.3)	(9.0)	(0.032)	(12.0)	(0.044)
Question-version controls	✓	✓	✓	✓	✓
FDR q-value	0.16	0.06	0.00	0.13	0.07
Observations	238	238	238	238	233
Control mean (west side)	344.6	244.6	0.000	251.1	−0.015
East-side mean	371.8	262.9	0.166	265.6	0.053
R^2	0.074	0.087	0.112	0.131	0.156

Panel B. Localized network benefits and the cost of leaving	Times helped by neighbors, past 12 months	Share of non-market time with neighbors	Share of non-market time supporting children's studies	Vignette: the stayer's family gets more help	Vignette: leaving for university strains relations
	(12)	(13)	(14)	(15)	(16)
$\mathbb{1}\{East_i\}$	1.10**	0.040**	−0.015	0.17	0.16
	(0.44)	(0.018)	(0.044)	(0.23)	(0.25)
Weekly work hours		✓	✓		
FDR q-value	0.07	0.07	0.78	0.63	0.63
Observations	238	234	234	238	238
Control mean (west side)	0.97	0.041	0.333	0.52	0.59
East-side mean	1.96	0.073	0.319	0.60	0.72
R^2	0.055	0.078	0.237	0.056	0.065

Panel C. Aspirations and expectations	Aspires to high school or more	Aspires to university	Expects high school or more	Expects university
	(17)	(18)	(19)	(20)
$\mathbb{1}\{East_i\}$	−0.007	0.027	0.010	0.020
	(0.007)	(0.036)	(0.027)	(0.049)
Focal-child controls	✓	✓	✓	✓
FDR q-value	1.00	1.00	1.00	1.00
Observations	238	238	238	238
Control mean (west side)	1.000	0.935	0.957	0.870
East-side mean	0.990	0.948	0.953	0.850
R^2	0.167	0.163	0.224	0.252

Notes: Survey modules of the incentivized choice experiment, for the 239 households. Each column regresses the outcome in its header on the east-side indicator, controlling for Map-First assignment, respondent sex, age and age squared, and respondent-education fixed effects, and for the row-specific controls marked in each panel. Panel A controls for the randomized order of the allocation questions. In Panel B, non-market time is the sum of weekly hours with neighbors, supporting the children's studies, and housework; the two share columns also control for weekly work hours and exclude respondents who report no non-market time. Panel C also controls for the focal child's sex, age, and current school attendance, as does the aspirations index. Each summary index is the first principal component of its measures over the 239 households, standardized over the 239, so its coefficient is in standard deviations; its q-values are computed over the six indices. The neighborhood-ties index combines the number of times the household received help from neighbors in the past twelve months with child care or care when sick, with loans, and with finding work, coded from the answer bins 1, 2, 3, 4, and 5 as 0, 1.5, 4, 8, and 12 times, and time with neighbors relative to work time. Table D.10 reports a nine-measure version that adds the two allocations, the reservation salary, and the two vignettes. The aspirations index combines aspiring to and expecting university for the focal child; the returns index, expected earnings with high school and with university (logs), the perceived return, and the belief that an excluded-side graduate earns less; the norms index, the personal and the perceived work-over-study norm; the attachment index, belonging, the preference for another colonia, and the effort to have children know the neighbors; the patience and risk index, the two items of Panel H in Table C.5. All columns of this table are estimated by OLS in levels. The rows *Control mean (west side)* and *East-side mean* report unadjusted means for west-side and east-side respondents; the coefficient is the adjusted difference and need not equal their difference. The FDR q-value row reports sharpened false-discovery-rate values (Anderson, 2008), computed separately within each panel. The number of observations varies across columns because some items are not answered. Panels D to F and H continue in Table C.5, and household characteristics of the sample are in Table C.7. All estimates are observational cross-boundary comparisons. HC1 robust standard errors are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

↩ Back to text

Table C.5: Neighborhood ties and alternative mechanisms (continued)

Panel D. Returns to schooling and investment threshold

	Expected earnings, high-school graduate	Expected earnings, university graduate	Perceived return to university (uni. − HS)	Perceives excluded-side graduate earns less	Reservation salary to justify university
	PPML	PPML	OLS	OLS	PPML
	(1)	(2)	(3)	(4)	(5)
$\mathbb{1}\{East_i\}$	−0.045	0.021	1,051	0.091	0.166**
	(0.081)	(0.078)	(1,657)	(0.177)	(0.071)
Question-version controls			✓	✓	✓
FDR q-value	1.00	1.00	1.00	1.00	0.10
Observations	238	238	238	238	238
Control mean (west side)	12,924	24,174	11,250	0.20	31,043
East-side mean	12,690	24,503	11,812	0.23	36,363
R^2	0.110	0.066	0.100	0.087	0.083

Panel E. Perceived neighbor norms (pluralistic ignorance)

	Personal norm: youth should work, not study (1–7)	Perceived neighbors sharing that norm (0–10)	Pluralistic ignorance (perceived − own, 0–10)
	(6)	(7)	(8)
$\mathbb{1}\{East_i\}$	−0.237	0.163	0.558
	(0.311)	(0.355)	(0.636)
FDR q-value	1.00	1.00	1.00
Observations	238	238	238
Control mean (west side)	2.61	5.57	2.88
East-side mean	2.55	5.70	3.11
R^2	0.163	0.044	0.112

Panel F. Stated attachment to the neighborhood

	Feels part of this colonia (1–7)	Would prefer to live in another colonia (1–7)	Makes an effort so children know the neighbors (1–7)
	(9)	(10)	(11)
$\mathbb{1}\{East_i\}$	−0.137	−0.077	0.258
	(0.253)	(0.437)	(0.377)
FDR q-value	1.00	1.00	1.00
Observations	238	238	238
Control mean (west side)	5.76	3.22	4.48
East-side mean	5.72	3.16	4.92
R^2	0.113	0.055	0.073

Panel H. Patience and willingness to take risks

	Patience: gives up something today for more later (1–7)	Willingness to take risks (1–7)
	(12)	(13)
$\mathbb{1}\{East_i\}$	0.581*	−0.010
	(0.324)	(0.351)
FDR q-value	0.18	0.96
Observations	238	238
Control mean (west side)	4.91	4.74
East-side mean	5.53	4.86
R^2	0.077	0.111

Notes: Continuation of Table C.4, with the same households, specification, and rows. In Panel D the question-version control is the randomized version of the reservation-salary question. In Panel E the personal norm is elicited on a 1–7 agreement scale and the perceived neighbor norm as a count out of ten; the personal norm is linearly rescaled to 0–10 before the difference is taken, so both enter the pluralistic-ignorance column on a common metric. Because the two questions use different response formats, the level of that difference depends on the mapping between them, and only its comparison across sides is interpretable. Panel H reports the two preference items, each on a 1–7 agreement scale. Columns marked PPML are estimated by Poisson pseudo-maximum-likelihood, so their coefficients are semi-elasticities while the means are in MXN, and their R^2 is the pseudo-R^2; all other columns are estimated by OLS in levels. The FDR q-value row is computed separately within each panel. All estimates are observational cross-boundary comparisons. HC1 robust standard errors are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

↩ Back to text

Table C.6: The east-side reservation salary, conditioning on neighborhood ties and on private-cost measures

	Reservation salary for university (MXN per month at age 30)					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. OLS, MXN						
East side	5691**	4737**	5155**	5951**	5546**	4987*
	(2346)	(2308)	(2324)	(2684)	(2391)	(2608)
Neighborhood-ties index		2163				2107
		(1358)				(1336)
Help-received index			1524			
			(1348)			
p-value, east side	0.016	0.041	0.028	0.028	0.021	0.057
Share of the east-side coefficient absorbed		0.17	0.09	−0.05	0.03	0.12
95% bootstrap interval		[−0.01, 0.61]	[−0.08, 0.44]	[−0.57, 0.52]	[−0.13, 0.37]	[−0.34, 0.86]
Observations	238	238	238	238	238	238
R^2	0.074	0.087	0.080	0.090	0.084	0.102
Panel B. Poisson pseudo-maximum likelihood, log points						
East side	0.166**	0.140**	0.151**	0.172**	0.161**	0.146*
	(0.071)	(0.070)	(0.071)	(0.079)	(0.072)	(0.078)
Neighborhood-ties index		0.054*				0.053*
		(0.032)				(0.032)
Help-received index			0.039			
			(0.033)			
p-value, east side	0.019	0.047	0.032	0.030	0.026	0.062
Share of the east-side coefficient absorbed		0.16	0.09	−0.04	0.03	0.12
95% bootstrap interval		[−0.01, 0.62]	[−0.08, 0.42]	[−0.56, 0.52]	[−0.13, 0.36]	[−0.34, 0.91]
Observations	238	238	238	238	238	238
Pseudo R^2	0.083	0.096	0.089	0.101	0.093	0.113
Neighborhood-ties index		✓				✓
Help-received index			✓			
Location measures				✓		✓
Income bracket					✓	
Map-First, sex, age, age^2, question version	✓	✓	✓	✓	✓	✓
Respondent-education FE	✓	✓	✓	✓	✓	✓
Dep. var. mean (MXN)	35,361	35,361	35,361	35,361	35,361	35,361

Notes: OLS (Panel A) and Poisson pseudo-maximum-likelihood (Panel B) estimates from the parent survey, 193 households on the east side and 46 on the west, of which one in a single-member respondent-education cell drops. The dependent variable is the monthly salary at age 30 that the respondent says a young person would need to earn for university to have been worthwhile, in MXN. Every column controls for Map-First assignment, respondent sex, age and age squared, and the randomized question version, with respondent-education fixed effects, the specification of Panel D of Table C.5, whose estimate column (1) of Panel B reproduces. The neighborhood-ties index is that of Table 3, and the help-received index is the first principal component of its three help counts, both standardized over the 239 households, so their coefficients are per standard deviation. The location measures are the distance from the home to the boundary, the walking distance from the household's block to the nearest upper-secondary school (Table B.9), and the share of the household's school-attending children aged 6 to 20 whose school is outside its own *colonia* (Table B.10), with an indicator for the 17 households without such a child. The income bracket (1 to 10) enters linearly, with an indicator for the 25 households that did not report it. The share absorbed is $1 - \beta_{\text{with}}/\beta_{\text{without}}$, where β is the east-side coefficient with and without the variables of the column on the same sample, and its interval is the 95 percent percentile interval from 999 bootstrap draws over households that rebuild the indices in each draw. The survey has no item on commute time or transport mode. The log specification is not reported, because a few answers near zero (the minimum is MXN 3) make its east-side coefficient indistinguishable from zero. Heteroskedasticity-robust standard errors are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

↩ Back to text

Open-ended descriptions of the neighborhood. Most answers on both sides refer to streets, landmarks, and everyday routes. A minority define the neighborhood by the people in it, and this minority is larger on the east side, 15% against 7% on the west side (28 of 193 and 3 of 46 answers), although the difference is imprecise. The original answers quoted in the main text are "Dónde me conocen y conozco a los vecinos," "Hasta donde la gente se conoce del barrio," and "Dónde hay más gente de confianza." Other east-side respondents draw the neighborhood along "the streets that count for organizing the patron-saint festivities" ("Son las calles que se toman en cuenta para la organización de las fiestas patronales"). The answers were given after drawing in both treatment arms, and these examples were selected to illustrate the category.

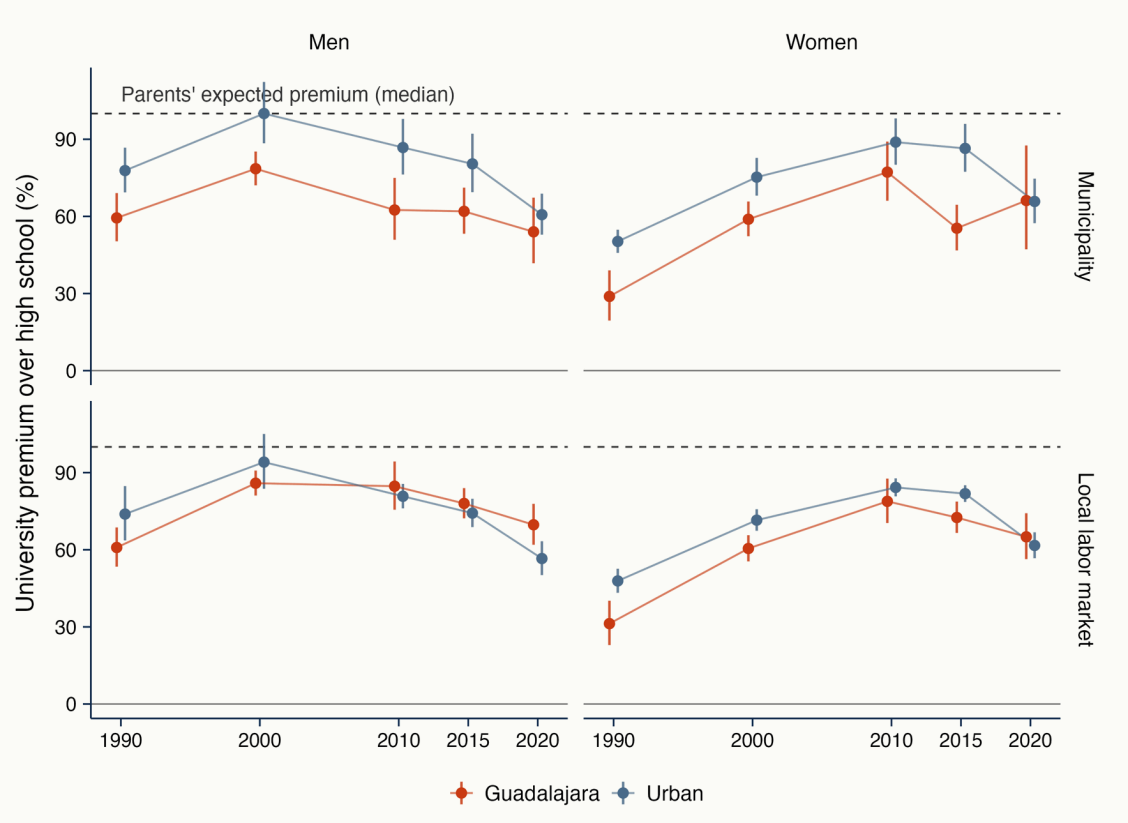
Allocation exercises. The allocations follow Enke et al. (2023) and Cappelen et al. (2025). In the first, respondents divide MXN 500 between two equally well-off strangers, one from their own *barrio* and one from elsewhere in Guadalajara. In the second, they divide MXN 500 between a Guadalajara resident and someone from outside the city. The respondent keeps nothing, so each allocation reflects the relative weight placed on the two recipients. A third question asks how much of an unexpected MXN 500 windfall the respondent would donate to a good cause. The relative *barrio* premium in Panel A of Table C.4, which scales the first allocation by the second, is driven by a few respondents who give little to the Guadalajara stranger, and symmetric versions of the same comparison are small and imprecise.

Aspirations, norms, and attachment. Aspirations and expectations refer to a randomly selected focal child and control for the child's sex, age, and current school attendance (Panel C). Following Bursztyn and Yang (2022), respondents report their own and their perceived neighbors' agreement with the statement that young people should work rather than continue studying; neither differs across the boundary, nor does pluralistic ignorance (Panel E). Stated attachment covers feelings of belonging, preferences over living elsewhere, and effort to have children know the neighbors (Panel F).

C.3 Realized university wage premium, 1990–2020

Figure C.1 estimates the university wage premium over completed high school in Guadalajara and in other urban municipalities and local labor markets, using census microdata from 1990 to 2020. The premium rises until 2000 for men and 2010 for women and declines afterward, ending between 54 and 70 percent in 2020. In almost every year, at both geographic levels, and for both sexes, the realized premium lies below the 100 percent premium that parents in the survey of Section 4 expect at age 30. The exception is men in other urban municipalities in 2000, whose estimated premium reaches 100 percent with a confidence interval extending above it.

Figure C.1: University wage premium in Guadalajara and urban Mexico, 1990–2020



Notes: Each point reports the university wage premium implied by a Mincer regression of log monthly labor earnings on a university indicator and a quadratic in age, estimated separately by census year, sex, and geographic group on IPUMS International microdata from the 1990, 2000, 2010, and 2020 censuses and the 2015 Intercensal Survey. The sample covers employed workers aged 25–55 whose highest attainment is completed high school or a university degree, restricted to at least 30 weekly hours where hours are recorded; the premium is $100(e^{\beta} - 1)$. The top row compares the municipality of Guadalajara with all other urban municipalities, defined by an urban population share of at least 50 percent in each year; the bottom row repeats the exercise at the level of the local labor markets delimited by Aldeco et al. (2024), comparing Guadalajara's market with other urban markets. Vertical bars show 95 percent confidence intervals, based on heteroskedasticity-robust standard errors for Guadalajara and standard errors clustered by municipality or labor market otherwise. The dashed line marks the median university premium that surveyed parents on both banks expect at age 30 (Section 4). Restricting the 2020 Guadalajara estimate to workers aged 25–35, the age in the survey vignette, yields a realized premium of 57 percent with a 95 percent confidence interval of 44 to 72 percent.

↩ Back to text

C.4 Household characteristics of the experimental sample

Table C.7 compares the households on the two sides of the boundary. Household income, household size, parental and grandparental schooling, the share of close relatives living in the *colonia*, years of residence, and whether a parent lived there show no detectable difference after the correction for multiple comparisons. Among respondents who work, a written contract is, if anything, more common on the east side, and self-employment is similar on the two sides; these two rows also control for the respondent's schooling.

Table C.7: Household characteristics of the experimental sample

Panel G. Household income, family schooling, and roots in the colonia	Monthly household income PPML	Monthly household income per member PPML	Household size OLS	Years of schooling, more-schooled parent OLS	A parent completed high school OLS	Years of schooling, respondent's father OLS	Years of schooling, respondent's mother OLS
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$\mathbb{1}\{East_i\}$	−0.100	−0.065	0.10	−0.69	−0.050	−1.83**	0.16
	(0.140)	(0.148)	(0.24)	(0.52)	(0.071)	(0.90)	(0.71)
Respondent-education FE							
FDR q-value	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Observations	214	214	239	239	239	203	229
Control mean (west side)	17,400	4,149	4.48	13.15	0.739	9.58	7.77
East-side mean	15,705	3,824	4.67	12.40	0.679	8.08	8.08
R^2	0.135	0.137	0.054	0.101	0.051	0.143	0.137

	Schooling gain over the more-schooled grandparent	Share of close relatives living in the colonia	Years lived in the colonia	A parent lived in the colonia	Works with a written contract	Self-employed
	(8)	(9)	(10)	(11)	(12)	(13)
$\mathbb{1}\{East_i\}$	0.04	0.050	−0.74	0.091	0.142	−0.060
	(0.77)	(0.036)	(2.74)	(0.083)	(0.098)	(0.101)
Respondent-education FE					✓	✓
FDR q-value	1.00	1.00	1.00	1.00	1.00	1.00
Observations	234	239	239	239	161	161
Control mean (west side)	3.52	0.141	25.70	0.478	0.294	0.412
East-side mean	3.27	0.196	25.17	0.580	0.433	0.370
R^2	0.081	0.025	0.129	0.021	0.248	0.139

Notes: Survey modules of the incentivized choice experiment, for the 239 households. Each column regresses the outcome in its header on the east-side indicator, controlling for Map-First assignment and respondent sex, age, and age squared. The household columns omit the respondent-education fixed effects, which would absorb parental schooling mechanically; the two informality columns, estimated among respondents who work, include them. Income is the midpoint of the reported bracket; years of schooling are assigned to completed levels; the more-schooled parent is the respondent or the partner. The share of close relatives (siblings, uncles and aunts, cousins) living in the colonia maps the five answer categories to 0, 0.25, 0.5, 0.75, and 1; the colonia is the area the respondent drew on the map. A written contract and self-employment come from the question "Do you have a formal work contract?", whose answers distinguish a written contract, work without a written contract, and self-employment. Columns marked PPML are estimated by Poisson pseudo-maximum-likelihood, so their coefficients are semi-elasticities while the means are in MXN, and their R^2 is the pseudo-R^2; all other columns are estimated by OLS in levels. The rows *Control mean (west side)* and *East-side mean* report unadjusted means for west-side and east-side respondents. The FDR q-value row reports sharpened false-discovery-rate values (Anderson, 2008) computed across the columns of the table. The number of observations is smaller where income, a grandparent's schooling, or employment is not reported. HC1 robust standard errors are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

↩ Back to text

C.5 Neighbor trust and the educational gap: heterogeneity and falsification

The mediation decomposition rested on a mediator that is neither randomized nor plausibly exogenous, so I no longer report it. What the estimates can do is describe where the educational gap is larger and weigh against two accounts of it. Table C.8 reports them; the causal evidence on bringing the neighborhood to mind is the experiment of Section 4.4.

Individual mediation. Columns (4)–(6) interact the east-side indicator with candidate mediators, so the reported main effect is the western slope and the interaction is the difference between the two sides rather than the eastern slope itself. In Column (4) neighbor trust is modestly positively associated with schooling on the western side, at 0.796, and the interaction of −1.74 implies an eastern slope of about −0.94: among east-side residents, higher trust in neighbors goes with roughly one fewer year of schooling. The interaction has a robust p-value of 0.067 and is not statistically significant when standard errors are clustered by census tract ($p = 0.16$); it falls to −0.80 within three kilometers of the boundary. The neighbor-vs-nonlocals premium in

Table C.8: Neighbor trust and educational attainment: heterogeneity and falsification

	Years of schooling	Trust in neighbors	Neighbor vs. nonlocals trust	Years of schooling					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$\mathbb{1}\{East_i\}$	−1.86	0.368**	0.093	−0.798	0.172	−1.28	−2.32**	−0.761	−1.81
	(1.13)	(0.151)	(0.064)	(1.23)	(1.73)	(1.25)	(1.17)	(1.07)	(1.12)
Trust in neighbors				0.796					
				(0.691)					
Trust in neighbors × $\mathbb{1}\{East_i\}$				−1.74*					
				(0.945)					
Neighbor vs. nonlocals trust					0.703				
					(1.65)				
Neighbor vs. nonlocals trust × $\mathbb{1}\{East_i\}$					−3.14				
					(2.19)				
Institutional trust index						1.67			
						(1.68)			
Institutional trust index × $\mathbb{1}\{East_i\}$						−1.89			
						(2.15)			
Tract-agg. neighbor trust (z)							0.775**		
							(0.310)		
Tract-agg. neighbor trust (z) × $\mathbb{1}\{East_i\}$							−1.36***		
							(0.464)		
Tract-agg. schooling (z)								1.50***	
								(0.268)	
Tract-agg. schooling (z) × $\mathbb{1}\{East_i\}$								0.042	
								(0.572)	
Tract-agg. land value (log, z)									1.22***
									(0.258)
Tract-agg. land value (log, z) × $\mathbb{1}\{East_i\}$									−0.720
									(0.719)
Age-by-sex FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Distance to boundary	✓	✓	✓	✓	✓	✓	✓	✓	✓
Schooling control		✓	✓						
Survey weights	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observations	289	289	289	289	289	289	289	289	289
Dep. var. mean	11.33	0.655	0.595	11.33	11.33	11.33	11.33	11.33	11.33
R^2	0.225	0.089	0.081	0.234	0.234	0.228	0.248	0.325	0.277

Notes: Estimates from the 2022 wave of the local Life Quality Survey (*Jalisco Cómo Vamos*) for Guadalajara. The dependent variable is years of schooling in columns (1) and (4)–(9), and columns (2) and (3) report the east-side gap in the two trust measures themselves. All models include age-by-sex fixed effects and control for distance to the boundary and its interaction with the east-side indicator. Columns (2) and (3) also control linearly for the respondent's years of schooling. Columns (4)–(6) interact each trust measure with the east-side indicator, so the main effect is the western slope and the interaction the difference between the two sides. Columns (7)–(9) interact the east-side indicator with tract averages of neighbor trust, completed schooling, and log land values. Estimates are weighted by the survey's expansion factors. The trust measures are not randomized, so these estimates describe heterogeneity and bound two accounts of the gap; they do not identify a channel. Dep. var. mean is the survey-weighted mean of the outcome. Heteroskedasticity-robust standard errors are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

↩ Back to text

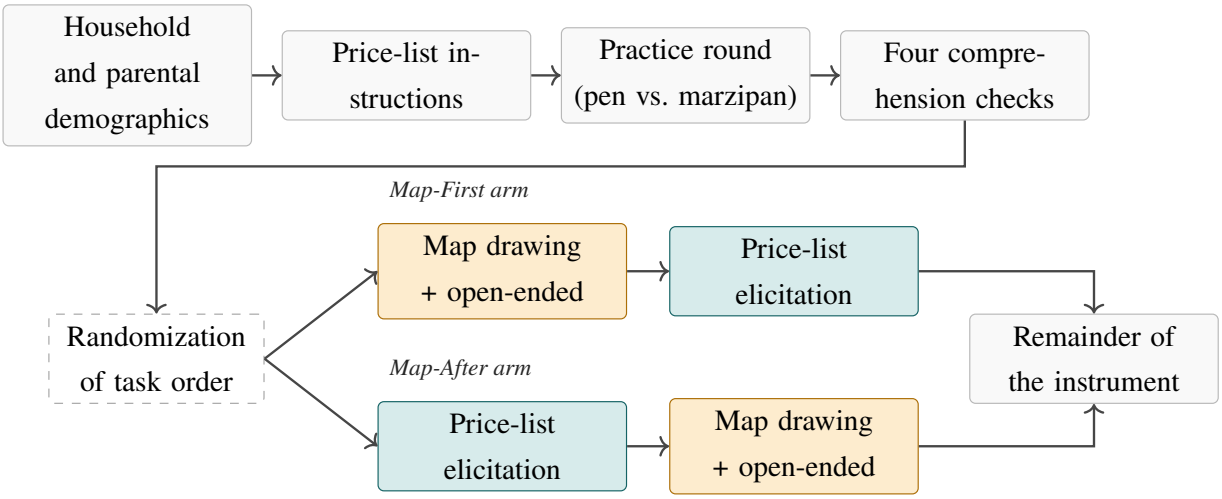
Column (5) gives a larger difference, −3.14 against a western slope of 0.703, for an eastern slope near −2.44. The institutional-trust index in Column (6) gives an interaction of −1.89 that is not statistically distinguishable from zero, and a western slope of 1.67, so the implied eastern slope is close to zero. The indirect effects reported below are computed from the side-specific slopes rather than from the interactions.

Moderated mediation. Columns (2) and (3) report the east-side gap in the two trust measures themselves, which is the association documented in Section 4.1: the estimated discontinuity is positive for trust in neighbors (0.368) and positive but imprecise for the neighbor-versus-nonlocals premium (0.093).

Contextual moderators. Columns (7) to (9) interact the east-side indicator with three tract-level variables. The gap is larger where average neighbor trust in the tract is higher (interaction −1.36), so the pattern within individuals also appears between neighborhoods. The remaining two columns interact the east-side indicator with average completed schooling in the tract (interaction 0.04) and with log land values (interaction −0.72). The slopes do not differ detectably across sides, which weighs against an account in which the east side converges faster toward neighbors' schooling or wealth, and does not exclude one in which the same slope operates on different compositions. These are tract averages of realized outcomes rather than measurements of the state variables of Section 5, and the nulls bound the two accounts weakly given the size of the Life Quality Survey. Tract-level confounders correlated with both neighbor trust and schooling could produce the first pattern.

D The Incentivized Choice Experiment: Supporting Material

Figure D.1: Order of tasks in the incentivized choice experiment



Notes: Task order in the survey instrument. The price-list instructions, the practice round with a pen and marzipan candies, and the four comprehension checks precede the randomization and are therefore identical in both arms. The randomization determines only the order of the two remaining blocks: in the Map-First arm respondents draw their neighborhood and answer the open-ended question about the boundary they drew, and then complete the price list; in the Map-After arm they complete the price list first. Nothing unrelated to the neighborhood intervenes between the map-drawing module and the price list in the Map-First arm.

↩ Back to text

D.1 Confirmatory results

Table D.1 reports the full regression table behind the main-text figure: both outcomes, the pooled and interacted specifications, and the east-side subsample.

The confirmatory estimates of H1 (average effect) and H2 (heterogeneity by side) are reported in the main text, Table D.1.

Inference. Map-First was randomized within side of the historical boundary, so the permutation distribution holds fixed the number treated on each side. The reported randomization-inference p-values are the share of 50,000 permutations producing an estimate at least as large in magnitude as the observed one; Table D.1 also reports conventional robust standard errors.

Magnitude. Using Glass's Δ, the east-side effects correspond to 0.33 standard deviations for the tutoring share and 0.30 for willingness to pay. The MXN 75 decline in willingness to pay is about one quarter of Mexico's 2026 general daily minimum wage, set by the *Comisión Nacional de los Salarios Mínimos* at MXN 315.04 from January 1, 2026 and published in the *Diario Oficial de la Federación* on December 9, 2025. Guadalajara lies in the general minimum-wage zone rather than the northern-border free zone.

Table D.1: Effect of the map on parents' educational investment

	Share of tutoring choices			Willingness to pay (MXN pesos)		
	OLS			Jack et al. (2022), two-step latent WTP		
	(1)	(2)	(3)	(4)	(5)	(6)
$\mathbb{1}\{Map\text{-}First_i\}$	−0.077*	0.144	−0.117**	−44.7	97.6	−70.9**
Robust	(0.049)	(0.126)	(0.053)	(36.8)	(93.4)	(39.6)
Randomization inference	(0.046)	(0.117)	(0.051)	(30.7)	(80.8)	(33.4)
$\mathbb{1}\{East_i\}$		0.006			−18.4	
Robust		(0.219)			(172.7)	
Randomization inference		—			—	
$\mathbb{1}\{Map\text{-}First_i\} \times \mathbb{1}\{East_i\}$		−0.267**			−172.2*	
Robust		(0.137)			(102.9)	
Randomization inference		(0.129)			(88.4)	
Effect on the west side		0.144			97.6	
Robust		(0.126)			(93.4)	
Randomization inference		(0.117)			(80.8)	
Effect on the east side		−0.123**			−74.6**	
Robust		(0.052)			(39.6)	
Randomization inference		(0.050)			(33.2)	
p-value: east-side effect = west-side effect (robust / RI)		0.052 / 0.037			0.094 / 0.051	
Sample	Full	Full	East side	Full	Full	East side
Enumerator FE	✓	✓	✓	✓	✓	✓
Day FE	✓	✓	✓	✓	✓	✓
LASSO controls	✓	✓	✓	✓	✓	✓
Distance to boundary	✓	✓	✓	✓	✓	✓
Dep. var. mean	0.548	0.548	0.538	164.0	164.0	155.6
Control mean (Map-After)	0.561	0.561	0.573	171.6	171.6	176.3
Control SD (Map-After)	0.384	0.384	0.372	255.9	255.9	249.5
Participants	239	239	193	239	239	193
Within R^2	0.235	0.251	0.216	0.222	0.238	0.208

Notes: Estimates from the pre-registered incentivized choice experiment near the historical boundary (AEA RCT Registry AEARCTR-0018600). *Map-First* equals one when the map-drawing module precedes the incentivized tutoring choice and zero when it follows; $\mathbb{1}\{East_i\}$ identifies households east of the historical boundary. Columns (1)–(3) report the share of eleven multiple-price-list rows in which the parent chose one hour of online math tutoring over an Amazon gift card. Columns (4)–(6) report latent willingness to pay (WTP) in MXN, estimated following the two-step procedure of Jack et al. (2022): a random-effects probit recovers each participant's WTP as the conditional mode of the random intercept, from the ten choices with a positive card value, and the resulting WTP is regressed on the treatment indicators. All specifications include enumerator and day fixed effects, side-specific linear slopes in distance to the boundary, and controls selected by LASSO. Columns (1) and (4) report the pooled treatment effect; columns (2) and (5) allow it to differ by side, with the reported p-value testing equality of the two effects; columns (3) and (6) restrict the sample to the east side. The rows *Effect on the west side* and *Effect on the east side* report the Map-First effect implied by columns (2) and (5): the coefficient on Map-First on the west side, and its sum with the interaction on the east side. The *Robust* rows report standard errors in parentheses, using HC1 with a finite-sample correction for the share outcome and a subject-level Bayesian bootstrap with 1,000 replications (Rubin, 1981) for WTP. The *Randomization inference* rows report the standard deviation of the estimate across 50,000 permutations of treatment within the original randomization strata, holding the number treated on each side fixed. The equality test imposes a constant common effect before permutation. No randomization-inference statistic is reported for $\mathbb{1}\{East_i\}$ because side is not randomized. Significance stars use randomization-inference p-values where available and robust p-values otherwise. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

↩ Back to text

Construction of the main-text figure. In Figure 8, the Map-After bar is the unadjusted control mean on each side. The Map-First bar adds to it the estimated Map-First effect on that side from the interacted specification in columns (2) and (5) of Table D.1: the Map-First coefficient on the

west side, and the sum of the Map-First coefficient and its interaction with the east-side indicator on the east side. The whiskers are the randomization-inference interval for that effect, shifted by the control mean, which is treated as fixed. The bracket between the two Map-After bars compares the unadjusted control means across sides. The side of the boundary is not randomized, so its p-value does not come from randomization inference: it uses robust standard errors for the tutoring share and the respondent-level bootstrap of both steps for willingness to pay. On the east side, the implied Map-First levels are 45.0 percent and MXN 102, the values reported in the main text; the specification estimated on the east side alone, in columns (3) and (6), gives effects of -11.7 percentage points and MXN -71.

Latent willingness to pay. The pre-registered willingness-to-pay measure (Table D.1) is the two-step latent estimator of Jack et al. (2022), and Table D.2 compares it with two alternatives. The bounded measure is the area under each respondent's demand curve within the price grid, which is the share outcome in pesos. The latent measure comes from a random-intercept probit that places respondents who never switch beyond the grid, so it differs from the bounded one mainly through those values, and an interval regression treats them as censored instead. All three give a negative priming effect on willingness to pay concentrated on the east side.

D.2 Opening the black box: what the neighborhood priming activates

A standard objection to priming manipulations is that they are "black boxes": they change behavior without revealing which mental representations they activate (Cohn and Maréchal, 2016). The map-drawing module is unusually transparent on this point, because the object the respondent produces, a hand-traced neighborhood, *is* the activated representation. I characterize it in two ways: the geometry of the drawn barrio and what respondents say they had in mind. Both are descriptive. Both arms draw and answer immediately afterward, so a contrast between them would reflect what preceded the drawing rather than the drawing itself.

The geometry of the drawn barrio. Table D.3 summarizes the polygons. Each traced barrio is represented by the convex hull of its vertices. The neighborhoods people draw are small (a median of just over half a square kilometer, a few blocks), and home-centered: for 88% of respondents the geolocated home falls *inside* their own drawing. Only 11% of the drawn barrios cross the historical boundary, the SJdD creek, today Calzada Independencia (Figure 9). That figure should not be read as avoidance of the line, because most respondents live too far from it for a drawing of this size to reach it. A polygon of the median area corresponds to a circle of about 430 metres in radius, while the median distance from home to the boundary is close to a kilometre, so a home-centred drawing of typical size falls several hundred metres short of the avenue before any question of identity arises. The exercise reports no opportunity-to-cross adjustment and no placebo line elsewhere in the city against which to compare the rate, and a drawing that stops at a

Table D.2: Willingness-to-pay treatment effects: bounded area vs. latent scaled-probit measure

Panel A. Treatment effects on the two measures

	Latent WTP (MXN pesos) Jack et al. (2022), two-step		Area under demand (MXN pesos) Participant random effects	
	(1)	(2)	(3)	(4)
$\mathbb{1}\{Map\text{-}First_i\}$	−44.7	97.6	−18.4	37.5
Robust	(36.8)	(93.4)	(12.7)	(32.5)
Randomization inference	(30.7)	(80.8)	(12.1)	(31.1)
$\mathbb{1}\{Map\text{-}First_i\} \times \mathbb{1}\{East_i\}$		−172.2*		−67.7**
Robust		(102.9)		(35.2)
Randomization inference		(88.4)		(34.1)
Effect on the east side		−74.6**		−30.2**
Robust		(39.6)		(13.5)
Randomization inference		(33.2)		(13.2)
Sample	Full	Full	Full	Full
Enumerator FE	✓	✓	✓	✓
Day FE	✓	✓	✓	✓
LASSO controls	✓	✓	✓	✓
Distance to boundary	✓	✓	✓	✓
Effect on the east side, % of east-side Map-After mean		−42.3		−22.1
Randomization inference, 95% interval		[−79.2, −5.7]		[−41.0, −3.3]
Participants	239	239	239	239
Dep. var. mean	164.0	164.0	130.6	130.6
East-side Map-After mean	176.3	176.3	136.5	136.5
Within R^2	0.222	0.238	—	—

Panel B. Effect on the east side under other response models and samples

	Latent WTP (MXN pesos), two-step					Interval regression (MXN pesos)	
	Paper's estimator	Capped at 0 and 250	Logit first step	Without multiple switchers	Bounded area	All responses	Without multiple switchers
	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Effect on the east side	−74.6	−30.8	−76.1	−62.7	−30.2	−51.9	−82.9
Bootstrap	(39.6)	(14.4)	(41.0)	(32.2)	(12.9)	(21.8)	(32.7)
% of east-side Map-After mean	−42.3	−21.1	−43.0	−36.0	−22.1	−32.2	−45.9
Bootstrap, 95% interval	[−85.2, −1.0]	[−39.4, −2.9]	[−87.3, −0.4]	[−67.6, −1.9]	[−39.4, −5.0]	[−56.3, −8.9]	[−135.7, −13.3]
First step	Probit	Probit	Logit	Probit	—	—	—
Capped at the ends of the price grid		✓					
Sample	Full	Full	Full	Without MSB	Full	Full	Without MSB
Controls of Panel A	✓	✓	✓	✓	✓	✓	✓
p-value: East = West (randomization inference)	0.051	0.034	0.052	0.055	0.047	0.027	0.030
Participants	239	239	239	187	239	239	187
East-side Map-After mean	176.3	146.0	176.9	174.1	136.5	160.8	180.5

Notes: Treatment effects on willingness to pay for tutoring in the incentivized choice experiment, for the 239 participants, with two measures. Columns (1) and (2) reproduce the pre-registered latent measure of Table D.1, in MXN, from the two-step procedure of Jack et al. (2022). Step one recovers each participant's latent willingness to pay as the conditional mode of the random intercept of a scaled random-effects probit of the ten binary choices with a positive card value, with row- and column-order controls, estimated *without* the treatment, and step two regresses that individual willingness to pay on the treatment indicators. Columns (3) and (4) use the area under the step demand curve over the ten positive-price rows, which is bounded at 250 pesos, estimated on the ten choices of each participant with a participant-level random intercept. This area is 25 pesos times the number of tutoring choices in those rows, so it is the share outcome of Table D.1 in pesos, without the row in which the card is worth nothing. The latent measure places the respondents who never switch beyond the grid, at about 480 pesos for those who always choose tutoring and about −180 pesos for those who always choose the gift card, and 156 of the 239 values lie outside the grid, so its scale and its effects are wider. All specifications include enumerator and day fixed effects, side-specific linear slopes in distance to the boundary, and the LASSO-selected controls. Columns (1) and (3) report the pooled effect (H1), and columns (2) and (4) let it differ by side. The row *Effect on the east side* is the sum of the Map-First coefficient and its interaction with the east-side indicator, so it differs slightly from the east-side subsample estimate reported in Tables D.1 and D.7. The row *Effect on the east side, % of east-side Map-After mean* divides that effect by the mean of the measure among east-side Map-After participants, and its interval divides the randomization-inference 95% interval of the effect, obtained by inverting the test, by the same mean. Both measures give a negative priming effect concentrated on the east side. The mixed model of columns (3) and (4) has no within R^2. The *Robust* rows report standard errors in parentheses: for the latent measure, a subject-level Bayesian bootstrap with 1,000 replications (Rubin, 1981) that reweights subjects and repeats both steps, which corrects for the generated-regressand problem; for the area measure, CR2 standard errors clustered on the participant. The *Randomization inference* rows report the standard deviation of the estimate across 50,000 permutations of treatment within side of the historical boundary, holding the number treated on each side fixed; for the area measure the permutations use each participant's mean over the ten rows, which gives the same coefficients. The test of the interaction imposes a constant common effect before permutation. Panel B reports the effect on the east side of column (2) under other response models and samples. Column (5) repeats column (2), and column (9) repeats column (4) with the area measured at the participant level, which gives the same coefficient. Column (6) caps each participant's latent willingness to pay at 0 and 250 pesos, the card values of the first and the last row, so that no value lies beyond the price grid, and its effect, −30.8 pesos, is close to the −30.2 of the bounded measure in column (9). Column (7) estimates step one with a logit link in place of the probit. Column (8) drops the 52 participants who switch more than once over the eleven rows (MSB) and repeats both steps without them, a change of sample reported as a sensitivity check. Columns (10) and (11) estimate an interval regression with normal errors on the specification of column (2), in which each respondent's choices place willingness to pay between the card value before the first switch and the card value after the last one, respondents who always choose tutoring are right-censored at 250 pesos, and respondents who always choose the gift card are left-censored at 0. For the 52 multiple-switchers this interval covers all their switches, with an average width of 184 pesos, and column (11) drops them. This estimator extrapolates beyond the grid only through the normal distribution of the error, so never-switchers enter as censored observations and no value is assigned to them. Its east-side Map-After mean is the mean predicted willingness to pay of those participants. The percentages in Panel B divide each column's effect by the east-side Map-After mean of its own measure. Columns (6) to (8), (10), and (11) vary the response model and the sample, and the bootstrap intervals capture sampling uncertainty only, so they do not reflect these modeling choices. In Panel B, parentheses report the standard deviation of the estimate across the 1,000 Bayesian bootstrap draws of Panel A, with the same weights, which repeat both steps for the latent measures and reweight the participant-level regression for the area and the interval regression, and brackets report the 2.5th and 97.5th percentiles of the percentage across the same draws. In column (8), step one cannot be estimated in 3 of the 1,000 draws, which are dropped. The row *p-value: East = West* tests equal effects on the two sides by randomization inference with 50,000 permutations under a constant common effect, as for the interaction in Panel A, and for the interval regression each permutation re-estimates the model. Significance stars in Panel A use the randomization-inference p-values, and Panel B carries no stars. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

↩ Back to text

major avenue may simply be stopping at a salient physical landmark. What the polygons establish is narrower and still useful: when asked for their neighborhood, people draw a compact area around their own home rather than a district or a side of the city. The table reports the geometry overall and by side of the boundary.

Table D.3: Geometry of the neighborhood respondents drew

	All	East side	West side
Barrio area (km^2), median	0.58	0.56	0.71
Barrio area (km^2), mean	0.79	0.77	0.91
Home inside the drawn barrio	0.876	0.862	0.933
Home off-centering (dist./$\sqrt{\text{area}}$)	0.40	0.39	0.43
Crosses the historical boundary	0.112	0.122	0.067
Vertices drawn	4.3	4.3	4.4
Participants	233	188	45

Notes: Geometry of the neighborhood each respondent traced on the interactive map, for the 233 of 239 participants who drew a valid polygon of at least three vertices (3 far-vertex drawing errors, with a vertex over 5 km from the home, are dropped). Each traced neighborhood is represented by the convex hull of its vertices, so that self-intersecting traces still define a well-behaved convex region. The barrio area is that hull's area, reported as both the median and the mean because the distribution is right-skewed; the remaining rows are means. *Home inside* is the share whose geolocated home falls within their own drawing; *home off-centering* is the distance from the home to the polygon centroid divided by the square root of the area, so lower values are more centered; *crosses the historical boundary* is the share whose drawing intersects the SJdD creek, today Calzada Independencia. The drawings are small, home-centered, and rarely cross the historical boundary. The table is descriptive. Both experimental arms draw their neighborhood and answer immediately afterward, so treatment-arm comparisons are not reported.

↩ Back to text

What respondents said they thought about. The open-ended companion question describes the same object. I classify the free-text answers with `claude-opus-4-8` against a fixed codebook (Section D.8.1), and a second, independent classification of the same answers, formed without access to the first, agrees closely: mean Cohen's $\kappa = 0.93$ across the six themes, and 0.98 for social ties. Table D.4 reports the shares by side of the boundary.

The neighborhood is conceived as a concrete place that respondents walk and use. Physical extent is the most common theme, at 62%: streets, avenues, landmarks, and how far the area reaches. Everyday use follows at 49%, pooling lived familiarity and daily routine ("where I move," "what

I know," "as far as I can walk") with errands, shops, work and school. People and belonging appear in 13% of answers and safety in 6%, with social ties named by one respondent in ten. Respondents describe the neighborhood as a place more often than as a group of people, which is what the polygons also show; the descriptions do not say which feature of that place enters the decision, and Section 4.6 relates the response to the survey measures of neighborhood ties. That is the object urban sociology describes, a neighborhood constituted by routine interaction grounded in shared residence and the common use of local facilities rather than by dense, intimate bonds (Smith, 2010; Sampson et al., 2002), as Section 4.4 sets out. What the exercise does not support is a claim about local identity.

The table reports no test. The experimental arm is absent by design: both arms answer this question immediately after drawing, so a contrast between them would identify the effect of the preceding multiple-price list on the description rather than what the mapping task brings to mind. Differences by side are small, and none of the six themes is distinguishable from zero once the multiplicity of the comparisons is taken into account.

Table D.4: What respondents had in mind when drawing their neighborhood

	All	East side	West side
Physical extent	0.62	0.63	0.61
Everyday use	0.49	0.50	0.48
Daily routine or familiarity	0.37	0.37	0.39
Errands, shops, work, or school	0.19	0.20	0.13
People or belonging	0.13	0.15	0.07
Social ties	0.10	0.11	0.04
Long residence or family roots	0.04	0.04	0.02
Safety or calm	0.06	0.05	0.11
Participants	239	193	46

Notes: Share of respondents whose open-ended answer to "what did you have in mind when drawing the limits of your neighborhood?" mentions each theme. Themes are coded by `claude-opus-4-8` using a fixed codebook and are not mutually exclusive. *Physical extent* includes streets, avenues, landmarks, and the reach of the area; *Everyday use* includes daily routines and places respondents regularly use; *People or belonging* includes social ties, long residence, and family roots. An independent classification yields mean Cohen's $\kappa = 0.93$ across themes. The table is descriptive. Both experimental arms answer this question immediately after drawing their neighborhood, so treatment-arm comparisons are not reported.

↩ Back to text

D.3 Balance, diagnostics, and influential observations

Covariate balance. Table D.5 reports balance across the Map-First (treatment) and Map-After (control) arms, in two panels. Panel A covers the twenty-six variables recorded before the randomized block — side of the boundary and absolute distance to it, parent and grandparent education, household composition, income bracket, and price-list comprehension — and none shows a detectable difference across arms after a sharpened false-discovery-rate correction (Anderson, 2008). Individually, only the absolute distance to the boundary differs at the 5 percent level, by 209 meters ($p = 0.049$), and the specifications of Table D.1 already control for that distance with a separate slope on each side. Side of the boundary is the randomization stratum: it is identical under every admissible permutation and therefore carries no test. Panel B reports the focal child's characteristics and the stated reason for choosing that child. The household roster is recorded before treatment, but which child becomes the focal one is decided afterwards, so these are not balance variables and the treatment could in principle have shifted the choice. None of the six shows a detectable difference across arms either, so parents in the two arms are making the tutoring decision about comparable children.

Implementation diagnostics. Table D.6 reports response-quality diagnostics by arm. Comprehension is high — 87% of respondents answer all four price-list comprehension questions correctly — and none of the quality metrics shows a detectable difference across arms except the share of west-side respondents who always choose the gift card, 5 of 17 in Map-After and none of 29 in Map-First, so the priming manipulation did not detectably degrade response quality; treatment assignment is also orthogonal to enumerator identity ($F = 1.45$, $p = 0.21$). As is common in the multiple-price-list literature (Jack et al., 2022), a non-trivial share of respondents exhibit *never-switching* behaviour (43% choose the same option in all eleven MPL rows) or *multiple-switching* behaviour (22% switch more than once); Figure D.2 plots the distribution of switches in the tutoring MPL separately for the treatment and control arms, which show no detectable difference, so the manipulation did not detectably affect elicitation quality. This is why non-monotonic switching is treated as a comprehension flag rather than a discarded observation: the pre-registered latent estimator of Jack et al. (2022) accommodates never- and multiple-switchers, as does the bounded area under each respondent's demand curve reported alongside it in Table D.2.

Randomization inference and influential observations. Tables D.1 and D.7 probe the two pre-registered results against the small sample and the possibility that a few households drive them. Randomization inference is reported in Table D.1, which permutes treatment 50,000 times within the original randomization strata, holding the number treated on each side fixed; the significance stars in that table are computed from those permutations. The Map-First × East interaction (H2) is significant on both outcomes, with randomization-inference p-values of 0.037

Table D.5: Covariate balance across the Map-First treatment

Panel A. Measured before the randomized block

	East side	Absolute distance to boundary (m)	Parent age	Female respondent	Parent completed high school	Father completed high school	Mother completed high school	Has partner
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\mathbb{1}\{Map\text{-}First_i\}$	−0.058	209**	−1.40	−0.065	−0.008	−0.026	0.019	0.024
	(0.051)	(106)	(1.43)	(0.058)	(0.064)	(0.067)	(0.059)	(0.061)
p-value (randomization inference)	—	0.049	0.327	0.302	1.000	0.766	0.767	0.783
FDR q-value	—	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Participants	239	239	239	239	239	203	229	239
Control mean (Map-After)	0.840	1465	44.93	0.764	0.594	0.344	0.265	0.660

	Adults in household	Minors in household	Rooms in dwelling	Crowding (persons per room)	Children aged 6–20	Respondent works	Main income is a formal salary	Price-list comprehension (0–4)
	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
$\mathbb{1}\{Map\text{-}First_i\}$	−0.05	0.01	0.00	0.10	−0.01	0.041	−0.025	0.05
	(0.16)	(0.16)	(0.13)	(0.10)	(0.11)	(0.061)	(0.057)	(0.06)
p-value (randomization inference)	0.800	0.957	1.000	0.311	0.937	0.574	0.771	0.423
FDR q-value	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Participants	239	239	239	239	239	239	239	239
Control mean (Map-After)	2.87	1.78	2.73	1.79	1.66	0.651	0.755	3.81

	Monthly household income bracket (MXN)									
	Under 6,000	6,001–9,000	9,001–12,000	12,001–15,000	15,001–18,000	18,001–22,000	22,001–27,500	27,501–35,000	35,001–55,000	Over 55,000
	(17)	(18)	(19)	(20)	(21)	(22)	(23)	(24)	(25)	(26)
$\mathbb{1}\{Map\text{-}First_i\}$	−0.032	−0.024	0.032	0.039	−0.002	0.084*	−0.056	−0.010	−0.037	0.006
	(0.047)	(0.048)	(0.052)	(0.049)	(0.038)	(0.043)	(0.038)	(0.038)	(0.026)	(0.016)
p-value (randomization inference)	0.541	0.689	0.597	0.446	1.000	0.088	0.185	0.798	0.253	1.000
FDR q-value	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Participants	214	214	214	214	214	214	214	214	214	214
Control mean (Map-After)	0.149	0.149	0.160	0.128	0.085	0.074	0.106	0.085	0.053	0.011

Panel B. Measured after the randomized block

	Focal child female	Focal child age	Focal child attends school	Chose child: needs academic help	Chose child: ability or aspiration	Chose child: age or only eligible
	(27)	(28)	(29)	(30)	(31)	(32)
$\mathbb{1}\{Map\text{-}First_i\}$	−0.011	−0.40	0.029	−0.004	−0.042	0.108
	(0.065)	(0.53)	(0.038)	(0.038)	(0.035)	(0.063)
p-value (randomization inference)	0.895	0.447	0.502	1.000	0.304	0.105
FDR q-value	1.000	1.000	1.000	1.000	1.000	1.000
Participants	239	239	239	239	239	239
Control mean (Map-After)	0.425	12.80	0.896	0.094	0.094	0.321

Notes: Covariate balance across the Map-First (treatment) and Map-After (control) arms of the pre-registered incentivized choice experiment. Panel A reports variables recorded before the randomized block of the instrument, items (i) to (iv) of Section D.8.1, including the price-list comprehension checks, which are administered before assignment; these are the balance tests proper. Panel B reports the characteristics of the focal child and the stated reason for choosing that child, recorded at item (vi), after the randomized block. The household roster from which the focal child is drawn is collected before treatment, but the choice of which child becomes focal is not, so Panel B does not test randomization and instead asks whether the treatment shifted that choice. Each column regresses the variable in its header on the Map-First indicator without controls, so the coefficient is the difference in arm means (treatment minus control). The distance in column (2) is in absolute value; the specifications of Table D.1 control for that distance with a separate slope on each side of the boundary. The randomization-inference p-value comes from 10,000 permutations of the Map-First label within side of the historical boundary, the randomization stratum of the experiment, holding fixed the number of treated households on each side. The side indicator in column (1) is the stratum itself, so it is identical under every admissible permutation and admits no test, and its p- and q-value cells are marked with a dash. The FDR q-value row reports sharpened false-discovery-rate q-values (Anderson, 2008) across all tests in the table, and no individual difference in either panel survives that correction. The number of participants varies across columns because some parental-education and income items are missing for a minority of respondents. Heteroskedasticity-robust (HC1) standard errors are in parentheses, and stars use the randomization-inference p-values. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

↩ Back to text

Table D.6: Incentivized choice experiment: implementation diagnostics

	Map-After (control)	Map-First (treatment)	All	Difference p-value
Comprehension score (0–4)	3.81	3.86	3.84	0.423
Passed all comprehension checks	0.840	0.902	0.874	0.184
Consistent on the practice MPL	0.547	0.624	0.590	0.250
Never-switching (NSB)	0.434	0.429	0.431	1.000
Always tutoring, west side	0.353	0.414	0.391	0.761
Always tutoring, east side	0.292	0.260	0.275	0.632
Always gift card, west side	0.294	0.000	0.109	0.004
Always gift card, east side	0.101	0.173	0.140	0.212
Multiple-switching (MSB)	0.226	0.211	0.218	0.874
Takes the \$0 card at a zero price	0.179	0.203	0.192	0.752
Enumerators / field days	6 enumerators, 15 days; treatment $\perp$ enumerator: $F = 1.45$, $p = 0.215$			

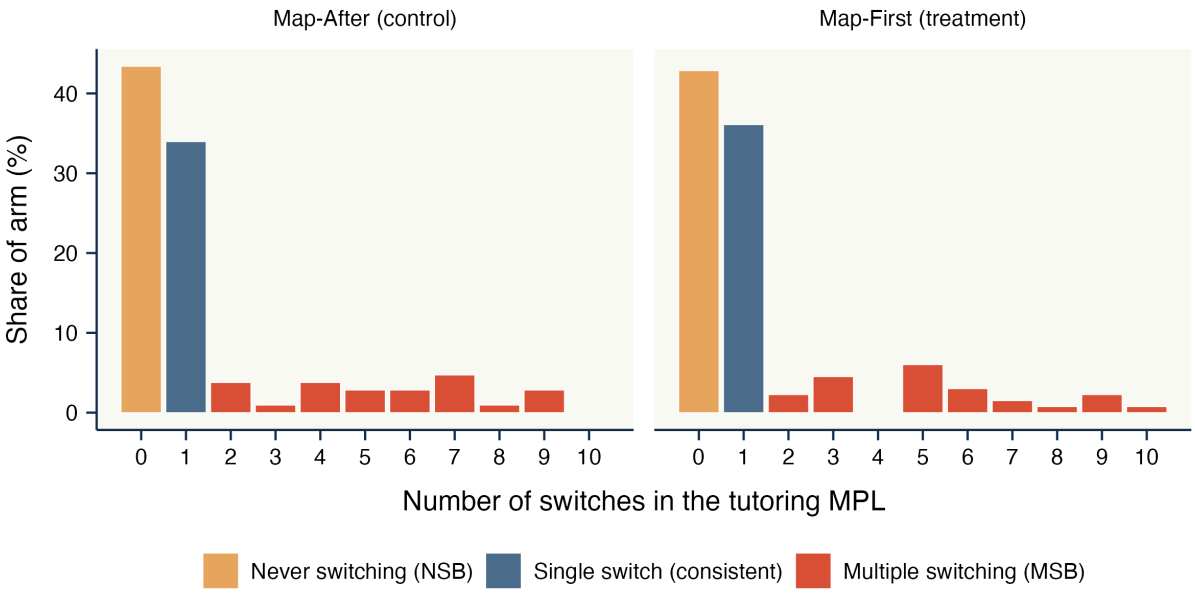
Notes: Response-quality diagnostics for the incentivized choice experiment, by treatment arm. Rows report the mean of each indicator among control (Map-After) and treatment (Map-First) respondents and overall, with the randomization-inference p-value (Map-First permuted within side of the historical boundary, the randomization stratum). The comprehension score counts correct answers to the four price-list comprehension questions; "passed" requires all four. "Consistent on the practice MPL" flags respondents who switch at most once (and not into a dominated choice) on the pen-versus-candy practice list. Never-switching (NSB) respondents choose the same option in all eleven MPL rows; multiple-switching (MSB) respondents switch more than once; the last row reports the share who take the \$0 gift card when tutoring is offered at a price of zero. That choice is not treated as an error: reading it as dominated would require assuming that tutoring carries no non-monetary cost and weakly positive value for every household, and online tutoring involves time, scheduling and supervision, so a zero or negative valuation is coherent. The row is reported as a rate and no respondent is excluded on its basis. The four indented rows split the never-switchers of each side by the option they always choose, as shares of that side's respondents in each arm, so on each side the two rows add up to that side's never-switching share, and their p-values permute Map-First within that side. In the latent measure of Table D.2, every respondent who always chooses tutoring has a willingness to pay above 250 pesos and every respondent who always chooses the gift card has one below zero, beyond the ends of the price grid. No metric differs across arms except the share of west-side respondents who always choose the gift card (5 of 17 in Map-After and 0 of 29 in Map-First), and a joint test finds treatment assignment orthogonal to enumerator identity, indicating the priming manipulation did not degrade response quality.

↩ Back to text

for the tutoring share and 0.051 for willingness to pay, and the implied priming effect on the east side is significant at the 5 percent level on both. The pooled effect (H1) is significant at the 10 percent level for the tutoring share and is not significant for willingness to pay. Table D.7 re-estimates each effect dropping one participant at a time. The east-side priming effect stays within a narrow band that never approaches zero, $[-0.130, -0.103]$ for the tutoring share and $[-78.0, -62.3]$ pesos for willingness to pay, so no single household drives the result.

Interviews before registration, distance, and the west side. The registration is dated May 14, 2026, and 23 households, 17 on the east side and 6 on the west, were interviewed before that date. Excluding them gives an east-side effect of -0.124 on the tutoring share (randomization-inference $p = 0.014$) and an interaction of -0.297 ($p = 0.039$), against -0.123 and -0.267 in the full

Figure D.2: Distribution of tutoring-MPL switches by treatment arm: never-switching and multiple-switching behaviour



Notes: Number of switches between the two options across the eleven rows of the incentivized tutoring multiple price list (MPL), following the diagnostics of Jack et al. (2022), shown as the share of each arm. *Never-switching* respondents (0 switches, amber) choose the same option throughout; a single switch (teal) is the consistent pattern; *multiple-switching* respondents (> 1 switch, red) make inconsistent choices. The left panel is the Map-After (control) arm and the right panel the Map-First (treatment) arm; the two distributions are similar, indicating the priming manipulation did not change elicitation quality.

↩ Back to text

sample. The specification of Table D.1 controls for absolute distance to the boundary with a separate slope on each side, because the signed distance interacted with the east-side indicator spans it exactly. Replacing the linear terms with side-specific quadratics or with side-by-500-meter-bin fixed effects leaves the east-side effect between −0.119 and −0.127. Restricting the sample to households within two kilometers of the boundary, 126 of them on the east side, gives −0.101 ($p = 0.14$). On the west side, Map-First households have lived in their *colonia* 10.6 fewer years than Map-After households (randomization-inference $p = 0.040$) and report belonging 1.4 points lower on the seven-point scale ($p = 0.014$), while no local-ties measure differs across arms on the east side. The west-side estimate is 0.144 in the full sample, 0.143 without the respondents who inferred the purpose of the study, and 0.099 when that inference is controlled for, and it is never distinguishable from zero. The randomization-inference p-value of the difference between sides ranges from 0.039 to 0.11 across these variants.

Table D.7: Robustness of the incentivized-choice-experiment estimates: leave-one-out

	Share of tutoring choices			Willingness to pay (MXN pesos)		
	OLS			Jack et al. (2022), two-step latent WTP		
	(1)	(2)	(3)	(4)	(5)	(6)
$\mathbb{1}\{Map\text{-}First_i\}$	−0.077		−0.117	−44.7		−70.9
	[−0.090, −0.067]		[−0.130, −0.103]	[−53.1, −38.5]		[−78.0, −62.3]
$\mathbb{1}\{Map\text{-}First_i\} \times \mathbb{1}\{East_i\}$		−0.267			−172.2	
		[−0.297, −0.195]			[−195.3, −127.8]	
Sample	Full	Full	East side	Full	Full	East side
Enumerator FE	✓	✓	✓	✓	✓	✓
Day FE	✓	✓	✓	✓	✓	✓
LASSO controls	✓	✓	✓	✓	✓	✓
Distance to boundary	✓	✓	✓	✓	✓	✓
Participants	239	239	193	239	239	193
Dep. var. mean	0.548	0.548	0.538	164.0	164.0	155.6
Within R^2	0.235	0.251	0.216	0.222	0.238	0.208

Notes: Leave-one-out robustness of the two pre-registered outcomes of the incentivized choice experiment, for the 239 participants. Columns (1) to (3) use the share of tutoring choices and columns (4) to (6) the pre-registered two-step latent willingness to pay of Jack et al. (2022) reported in Table D.1, in MXN. Each column repeats the corresponding column of Table D.1: the pooled Map-First effect (H1) in columns (1) and (4), the Map-First × East interaction (H2) in columns (2) and (5), and the Map-First effect estimated on the east-side subsample in columns (3) and (6). All specifications carry the enumerator and day fixed effects, the LASSO-selected controls, and the east-side-by-distance terms of the main table. Below each estimate, the range in brackets is the minimum and maximum of the estimate across the 239 re-estimations that each drop one participant, and each re-estimation repeats both steps of the latent estimator. The estimates reproduce those of Table D.1, which reports their inference, conventional and by randomization.

↩ Back to text

D.4 Experimenter-demand effects

A priming experiment invites the concern that respondents guess the hypothesis and answer accordingly. After the elicitation, an open-ended question (Q88) asks each respondent what they think the study is about; I classify these guesses with `claude-opus-4-8` against a fixed codebook (Section D.8.1) — the same procedure used for the map-drawing responses — coding whether each guess refers to the neighborhood, education, economic conditions, or security, and whether it connects the neighborhood to education ("guessed the mechanism"); the coding agrees closely with an independent keyword dictionary on the flags central to the demand check (Cohen's κ between 0.80 and 0.97 for the neighborhood, education, security, and guessed-mechanism indicators). Only a small minority — 30 of 239 (13%) — guess the mechanism. Panel B of Table D.8 regresses these indicators on the treatment. The Map-First coefficient on guessing the mechanism is +0.22 with a randomization-inference p-value of 0.12, so the manipulation does not significantly raise the inference, and the point estimate sits on the west side rather than on the east side. The Map-First×East interaction is −0.22, which leaves the probability of guessing essentially unchanged on the excluded side, and the effect on merely mentioning the neighborhood is smaller and further from significance. Whatever rise there is does not track the side on which the treatment moves behavior. Panel A re-estimates the east-side priming effect on both outcomes and on both willingness-to-pay measures, the pre-registered latent one and the bounded area. Controlling for the inferred themes gives −0.124 for the tutoring share, −75.3 pesos for latent willingness to pay and −30.6 for the bounded area; dropping the 30 guessers

gives −0.124, −78.8 and −30.6. Against baselines of −0.117, −71.1 and −28.6, every estimate is unchanged or slightly stronger. These checks are reassuring rather than decisive. Question Q88 is asked after both arms have completed the tasks, so it records what respondents infer at the end of the interview and not what they inferred at the moment of choice. The inferred themes and guesser status are themselves measured after treatment, so controlling for them or selecting on them shows that the estimate is stable rather than establishing that the intention-to-treat effect is free of demand. Panel C of Table D.8 re-estimates the interaction specification of Table D.1 under the same checks. The east-side effect is stable, and the cross-side contrast, including its western component, moves by 4.2 percentage points in the tutoring share with guessed-theme controls, from −26.7 to −30.9, and by 0.6 without the guessers, with randomization-inference intervals that include the headline contrast. These checks assess robustness to observed recognition of the study's theme, and they do not establish that the contrast is unaffected by experimenter demand, since guessing is measured after treatment and after the choices.

D.5 Heterogeneity within the east side, and the variation available

If the effect ran through household resources or inherited human capital rather than through the neighborhood, it should be concentrated among poorer or less-educated households. Table D.9 adds each moderator and its interaction with Map-First to the east-side-subsample specification. No interaction is significant. The effect is present among households with above-median income (−0.176), among those where a parent completed high school (−0.129), and among those that are both (−0.207), and in each case it is at least as large as among the households without those advantages.

A null interaction has two possible sources: the effect may not vary with the moderator, or the moderator may not vary in the sample. Table D.9 reports the mean and dispersion of each moderator among east-side households to separate them. Income, parental schooling, and grandparental schooling are dispersed on the eastern side, and about as dispersed as on the western side. The local-ties index varies more on the east side than in the west, with a standard deviation of 1.09 against 0.26, so the eastern sample is not a homogeneous block of households whose social relations are concentrated locally.

What the design cannot do is detect moderate interactions. With 193 east-side households, the interaction terms carry standard errors around 0.11 for a median split, so the smallest interaction detectable with 80 percent power is about 0.30, roughly two and a half times the effect being split. These rows weigh against the case where the effect exists only among constrained households, and they do not establish that the effect is uniform.

The individual measures of neighborhood ties behave like the index. Interacting Map-First with help received, hours with neighbors, relatives in the *colonia*, years of residence, the relative *barrio* premium, and belonging gives a joint randomization-inference p-value of 0.91, and interacting

Table D.8: Robustness to experimenter-demand effects

Panel A. Robustness of the priming effect on the east side

	Share of tutoring choices			WTP, latent (MXN pesos)			WTP, bounded area (MXN pesos)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Effect on the east side	−0.117**	−0.124**	−0.124**	−71.1**	−75.3**	−78.8**	−28.6**	−30.6**	−30.6**
	(0.052)	(0.051)	(0.056)	(33.9)	(33.3)	(36.6)	(13.6)	(13.3)	(14.7)
Sample	All	All	Non-guessers	All	All	Non-guessers	All	All	Non-guessers
Guess-theme controls		✓			✓			✓	
Enumerator FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Day FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
LASSO controls	✓	✓	✓	✓	✓	✓	✓	✓	✓
Distance to boundary	✓	✓	✓	✓	✓	✓	✓	✓	✓
p-value: east-side effect = west-side effect (robust / RI)	0.085 / 0.058	0.043 / 0.029	0.125 / 0.097	0.088 / 0.071	0.042 / 0.036	0.124 / 0.107	0.088 / 0.071	0.043 / 0.035	0.133 / 0.115
Participants	239	239	209	239	239	209	239	239	209
Dep. var. mean	0.548	0.548	0.538	164.0	164.0	150.7	130.6	130.6	127.9
R^2	0.371	0.396	0.377	0.372	0.394	0.386	0.366	0.390	0.375

Panel B. Effect of the treatment on the inferred topic (linear probability models)

	Guessed the mechanism	Mentioned the neighborhood
	(10)	(11)
$\mathbb{1}\{\textit{Map-First}_i\}$	0.217	0.178
	(0.102)	(0.139)
$\mathbb{1}\{\textit{Map-First}_i\} \times \mathbb{1}\{\textit{East}_i\}$	−0.224*	−0.131
	(0.111)	(0.155)
Sample	All	All
Participants	239	239
Dep. var. mean	0.126	0.360
R^2	0.030	0.008

Panel C. Effects by side in the specification of Table D.1

	Share of tutoring choices			WTP, latent (MXN pesos)		
	(12)	(13)	(14)	(15)	(16)	(17)
Effect on the east side	−0.123**	−0.129***	−0.130**	−74.6**	−77.9**	−82.5**
	(0.052)	(0.051)	(0.056)	(39.6)	(38.8)	(42.2)
	[−0.221, −0.025]	[−0.227, −0.031]	[−0.241, −0.020]	[−139.1, −10.2]	[−142.8, −13.1]	[−154.8, −10.7]
Effect on the west side	0.144	0.180	0.143	97.6	121.8	94.9
	(0.126)	(0.126)	(0.135)	(93.4)	(91.5)	(97.2)
	[−0.089, 0.373]	[−0.056, 0.411]	[−0.123, 0.399]	[−62.9, 255.7]	[−41.4, 280.9]	[−86.4, 267.3]
Difference, east − west	−0.267**	−0.309**	−0.274*	−172.2*	−199.7**	−177.4*
	(0.137)	(0.136)	(0.148)	(102.9)	(100.6)	(108.3)
	[−0.529, −0.013]	[−0.579, −0.051]	[−0.570, 0.010]	[−352.2, 1.5]	[−384.1, −22.6]	[−377.6, 14.0]
Sample	All	All	Non-guessers	All	All	Non-guessers
Guess-theme controls		✓			✓	
Enumerator FE	✓	✓	✓	✓	✓	✓
Day FE	✓	✓	✓	✓	✓	✓
LASSO controls	✓	✓	✓	✓	✓	✓
Distance to boundary, slope by side	✓	✓	✓	✓	✓	✓
Participants	239	239	209	239	239	209

Notes: The open-ended guesses to question Q88 ("what do you think the study is about?") are classified with `claude-opus-4-8` at temperature 0, one independent call per response, against a fixed codebook, and the structured outputs are archived with the replication package (`llm-coding/q88_opus_coded.csv`). Overall, 30 of 239 respondents (13%) guess the mechanism. Panel A reports the implied priming effect on the east side, the sum of the Map-First coefficient and its interaction with the east-side indicator, from the interaction specification of Table D.1 with a single linear slope in distance to the boundary, for the share of tutoring choices and for willingness to pay. Both willingness-to-pay measures are reported, the pre-registered latent two-step measure of Table D.1 and the bounded area under the respondent's demand curve, which stays within the price grid and so is smaller in magnitude. Columns (1), (4) and (7) are the baseline, columns (2), (5) and (8) add controls for the themes the guess mentions (neighborhood, education, economic conditions, security), and columns (3), (6) and (9) drop respondents who guessed the mechanism; for the latent measure, column (6) re-estimates its first step without them. In Panel A the p-value row tests that the east-side effect equals the west-side effect, the coefficient on the Map-First × East interaction of the same regression, with the robust standard error and by randomization inference that imposes a constant common effect before permutation. Panel B regresses, by linear probability model, an indicator for whether the guess named the neighborhood–education mechanism (column (10)) or mentioned the neighborhood (column (11)) on the Map-First treatment, the east-side indicator, and their interaction, without other controls. In Panel B the interaction row is itself that comparison. The treatment does not significantly raise the probability of guessing the mechanism, and the point estimate sits on the west side rather than on the east side, while the east-side priming effect is unchanged when the inferred themes are controlled for or the guessers dropped. Q88 is asked after both arms have completed the tasks, and the inferred themes and guesser status are measured after treatment, so these columns show that the estimate is stable without establishing that it is free of experimenter demand. Panel C re-estimates the interaction specification of Table D.1, columns (2) and (5), with the distance to the boundary on a separate slope on each side, under the same three checks, and reports the effect on each side and their difference; Panel A uses a single linear slope in distance, so its estimates differ slightly from the east-side row of Panel C. In Panels A and B, HC1 heteroskedasticity-robust standard errors are in parentheses. In Panel C, parentheses report HC1 standard errors for the share and, for the latent measure, subject-level Bayesian bootstrap standard errors that repeat both steps with 1,000 draws, as in Table D.1, and brackets report 95 percent randomization-inference intervals obtained by inverting the test; the interval for the difference holds the west-side effect at the pooled effect with the same controls, as the equality test does. Significance is assessed by randomization inference with 10,000 permutations of the Map-First label within side of the historical boundary, the randomization stratum. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

↩ Back to text

it with the average of each measure among the other sampled households of the same *colonia* gives 0.25. Outside the interaction, help received from neighbors is associated with a tutoring share 3.4 percentage points lower per standard deviation in the full east-side sample ($p = 0.074$) and 5.4 points lower in the Map-After arm ($p = 0.19$); no other measure predicts demand. The measures barely cluster by *colonia*, with intraclass correlations between −0.06 and 0.02 for all but the share of time spent with neighbors, so the sample cannot separate a payoff attached to the place from one attached to the household.

Table D.9: Heterogeneity of the Map-First effect within the east side, and the variation available

Moderator Z_i:	Share of tutoring choices							
	Local-ties index (per SD)	Local-ties index above median	Income bracket (1–10)	Income bracket above median	Parental years of schooling	A parent completed high school	A grandparent completed high school	Above-median income and a parent with high school
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\mathbb{1}\{Map\text{-}First_i\}$	−0.117	−0.116	−0.046	−0.072	−0.174	−0.093	−0.123	−0.059
	(0.054)	(0.087)	(0.122)	(0.074)	(0.205)	(0.096)	(0.065)	(0.071)
$\mathbb{1}\{Map\text{-}First_i\} \times Z_i$	−0.004	−0.007	−0.014	−0.104	0.005	−0.035	0.008	−0.148
	(0.045)	(0.111)	(0.026)	(0.114)	(0.015)	(0.107)	(0.107)	(0.116)
Effect when $Z_i = 1$		−0.124		−0.176		−0.129	−0.115	−0.207
		(0.067)		(0.090)		(0.059)	(0.089)	(0.093)
Sample	East side	East side	East side	East side	East side	East side	East side	East side
Moderator Z_i	✓	✓	✓	✓	✓	✓	✓	✓
Enumerator FE	✓	✓	✓	✓	✓	✓	✓	✓
Day FE	✓	✓	✓	✓	✓	✓	✓	✓
LASSO controls	✓	✓	✓	✓	✓	✓	✓	✓
Distance to boundary	✓	✓	✓	✓	✓	✓	✓	✓
p-value: interaction (randomization inference)	0.942	0.953	0.573	0.347	0.774	0.752	0.931	0.193
Minimum detectable interaction	0.126	0.310	0.073	0.320	0.043	0.300	0.300	0.324
Mean of Z_i, east side	0.08	0.54	4.27	0.41	12.40	0.68	0.37	0.37
SD of Z_i, east side	1.09		2.29		3.45			
Participants	193	193	174	174	193	193	190	174
Dep. var. mean	0.538	0.538	0.513	0.513	0.538	0.538	0.543	0.513
R^2	0.393	0.394	0.377	0.381	0.393	0.394	0.392	0.386

Notes: East-side households of the incentivized choice experiment. The dependent variable is the share of tutoring choices. Each column adds a moderator Z_i, named in the header, and its interaction with Map-First to the east-side-subsample specification of Table D.1, with enumerator and day fixed effects and the controls selected by LASSO, which include the distance to the boundary. For binary moderators the Map-First row is the effect at $Z_i = 0$ and the row *Effect when* $Z_i = 1$ adds the interaction; for continuous moderators the Map-First row is the effect at $Z_i = 0$, which for the local-ties index is the full-sample mean. The mean of the moderator among east-side households, with its standard deviation for continuous moderators, shows how much variation the interaction can use. The minimum detectable interaction is the interaction the design would detect with 80% power at the 5% level, 2.8 times its standard error. Randomization-inference p-values redraw Map-First within the east-side subsample, holding the moderator and the controls fixed, with 2,000 permutations. The local-ties index is the first principal component of the survey-module measures that differ across the boundary: allocation to the *barrio* stranger, giving strictly more to that stranger, the relative *barrio* premium, times helped by neighbors, hours with neighbors, and the reservation salary for university. Higher values indicate social relations more concentrated locally. It is measured after the randomized block, so its interaction is descriptive. HC1 standard errors are in parentheses. Stars appear only on the interaction and use its randomization-inference p-value. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

↩ Back to text

D.6 Neighborhood ties, placebo indices, and willingness to pay

Table D.11 repeats Table 3 with latent willingness to pay for tutoring as the dependent variable. Table D.10 reports the east-west difference and the slopes of both outcomes with other indices of neighborhood ties, among them the nine-measure index that adds the two allocations, the reservation salary, and the two vignettes.

D.7 Bringing the historical divide to mind: how schooling is explained

A second experiment embedded in the same survey, with a pre-registered secondary outcome, brings the historical boundary to mind through the expression associated with it rather than through the neighborhood as a place, and asks what changes in the explanations respondents give. They explain, in their own words, why a young person from their colonia would not continue studying after high school (question Q91), and whether Q90 — how often they hear or use *de la Calzada para allá* — is asked before or after that question is randomized, without stratification.

Table D.10: Neighborhood ties and the demand for tutoring, with other indices

Index built from	East − West (SD)	Share choosing tutoring (pp)			Willingness to pay (MXN)		
		East	West	p, equal slopes	East	West	p, equal slopes
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Help received, time with neighbors relative to work (Table 3)	0.44***	−4.0**	7.0	0.070	−27.0**	59.5	0.039
	(0.11)	(1.9)	(13.6)		(12.3)	(84.9)	
Nine measures, adding giving, the reservation salary, and the vignettes	0.53***	−4.2**	15.8	0.012	−28.1**	120.6	0.005
	(0.11)	(2.0)	(13.0)		(13.2)	(83.2)	
Help received, share of non-market time with neighbors	0.37***	−2.6	2.0	0.176	−18.8	28.0	0.119
	(0.11)	(1.9)	(12.3)		(12.2)	(76.4)	
Help received, time relative to work, giving	0.50***	−3.9**	6.9	0.072	−26.6**	58.9	0.041
	(0.11)	(2.0)	(13.7)		(12.9)	(85.9)	

Notes: Each index is the first principal component of the listed measures, standardized over the 239 households. Help received is the three items on help with child care, loans, and finding work, and the first row is the neighborhood-ties index of Table 3. Giving is the two allocations between a stranger from the respondent's own neighborhood and one from elsewhere, and the nine-measure index adds them, the reservation salary, and the two vignettes to the first row. Column (1) reports the east-minus-west difference in the index with the specification of Table C.4. Columns (2), (3), (5), and (6) report the slope of the outcome on the index, per standard deviation, on each side, with the specification of Table 3, and columns (4) and (7) test that the two slopes are equal in a model on both sides that interacts the index with the east-side indicator and lets the slope in distance to the boundary differ by side. Observations are 193 households on the east side and 46 on the west. Heteroskedasticity-robust standard errors are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

↩ Back to text

The randomization identifies the effect of bringing that expression to mind. It does not identify what the expression carries, which may be identity, stigma, geography, or perceived inequality, and the exercise does not separate them or claim to have activated a neighborhood identity.

Each answer is classified against a fixed codebook into the reasons it gives: the local place, the surrounding social environment, the family, material constraints, and the young person's own choices. Answers typically give more than one reason, so the categories are not exclusive. A second, independent classification of the same answers, blind to the arm and formed without access to the first, agrees closely, and reproduces the estimate below.

Table D.12 reports the results. On the excluded side the prime displaces the material explanation: the share of east-side respondents attributing non-continuation to money or the need to work falls by 15 percentage points, which survives a false-discovery-rate correction over the five outcomes of the panel, the four categories and the derived neighborhood-minus-economic contrast. On the west side the same displacement is 2 percentage points and indistinguishable from zero. The number of reasons respondents give shows no detectable difference across arms, so the shift is not an artifact of shorter answers. The table reports the two sides separately and no cross-side test. A displacement on the east side alongside a null on the west side does not establish that the two sides differ, and with 46 west-side respondents the design has little power to tell them apart.

Nearly 80% of respondents report hearing or using the expression. Among east-side respondents, references to money or the need to work fall from 91 to 76%. References to the neighborhood rise from 8 to 14%, and references to family from 10 to 14%, although these increases are individually imprecise. Combining the neighborhood and economic categories, the within-response difference shifts by 21 percentage points toward neighborhood explanations. Controls for the earlier map

Table D.11: Neighborhood ties and willingness to pay for tutoring

	Willingness to pay for tutoring (MXN)										
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Neighborhood-ties index	−27.0**								−32.7**	59.5	82.1
	(12.3)								(13.3)	(84.9)	(80.0)
Aspirations index		−4.9							−12.6		8.7
		(17.3)							(18.7)		(62.8)
Returns index			9.9						10.8		113.5*
			(18.7)						(20.1)		(58.6)
Work-over-study norms index				6.3					4.9		46.9
				(17.6)					(17.4)		(57.5)
Attachment index					14.8				18.5		−52.2
					(15.9)				(16.6)		(61.4)
Patience and risk index						5.6			8.0		10.7
						(15.9)			(16.1)		(48.6)
Socioeconomic index							−2.3		3.9		−150.7**
							(19.1)		(27.5)		(70.5)
Human capital index								−8.4	−7.9		99.0
								(18.5)	(25.0)		(60.2)
Sample	East side	East side	East side	East side	East side	East side	East side	East side	East side	West side	West side
Enumerator FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Day FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
LASSO controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Distance to boundary	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Map-First	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
East − West difference in the index (SD)	0.44***	0.10	−0.04	−0.03	0.02	0.18	−0.11	−0.18		0.44***	
	(0.11)	(0.15)	(0.15)	(0.13)	(0.14)	(0.16)	(0.16)	(0.15)		(0.11)	
p-value: east-side slope = west-side slope	0.039	0.857	0.139	0.850	0.909	0.624	0.825	0.738	0.012		
Observations	193	193	193	193	193	193	193	193	193	46	46
Dep. var. mean	155.6	155.6	155.6	155.6	155.6	155.6	155.6	155.6	155.6	199.3	199.3
R^2	0.412	0.401	0.402	0.401	0.404	0.401	0.401	0.401	0.421	0.521	0.641

Notes: The dependent variable is latent willingness to pay for one hour of tutoring in MXN, the pre-registered two-step estimator of Table D.1, and each coefficient is per standard deviation of the index. Columns (1) to (9) use east-side households and columns (10) and (11) west-side households; columns (9) and (11) include the eight indices together. Every column includes enumerator and survey-day fixed effects, Map-First assignment, the signed distance to the boundary, and the LASSO-selected controls of Table D.1. The row East − West reports, for the index that enters the column alone, the difference between the sides with the specification of Table C.4, without the respondent-education fixed effects for the socioeconomic and human capital indices, which would absorb their schooling measures. The p-value row tests that the slope of the index in the column (in column (9), the neighborhood-ties index) is equal on both sides, in a model on both sides that interacts each index with the east-side indicator and lets the slope in distance to the boundary differ by side. Each index is the first principal component of its measures over the 239 households, standardized over the 239. The neighborhood-ties index combines the number of times the household received help from neighbors in the past twelve months with child care or care when sick, with loans, and with finding work, coded from the answer bins 1, 2, 3, 4, and 5 as 0, 1.5, 4, 8, and 12 times, and time with neighbors relative to work time. Table D.10 reports the slopes with the nine-measure index that adds the two allocations, the reservation salary, and the two vignettes, and with other variants. The placebo indices combine aspiring to and expecting university for the focal child; expected earnings with high school and with university, the perceived return, and the belief that an excluded-side graduate earns less; the personal and the perceived work-over-study norm; belonging, the preference for another *colonia*, and the effort to have children know the neighbors; and patience and the willingness to take risks. The socioeconomic index combines log household income, log income per household member, years of schooling of the more-schooled parent and of the more-schooled grandparent, a formal salary as main income, and rooms per household member, and the human capital index combines years of schooling of the more-schooled parent, of the respondent's father, and of the respondent's mother (Table C.7). Heteroskedasticity-robust standard errors are in parentheses and stars use them; they do not correct for the generated regressand. All index measures were taken after the experimental block, and Map-First does not change the neighborhood-ties index. These analyses were not pre-registered. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

↩ Back to text

assignment, interview length, and respondent demographics leave the estimates similar. The exercise identifies an effect of bringing the historical divide to mind on explanations. It does not identify the historical assignment's effect on those explanations or any component of the opportunity cost in Section 4.4, and it is not a replication of the tutoring experiment, whose treatment and outcome differ.

Table D.12: The Calzada expression and explanations of educational non-continuation

	Economic		Neighborhood		Family		Individual		Neighborhood − economic	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Panel A. East side, historically excluded bank										
$\mathbb{1}\{Boundary\ primed_i\}$	−0.150***	−0.170***	0.059	0.067	0.048	0.039	0.041	0.044	0.209***	0.237***
	(0.054)	(0.055)	(0.046)	(0.046)	(0.047)	(0.047)	(0.068)	(0.070)	(0.079)	(0.079)
FDR q-value	0.020	0.005	0.350	0.170	0.395	0.320	0.637	0.320	0.020	0.005
Participants	193	193	193	193	193	193	193	193	193	193
Control mean (not primed)	0.905	0.905	0.084	0.084	0.095	0.095	0.316	0.316		
R^2	0.040	0.090	0.009	0.054	0.006	0.037	0.002	0.011	0.037	0.095
Panel B. West side										
$\mathbb{1}\{Boundary\ primed_i\}$	−0.023	−0.107	−0.057	−0.129	0.078	0.126	−0.194	−0.149	−0.034	−0.022
	(0.101)	(0.105)	(0.113)	(0.120)	(0.119)	(0.131)	(0.146)	(0.160)	(0.163)	(0.183)
FDR q-value	1.000	0.791	1.000	0.791	1.000	0.791	1.000	0.791	1.000	0.791
Participants	46	46	46	46	46	46	46	46	46	46
Control mean (not primed)	0.880	0.880	0.200	0.200	0.160	0.160	0.480	0.480		
R^2	0.001	0.173	0.006	0.141	0.010	0.067	0.039	0.071	0.001	0.012
Adjustment controls		✓		✓		✓		✓		✓

Notes: At the end of the survey, respondents explain in their own words why a young person from their *colonia* might not continue studying after high school. I randomize whether respondents are first asked how often they hear or use the expression "on the other side of the Calzada" (*de la Calzada para allá*), associated with the historical divide. *Boundary primed* equals one for respondents who receive this item before the open-ended education question and zero for those who receive it afterward (*Not primed*). The order is randomized without stratification. Each answer is classified by `claude-opus-4-8` using a fixed codebook. Categories are not mutually exclusive because respondents can give multiple reasons. *Economic* includes money and the need to work. *Neighborhood* includes explanations referring to the local place or surrounding social environment, including friends and what other young people nearby do. *Family* and *Individual* capture explanations referring to the family and the young person's own choices, respectively. In columns (1) to (8) the dependent variable is an indicator for mentioning the category. In columns (9) and (10) it is the within-respondent difference between the neighborhood and economic indicators, which takes the values −1, 0 and 1. That difference answers whether the prime shifts the relative weight of the two explanations rather than the level of either, so it is treated as an outcome in its own right and enters the false-discovery-rate correction; it is not a share, so its control mean is omitted. Odd columns report the difference between the Boundary-primed and Not-primed groups. Even columns report the same difference after partialling out the adjustment controls: the randomized Map-First assignment from the earlier module, the duration of the interview in minutes, and the respondent's sex, age and age squared. Q91 is asked at the end of the survey, so the adjustment absorbs both the earlier experimental arm and respondent fatigue; duration is winsorized at its 95th percentile because a few sessions were left open. The FDR q-value row reports sharpened false-discovery-rate values (Anderson, 2008) within each panel, computed separately for the unadjusted and adjusted estimates, over the five outcomes of the panel. Inference is by randomization inference using 50,000 permutations of question order, conducted within side. The values in parentheses are the standard deviation of the estimate across those permutations, and stars use the randomization-inference p-values. Because the reference distribution is generated under the sharp null, the adjusted p-values are exact whether or not the adjustment model is correct, and the controls affect only precision. Counting the four categories, respondents give a similar number of reasons in the two treatment arms (1.47 versus 1.42). An independent classification of the same answers, conducted without access to the first, yields mean Cohen's $\kappa = 0.83$ across categories and $\kappa = 0.94$ for the economic category. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

↩ Back to text

D.8 Incentivized choice experiment: additional design detail

D.8.1 Survey instrument

The interview has a median duration of 26.5 minutes, and the instrument includes, in order: (i) household and parental demographics; (ii) a child-by-child loop for each child aged 6–20; (iii) parental work and time use; (iv) the price-list instructions, the practice round, and the four comprehension checks, all administered before the randomization and therefore identical across arms; (v) the randomized block, in which the price-list elicitation and the interactive map-drawing module with its open-ended follow-up are administered in the order given by treatment assignment, with the 1-in-10 ex-post lottery resolved after the price list (Figure D.1); (vi) focal-child selection; (vii) subjective lifetime returns to schooling; (viii) aspirations and expectations; (ix) the misperceptions module (peer beliefs at the $100 and $200 reference prices); (x) personal

and peer norms about work-versus-study; (xi) trust and participation items in JCV format; (xii) the Martínez–Ramírez vignettes; (xiii) network benefits (emergencies, loans, job-finding); (xiv) hours-with-neighbors; (xv) belonging and identity; (xvi) residential stability and roots, including 1992-explosion exposure; (xvii) preferences (qualitative GPS items); (xviii) disinterested dictator games and an altruism item; (xix) the demand-effects open-ended question; and (xx) the secondary priming experiment on *"de la Calzada para allá."* The complete instrument is reproduced in the supplementary materials.

D.8.2 Randomization protocol

All randomizations are implemented within SurveyToGo at survey start. Three independent randomizations are conducted: (1) the map treatment (Map-First vs Map-After), a fair binary draw stratified through the sampling quotas; (2) the local-expression priming (secondary), an independent fair binary draw; and (3) survey length (core 35-minute vs extended 42-minute versions), with all primary outcomes present in both. Treatment assignment is logged at survey start and exported with the response data; enumerators learn the assignment only when the software prompts the corresponding module.

D.8.3 Price-list comprehension protocol

The protocol consists of (a) a six-row practice round (BIC pen vs. marzipan candies); (b) four comprehension checks covering the random-row mechanism, the binding choice, the inability to revise ex post, and the 1-in-10 disbursement; and (c) a re-explanation step if any check fails: an incorrect answer triggers the relevant instructions before the question is re-administered, and the original answer remains recorded. In the main sample the mean score is 3.8 out of 4. The binding row is determined by a numbered slip that the respondent draws from a sealed envelope; the binding-row procedure and the one-in-ten payment probability are explained before the task, and all households retain the MXN 100 participation payment regardless of the experimental outcome. Pilot results show 92% comprehension on the first pass and 100% after one round of clarification.

E The Framework: Derivations

E.1 The Decision to Educate

Setup. Household i lives in neighborhood ℓ and has a child aged n. A share q_ℓ of households in the neighborhood take part in reciprocal exchange with neighbors, and Z_ℓ is the quality of the outside channels, markets and public services, through which households obtain the same goods when they do not. A household that takes part obtains R when a neighbor can help, with probability q_ℓ, and Z_ℓ from outside channels otherwise, where R is the value of a unit of help from a neighbor. The expected payoff from taking part is $q_\ell R + (1 - q_\ell)Z_\ell$, and its value over the outside alternative, Z_ℓ, is

$$\Delta_\ell = q_\ell\,(R - Z_\ell), \qquad \frac{\partial \Delta_\ell}{\partial q_\ell} = R - Z_\ell, \qquad \frac{\partial \Delta_\ell}{\partial Z_\ell} = -q_\ell < 0, \tag{E.1}$$

a linear version of the participation externality in Kranton (1996). Exchange is worth more when more neighbors take part, as long as help from a neighbor is worth more than the outside channel, $R > Z_\ell$, and less when outside channels are better. Household i draws on exchange with intensity $\beta_i \geq 0$, so it obtains $\beta_i \Delta_\ell$ from taking part, and $\beta_i = 0$ for households outside exchange.

Educating the child yields a return ρ, common to all households, at a private cost C_i that falls with school provision, parental human capital, and household resources. Education also puts a share $\psi_n \in [0, 1)$ of the value of exchange at risk, because a child in school is less available to the neighbors, delays the earnings with which the household reciprocates, and is more likely to leave the neighborhood when grown, and ψ_n rises with the age n of the child.[33] When parents choose, they weigh everything that concerns the neighborhood by $a(s) \in (0, 1]$, which rises with how present the neighborhood is in their minds, s (Gabaix, 2019).

The investment condition. If the household educates the child, it obtains $\rho - C_i$ from education and keeps $(1 - \psi_n)\beta_i \Delta_\ell$ from exchange, weighted by $a(s)$. If it does not, it obtains nothing from education and keeps $\beta_i \Delta_\ell$. It educates if

$$\rho - C_i + a(s)(1 - \psi_n)\beta_i \Delta_\ell \;\geq\; a(s)\beta_i \Delta_\ell,$$

that is, if

$$\rho - C_i \;\geq\; a(s)\,\Pi_i, \qquad \Pi_i \equiv \psi_n\,\beta_i\,\Delta_\ell,$$

which is (3) in Section 5. Π_i is the neighborhood cost of schooling, an opportunity cost that the household bears through its child, the part of the value of exchange that schooling puts at risk. It is zero for households outside exchange and larger when the child is older, the household relies more on neighbors, more neighbors take part, and outside channels are worse.

[33] Neighbors can also support schooling, by sharing information or looking after children. If this support is worth γ_i, every Π_i below should be read net of it, as $\psi_n \beta_i \Delta_\ell - \gamma_i$, and education and exchange are complements when the support exceeds what education puts at risk. None of the derivations change.

Four predictions. Each follows from (3) in one step.

Bringing the neighborhood to mind. Let $a_0 < a_1$ be the weights in the Map-After and Map-First arms. A household educates in Map-After but not in Map-First if $a_0\Pi_i \leq \rho - C_i < a_1\Pi_i$. This interval is empty when $\Pi_i = 0$ and widens with Π_i, so bringing the neighborhood to mind lowers demand among households in exchange and not among the rest, and lowers it more where Δ_ℓ is larger. For a marginal investment such as an hour of tutoring, willingness to pay is proportional to $\rho - C_i - a(s)\Pi_i$, and its drop from Map-After to Map-First is $(a_1 - a_0)\Pi_i$.

Reliance on neighbors. At any $a(s)$, the right-hand side of (3) rises with β_i, so among households in exchange, demand falls with reliance on neighbors in both arms.

Age of the child. Π_i rises with ψ_n, so the gap between households in exchange and the rest is larger for older children.

Required return. The return at which household i is indifferent is $\bar{\rho}_i = C_i + a(s)\Pi_i$. Two households with the same cost and the same expected return require different returns if their neighborhood costs differ.

The gap and provision. Suppose the return and the distribution of C_i are the same on both sides of the boundary, and let F be the distribution of $\rho - C_i$. A household outside exchange educates with probability $1 - F(0)$ and a household in exchange with probability $1 - F(a_0\Pi)$, where Π is the neighborhood cost of a household in exchange on that side. Attendance on side ℓ is

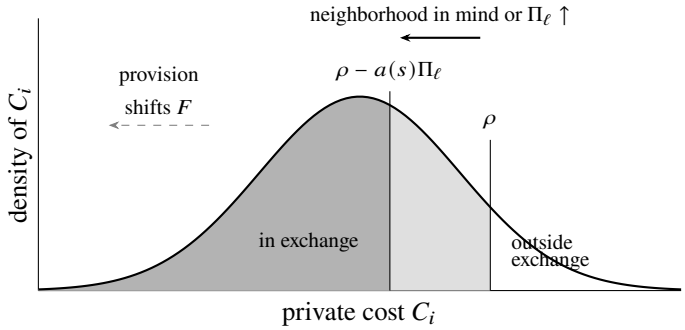
$$A_\ell = 1 - F(0) - q_\ell\big[F(a_0\Pi_\ell) - F(0)\big], \tag{E.2}$$

and the gap between west and east is

$$A_W - A_E = q_E\big[F(a_0\Pi_E) - F(0)\big] - q_W\big[F(a_0\Pi_W) - F(0)\big]. \tag{E.3}$$

The gap rises with q_E and with Π_E, and so with Δ_E. A common reduction in private costs shifts the distribution of investment margins on both sides. Its effect on the attendance gap depends on the probability mass between the two thresholds, so the gap can widen, narrow, or remain unchanged. Persistent neighborhood costs need not disappear when provision improves, although a sufficiently large improvement can take every household past both thresholds and close the attendance gap while the cost remains. Provision convergence lowers C_i, and the standing of residents in formal institutions, which enters Z_ℓ, is what did not converge, so q_ℓ and Δ_ℓ need not fall with C_i, and the framework does not derive the time path of the gap from either. What is left once provision has converged is the part of Δ_E that comes from the participation of neighbors and from the social part of outside channels, and Appendix E.3 shows why that part need not converge on its own.

Figure E.1: The decision to educate



Notes: The figure shows the distribution F of private costs and the two thresholds of condition (3): households outside exchange educate when $C_i \leq \rho$ and households in exchange when $C_i \leq \rho - a(s)\Pi_\ell$, and attendance in each group is the shaded mass to the left of its threshold. Bringing the neighborhood to mind or a larger neighborhood cost moves the second threshold left; better provision shifts F and leaves the distance between thresholds unchanged.

↩ Back to text

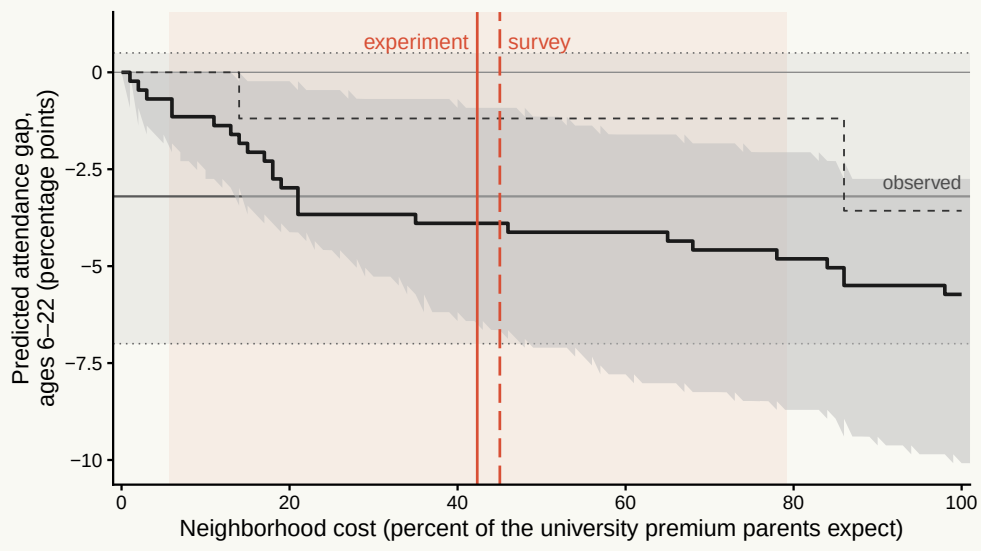
E.2 A Back-of-the-Envelope Calculation of the Neighborhood Cost

Inputs. For each parent, the survey records the monthly salary they expect a university graduate to earn at age 30 and the lowest salary at which they would consider university worthwhile for a young person from their neighborhood; both questions were asked after the mapping task in both arms. The margin is the difference between the two, and the investment passes the condition in (3) when the margin is positive. East-side parents expect a salary that is not detectably different and require MXN 5,319 more, 18 percent, and this difference is the neighborhood cost.

A lower bound from the experiment. Under the assumptions below, the experiment places a lower bound on the neighborhood cost of tutoring as a share of its value, since the prime lowers east-side willingness to pay for tutoring by between 21 percent on the choices within the grid and 42 percent under the choice model, from MXN 176 to MXN 102 (32 percent with explicit censoring), with a randomization-inference interval from 6 to 79 percent for the model-based figure. Figure E.2 places this bound and the survey's 45 percent on one axis, the share of each household's stated premium, and reports the predicted gap at every value of the cost.

Assumptions behind the experimental bound and the survey measure. Willingness to pay for an hour of tutoring is proportional to $\rho - C_i - a(s)\Pi_i$, and the weight without the prime, written a_{0i} here, may differ across households. With a_1 close to one and $a_{0i} \geq 0$, the proportional drop in willingness to pay, $\mathbb{E}[(a_1 - a_{0i})\Pi_i]/\mathbb{E}[\rho - C_i - a_{0i}\Pi_i]$, is at most $\mathbb{E}[\Pi_i]/\rho$ when C_i is close to zero, as for an hour of free online tutoring, and the average cost is below the return, and it is strictly below it if $a_{0i}\Pi_i$ is positive for some households. Its denominator is the net value of tutoring. If attention without the prime rises with reliance on neighbors, as the pattern in Section 4.6 suggests, the households with the largest costs contribute least to the drop, and

Figure E.2: The predicted attendance gap as a function of the neighborhood cost



Notes: The horizontal axis is the neighborhood cost as a share of each household's stated university premium, and each point on the line imposes a cost equal to that share of each household's stated premium. The line reports the attendance gap at the boundary among residents aged 6–22 that the calculation of Appendix E.2 implies at each such cost, using the margins of all 239 households computed with the 57 percent premium the census pays, after recentering east-side reservation salaries by the MXN 5,319 gap; the band is the 95 percent bootstrap interval over survey households. At 45 percent the curve gives the predictions of Figure 10. The dashed line uses the 46 west-side households only. The horizontal line and band are the observed gap and its 95 percent interval (Section 3). The vertical lines mark the model-based 42 percent, the experimental lower bound on the neighborhood cost of an hour of tutoring, with its randomization-inference interval shaded, and 45 percent, the survey measure from the reservation salary. On this axis the experiment bounds the share from below and the survey measures it, and the calibration uses the survey measure; Appendix E.2 states the assumptions behind each.

↩ Back to text

each of the 21 to 42 percent estimates is, under these assumptions, a lower bound on the average neighborhood cost of the tutoring good and not its level. A weight with the prime below one lowers the drop further, and a positive private cost of tutoring would raise it, which is why the bound requires that cost to be close to zero. The bound also requires that the prime leaves the value of the gift card unchanged, since if the prime raised it, part of the drop would measure that change and not the neighborhood cost.

The reservation salary of household i is the return at which it is indifferent, $\bar{\rho}_i = C_i + a_1 \Pi_i$, since the question came after the mapping task in both arms. The difference in average thresholds between the east and the west is

$$(C_E - C_W) + a_1 \, (\Pi_E - \Pi_W),$$

with C_ℓ and Π_ℓ the averages on side ℓ. With private costs drawn from the same distribution F on both sides, as in (4), the first term is zero on average, and with a_1 approximately one and Π_W small, the difference identifies Π_E. Divided by the university premium parents state, it gives the share, about 45 percent. This restriction on private costs concerns university and is separate from

the negligible private cost in the experimental bound, which concerns an hour of tutoring. The reservation-salary prompt includes tuition, materials, transportation, and forgone earnings, so part of the east-side gap could reflect private costs rather than the neighborhood cost, for instance longer commutes on the east side (Table B.5). Conditioning on the survey's measures does not identify the split, as Section 5.1 reports (Table C.6). The survey does not measure commute time, so that channel is not tested.

The two ratios also refer to different investments. The experiment prices an hour of online tutoring, which does not reduce a child's availability, delay earnings, or move the child away in the way continued enrollment does, so its neighborhood cost need not equal that of schooling. The calibration uses the survey, which asks about university, and the experiment enters as evidence that the channel operates on a marginal educational investment and as a bound on its size for that good.

Returns. Parents on both sides expect a university premium of about 100 percent over upper-secondary schooling, against 57 percent in the 2020 census for Guadalajara. I replace the expected graduate salary of every parent with the upper-secondary salary they expect times 1.57, which lowers the share of west-side parents for whom university passes the condition from 30 to 11 percent. Across respondents, with the controls of Panel D of Table C.5, the elasticity of the reservation salary to the expected university salary is 0.66 (standard error 0.07, 238 households), 0.65 on the east side and 0.65 on the west, and its elasticity to the expected upper-secondary salary is 0.48 (0.08). The survey does not distinguish whether the thresholds are anchored in pesos or in beliefs, so keeping them fixed while correcting the expected earnings is an assumption of the calculation. Letting the thresholds move with the corrected earnings with an elasticity of 0.66 widens the implied gap to 7.6 points, with a 95 percent bootstrap interval from 1.8 to 12.6, against 3.9 with the thresholds fixed (7.3 to 8.2 points over the interval of the elasticity). A proportional correction for each parent leaves no west-side parent above the threshold and cannot be used.

From margins to attendance. With the corrected returns, the headline cost moves 17 of the 239 households, 7.1 percent, below the threshold, those whose margin lies between zero and the cost (63 households, 26.4 percent, with the returns parents state). That share is the predicted attendance gap at the ages the neighborhood cost affects, so the west side is the anchor and the framework predicts only the difference, as the boundary design does, and the calculation uses the distribution of margins in the survey without assuming its shape. The headline predictions use the margins of all 239 households, with each east-side reservation salary net of the MXN 5,319 gap, which recovers the margin the household would have without the neighborhood cost under the assumption that the cost is the same across the east side; Table E.1 also reports a fixed cost of MXN 5,319 for every household, which moves 31 of the 239, and the version that uses the 46 west-side households alone, where 4 households near the threshold drive the prediction. For ages below 15 the predicted gap is zero, since the neighborhood cost rises with the age of the child;

for years of schooling, each year of age with a predicted attendance gap g subtracts g years, as in a steady state.

Results. Figure 10 reports each comparison, and Table E.1 the numbers.

Table E.1: Gaps implied by the neighborhood cost and gaps observed

	Observed		Predicted, headline (45% of each household's stated premium, census returns)			Predicted (fixed MXN 5,319, census returns)			Predicted (45% of each household's stated premium, returns parents state)			Predicted, west side only (46 households, fixed MXN 5,319)		
	Estimate	95% interval	Estimate	95% bootstrap interval	p-value: predicted = observed	Estimate	95% bootstrap interval	p-value: predicted = observed	Estimate	95% bootstrap interval	p-value: predicted = observed	Estimate	95% bootstrap interval	p-value: predicted = observed
Panel A. School attendance (percentage points)														
At the boundary, ages 6–22 (Census 2020)	−3.2	[−7.0, 0.5]	−3.9	[−6.7, −0.9]	0.938	−7.1	[−14.2, −0.5]	0.416	−14.4	[−15.4, −6.6]	0.011	−4.8	[−9.5, 0.0]	0.834
At the boundary, comparable households, ages 12–18 (Intercensal)	−5.1	[−9.6, −0.6]	−4.2	[−7.2, −1.0]	0.620	−7.7	[−15.5, −0.5]	0.680	−15.7	[−16.7, −7.2]	0.053	−5.2	[−10.4, 0.0]	0.836
Southern point 4, ages 6–22 (rd2d)	−12.6	[−18.5, −6.7]	−3.9	[−6.7, −0.9]	0.007	−7.1	[−14.2, −0.5]	0.196	−14.4	[−15.4, −6.6]	0.639	−4.8	[−9.5, 0.0]	0.047
Southern point 5, ages 6–22 (rd2d)	−11.8	[−18.9, −4.7]	−3.9	[−6.7, −0.9]	0.033	−7.1	[−14.2, −0.5]	0.296	−14.4	[−15.4, −6.6]	0.819	−4.8	[−9.5, 0.0]	0.106
Panel B. Years of schooling														
At the boundary, ages 6+ (Census 2020)	−0.79	[−1.10, −0.49]	−0.58	[−0.99, −0.14]	0.304	−1.06	[−2.13, −0.07]	0.758	−2.16	[−2.30, −0.99]	0.025	−0.71	[−1.42, 0.00]	0.702
Households the cost moves below the threshold				17 of 239			31 of 239			63 of 239			4 of 46	

Notes: Observed estimates come from Table B.1 (Census 2020), column 5 of Table 1 (Intercensal), and Table B.3 (rd2d, pointwise intervals from the reported standard errors). The headline schedule imposes on each household a cost equal to 45 percent of the university premium it states, which averages MXN 5,267, on margins computed with the 57 percent premium the census pays for all 239 households, and the share of households it moves below the threshold is passed through the age structure of each estimate. Before any cost is applied, east-side reservation salaries are recentered by the MXN 5,319 east-west gap, so that the bars of the two sides are measured from the same baseline, and this recentering is separate from the cost. The fixed schedule subtracts MXN 5,319 from the margin of every household, with the same households, returns, and recentering. The third predicted column applies the headline schedule to margins computed with the premium parents state. The west-side column changes both the sample, the 46 west-side households alone, and the cost, a fixed MXN 5,319, with the 57 percent premium the census pays. MXN 5,319 is about 73 percent of the premium the census pays as a ratio of means. Bootstrap intervals come from 1,000 resamples of the survey households within each side, with the census inputs held fixed. The p-value tests that the predicted gap equals the observed one, treating the census estimate and the survey prediction as independent: each bootstrap draw of the prediction is paired with draws of the observed estimate from a normal distribution with its standard error, and the p-value is twice the smaller tail probability of their difference. The predictions use the survey measure of the cost, and the experimental estimate, a lower bound on the neighborhood cost of an hour of tutoring, does not enter them; Appendix E.2 states the assumptions behind each.

↩ Back to text

Units. The MXN 5,319 reservation-salary gap is about 45 percent of respondents' stated university premium and about 73 percent of the corrected premium as a ratio of means, and about 93 percent of the corrected premium at the median household. These are normalizations of one monetary amount, and changing the denominator alone leaves predicted decisions unchanged. The calibration imposes a cost proportional to each household's stated premium, 45 percent of it, which averages MXN 5,267, and a fixed cost of MXN 5,319 for every household is a different specification, whose predictions Table E.1 reports separately. A point on the pooled curve of Figure E.2 is a cost equal to that share of each household's stated premium, so the curve at 73 percent, which gives −4.6 percentage points, is a different assumed cost and not the fixed MXN 5,319 cost expressed on another axis.

What the calculation assumes. The cost imposed is a share of each household's stated premium rather than a fixed amount, so households that expect larger premia bear larger costs in pesos, as the framework's $\Pi_i = \psi_n \beta_i \Delta_\ell$ implies for households that rely more on exchange. Parents decide with the neighborhood in mind, as when they answered the reservation-salary question; the two sides share the distribution of margins except for the neighborhood cost; every parent expects the census premium; years of schooling are in steady state, so the schooling prediction is a long-run benchmark rather than a decomposition of the 2020 stock; the east-side gap that the recentering removes is the same for every east-side household; and, for the experimental bound, that willingness to pay for one hour of tutoring is proportional to the net value of education, as Appendix E.1 assumes. It also rests on the assumptions behind the experimental bound and the survey measure above and on the three listed at the end of Section 5.1, that the university margin stands in for the decision to stay in school at the transition to upper secondary, that the reservation salary concerns a young person from the respondent's neighborhood and not the respondent's own child, and that the experimental bound is imprecise, with a randomization-inference interval from 6 to 79 percent. Each of these can be relaxed, and the one that matters most is the return: with the returns parents state, the predicted attendance gap is three to five times the observed one.

E.3 Persistence of Participation

The static framework takes q_ℓ as given. The disposition to take part in exchange passes from parents to children, following Bisin and Verdier (2001), and the question is whether q_ℓ converges once outside channels do. I drop the subscript ℓ and write q_t for the share of adults in generation t who take part in exchange.

Step 1: How a child acquires the disposition. Each adult has one child. A parent who takes part in exchange tries to pass the disposition on and succeeds with probability τ_1, which the parent chooses. If the attempt fails, the child adopts the disposition of an adult drawn at random from the neighborhood, who takes part with probability q_t. The child of a participating parent takes

part with probability

$$P^{11} = \tau_1 + (1 - \tau_1)\, q_t.$$

In the same way, a parent who does not take part passes on that disposition with probability τ_0, and otherwise the child adopts the disposition of a random adult, so the child of a non-participating parent takes part with probability

$$P^{01} = (1 - \tau_0)\, q_t.$$

Step 2: What parents gain from passing on the disposition. Parents value their child's disposition with their own preferences. A participating parent values a child who takes part over one who does not by $V^1 = \lambda\Delta(q_t, Z)$, the value of keeping access to exchange, with $\lambda > 0$ and Δ as in (E.1), evaluated at the current share q_t. A non-participating parent values a child who relies on outside channels over one who takes part by $V^0 = b(Z)$, which rises with the quality of outside channels.

Step 3: How much effort parents put in. Passing on the disposition costs $\tau^2/2k$. A participating parent chooses τ_1 to maximize

$$P^{11}V^1 - \frac{\tau_1^2}{2k} = \big[q_t + \tau_1(1 - q_t)\big]V^1 - \frac{\tau_1^2}{2k},$$

and the first-order condition $(1 - q_t)V^1 - \tau_1/k = 0$ gives

$$\tau_1 = k\,(1 - q_t)\,\lambda\,\Delta(q_t, Z).$$

By the same steps, a non-participating parent chooses

$$\tau_0 = k\, q_t\, b(Z).$$

A non-participating parent tries harder when its own disposition is rare in the neighborhood, since the random adult is then less likely to pass it on. A participating parent weighs that same force against the value of exchange, which by (E.1) is small when few neighbors take part, so its effort, $k\lambda q_t(1 - q_t)(R - Z)$, is largest at intermediate shares and rises with $R - Z$.

Step 4: The share of participants in the next generation. The share of children who take part is the share of participating parents times P^{11} plus the share of non-participating parents times P^{01},

$$q_{t+1} = q_t\big[\tau_1 + (1 - \tau_1)q_t\big] + (1 - q_t)(1 - \tau_0)\, q_t.$$

Expanding and collecting terms,

$$q_{t+1} = q_t + q_t(1 - q_t)\,(\tau_1 - \tau_0),$$

and substituting the efforts from Step 3,

$$q_{t+1} - q_t = k\, q_t(1 - q_t)\Big[(1 - q_t)\,\lambda\,\Delta(q_t, Z) - q_t\, b(Z)\Big]. \tag{E.4}$$

Participation grows when the term in brackets is positive, that is, when the effort of participating parents, weighted by how many non-participants their children may meet, exceeds the effort of non-participating parents, weighted by how many participants their children may meet. Substituting $\Delta(q_t, Z) = q_t(R - Z)$ from (E.1),

$$q_{t+1} - q_t = k\, q_t^2 (1 - q_t) \Big[(1 - q_t)\, \lambda\, (R - Z) - b(Z) \Big]. \tag{E.5}$$

Step 5: Where participation settles. The share stops moving at $q = 0$, at $q = 1$, and where the last bracket in (E.5) is zero,

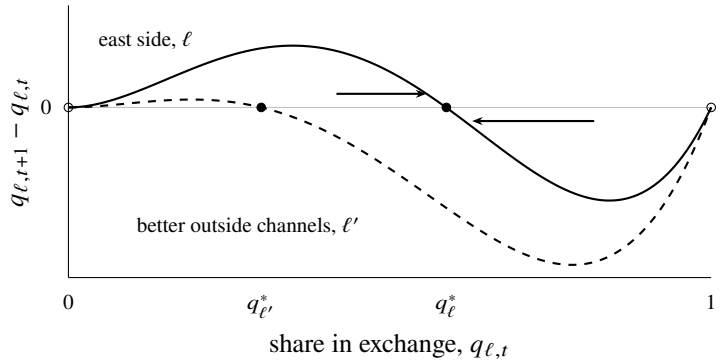
$$q^* = 1 - \frac{b(Z)}{\lambda\, (R - Z)}, \tag{E.6}$$

which lies strictly between zero and one when $\lambda(R-Z) > b(Z)$. An interior share of participating households exists when the value of exchange, scaled by λ, exceeds the stake of a non-participating parent in transmitting the opposite disposition, and otherwise the bracket is negative at every positive share and participation goes to zero. If $R < Z$, exchange is worth less than the outside channel at any share, $\Delta < 0$, and no household takes part in the long run. Under $\lambda(R-Z) > b(Z)$ the last bracket falls with q, so the share rises below q^* and falls above it. The slope of the right-hand side of (E.5) at q^* is $-k\, q^{*2}\, b(Z)$, so q^* is stable when $k\, q^{*2}\, b(Z) < 2$, and the share approaches it without overshooting when $k\, q^{*2}\, b(Z) < 1$, both of which hold for a small enough k. Participation then returns toward q^* after a temporary shock. The rest point at zero is unstable when q^* exists, although the share moves away from it slowly, since the right-hand side of (E.5) is of order q^2 near zero. By (E.6), q^* falls when outside channels improve, since $\partial q^*/\partial Z = -[b'(Z)(R - Z) + b(Z)]/[\lambda(R - Z)^2] < 0$ with $b'(Z) > 0$, and it rises when help from neighbors gains value relative to outside channels, since $\partial q^*/\partial R = b(Z)/[\lambda(R - Z)^2] > 0$.

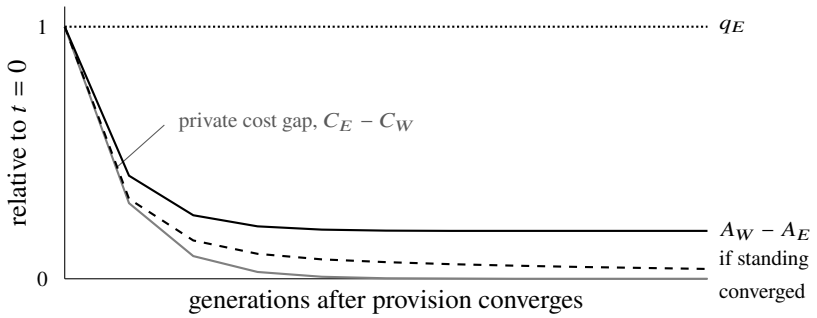
Step 6: Why the social legacy outlasts the gap in access. Consider the east side, where exclusion left poor outside channels, Z_E low, and so a high q_E^* by (E.6). Now let public provision converge, which lowers the private cost of education C_i, since it depends on the schools that exist today. The share of participants and the value of exchange move only if Z_E moves, and Z_E includes the standing of residents in formal institutions, which did not converge. If Z_E stays where it was, q_E stays at q_E^*, Δ_E stays high by (E.1), and so does the neighborhood cost $\Pi_E = \psi_n \beta \Delta_E$ in (3). If standing improves, q_E moves toward a lower rest point only through (E.5), one generation at a time, at a rate that depends on k. By (E.3), the fall in C_i moves the attendance gap through the mass of households between the two thresholds, and a gap remains while q_E and Δ_E stay above their western values and some households lie between the thresholds. This is consistent with the pattern in Section 3, a gap much larger in 1930 that has not narrowed detectably since 2000, and the framework does not derive that path.

Figure E.3b illustrates one parameterization of this sequence, in which the private cost gap closes within a generation, the share in exchange does not move, and the attendance gap implied by (4) falls to the level that participation and the value of exchange on the east side sustain. The

Figure E.3: The transmission of participation and the persistence of the social legacy



(a) Transmission of participation



(b) Access and the social legacy over time

Notes: Panel (a) plots the law of motion (5) with $\Delta_\ell = q_\ell(R - Z_\ell)$ for the east side (solid) and for a neighborhood with better outside channels (dashed); filled circles are the interior rest points, hollow circles the rest points at zero and one, and arrows show the return after a one-time change in the share in exchange. In panel (a), $\lambda(R - Z_\ell)$ is 0.8 and $b(Z_\ell)$ is 0.33 on the east side, 0.6 and 0.42 in the other neighborhood, and k is 1.2. Panel (b) illustrates one parameterization of access and the social legacy after provision converges, with the panel (a) values for the two sides. The private cost gap closes within a generation, the share in exchange does not move while the standing of east-side residents in formal institutions does not converge, and the attendance gap implied by (4) falls to the level that participation and the value of exchange on the east side sustain; the dashed line lets standing converge as well. Each series is divided by its value at $t = 0$. Parameters are illustrative and not estimated, and other parameterizations give other paths, since the response of the gap depends on the mass of households near the two thresholds.

↩ Back to text

dashed line lets standing converge as well, so that q_E moves toward the western rest point one generation at a time and the gap keeps closing, and in this parameterization what holds the gap open is participation together with the standing residents have outside the neighborhood.

Step 7: Residential turnover. If a share μ of adults is replaced each generation by entrants of whom a share q^M take part, the law of motion becomes

$$q_{t+1} = (1 - \mu)\,\tilde{q}_{t+1} + \mu\, q^M, \tag{E.7}$$

where $\tilde{q}_{t+1}$ is given by (E.4). Entrants who take part less lower q_t and, by (E.3), raise attendance. A one-time replacement of residents in a small part of the neighborhood, as after the 1992

explosions, moves q_t once and is then followed by a return toward q^* when $k\, q^{*2}\, b(Z) < 2$, so its effect on attendance fades. A sustained inflow with $q^M < q^*$ lowers the rest point itself.

E.4 Other Mechanisms

Table E.2 writes each alternative mechanism of Section 5 as a change in one term of the investment condition, states what it predicts for the experiment and the survey, and records what the evidence shows.

Table E.2: Mechanisms, predictions, and evidence

Mechanism	Enters through	Predictions	Evidence
Reciprocal exchange	$\Pi_i > 0$	Bringing the neighborhood to mind lowers demand, more so with reliance and age; demand falls with reliance only where exchange is valuable; same expected return, higher required return	Consistent, except that within the east side the response to the map is largest among households with the fewest ties
Culture of poverty, low aspirations, misinformed beliefs	Lower ρ	No response to the map; lower aspirations or expected returns; same required return	Aspirations, expected returns, and work-study norms not detectably different; required return higher; weighs against it
Penalty for getting ahead (image of limited good, zero-sum thinking, tall poppy syndrome)	Π_i, through beliefs and norms	Shares the core of reciprocal exchange, since a member who leaves or rises takes something from those who stay; the penalty lies in beliefs and norms, which should differ across sides; lower well-being if beliefs are zero-sum	Beliefs, norms, and well-being similar across sides; open
Credit, school quality, discrimination	Higher C_i or lower ρ	No response to the map; differences in resources, provision, or returns	Measured differences in resources, provision, and returns do not account for the gap; liquidity and collateral are not measured
Identity	Identity cost of investing	Bringing the neighborhood to mind lowers demand, independently of reliance; stronger belonging	No detectable difference in attachment; compatibility of schooling with neighborhood identity not measured; open
Signaling, acting white	Π_i, only if observed	No response in a private choice; possible effect on attendance, which neighbors observe	Signaling through the private choice excluded; costs from tutoring or attendance that neighbors later observe open
Redistributive pressure	Value of cash	Bringing the neighborhood to mind raises demand for in-kind tutoring	Opposite sign

Notes: Each row names where a mechanism enters the investment condition (3), what it predicts for the experiment and the survey, and what Sections 3 and 4 find.

↩ Back to text

F External Validity: Supporting Material

F.1 The World Values Survey sample

Table F.1 reports the countries available for the individual-level analysis of Section 5.4. Forty-two countries have individual-level data in Wave 7 of the World Values Survey (2017–2023) with the variables required to construct trust concentrated in neighbors and the educational outcomes non-missing, and a country-level ethnic segregation index in Alesina and Zhuravskaya (2011). Countries marked with an asterisk are classified as *high ethnic segregation* (i.e., their score on that ethnic segregation index lies at or above the cross-country mean). Seven of the forty-two, marked with a dagger, do not enter the estimates of Table 4: the World Values Survey does not record ethnic group in Germany, Japan, Jordan, New Zealand, or Turkey, and does not record parental education in Guatemala or the United Kingdom, so the corresponding fixed effects absorb those countries entirely. The estimation sample is therefore thirty-five countries, twelve of them classified as high ethnic segregation. Table F.2 re-estimates the specification without those two sets of fixed effects, which restores the full forty-two-country sample, and shows that the contrast between segregation regimes is unaffected.

Specification. The specifications include region-by-year, region-by-settlement-size, sex-by-age, country-of-birth, religion, ethnic-group, and parental-education fixed effects, so they control flexibly for mother's and father's education. They exclude respondent occupation, which is realized after schooling. For educational mobility, forty of the forty-two countries have the outcome, and the same seven countries leave the main estimates. Both levels of trust are associated with better outcomes: out-group trust predicts higher attainment, and trust in close ties does too, with a smaller coefficient in each of the three outcomes. The larger out-group coefficient is related to Chetty et al. (2022), who find that upward mobility is associated with connectedness to higher-SES contacts.

F.2 The Social Connectedness Index

The localization of social ties is measured from the Facebook Social Connectedness Index (Bailey et al., 2018) as the share of a place's friendship links that stay within a short distance; the complementary distant-ties share is the fraction reaching far. Both measure how geographically local a place's friendship ties are, although only for ties recorded on Facebook. Table F.4 covers the world's subnational regions (schooling from the Global Data Lab), Table F.5 the NUTS2 regions of Europe (tertiary attainment from Eurostat, the developed-economy counterpart absent from the Global Data Lab sample), and Table F.6 the municipalities of Mexico (2020 Census), with the scope condition estimated against, respectively, ethnic segregation and block-level SES segregation. Table F.7 reports U.S. ZIP-code areas (ACS 5-year), where county residential

Table F.1: Countries included in the World Values Survey Wave 7 sample

Continent	Countries
Africa	Ethiopia*, Kenya*, Morocco*, Zimbabwe*
Americas	Argentina, Bolivia, Brazil, Canada, Chile, Colombia*, Ecuador*, Guatemala*,†, Mexico*, Peru*, United States
Asia	Armenia, Bangladesh, China, India, Indonesia*, Jordan†, Japan†, Kazakhstan, Kyrgyzstan, Pakistan*, Philippines*, South Korea, Taiwan, Tajikistan, Turkey*,†, Uzbekistan, Vietnam
Europe	Czech Republic, Germany†, Greece, Netherlands, Russia*, Slovakia, Ukraine, United Kingdom†
Oceania	Australia, New Zealand†

Notes: The table lists the 42 countries for which the Alesina and Zhuravskaya (2011) country-level ethnic segregation index is available and the World Values Survey Wave 7 individual-level variables required for the analysis of Table 4 are non-missing. Countries marked with an asterisk (*) are classified as high ethnic segregation, i.e., their score on that ethnic segregation index lies at or above the cross-country mean. Countries marked with a dagger (†) are absorbed by the ethnic-group or parental-education fixed effects, which the World Values Survey does not record for them, so they do not contribute to Table 4; the estimation sample there is the remaining 35 countries, 12 of them high segregation. Table F.2 reports the estimates without those fixed effects, on all 42. Countries are assigned to continents following the United Nations M49 standard, with the exception that the Russian Federation is grouped with Europe and Taiwan is grouped with Asia to follow common practice in cross-country economics applications. Sampling weights provided by the WVS are applied throughout.

↩ Back to text

segregation moderates the gradient.

Table F.2: Neighborhood trust premium by segregation regime: sensitivity to the fixed-effect set

	(1)	(2)	(3)	(4)	(5)
Panel A. Years of education					
Neighborhood trust premium × Low ethnic segregation	−0.011**	−0.011**	−0.009*	−0.009*	−0.011**
	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
Neighborhood trust premium × High ethnic segregation	−0.056***	−0.054***	−0.041***	−0.054***	−0.038***
	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)
Observations	64,147	63,194	53,730	55,900	46,952
Countries	42	42	40	37	35
Dep. var. mean	11.67	11.65	11.71	11.61	11.68
R^2	0.3786	0.3841	0.4953	0.4022	0.5095
Panel B. Upward educational mobility					
Neighborhood trust premium × Low ethnic segregation	−0.012*	−0.012*	−0.014**	−0.013*	−0.014**
	(0.006)	(0.006)	(0.006)	(0.007)	(0.006)
Neighborhood trust premium × High ethnic segregation	−0.032***	−0.032***	−0.035***	−0.030***	−0.034***
	(0.008)	(0.008)	(0.008)	(0.008)	(0.008)
Observations	56,592	55,826	53,712	48,906	46,934
Countries	40	40	40	35	35
Dep. var. mean	0.673	0.673	0.677	0.671	0.674
R^2	0.1438	0.1473	0.3037	0.1595	0.3163
Panel C. Dropout before upper secondary					
Neighborhood trust premium × Low ethnic segregation	0.004	0.004	0.004	0.003	0.005
	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
Neighborhood trust premium × High ethnic segregation	0.042***	0.041***	0.030***	0.040***	0.027***
	(0.007)	(0.007)	(0.007)	(0.007)	(0.008)
Observations	64,147	63,194	53,730	55,900	46,952
Countries	42	42	40	37	35
Dep. var. mean	0.305	0.308	0.297	0.313	0.302
R^2	0.3334	0.3381	0.4239	0.3551	0.4372
Region×year, region×settlement size FE	✓	✓	✓	✓	✓
Sex×age FE	✓	✓	✓	✓	✓
Country of birth, religion FE		✓	✓	✓	✓
Parental education FE			✓		✓
Ethnic group FE				✓	✓
Survey weights	✓	✓	✓	✓	✓

Notes: OLS estimates from the World Values Survey / European Values Study integrated time series, wave 7 (2017–2023), as in Table F.3. Each panel takes one standardized outcome, and each column re-estimates the fully interacted specification of Table F.3 under a different set of fixed effects and reports the slopes of the neighborhood trust premium (trust concentrated in neighbors in the text) in low- and high-segregation countries. Column (5) is the specification reported there. The interactions of out-group trust and of trust in close ties with the two segregation indicators are included throughout and omitted from the displayed rows. Occupation is excluded throughout because it is realized after schooling. The ethnic-group and parental-education fixed effects drop entire countries in which the World Values Survey does not record those variables, which is why the country count falls as they are added. All specifications are weighted by survey weights. Countries is the number of countries in the estimation sample, and Dep. var. mean is the survey-weighted mean of the outcome before standardization, in years for years of education and as a share for the two binary outcomes. Heteroskedasticity-robust standard errors are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

↩ Back to text

Table F.3: The neighborhood trust premium and human capital accumulation, World Values Survey Wave 7

	Years of education				Upward educational mobility				Dropout before upper secondary			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Neighborhood trust premium (vs. close ties)	−0.018***	−0.025***	−0.022***		−0.020***	−0.026***	−0.022***		0.014***	0.019***	0.014***	
	(0.004)	(0.004)	(0.004)		(0.005)	(0.005)	(0.005)		(0.004)	(0.004)	(0.004)	
	[0.005]	[0.005]	[0.006]		[0.004]	[0.004]	[0.005]		[0.004]	[0.004]	[0.005]	
Out-group trust		0.051***	0.047***			0.044***	0.039***			−0.041***	−0.034***	
		(0.005)	(0.005)			(0.005)	(0.006)			(0.005)	(0.005)	
		[0.007]	[0.008]			[0.008]	[0.009]			[0.007]	[0.007]	
Trust in close ties (family + known)			0.011**				0.014***				−0.018***	
			(0.005)				(0.006)				(0.005)	
			[0.007]				[0.006]				[0.006]	
Neighborhood trust premium × Low ethnic seg.				−0.011**				−0.014**				0.005
				(0.005)				(0.006)				(0.005)
				[0.006]				[0.006]				[0.006]
Neighborhood trust premium × High ethnic seg.				−0.038***				−0.034***				0.027***
				(0.007)				(0.008)				(0.008)
				[0.014]				[0.011]				[0.009]
Region×year, region×settlement size FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Sex×age FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Country of birth, religion FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Parental education FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Ethnic group FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Survey weights	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observations	47,072	46,952	46,952	46,952	47,054	46,934	46,934	46,934	47,072	46,952	46,952	46,952
Dep. var. mean	11.68	11.68	11.68	11.68	0.674	0.674	0.674	0.674	0.302	0.302	0.302	0.302
R^2	0.5072	0.5092	0.5093	0.5095	0.3145	0.3160	0.3162	0.3163	0.4355	0.4369	0.4371	0.4372
Within R^2	0.0006	0.0042	0.0044	0.0049	0.0005	0.0024	0.0026	0.0028	0.0003	0.0023	0.0027	0.0029

Notes: OLS estimates from the World Values Survey / European Values Study integrated time series, wave 7 (2017–2023). The unit of observation is the respondent. All outcomes are standardized. The neighborhood trust premium (trust concentrated in neighbors in the text) is standardized trust in neighbors minus average trust in family and in people known personally, so it measures how much trust is concentrated on neighbors relative to close ties rather than the level of trust. A country is classified as high ethnic segregation if its Alesina and Zhuravskaya (2011) ethnic-segregation index is at or above the cross-country mean, and as low ethnic segregation otherwise. Each outcome block reports a stepwise sequence, with the neighborhood trust premium alone in its first column, out-group trust added in the second, and trust in close ties added in the third. The fourth column of each block interacts all three trust terms with the low- and high-segregation indicators, so it reports separate slopes by segregation regime with no omitted category. The displayed rows of that column give the low- and high-segregation slopes of the neighborhood trust premium, and the corresponding interactions for out-group trust and for trust in close ties are included in the regression and omitted from the displayed rows. All specifications include region-by-year, region-by-settlement-size, sex-by-age, country-of-birth, ethnicity, religion, and father-by-mother-education fixed effects. Observations are weighted by survey weights. Dep. var. mean is the survey-weighted mean of the outcome in the estimation sample before standardization, in years for years of education and as a share for the two binary outcomes. Heteroskedasticity-robust standard errors, without clustering, are in parentheses. Standard errors from a cluster bootstrap by country are in brackets, with 999 draws that each resample the 35 countries with replacement within each segregation regime, so that every draw keeps 12 high-segregation countries. Stars use the heteroskedasticity-robust standard errors. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

↩ Back to text

Table F.4: Localization of social ties and schooling across subnational regions

	Mean yrs. 20+				Mean yrs. 20–39				Expected yrs.			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Localization of social ties (share ≤50 km)	−2.650***	−2.806***	−1.706***	−3.130***	−2.442***	−2.583***	−1.473***	−2.752***	−1.776***	−1.880***	−1.173***	−2.292***
Robust	(0.236)	(0.350)	(0.354)	(0.487)	(0.235)	(0.348)	(0.369)	(0.511)	(0.258)	(0.405)	(0.440)	(0.543)
Clustered by country	[0.339]	[0.452]	[0.438]	[0.540]	[0.347]	[0.437]	[0.429]	[0.566]	[0.368]	[0.460]	[0.464]	[0.560]
Distant social ties (share ≥500 km)		−0.327	0.792	−0.298		−0.295	0.789	−0.132		−0.217	0.338	−0.471
Robust		(0.570)	(0.511)	(0.643)		(0.562)	(0.532)	(0.687)		(0.592)	(0.576)	(0.664)
Clustered by country		[0.903]	[0.771]	[0.804]		[0.796]	[0.717]	[0.724]		[0.802]	[0.765]	[0.715]
Localization of social ties × Ethnic segregation				−0.400*				−0.398				0.082
Robust				(0.240)				(0.276)				(0.284)
Clustered by country				[0.264]				[0.310]				[0.322]
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Log population			✓	✓			✓	✓			✓	✓
Log area			✓	✓			✓	✓			✓	✓
Observations	1,145	1,145	901	474	1,145	1,145	901	474	1,080	1,080	836	441
Dep. var. mean	7.07	7.07	7.20	7.63	8.34	8.34	8.46	8.93	11.86	11.86	11.96	12.12
R^2	0.840	0.840	0.895	0.891	0.843	0.843	0.897	0.893	0.734	0.734	0.808	0.830
Within R^2	0.134	0.135	0.311	0.386	0.111	0.111	0.283	0.351	0.055	0.055	0.214	0.282

Notes: Region-level regressions across the subnational regions of the Global Data Lab (GDL), with country fixed effects. The outcomes are mean years of schooling at ages 20 and older and at ages 20–39, and expected years of schooling, from the Global Data Lab. Localization of social ties is the share of a region's Facebook friendship links to places within 50 km, and distant ties the share to places 500 km or farther (haversine distance between GADM1 centroids). Within each outcome, the first column includes localization only, the second adds distant ties, the third adds log population and log area, and the fourth adds the interaction with the ethnic-segregation index of Alesina and Zhuravskaya (2011), standardized. The segregation index is unavailable for many countries, so column (4) is estimated on 474 regions against 901 in column (3), the two columns are not comparable, and the same holds for columns (8) and (12). Re-estimating column (3) on the 474-region sample gives −3.230 (0.488) for mean years at 20+, against −1.706 (0.354) on the full sample and −3.130 (0.487) in column (4), so the change between columns (3) and (4) comes from the change of sample. In columns (4), (8), and (12) the localization coefficient is the slope at zero standardized segregation. Dep. var. mean is the mean of the outcome in the estimation sample of each column. HC1 robust standard errors are in parentheses and, below them in brackets, standard errors clustered on country. The segregation moderator varies only across countries, so the clustered errors are the relevant ones for the interaction, and under them the interaction is not distinguishable from zero for any of the three outcomes. The moderation result of Section 5.4 rests on Table F.7. Stars use the robust standard errors. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

↩ Back to text

Table F.5: Localization of social ties and tertiary attainment across European regions (NUTS2)

	% tertiary (25–64)		
	(1)	(2)	(3)
Localization of social ties (within-NUTS2 share)	−26.33***	−30.85***	−50.92***
	(2.39)	(4.26)	(6.31)
Distant social ties (international share)		−14.61	−12.28
		(9.79)	(8.61)
Country FE	✓	✓	✓
Log population			✓
Observations	281	281	281
Dep. var. mean	33.00	33.00	33.00
R^2	0.753	0.755	0.792
Within R^2	0.389	0.394	0.485

Notes: NUTS2-region regressions with country fixed effects. The outcome is the share of the population aged 25–64 with tertiary education (ISCED 5–8), in percent, from Eurostat for 2023. Localization of social ties is the within-NUTS2 share of Facebook connectedness, and distant ties the share to other countries. Because this measure is defined by the region's own boundary, part of it reflects how large and how compact the region is, since a bigger or rounder NUTS2 unit retains more of its links mechanically. Country fixed effects absorb differences in how each country draws its regions but not variation in extent within a country, and the table includes no area control. The fixed-radius measures of Table F.4 and Table F.7, which count links within 50 and 25 kilometres rather than within the unit, do not have that property and give the same negative relationship with log population and log area held fixed. The table is the developed-economy counterpart to the global region-level analysis, which lacks schooling data for high-income countries. Dep. var. mean is the mean of the outcome in the estimation sample of each column. HC1 robust standard errors are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

↩ Back to text

Table F.6: Localization of social ties and schooling across Mexican municipalities

	Mean yrs. schooling				Attendance 15–17 (%)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Localization of social ties (within-muni share)	−3.542***	−3.373***	−1.487***	−1.421***	−10.033***	−12.815***	−12.548***	−9.486***
	(0.151)	(0.188)	(0.198)	(0.204)	(1.301)	(1.632)	(2.220)	(2.212)
Distant social ties (international share)		1.807*	−2.704***	−3.341***		−29.789***	−20.656**	−25.281***
		(0.986)	(0.862)	(0.840)		(8.001)	(8.732)	(8.748)
SES segregation (block)				0.376***				2.610***
				(0.060)				(0.553)
Localization of social ties × SES segregation (block)				−0.621***				0.002
				(0.122)				(1.215)
State FE	✓	✓	✓	✓	✓	✓	✓	✓
Log population			✓	✓			✓	✓
Log area			✓	✓			✓	✓
Observations	2,207	2,207	2,207	2,191	2,207	2,207	2,207	2,191
Dep. var. mean	7.91	7.91	7.91	7.92	67.57	67.57	67.57	67.61
R^2	0.472	0.473	0.569	0.582	0.180	0.184	0.194	0.227
Within R^2	0.250	0.252	0.388	0.406	0.024	0.029	0.041	0.079

Notes: Municipality-level regressions for Mexico with state fixed effects. The outcomes are mean years of schooling (GRAPROES) and the school attendance rate at ages 15–17, in percent, from the 2020 Census, so the attendance coefficients are in percentage points. Localization of social ties is the share of a municipality's Facebook connectedness that stays within the municipality, so like the NUTS2 measure of Table F.5 it depends on the size and shape of the unit, and distant ties are the share to other countries. Columns (4) and (8) add the scope condition, the interaction with municipal SES segregation, measured as the population-weighted standard deviation of mean years of schooling across the municipality's urban city blocks (*manzanas*) in the 2020 Census and standardized. Dep. var. mean is the mean of the outcome in the estimation sample of each column. HC1 robust standard errors are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

↩ Back to text

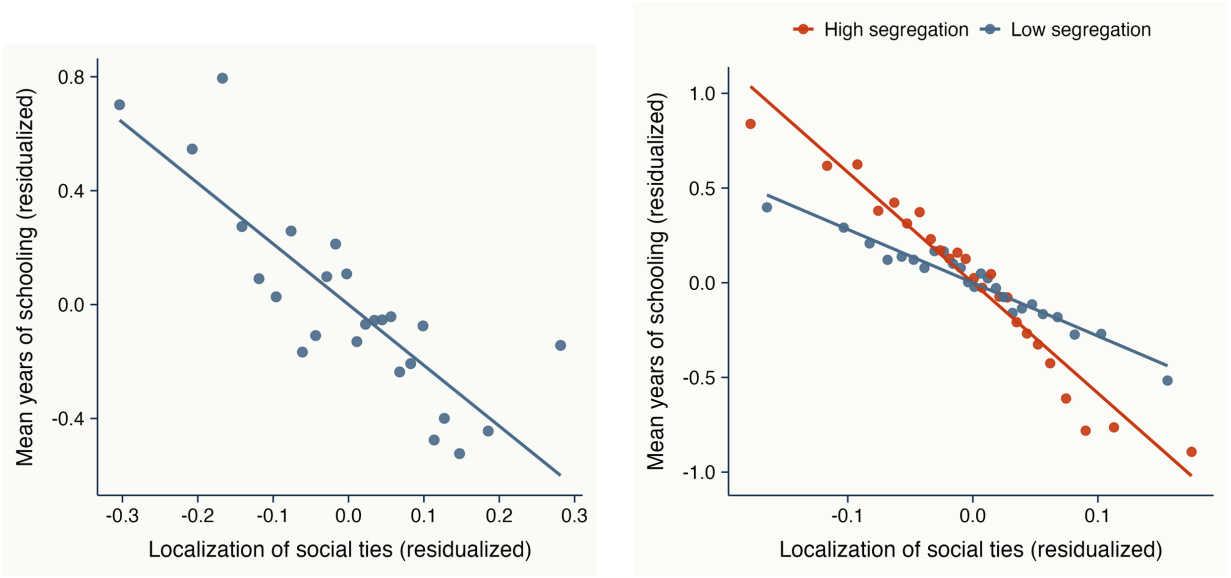
Table F.7: Localization of social ties and schooling across U.S. ZIP-code areas (ZCTA)

	Mean yrs. schooling				Share BA+			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Localization of social ties (share ≤25 km)	−3.539***	−1.751***	−2.500***	−2.376***	−0.536***	−0.254***	−0.344***	−0.318***
Robust	(0.082)	(0.110)	(0.125)	(0.123)	(0.011)	(0.016)	(0.018)	(0.018)
Clustered by county	[0.186]	[0.220]	[0.236]	[0.215]	[0.027]	[0.033]	[0.037]	[0.032]
Distant social ties (share ≥500 km)		4.055***	4.306***	4.447***		0.640***	0.626***	0.653***
Robust		(0.212)	(0.243)	(0.243)		(0.032)	(0.037)	(0.037)
Clustered by county		[0.425]	[0.458]	[0.449]		[0.063]	[0.067]	[0.065]
Localization of social ties × Racial segregation				−0.789***				−0.165***
Robust				(0.076)				(0.011)
Clustered by county				[0.149]				[0.025]
County FE	✓	✓	✓	✓	✓	✓	✓	✓
Log population			✓	✓			✓	✓
Log area			✓	✓			✓	✓
Observations	19,561	19,561	19,561	19,433	19,561	19,561	19,561	19,433
Dep. var. mean	13.47	13.47	13.47	13.48	0.298	0.298	0.298	0.299
R^2	0.499	0.520	0.533	0.534	0.584	0.607	0.612	0.617
Within R^2	0.115	0.152	0.175	0.182	0.133	0.181	0.192	0.206

Notes: ZCTA-level regressions with *county* fixed effects. The outcomes are mean years of schooling, from ACS educational attainment, and the share with a bachelor's degree or more. Localization of social ties is the share of a ZCTA's Facebook friendship links within 25 km, and distant ties the share 500 km or farther. Columns (4) and (8) add the scope condition, the interaction with county-level white/non-white residential dissimilarity across ZCTAs, whose main effect is absorbed by the county fixed effects. Dep. var. mean is the mean of the outcome in the estimation sample of each column. HC1 robust standard errors are in parentheses and, below them in brackets, standard errors clustered on county. The segregation moderator varies only across counties, so the clustered errors are the relevant ones for the interaction; they run about twice the robust errors and leave every coefficient in the table significant at the 1 percent level. Stars use the robust standard errors. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

↩ Back to text

Figure F.1: Localization of social ties and schooling



(a) Subnational regions, within country

(b) U.S. ZIP codes, by county segregation

Notes: Binned scatters of mean years of schooling against the localization of a place's Facebook friendship network (Bailey et al., 2018), with both axes residualized on fixed effects and controls. Panel (a) compares subnational regions within country, controlling for log population and log area; schooling is from the Global Data Lab. Panel (b) compares U.S. ZIP codes within county, with the same controls and schooling from the ACS. ZIP codes are shown separately for counties in the bottom and top terciles of the white/non-white residential-dissimilarity index. The schooling gradient is negative in both groups and roughly twice as steep in high-segregation counties. Full regressions are in Tables F.4 and F.7.

↩ Back to text

Appendix References

Aldeco, Lucía, Mariana Calderón, Daniel Chiquiar, Gordon Hanson, Jorge Pérez Pérez, and Carlos Velázquez, "Local Labor Markets in Mexico: Definition, Databases, and Analysis," Working Paper, Banco de México, Mexico City 2024. (cited on pp. 8 and 21).

Alesina, Alberto and Ekaterina Zhuravskaya, "Segregation and the Quality of Government in a Cross Section of Countries," *American Economic Review*, 2011, *101* (5), 1872–1911. (cited on p. 45).

Ananat, Elizabeth Oltmans, "The Wrong Side(s) of the Tracks: The Causal Effects of Racial Segregation on Urban Poverty and Inequality," *American Economic Journal: Applied Economics*, 2011, *3* (2), 34–66. (cited on p. 11).

Anderson, Michael L., "Multiple Inference and Gender Differences in the Effects of Early Intervention: A Reevaluation of the Abecedarian, Perry Preschool, and Early Training Projects," *Journal of the American Statistical Association*, 2008, *103* (484), 1481–1495. (cited on pp. 18 and 25).

Anderson, Rodney D. and Tamara Spike, "Making History Count: The Guadalajara Census Project (1791–1930)," *Hispanic American Historical Review*, 2007, *87* (2), 327–351. (cited on p. 12).

Bagagli, Sara, "The (Express)Way to Segregation: Evidence from Chicag," Technical Report 2025. (cited on p. 11).

Bailey, Michael, Rachel Cao, Theresa Kuchler, Johannes Stroebel, and Arlene Wong, "Social Connectedness: Measurement, Determinants, and Applications in Economic Research," *Journal of Economic Perspectives*, 2018, *32* (3), 259–280. (cited on p. 170).

Baldomero-Quintana, Luis, ⓡ Woo-Mora, L. Guillermo, and ⓡ De la Rosa-Ramos, Enrique, "Infrastructures of Race? Colonial Indigenous Segregation and Contemporary Land Values," *Regional Science and Urban Economics*, 2025, *110*, 104065. (cited on pp. 6 and 21).

Borusyak, Kirill and Peter Hull, "Nonrandom Exposure to Exogenous Shocks," *Econometrica*, 2023, *91* (6), 2155–2185. (cited on p. 24).

Buringh, E., "Urbanisation Hub: The Clio-Infra Database on Urban Settlement Sizes: 1500–2000," 2018. (cited on p. 10).

Bursztyn, Leonardo and David Y. Yang, "Misperceptions About Others," *Annual Review of Economics*, 2022, *14*, 425–452.

Calonico, Sebastian, Matias D. Cattaneo, and Rocío Titiunik, "Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs," *Econometrica*, 2014, *82* (6), 2295–2326. (cited on pp. 13 and 17).

Cappelen, Alexander W., Benjamin Enke, and Bertil Tungodden, "Universalism: Global Evidence," *American Economic Review*, 2025, *115*, 43–76. (cited on p. 28).

Cattaneo, Matias D., Brigham R. Frandsen, and Rocío Titiunik, "Randomization Inference in the Regression Discontinuity Design: An Application to Party Advantages in the U.S. Senate," *Journal of Causal Inference*, 2015, *3* (1), 1–24.

_ , Michael Jansson, and Xinwei Ma, "Manipulation Testing Based on Density Discontinuity," *The Stata Journal*, 2018, *18* (1), 234–261. (cited on p. 14).

_ , Rocío Titiunik, and Ruiqi Yu, "Estimation and Inference in Boundary Discontinuity Designs: Location-Based Methods," *arXiv Working Paper 2505.05670*, 2025. (cited on pp. 14, 18, and 19).

Chen, Jiafeng and Jonathan Roth, "Logs with Zeros? Some Problems and Solutions," *Quarterly Journal of Economics*, 2024, *139* (2), 891–936.

Chetty, Raj, Matthew O. Jackson, Theresa Kuchler, Johannes Stroebel, Nathaniel Hendren, Robert B. Fluegge, Sara Gong, Federico Gonzalez, Armelle Grondin, Matthew Jacob, Drew Johnston, Martin Koenen, Eduardo Laguna-Muggenburg, Florian Mudekereza, Tom Rutter, Nicolaj Thor, Wilbur Townsend, Ruby Zhang, Michael Bailey, Pablo Barberá, Monica Bhole, and Nils Wernerfelt, "Social Capital I: Measurement and Associations with Economic Mobility," *Nature*, 2022, *608* (7921), 108–121.

Chyn, Eric, Kareem Haggag, and Bryan A. Stuart, "The Effects of Racial Segregation on Intergenerational Mobility: Evidence from Historical Railroad Placement," *American Economic Journal: Applied Economics*, 2026, *18*, 34–60. (cited on p. 11).

Cohn, Alain and Michel André Maréchal, "Priming in Economics," *Current Opinion in Psychology*, 2016, *12*, 17–21.

Cunha, Flavio and James J. Heckman, "The Technology of Skill Formation," *American Economic Review*, 2007, *97* (2), 31–47. (cited on p. 20).

de Arregui, Domingo Lázaro, *Descripción de la Nueva Galicia*, Seville: Escuela de Estudios Hispano-Americanos, 1946. Edited by François Chevalier. (cited on pp. 9 and 10).

Enke, Benjamin, Ricardo Rodríguez-Padilla, and Florian Zimmermann, "Moral Universalism and the Structure of Ideology," *The Review of Economic Studies*, 2023, *90*, 1934–1962. (cited on p. 28).

Fryer, Roland G. and Steven D. Levitt, "The Causes and Consequences of Distinctively Black Names," *The Quarterly Journal of Economics*, 2004, *119* (3), 767–805.

Galor, Oded, "Inequality, Human Capital Formation, and the Process of Development," in Eric A. Hanushek, Stephen Machin, and Ludger Woessmann, eds., *Handbook of the Economics of Education*, Vol. 4, Elsevier, 2011, chapter 5, pp. 441–493. (cited on p. 20).

_ **and Daniel Tsiddon**, "The Distribution of Human Capital and Economic Growth," *Journal of Economic Growth*, 1997, *2* (1), 93–124.

_ **and Joseph Zeira**, "Income Distribution and Macroeconomics," *The Review of Economic Studies*, 1993, *60* (1), 35–52. (cited on p. 20).

Goldewijk, Kees Klein, Arthur Beusen, Jonathan Doelman, and Elke Stehfest, "Anthropogenic Land Use Estimates for the Holocene–HYDE 3.2," *Earth System Science Data*, 2017, *9* (2), 927–953. (cited on p. 10).

Haber, Stephen H., "The Industrialization of Mexico, 1890–1940: The Structure and Growth of Manufacturing in an Underdeveloped Economy," *The Journal of Economic History*, 1987, *47* (2), 493–495.

IMEPLAN, "Plan de Ordenamiento Territorial Metropolitano del Area Metropolitana de Guadalajara," Technical Report, Instituto Metropolitano de Planeación, Guadalajara 2016. (cited on p. 9).

Jack, B. Kelsey, Kathryn McDermott, and Anja Sautmann, "Multiple Price Lists for Willingness to Pay Elicitation," *Journal of Development Economics*, 2022, *159*, 102977. (cited on pp. 30 and 31).

McCrary, Justin, "Manipulation of the Running Variable in the Regression Discontinuity Design: A Density Test," *Journal of Econometrics*, 2008, *142* (2), 698–714. (cited on p. 14).

Rubin, Donald B., "The Bayesian Bootstrap," *The Annals of Statistics*, 1981, *9* (1), 130–134.

Sampson, Robert J., Jeffrey D. Morenoff, and Thomas Gannon-Rowley, "Assessing 'Neighborhood Effects': Social Processes and New Directions in Research," *Annual Review of Sociology*, 2002, *28*, 443–478. (cited on p. 7).

Smith, Michael E., "The Archaeological Study of Neighborhoods and Districts in Ancient Cities," *Journal of Anthropological Archaeology*, 2010, *29* (2), 137–154.

Valenzuela-Casasempere, Pablo, "Displacement and Infrastructure Provision: Evidence from the Interstate Highway System," Technical Report 2025. (cited on p. 11).